

My Project

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Chapter 1

Main Page

1.0.1 Libir SDK Introduction

version: V2.2.34

Here are the SDKs of infisense, including:

[error.h](#) : Error type in libraries.

[libircam.h](#) : The interface layer, using an inverted dependency, is co-dependent on the application and driver layers.

[libircmd.h](#) : send control commands about the image.

[libircmd_temp.h](#) : send control commands about the temperature.

[libirdfu.h](#) : update firmware.

[libiri2c.h](#) : send read and write commands via I2C channel.

[libirparse.h](#) : saperate the image and temperature,convert the rgb and yuv.

[libirtemp.h](#) : calculate the temperature,get the point,line,rect's temp.

[libiruart.h](#) : send read and write commands via serial port.

[libirupgrade.h](#) : support upgrade package update, single file data read/write, firmware download three functions.

[libiruvic.h](#) : uvc framework, encapsulating different ways of communication protocols.

[libirv4l2.h](#) : v4l2 framework, encapsulating different ways of communication protocols.

[libir_infoparse.h](#) : parse image information line and temperature line

Chapter 2

Class Index

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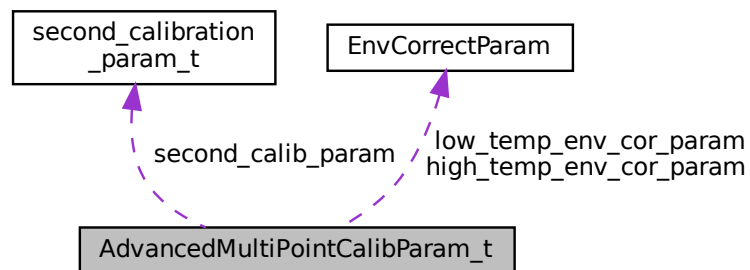
Chapter 4

Class Documentation

4.1 AdvancedMultiPointCalibParam_t Struct Reference

```
#include <libirtemp.h>
```

Collaboration diagram for AdvancedMultiPointCalibParam_t:



Public Attributes

- [EnvCorrectParam high_temp_env_cor_param](#)
- [EnvCorrectParam low_temp_env_cor_param](#)
- [second_calibration_param_t second_calib_param](#)
- [uint16_t cur_vtemp](#)

4.1.1 Member Data Documentation

4.1.1.1 cur_vtemp

uint16_t AdvancedMultiPointCalibParam_t::cur_vtemp

4.1.1.2 high_temp_env_cor_param

EnvCorrectParam AdvancedMultiPointCalibParam_t::high_temp_env_cor_param

4.1.1.3 low_temp_env_cor_param

EnvCorrectParam AdvancedMultiPointCalibParam_t::low_temp_env_cor_param

4.1.1.4 second_calib_param

second_calibration_param_t AdvancedMultiPointCalibParam_t::second_calib_param

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.2 Area_t Struct Reference

[Area_t](#) definition.

```
#include <libirtemp.h>
```

Public Attributes

- int [start_x](#)
The start horizontal coordinate of [Area_t](#).
- int [start_y](#)
The start vertical coordinate of [Area_t](#).
- int [width](#)
The width of [Area_t](#).
- int [height](#)
The height of [Area_t](#).

4.2.1 Detailed Description

[Area_t](#) definition.

4.2.2 Member Data Documentation

4.2.2.1 height

```
int Area_t::height
```

The height of [Area_t](#).

4.2.2.2 start_x

```
int Area_t::start_x
```

The start horizontal coordinate of [Area_t](#).

4.2.2.3 start_y

```
int Area_t::start_y
```

The start vertical coordinate of [Area_t](#).

4.2.2.4 width

```
int Area_t::width
```

The width of [Area_t](#).

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.3 CaliFactor_t Struct Reference

The normalization coefficient of KT curve.

```
#include <libirtemp.h>
```

Public Attributes

- int [Kn](#)
The K value of KT curve normalization coefficient.
- int [Bn](#)
The B value of KT curve normalization coefficient.

4.3.1 Detailed Description

The normalization coefficient of KT curve.

4.3.2 Member Data Documentation

4.3.2.1 Bn

```
int CaliFactor_t::Bn
```

The B value of KT curve normalization coefficient.

4.3.2.2 Kn

```
int CaliFactor_t::Kn
```

The K value of KT curve normalization coefficient.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.4 CaliPoint_t Struct Reference

The environment temperature point of primary calibration.

```
#include <libirtemp.h>
```

Public Attributes

- int [TH](#)
High environment temperature point of primary calibration.
- int [TL](#)
Low environment temperature point of primary calibration.

4.4.1 Detailed Description

The environment temperature point of primary calibration.

4.4.2 Member Data Documentation

4.4.2.1 TH

```
int CaliPoint_t::TH
```

High environment temperature point of primary calibration.

4.4.2.2 TL

```
int CaliPoint_t::TL
```

Low environment temperature point of primary calibration.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.5 CamDevParams_t Struct Reference

```
#include <libirv4l2.h>
```

Public Attributes

- int [width](#)
- int [height](#)
- int [format](#)
- int [fps](#)
- int [reserved](#) [48]

4.5.1 Member Data Documentation

4.5.1.1 format

```
int CamDevParams_t::format
```

4.5.1.2 fps

```
int CamDevParams_t::fps
```

4.5.1.3 height

```
int CamDevParams_t::height
```

4.5.1.4 reserved

```
int CamDevParams_t::reserved[48]
```

4.5.1.5 width

```
int CamDevParams_t::width
```

The documentation for this struct was generated from the following file:

- [libirv4l2.h](#)

4.6 CameraParam_t Struct Reference

Camera's one stream parameter.

```
#include <libiruvc.h>
```

Public Attributes

- char * [format](#)
format
- unsigned int [width](#)
resolution width
- unsigned int [height](#)
resolution height
- unsigned int [frame_size](#)
frame byte size
- unsigned int [fps](#)
frame per second
- unsigned int [timeout_ms_delay](#)
frame get's timeout timer(ms)

4.6.1 Detailed Description

Camera's one stream parameter.

4.6.2 Member Data Documentation

4.6.2.1 format

```
char* CameraParam_t::format
```

format

4.6.2.2 fps

```
unsigned int CameraParam_t::fps
```

frame per second

4.6.2.3 frame_size

```
unsigned int CameraParam_t::frame_size
```

frame byte size

4.6.2.4 height

```
unsigned int CameraParam_t::height
```

resolution height

4.6.2.5 timeout_ms_delay

```
unsigned int CameraParam_t::timeout_ms_delay
```

frame get's timeout timer(ms)

4.6.2.6 width

```
unsigned int CameraParam_t::width
```

resolution width

The documentation for this struct was generated from the following file:

- [libiruvc.h](#)

4.7 CameraStreamInfo_t Struct Reference

Camera's supported stream information.

```
#include <libiruvc.h>
```

Public Attributes

- char * [format](#)
format
- unsigned int [width](#)
resolution width
- unsigned int [height](#)
resolution height
- unsigned int [frame_size](#)
frame byte size
- unsigned int [fps](#) [[FPS_CNT_MAX](#)]
frame per second

4.7.1 Detailed Description

Camera's supported stream information.

4.7.2 Member Data Documentation

4.7.2.1 format

```
char* CameraStreamInfo_t::format
```

format

4.7.2.2 fps

```
unsigned int CameraStreamInfo_t::fps[FPS_CNT_MAX]
```

frame per second

4.7.2.3 frame_size

```
unsigned int CameraStreamInfo_t::frame_size
```

frame byte size

4.7.2.4 height

```
unsigned int CameraStreamInfo_t::height
```

resolution height

4.7.2.5 width

```
unsigned int CameraStreamInfo_t::width
```

resolution width

The documentation for this struct was generated from the following file:

- [libiruv.h](#)

4.8 CamStreamParams_t Struct Reference

```
#include <libirv4l2.h>
```

Public Attributes

- int [width](#)
- int [height](#)
- int [reserved](#) [56]

4.8.1 Member Data Documentation

4.8.1.1 height

```
int CamStreamParams_t::height
```

4.8.1.2 reserved

```
int CamStreamParams_t::reserved[56]
```

4.8.1.3 width

```
int CamStreamParams_t::width
```

The documentation for this struct was generated from the following file:

- [libirv4l2.h](#)

4.9 CorrectTable_t Struct Reference

dist correction table infomation definition

```
#include <libirtemp.h>
```

Public Attributes

- uint16_t [gain](#)
gain in the dist correction table
- uint16_t [version](#)
version of the dist correction table
- uint16_t [ems_cor_enable](#)
enable switch of ems correction
- uint16_t [ems_cor_version](#)
version of ems correction algorithm
- uint16_t [hum_cor_enable](#)
enable switch of hum correction
- uint16_t [hum_cor_version](#)
version of hum correction algorithm
- uint16_t [crosstalk_cor_enable](#)
enable switch of crosstalk correction
- uint16_t [crosstalk_cor_version](#)
version of crosstalk correction algorithm
- uint16_t [reflected_temp_cor_enable](#)
enable switch of reflected temperature correction
- uint16_t [reflected_temp_cor_version](#)
version of reflected temperature correction algorithm
- uint8_t [reserve](#) [236]

4.9.1 Detailed Description

dist correction table information definition

4.9.2 Member Data Documentation

4.9.2.1 crosstalk_cor_enable

```
uint16_t CorrectTable_t::crosstalk_cor_enable
```

enable switch of crosstalk correction

4.9.2.2 crosstalk_cor_version

```
uint16_t CorrectTable_t::crosstalk_cor_version
```

version of crosstalk correction algorithm

4.9.2.3 `ems_cor_enable`

```
uint16_t CorrectTable_t::ems_cor_enable
```

enable switch of ems correction

4.9.2.4 `ems_cor_version`

```
uint16_t CorrectTable_t::ems_cor_version
```

version of ems correction algorithm

4.9.2.5 `gain`

```
uint16_t CorrectTable_t::gain
```

gain in the dist correction table

4.9.2.6 `hum_cor_enable`

```
uint16_t CorrectTable_t::hum_cor_enable
```

enable switch of hum correction

4.9.2.7 `hum_cor_version`

```
uint16_t CorrectTable_t::hum_cor_version
```

version of hum correction algorithm

4.9.2.8 `reflected_temp_cor_enable`

```
uint16_t CorrectTable_t::reflected_temp_cor_enable
```

enable switch of reflected temperature correction

4.9.2.9 reflected_temp_cor_version

```
uint16_t CorrectTable_t::reflected_temp_cor_version
```

version of reflected temperature correction algorithm

4.9.2.10 reserve

```
uint8_t CorrectTable_t::reserve[236]
```

4.9.2.11 version

```
uint16_t CorrectTable_t::version
```

version of the dist correction table

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.10 DataInfo_t Struct Reference

```
#include <libirparse.h>
```

Public Attributes

- int [data_byte_size](#)
- uint8_t * [data](#)

4.10.1 Member Data Documentation

4.10.1.1 data

```
uint8_t* DataInfo_t::data
```

4.10.1.2 data_byte_size

```
int DataInfo_t::data_byte_size
```

The documentation for this struct was generated from the following file:

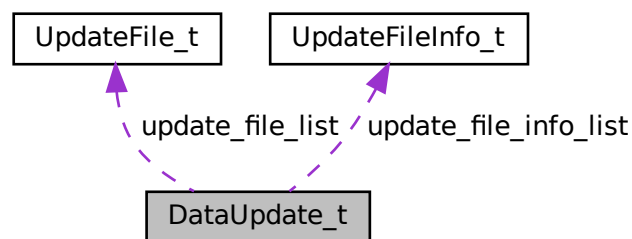
- [libirparse.h](#)

4.11 DataUpdate_t Struct Reference

Data difference information from the original version to the target version in the update package zip.

```
#include <libirupgrade.h>
```

Collaboration diagram for DataUpdate_t:



Public Attributes

- bool [update_flag](#)
- char * [data_version](#)
- int [file_number](#)
- int [file_info_number](#)
- [UpdateFileInfo_t](#) [update_file_info_list](#) [[UPDATE_FILE_INFO_LIST_MAX_NUM](#)]
- [UpdateFile_t](#) [update_file_list](#) [[UPDATE_FILE_LIST_MAX_NUM](#)]

4.11.1 Detailed Description

Data difference information from the original version to the target version in the update package zip.

4.11.2 Member Data Documentation

4.11.2.1 data_version

```
char* DataUpdate_t::data_version
```

4.11.2.2 file_info_number

```
int DataUpdate_t::file_info_number
```

4.11.2.3 file_number

```
int DataUpdate_t::file_number
```

4.11.2.4 update_file_info_list

```
UpdateFileInfo_t DataUpdate_t::update_file_info_list[UPDATE_FILE_INFO_LIST_MAX_NUM]
```

4.11.2.5 update_file_list

```
UpdateFile_t DataUpdate_t::update_file_list[UPDATE_FILE_LIST_MAX_NUM]
```

4.11.2.6 update_flag

```
bool DataUpdate_t::update_flag
```

The documentation for this struct was generated from the following file:

- [libirupgrade.h](#)

4.12 DevCfg_t Struct Reference

Device's configuration.

```
#include <libiruvc.h>
```

Public Attributes

- unsigned int [pid](#)
pid
- unsigned int [vid](#)
vid
- char * [name](#)
device name

4.12.1 Detailed Description

Device's configuration.

4.12.2 Member Data Documentation

4.12.2.1 name

```
char* DevCfg_t::name
```

device name

4.12.2.2 pid

```
unsigned int DevCfg_t::pid
```

pid

4.12.2.3 vid

```
unsigned int DevCfg_t::vid
```

vid

The documentation for this struct was generated from the following file:

- [libiruv.h](#)

4.13 Dot_t Struct Reference

[Dot_t](#) definition.

```
#include <libirtemp.h>
```

Public Attributes

- [int x](#)
The horizontal coordinate of [Dot_t](#).
- [int y](#)
The vertical coordinate of [Dot_t](#).

4.13.1 Detailed Description

[Dot_t](#) definition.

4.13.2 Member Data Documentation

4.13.2.1 x

```
int Dot_t::x
```

The horizontal coordinate of [Dot_t](#).

4.13.2.2 y

```
int Dot_t::y
```

The vertical coordinate of [Dot_t](#).

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.14 EnvCorrectParam Struct Reference

Temperature infomation definition.

```
#include <libirtemp.h>
```

Public Attributes

- float `dist`
The target distance.
- float `ems`
The target emissivity(range: 0-1)
- float `hum`
The environment humidity(range: 0-1)
- float `ta`
The atmospheric temperature(celcius temperature, unit:celcius, range: -273.15 < ta < 100)
- float `tu`
The reflection temperature(celcius temperature, unit:celcius, range: -273.15 < tu < 100)

4.14.1 Detailed Description

Temperature infomation definition.

4.14.2 Member Data Documentation

4.14.2.1 `dist`

```
float EnvCorrectParam::dist
```

The target distance.

4.14.2.2 `ems`

```
float EnvCorrectParam::ems
```

The target emissivity(range: 0-1)

4.14.2.3 `hum`

```
float EnvCorrectParam::hum
```

The environment humidity(range: 0-1)

4.14.2.4 ta

```
float EnvCorrectParam::ta
```

The atmospheric temperature(celcius temperature, unit:celcius, range: $-273.15 < ta < 100$)

4.14.2.5 tu

```
float EnvCorrectParam::tu
```

The reflection temperature(celcius temperature, unit:celcius, range: $-273.15 < tu < 100$)

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.15 EnvFactor_t Struct Reference

Environment factor correction coefficient.

```
#include <libirtemp.h>
```

Public Attributes

- int [K_E](#)
The K value of environment factor correction.
- int [B_E](#)
The B value of environment factor correction.

4.15.1 Detailed Description

Environment factor correction coefficient.

4.15.2 Member Data Documentation

4.15.2.1 B_E

```
int EnvFactor_t::B_E
```

The B value of environment factor correction.

4.15.2.2 K_E

```
int EnvFactor_t::K_E
```

The K value of environment factor correction.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.16 EnvParam_t Struct Reference

Environment factor correction parameter.

```
#include <libirtemp.h>
```

Public Attributes

- int [EMS](#)
Target emissivity.
- int [TAU](#)
Atmospheric transmittance.
- int [Ta](#)
Atmospheric temperature.
- int [Tu](#)
Reflection temperature.

4.16.1 Detailed Description

Environment factor correction parameter.

4.16.2 Member Data Documentation

4.16.2.1 EMS

```
int EnvParam_t::EMS
```

Target emissivity.

4.16.2.2 Ta

```
int EnvParam_t::Ta
```

Atmospheric temperature.

4.16.2.3 TAU

```
int EnvParam_t::TAU
```

Atmospheric transmittance.

4.16.2.4 Tu

```
int EnvParam_t::Tu
```

Reflection temperature.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.17 EnvTemp_t Struct Reference

The environment temperature in double point calibration process.

```
#include <libirtemp.h>
```

Public Attributes

- double [T_high](#)
The high environment temperature in double point calibration process.
- double [T_low](#)
The low environment temperature in double point calibration process.

4.17.1 Detailed Description

The environment temperature in double point calibration process.

4.17.2 Member Data Documentation

4.17.2.1 T_high

```
double EnvTemp_t::T_high
```

The high environment temperature in double point calibration process.

4.17.2.2 T_low

```
double EnvTemp_t::T_low
```

The low environment temperature in double point calibration process.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.18 FrameGetParams_t Struct Reference

```
#include <libirv4l2.h>
```

Public Attributes

- int [reserved](#) [32]

4.18.1 Member Data Documentation

4.18.1.1 reserved

```
int FrameGetParams_t::reserved[32]
```

The documentation for this struct was generated from the following file:

- [libirv4l2.h](#)

4.19 I2cConDevParams_t Struct Reference

```
#include <libiri2c.h>
```

Public Attributes

- int [fd](#)
- int [reserved](#) [63]

4.19.1 Member Data Documentation

4.19.1.1 fd

```
int I2cConDevParams_t::fd
```

4.19.1.2 reserved

```
int I2cConDevParams_t::reserved[63]
```

The documentation for this struct was generated from the following file:

- [libiri2c.h](#)

4.20 I2cParam_t Struct Reference

```
#include <libiri2c.h>
```

Public Attributes

- uint16_t [polling_time](#)
- uint16_t [reg_addr](#)
- uint16_t [reserved](#) [60]

4.20.1 Member Data Documentation

4.20.1.1 polling_time

```
uint16_t I2cParam_t::polling_time
```

4.20.1.2 reg_addr

```
uint16_t I2cParam_t::reg_addr
```

4.20.1.3 reserved

```
uint16_t I2cParam_t::reserved[60]
```

The documentation for this struct was generated from the following file:

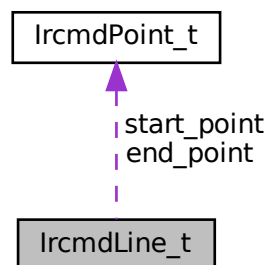
- [libiri2c.h](#)

4.21 IrcmdLine_t Struct Reference

Line struct, start from 1.

```
#include <libircmd_temp.h>
```

Collaboration diagram for IrcmdLine_t:



Public Attributes

- [IrcmdPoint_t start_point](#)
start point's position
- [IrcmdPoint_t end_point](#)
end point's position

4.21.1 Detailed Description

Line struct, start from 1.

4.21.2 Member Data Documentation

4.21.2.1 end_point

`IrcmdPoint_t IrcmdLine_t::end_point`

end point's position

4.21.2.2 start_point

`IrcmdPoint_t IrcmdLine_t::start_point`

start point's position

The documentation for this struct was generated from the following file:

- `libircmd_temp.h`

4.22 IrcmdPoint_t Struct Reference

Point struct, start from 0.

```
#include <libircmd.h>
```

Public Attributes

- `uint16_t x`
x position
- `uint16_t y`
y position

4.22.1 Detailed Description

Point struct, start from 0.

4.22.2 Member Data Documentation

4.22.2.1 x

```
uint16_t IrcmdPoint_t::x
```

x position

4.22.2.2 y

```
uint16_t IrcmdPoint_t::y
```

y position

The documentation for this struct was generated from the following file:

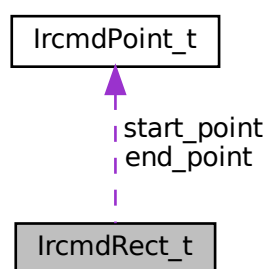
- [libircmd.h](#)

4.23 IrcmdRect_t Struct Reference

Rectangle struct, start from 1.

```
#include <libircmd.h>
```

Collaboration diagram for IrcmdRect_t:



Public Attributes

- [IrcmdPoint_t start_point](#)
start point's position
- [IrcmdPoint_t end_point](#)
end point's position

4.23.1 Detailed Description

Rectangle struct, start from 1.

4.23.2 Member Data Documentation

4.23.2.1 end_point

`IrcmdPoint_t IrcmdRect_t::end_point`

end point's position

4.23.2.2 start_point

`IrcmdPoint_t IrcmdRect_t::start_point`

start point's position

The documentation for this struct was generated from the following file:

- `libircmd.h`

4.24 IrControlHandle_t Struct Reference

```
#include <libircam.h>
```

Public Attributes

- `void * ir_control_handle`
- `int(* ir_control_open)(void *device_handle, void *params)`
- `int(* ir_control_init)(void *device_handle, void *params)`
- `int(* ir_control_read)(void *device_handle, void *cmd_param, uint8_t *data, uint32_t len)`
- `int(* ir_control_write)(void *device_handle, void *cmd_param, uint8_t *data, uint32_t len)`
- `int(* ir_control_write_without_read_return_status)(void *device_handle, void *cmd_param, uint8_t *data, uint32_t len)`
- `int(* ir_control_release)(void *device_handle, void *params)`
- `int(* ir_control_close)(void *device_handle)`
- `int(* ir_control_firmware_download)(void *device_handle, char *file_name, uint8_t *data, uint32_t file_len, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)`
- `int(* ir_control_detect_device_status)(void *device_handle, void *params, int *status)`
- `int(* ir_control_command_channel_type_get)(void *device_handle, int *command_channel_type)`
- `int(* ir_control_bootloader_download)(void *device_handle, char *file_name, uint8_t *data, uint32_t file_len, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)`

4.24.1 Member Data Documentation

4.24.1.1 ir_control_bootloader_download

```
int (* IrControlHandle_t::ir_control_bootloader_download) (void *device_handle, char *file_name,
uint8_t *data, uint32_t file_len, void *(*callback)(void *callback_data, void *priv_data), void
*priv_data)
```

4.24.1.2 ir_control_close

```
int (* IrControlHandle_t::ir_control_close) (void *device_handle)
```

4.24.1.3 ir_control_command_channel_type_get

```
int (* IrControlHandle_t::ir_control_command_channel_type_get) (void *device_handle, int *command←
_channel_type)
```

4.24.1.4 ir_control_detect_device_status

```
int (* IrControlHandle_t::ir_control_detect_device_status) (void *device_handle, void *params,
int *status)
```

4.24.1.5 ir_control_firmware_download

```
int (* IrControlHandle_t::ir_control_firmware_download) (void *device_handle, char *file_name,
uint8_t *data, uint32_t file_len, void *(*callback)(void *callback_data, void *priv_data), void
*priv_data)
```

4.24.1.6 ir_control_handle

```
void* IrControlHandle_t::ir_control_handle
```

4.24.1.7 ir_control_init

```
int (* IrControlHandle_t::ir_control_init) (void *device_handle, void *params)
```

4.24.1.8 ir_control_open

```
int (* IrControlHandle_t::ir_control_open) (void *device_handle, void *params)
```

4.24.1.9 ir_control_read

```
int (* IrControlHandle_t::ir_control_read) (void *device_handle, void *cmd_param, uint8_t *data,  
uint32_t len)
```

4.24.1.10 ir_control_release

```
int (* IrControlHandle_t::ir_control_release) (void *device_handle, void *params)
```

4.24.1.11 ir_control_write

```
int (* IrControlHandle_t::ir_control_write) (void *device_handle, void *cmd_param, uint8_t *data,  
uint32_t len)
```

4.24.1.12 ir_control_write_without_read_return_status

```
int (* IrControlHandle_t::ir_control_write_without_read_return_status) (void *device_handle,  
void *cmd_param, uint8_t *data, uint32_t len)
```

The documentation for this struct was generated from the following file:

- [libircam.h](#)

4.25 IrinfoAInfo_t Struct Reference

Autofocus Module.

```
#include <libir_infoparse.h>
```

Public Attributes

- `uint8_t af_info_en`
Autofocus INFO enable(0:off 1:on)
- `uint8_t af_sum [5]`
Autofocus detail evaluation value.

4.25.1 Detailed Description

Autofocus Module.

4.25.2 Member Data Documentation

4.25.2.1 af_info_en

```
uint8_t IrinfoAFInfo_t::af_info_en
```

Autofocus INFO enable(0:off 1:on)

4.25.2.2 af_sum

```
uint8_t IrinfoAFInfo_t::af_sum[5]
```

Autofocus detail evaluation value.

The documentation for this struct was generated from the following file:

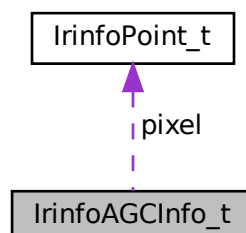
- [libir_infoparse.h](#)

4.26 IrinfoAGCInfo_t Struct Reference

AGC Module.

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoAGCInfo_t:



Public Attributes

- uint16_t [histogram_max_value](#)
Histogram Statistics High Throw Point.
- uint16_t [histogram_min_value](#)
Histogram Statistics Low Throw Point.
- uint16_t [frame_ave_value](#)
AGC input frame average.
- [IrinfoPoint_t](#) [pixel](#)
Output the coordinates of the point specified by AGC.
- uint16_t [pixel_value](#)
Output the value of the point specified by AGC.

4.26.1 Detailed Description

AGC Module.

4.26.2 Member Data Documentation

4.26.2.1 frame_ave_value

```
uint16_t IrinfoAGCInfo_t::frame_ave_value
```

AGC input frame average.

4.26.2.2 histogram_max_value

```
uint16_t IrinfoAGCInfo_t::histogram_max_value
```

Histogram Statistics High Throw Point.

4.26.2.3 histogram_min_value

```
uint16_t IrinfoAGCInfo_t::histogram_min_value
```

Histogram Statistics Low Throw Point.

4.26.2.4 pixel

```
IrinfoPoint_t IrinfoAGCInfo_t::pixel
```

Output the coordinates of the point specified by AGC.

4.26.2.5 pixel_value

```
uint16_t IrinfoAGCInfo_t::pixel_value
```

Output the value of the point specified by AGC.

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.27 IrinfoDeviceInfo_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- uint8_t [dev_pn](#) [4]
product name
- uint16_t [width](#)
Image output width.
- uint16_t [height](#)
Image output height.
- uint8_t [focal_length](#)
Physical focal length.
- uint8_t [Fnumber](#)
Aperture size.
- uint8_t [tpd_meter_range](#)
Temperature measurement range.
- uint8_t [fireware_version](#) [4]
fireware version

4.27.1 Member Data Documentation

4.27.1.1 dev_pn

```
uint8_t IrinfoDeviceInfo_t::dev_pn[4]
```

product name

4.27.1.2 fireware_version

```
uint8_t IrinfoDeviceInfo_t::fireware_version[4]
```

fireware version

4.27.1.3 Fnumber

```
uint8_t IrinfoDeviceInfo_t::Fnumber
```

Aperture size.

4.27.1.4 focal_length

```
uint8_t IrinfoDeviceInfo_t::focal_length
```

Physical focal length.

4.27.1.5 height

```
uint16_t IrinfoDeviceInfo_t::height
```

Image output height.

4.27.1.6 tpd_meter_range

```
uint8_t IrinfoDeviceInfo_t::tpd_meter_range
```

Temperature measurement range.

4.27.1.7 width

```
uint16_t IrinfoDeviceInfo_t::width
```

Image output width.

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.28 IrinfoDeviceSta_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- uint16_t [sensor_temp](#)
sensor temperature
- uint8_t [utc](#) [7]
UTC (utc[0]=year) year/month/day/hour/minute/second.
- uint8_t [alarm_bit](#)
Alarm bit(1:Ultra-high temperature operation 2:Ultra-low temperature operation)
- uint8_t [error_bit](#)
error bit(0:No abnormal)
- uint16_t [vtemp](#)
sensor temperature
- uint8_t [sun_protect_flag](#)
Sun detection(0:No-Sun-Detected 1:Sun-Detected)
- uint8_t [len_tpd_value](#)
Lens temperature information(celsius)

4.28.1 Member Data Documentation

4.28.1.1 alarm_bit

```
uint8_t IrinfoDeviceSta_t::alarm_bit
```

Alarm bit(1:Ultra-high temperature operation 2:Ultra-low temperature operation)

4.28.1.2 error_bit

```
uint8_t IrinfoDeviceSta_t::error_bit
```

error bit(0:No abnormal)

4.28.1.3 len_tpd_value

```
uint8_t IrinfoDeviceSta_t::len_tpd_value
```

Lens temperature information(celsius)

4.28.1.4 sensor_temp

```
uint16_t IrinfoDeviceSta_t::sensor_temp
```

sensor temperature

4.28.1.5 sun_protect_flag

```
uint8_t IrinfoDeviceSta_t::sun_protect_flag
```

Sun detection(0:No-Sun-Detected 1:Sun-Detected)

4.28.1.6 utc

```
uint8_t IrinfoDeviceSta_t::utc[7]
```

UTC (utc[0]=year) year/month/day/hour/minute/second.

4.28.1.7 vtemp

```
uint16_t IrinfoDeviceSta_t::vtemp
```

sensor temperature

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.29 IrinfoFrameSta_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- [uint8_t frame_fps](#)
frame fps
- [uint8_t frame_format](#)
Image data format(0:YUV422 1:Y14 2:NV12)
- [uint8_t frame_mode](#)
Output video stream splicing mode(0:image 1:temp 2:image+temp)
- [uint8_t frame_type](#)
Image type(0:Real-time Map 1:Freeze Map)
- [uint16_t frame_count](#)
ISP KBC module frame count.
- [uint16_t frame_cnt](#)
Video module frame count.
- [uint16_t temp_frame_cnt](#)
Temp module frame count.

4.29.1 Member Data Documentation

4.29.1.1 frame_cnt

```
uint16_t IrinfoFrameSta_t::frame_cnt
```

Video module frame count.

4.29.1.2 frame_count

```
uint16_t IrinfoFrameSta_t::frame_count
```

ISP KBC module frame count.

4.29.1.3 frame_format

```
uint8_t IrinfoFrameSta_t::frame_format
```

Image data format(0:YUV422 1:Y14 2:NV12)

4.29.1.4 frame_fps

```
uint8_t IrinfoFrameSta_t::frame_fps
```

frame fps

4.29.1.5 frame_mode

```
uint8_t IrinfoFrameSta_t::frame_mode
```

Output video stream splicing mode(0:image 1:temp 2:image+temp)

4.29.1.6 frame_type

```
uint8_t IrinfoFrameSta_t::frame_type
```

Image type(0:Real-time Map 1:Freeze Map)

4.29.1.7 temp_frame_cnt

```
uint16_t IrinfoFrameSta_t::temp_frame_cnt
```

Temp module frame count.

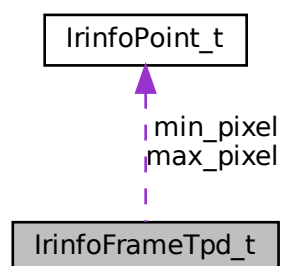
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.30 IrinfoFrameTpd_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoFrameTpd_t:



Public Attributes

- [uint8_t frame_info_en](#)
Enabling status(0:off 1 : on)
- [IinfoPoint_t max_pixel](#)
- [float max_pixel_value](#)
- [IinfoPoint_t min_pixel](#)
- [float min_pixel_value](#)

4.30.1 Member Data Documentation

4.30.1.1 frame_info_en

`uint8_t IinfoFrameTpd_t::frame_info_en`

Enabling status(0:off 1 : on)

4.30.1.2 max_pixel

`IinfoPoint_t IinfoFrameTpd_t::max_pixel`

4.30.1.3 max_pixel_value

`float IinfoFrameTpd_t::max_pixel_value`

4.30.1.4 min_pixel

`IinfoPoint_t IinfoFrameTpd_t::min_pixel`

4.30.1.5 min_pixel_value

`float IinfoFrameTpd_t::min_pixel_value`

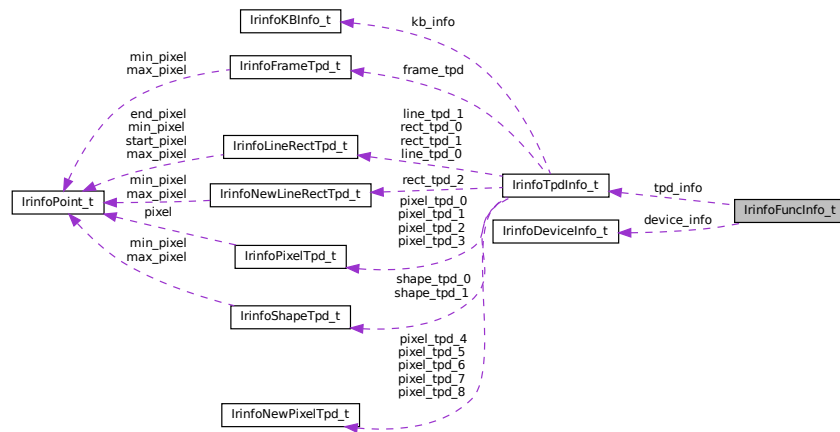
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.31 IrinfoFuncInfo_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoFuncInfo_t:



Public Attributes

- [uint8_t info_data_version](#)
info line data version
- [IrinfoDeviceInfo_t device_info](#)
- [IrinfoTpdInfo_t tpd_info](#)

4.31.1 Member Data Documentation

4.31.1.1 device_info

```
IrinfoDeviceInfo\_t IrinfoFuncInfo_t::device_info
```

4.31.1.2 info_data_version

```
uint8_t IrinfoFuncInfo_t::info_data_version
```

info line data version

4.31.1.3 tpd_info

```
IrinfoTpdInfo_t IrinfoFuncInfo_t::tpd_info
```

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.32 IrinfoImageSta_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- uint8_t [palette_idx](#)
- uint8_t [palette_mode](#)
- uint8_t [width_zoom_factor](#)
- uint8_t [height_zoom_factor](#)
- uint8_t [mirror_status](#)
- uint8_t [flip_status](#)

4.32.1 Member Data Documentation

4.32.1.1 flip_status

```
uint8_t IrinfoImageSta_t::flip_status
```

4.32.1.2 height_zoom_factor

```
uint8_t IrinfoImageSta_t::height_zoom_factor
```

4.32.1.3 mirror_status

```
uint8_t IrinfoImageSta_t::mirror_status
```


4.32.1.4 palette_idx

```
uint8_t IrinfoImageSta_t::palette_idx
```

4.32.1.5 palette_mode

```
uint8_t IrinfoImageSta_t::palette_mode
```

4.32.1.6 width_zoom_factor

```
uint8_t IrinfoImageSta_t::width_zoom_factor
```

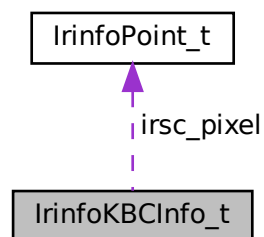
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.33 IrinfoKBCInfo_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoKBCInfo_t:



Public Attributes

- [IrinfoPoint_t irsc_pixel](#)
- `uint16_t` [irsc_pixel_value](#)

4.33.1 Member Data Documentation

4.33.1.1 irsc_pixel

[IrinfoPoint_t](#) [IrinfoKBCInfo_t::irsc_pixel](#)

4.33.1.2 irsc_pixel_value

[uint16_t](#) [IrinfoKBCInfo_t::irsc_pixel_value](#)

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.34 IrinfoKBCInfo_t Struct Reference

KB correction module.

```
#include <libir_infoparse.h>
```

Public Attributes

- [uint32_t](#) [k_e](#)
K environment variable correction parameter.
- [uint32_t](#) [b_e](#)
B environment variable correction parameter.
- [uint32_t](#) [k_d](#)
K distance correction parameter.
- [uint32_t](#) [b_d](#)
B distance correction parameter.
- [uint32_t](#) [k_us](#)
K non-stationary correction parameter.
- [uint32_t](#) [b_us](#)
B non-stationary correction parameter.
- [uint8_t](#) [dummy0](#)

4.34.1 Detailed Description

KB correction module.

4.34.2 Member Data Documentation

4.34.2.1 b_d

```
uint32_t IrinfoKBInfo_t::b_d
```

B distance correction parameter.

4.34.2.2 b_e

```
uint32_t IrinfoKBInfo_t::b_e
```

B environment variable correction parameter.

4.34.2.3 b_us

```
uint32_t IrinfoKBInfo_t::b_us
```

B non-stationary correction parameter.

4.34.2.4 dummy0

```
uint8_t IrinfoKBInfo_t::dummy0
```

4.34.2.5 k_d

```
uint32_t IrinfoKBInfo_t::k_d
```

K distance correction parameter.

4.34.2.6 k_e

```
uint32_t IrinfoKBInfo_t::k_e
```

K environment variable correction parameter.

4.34.2.7 k_us

```
uint32_t IrinfoKBInfo_t::k_us
```

K non-stationary correction parameter.

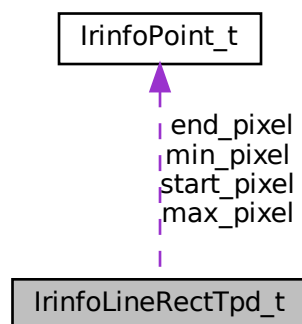
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.35 IrinfoLineRectTpd_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoLineRectTpd_t:



Public Attributes

- [uint8_t line_rect_info_en](#)
Enabling status(0:off 1 : on)
- [IrinfoPoint_t start_pixel](#)
- [IrinfoPoint_t end_pixel](#)
- [IrinfoPoint_t max_pixel](#)
- [IrinfoPoint_t min_pixel](#)
- float [max_pixel_value](#)
- float [min_pixel_value](#)
- float [ave_pixel_value](#)

4.35.1 Member Data Documentation

4.35.1.1 ave_pixel_value

```
float IrinfoLineRectTpd_t::ave_pixel_value
```

4.35.1.2 end_pixel

```
IrinfoPoint_t IrinfoLineRectTpd_t::end_pixel
```

4.35.1.3 line_rect_info_en

```
uint8_t IrinfoLineRectTpd_t::line_rect_info_en
```

Enabling status(0:off 1 : on)

4.35.1.4 max_pixel

```
IrinfoPoint_t IrinfoLineRectTpd_t::max_pixel
```

4.35.1.5 max_pixel_value

```
float IrinfoLineRectTpd_t::max_pixel_value
```

4.35.1.6 min_pixel

```
IrinfoPoint_t IrinfoLineRectTpd_t::min_pixel
```

4.35.1.7 min_pixel_value

```
float IrinfoLineRectTpd_t::min_pixel_value
```

4.35.1.8 start_pixel

```
IrinfoPoint_t IrinfoLineRectTpd_t::start_pixel
```

The documentation for this struct was generated from the following file:

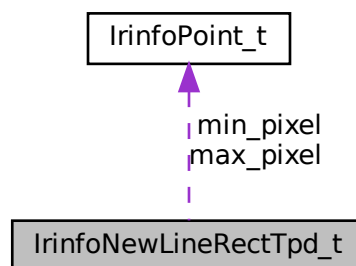
- [libir_infoparse.h](#)

4.36 IrinfoNewLineRectTpd_t Struct Reference

New temperature measurement rectangle at the software assignable information.

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoNewLineRectTpd_t:



Public Attributes

- `uint8_t line_rect_info_en`
Enabling status(0:off 1 : on)
- `IrinfoPoint_t max_pixel`
- `IrinfoPoint_t min_pixel`
- `float max_pixel_value`
- `float min_pixel_value`
- `float ave_pixel_value`

4.36.1 Detailed Description

New temperature measurement rectangle at the software assignable information.

4.36.2 Member Data Documentation

4.36.2.1 ave_pixel_value

```
float IrinfoNewLineRectTpd_t::ave_pixel_value
```

4.36.2.2 line_rect_info_en

```
uint8_t IrinfoNewLineRectTpd_t::line_rect_info_en
```

Enabling status(0:off 1 : on)

4.36.2.3 max_pixel

```
IrinfoPoint_t IrinfoNewLineRectTpd_t::max_pixel
```

4.36.2.4 max_pixel_value

```
float IrinfoNewLineRectTpd_t::max_pixel_value
```

4.36.2.5 min_pixel

```
IrinfoPoint_t IrinfoNewLineRectTpd_t::min_pixel
```

4.36.2.6 min_pixel_value

```
float IrinfoNewLineRectTpd_t::min_pixel_value
```

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.37 IrinfoNewPixelTpd_t Struct Reference

New temperature measurement points at the software assignable information.

```
#include <libir_infoparse.h>
```

Public Attributes

- uint8_t [pixel_info_en](#)
Enabling status(0:off 1 : on)
- float [pixel_value](#)

4.37.1 Detailed Description

New temperature measurement points at the software assignable information.

4.37.2 Member Data Documentation

4.37.2.1 [pixel_info_en](#)

```
uint8_t IrinfoNewPixelTpd_t::pixel_info_en
```

Enabling status(0:off 1 : on)

4.37.2.2 [pixel_value](#)

```
float IrinfoNewPixelTpd_t::pixel_value
```

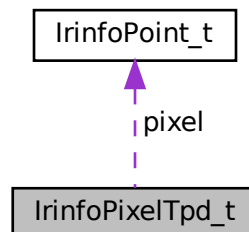
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.38 IrinfoPixelTpd_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoPixelTpd_t:



Public Attributes

- [uint8_t pixel_info_en](#)
Enabling status(0:off 1 : on)
- [IrinfoPoint_t pixel](#)
- [float pixel_value](#)

4.38.1 Member Data Documentation

4.38.1.1 pixel

```
IrinfoPoint_t IrinfoPixelTpd_t::pixel
```

4.38.1.2 pixel_info_en

```
uint8_t IrinfoPixelTpd_t::pixel_info_en
```

Enabling status(0:off 1 : on)

4.38.1.3 pixel_value

```
float IrinfoPixelTpd_t::pixel_value
```

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.39 IrinfoPoint_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- [uint16_t x](#)
x position
- [uint16_t y](#)
y position

4.39.1 Member Data Documentation

4.39.1.1 x

```
uint16_t IrinfoPoint_t::x
```

x position

4.39.1.2 y

```
uint16_t IrinfoPoint_t::y
```

y position

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.40 IrinfoRegSta_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- [uint16_t rmv_strength](#)
Rmcover strength.
- [uint8_t move_status](#)
Device testing movement status.
- [uint16_t noise_value](#)
Current sl noise judgment level.
- [uint16_t noise_self](#)
Current ide noise judgment level.
- [uint32_t regis_status](#)
Motion Alignment Status.

4.40.1 Member Data Documentation

4.40.1.1 move_status

```
uint8_t IrinfoRegSta_t::move_status
```

Device testing movement status.

4.40.1.2 noise_self

```
uint16_t IrinfoRegSta_t::noise_self
```

Current ide noise judgment level.

4.40.1.3 noise_value

```
uint16_t IrinfoRegSta_t::noise_value
```

Current sl noise judgment level.

4.40.1.4 regis_status

```
uint32_t IrinfoRegSta_t::regis_status
```

Motion Alignment Status.

4.40.1.5 rmv_strength

```
uint16_t IrinfoRegSta_t::rmv_strength
```

Rmcover strength.

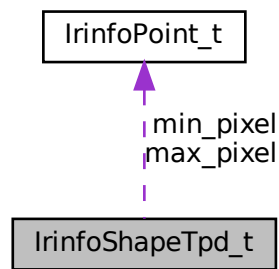
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.41 IrinfoShapeTpd_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoShapeTpd_t:



Public Attributes

- `uint8_t shape_info_en`
Enabling status(0:off 1 : on)
- `IrinfoPoint_t max_pixel`
- `IrinfoPoint_t min_pixel`
- `float max_pixel_value`
- `float min_pixel_value`
- `float ave_pixel_value`

4.41.1 Member Data Documentation

4.41.1.1 ave_pixel_value

```
float IrinfoShapeTpd_t::ave_pixel_value
```

4.41.1.2 max_pixel

```
IrinfoPoint_t IrinfoShapeTpd_t::max_pixel
```

4.41.1.3 max_pixel_value

float IrinfoShapeTpd_t::max_pixel_value

4.41.1.4 min_pixel

IrinfoPoint_t IrinfoShapeTpd_t::min_pixel

4.41.1.5 min_pixel_value

float IrinfoShapeTpd_t::min_pixel_value

4.41.1.6 shape_info_en

uint8_t IrinfoShapeTpd_t::shape_info_en

Enabling status(0:off 1 : on)

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.42 IrinfoShutterSta_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- uint8_t [ffc_countdown](#)
Update the shutter countdown.
- uint8_t [shutter_status](#)
shutter status(The current value is 0xFF)
- uint8_t [auto_shutter_status](#)
auto shutter status(0:off 1:on)
- float [shutter_tpd_value](#)
Shutter temperature information(celsius)
- uint8_t [Overexposure_interrupt_flag](#)
Overexposure interrupt flag(0:No-Interruption-Detected 1:Interruption-Detected)

4.42.1 Member Data Documentation

4.42.1.1 auto_shutter_status

`uint8_t IrinfoShutterSta_t::auto_shutter_status`

auto shutter status(0:off 1:on)

4.42.1.2 ffc_countdown

`uint8_t IrinfoShutterSta_t::ffc_countdown`

Update the shutter countdown.

4.42.1.3 Overexposure_interrupt_flag

`uint8_t IrinfoShutterSta_t::Overexposure_interrupt_flag`

Overexposure interrupt flag(0:No-Interruption-Detected 1:Interruption-Detected)

4.42.1.4 shutter_status

`uint8_t IrinfoShutterSta_t::shutter_status`

shutter status(The current value is 0xFF)

4.42.1.5 shutter_tpd_value

`float IrinfoShutterSta_t::shutter_tpd_value`

Shutter temperature information(celsius)

The documentation for this struct was generated from the following file:

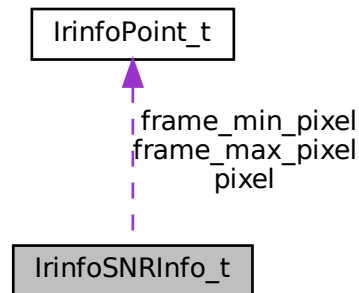
- [libir_infoparse.h](#)

4.43 IrinfoSNRInfo_t Struct Reference

SNR Module.

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoSNRInfo_t:



Public Attributes

- `uint16_t frame_max_value`
Maximum value of pixel points in the SNR module frame.
- `IrinfoPoint_t frame_max_pixel`
- `uint16_t frame_min_value`
Minimum value of pixel points in the SNR module frame.
- `IrinfoPoint_t frame_min_pixel`
- `IrinfoPoint_t pixel`
- `uint16_t pixel_value`
Value of the specified point in the SNR module frame.

4.43.1 Detailed Description

SNR Module.

4.43.2 Member Data Documentation

4.43.2.1 frame_max_pixel

```
IrinfoPoint_t IrinfoSNRInfo_t::frame_max_pixel
```

4.43.2.2 frame_max_value

```
uint16_t IrinfoSNRInfo_t::frame_max_value
```

Maximum value of pixel points in the SNR module frame.

4.43.2.3 frame_min_pixel

```
IrinfoPoint_t IrinfoSNRInfo_t::frame_min_pixel
```

4.43.2.4 frame_min_value

```
uint16_t IrinfoSNRInfo_t::frame_min_value
```

Minimum value of pixel points in the SNR module frame.

4.43.2.5 pixel

```
IrinfoPoint_t IrinfoSNRInfo_t::pixel
```

4.43.2.6 pixel_value

```
uint16_t IrinfoSNRInfo_t::pixel_value
```

Value of the specified point in the SNR module frame.

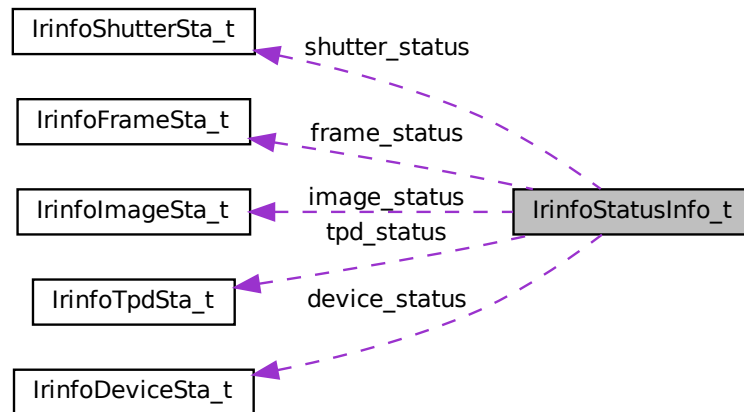
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.44 IrinfoStatusInfo_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoStatusInfo_t:



Public Attributes

- `uint8_t info_data_version`
info line data version
- `IrinfoFrameSta_t frame_status`
- `IrinfoImageSta_t image_status`
- `IrinfoTpdSta_t tpd_status`
- `IrinfoShutterSta_t shutter_status`
- `IrinfoDeviceSta_t device_status`

4.44.1 Member Data Documentation

4.44.1.1 device_status

```
IrinfoDeviceSta_t IrinfoStatusInfo_t::device_status
```

4.44.1.2 frame_status

```
IrinfoFrameSta_t IrinfoStatusInfo_t::frame_status
```

4.44.1.3 image_status

```
IrinfoImageSta_t IrinfoStatusInfo_t::image_status
```

4.44.1.4 info_data_version

```
uint8_t IrinfoStatusInfo_t::info_data_version
```

info line data version

4.44.1.5 shutter_status

```
IrinfoShutterSta_t IrinfoStatusInfo_t::shutter_status
```

4.44.1.6 tpd_status

```
IrinfoTpdSta_t IrinfoStatusInfo_t::tpd_status
```

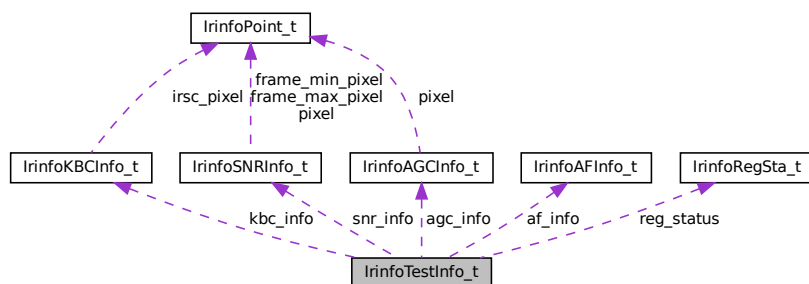
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.45 IrinfoTestInfo_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for IrinfoTestInfo_t:



Public Attributes

- [uint8_t info_data_version](#)
info line data version
- [IrinfoRegSta_t reg_status](#)
- [IrinfoKBCInfo_t kbc_info](#)
- [IrinfoAFInfo_t af_info](#)
- [IrinfoSNRInfo_t snr_info](#)
- [IrinfoAGCInfo_t agc_info](#)

4.45.1 Member Data Documentation

4.45.1.1 af_info

[IrinfoAFInfo_t](#) IrinfoTestInfo_t::af_info

4.45.1.2 agc_info

[IrinfoAGCInfo_t](#) IrinfoTestInfo_t::agc_info

4.45.1.3 info_data_version

[uint8_t](#) IrinfoTestInfo_t::info_data_version

info line data version

4.45.1.4 kbc_info

[IrinfoKBCInfo_t](#) IrinfoTestInfo_t::kbc_info

4.45.1.5 reg_status

[IrinfoRegSta_t](#) IrinfoTestInfo_t::reg_status

4.45.1.6 snr_info

`IrinfoSNRInfo_t IrinfoTestInfo_t::snr_info`

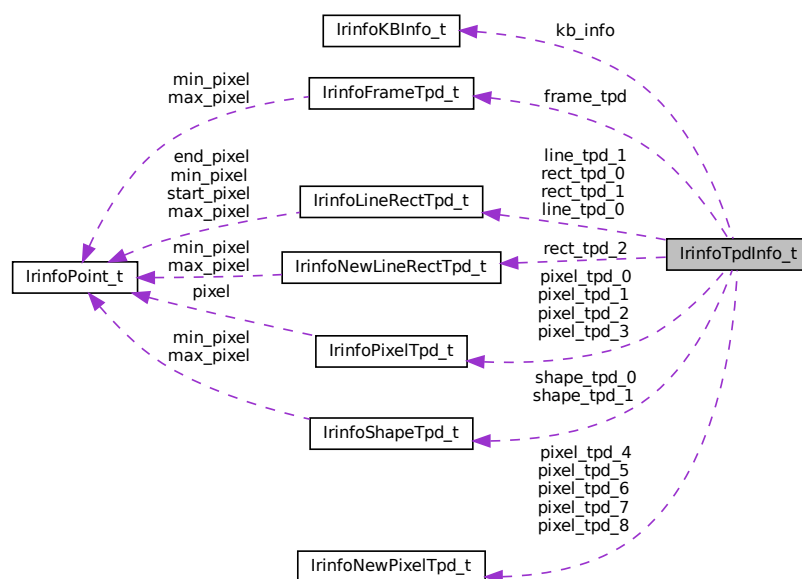
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.46 IrinfoTpdInfo_t Struct Reference

```
#include <libir_infoparse.h>
```

Collaboration diagram for `IrinfoTpdInfo_t`:



Public Attributes

- [IrinfoFrameTpd_t frame_tpd](#)
- [IrinfoPixelTpd_t pixel_tpd_0](#)
- [IrinfoPixelTpd_t pixel_tpd_1](#)
- [IrinfoPixelTpd_t pixel_tpd_2](#)
- [IrinfoPixelTpd_t pixel_tpd_3](#)
- [IrinfoNewPixelTpd_t pixel_tpd_4](#)
- [IrinfoNewPixelTpd_t pixel_tpd_5](#)
- [IrinfoNewPixelTpd_t pixel_tpd_6](#)
- [IrinfoNewPixelTpd_t pixel_tpd_7](#)
- [IrinfoNewPixelTpd_t pixel_tpd_8](#)
- [IrinfoLineRectTpd_t line_tpd_0](#)
- [IrinfoLineRectTpd_t line_tpd_1](#)
- [IrinfoLineRectTpd_t rect_tpd_0](#)
- [IrinfoLineRectTpd_t rect_tpd_1](#)
- [IrinfoNewLineRectTpd_t rect_tpd_2](#)
- [IrinfoShapeTpd_t shape_tpd_0](#)
- [IrinfoShapeTpd_t shape_tpd_1](#)
- [IrinfoKBInfo_t kb_info](#)

4.46.1 Member Data Documentation

4.46.1.1 frame_tpd

`IrinfoFrameTpd_t` `IrinfoTpdInfo_t::frame_tpd`

4.46.1.2 kb_info

`IrinfoKBInfo_t` `IrinfoTpdInfo_t::kb_info`

4.46.1.3 line_tpd_0

`IrinfoLineRectTpd_t` `IrinfoTpdInfo_t::line_tpd_0`

4.46.1.4 line_tpd_1

`IrinfoLineRectTpd_t` `IrinfoTpdInfo_t::line_tpd_1`

4.46.1.5 pixel_tpd_0

`IrinfoPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_0`

4.46.1.6 pixel_tpd_1

`IrinfoPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_1`

4.46.1.7 pixel_tpd_2

`IrinfoPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_2`

4.46.1.8 pixel_tpd_3

`IrinfoPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_3`

4.46.1.9 pixel_tpd_4

`IrinfoNewPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_4`

4.46.1.10 pixel_tpd_5

`IrinfoNewPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_5`

4.46.1.11 pixel_tpd_6

`IrinfoNewPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_6`

4.46.1.12 pixel_tpd_7

`IrinfoNewPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_7`

4.46.1.13 pixel_tpd_8

`IrinfoNewPixelTpd_t` `IrinfoTpdInfo_t::pixel_tpd_8`

4.46.1.14 rect_tpd_0

`IrinfoLineRectTpd_t` `IrinfoTpdInfo_t::rect_tpd_0`

4.46.1.15 rect_tpd_1

`IrinfoLineRectTpd_t` `IrinfoTpdInfo_t::rect_tpd_1`

4.46.1.16 rect_tpd_2

`IrinfoNewLineRectTpd_t` `IrinfoTpdInfo_t::rect_tpd_2`

4.46.1.17 shape_tpd_0

`IrinfoShapeTpd_t` `IrinfoTpdInfo_t::shape_tpd_0`

4.46.1.18 shape_tpd_1

`IrinfoShapeTpd_t` `IrinfoTpdInfo_t::shape_tpd_1`

The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.47 IrinfoTpdSta_t Struct Reference

```
#include <libir_infoparse.h>
```

Public Attributes

- `uint8_t` [gain_mode](#)
Real-time gain status(0:low gain 1:high gain)
- `uint8_t` [isotherm_switch](#)
Isotherm switch status(0:off 1:on)
- `uint8_t` [temp_type](#)
Output temp data type(0:degrees celsius temp 1:fahrenheit temp 2:kelfin)
- `uint8_t` [temp_area](#)
Temp area(0:high_temp_area 1:normal_temp_area 2:low_temp_area 3:fix_temp_area)

4.47.1 Member Data Documentation

4.47.1.1 gain_mode

`uint8_t` `IrinfoTpdSta_t::gain_mode`

Real-time gain status(0:low gain 1:high gain)

4.47.1.2 isotherm_switch

```
uint8_t IrinfoTpdSta_t::isotherm_switch
```

Isotherm switch status(0:off 1:on)

4.47.1.3 temp_area

```
uint8_t IrinfoTpdSta_t::temp_area
```

Temp area(0:high_temp_area 1:normal_temp_area 2:low_temp_area 3:fix_temp_area)

4.47.1.4 temp_type

```
uint8_t IrinfoTpdSta_t::temp_type
```

Output temp data type(0:degrees celsius temp 1:fahrenheit temp 2:kelvin)

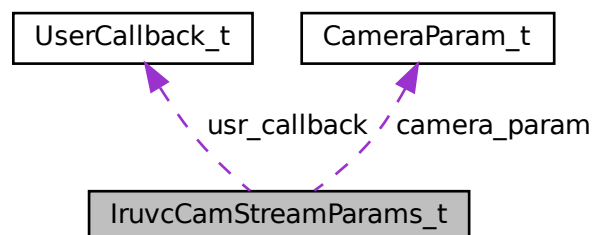
The documentation for this struct was generated from the following file:

- [libir_infoparse.h](#)

4.48 IruvcCamStreamParams_t Struct Reference

```
#include <libiruvvc.h>
```

Collaboration diagram for IruvcCamStreamParams_t:



Public Attributes

- [CameraParam_t](#) `camera_param`
- [UserCallback_t](#) `usr_callback`
- `uint8_t reserved` [48]

4.48.1 Member Data Documentation

4.48.1.1 camera_param

[CameraParam_t](#) `IruvcCamStreamParams_t::camera_param`

4.48.1.2 reserved

`uint8_t IruvcCamStreamParams_t::reserved[48]`

4.48.1.3 usr_callback

[UserCallback_t](#) `IruvcCamStreamParams_t::usr_callback`

The documentation for this struct was generated from the following file:

- [libiruv.h](#)

4.49 IruvcDevParam_t Struct Reference

Device's configuration.

```
#include <libiruv.h>
```

Public Attributes

- unsigned int [pid](#)
pid
- unsigned int [vid](#)
vid
- unsigned int [same_idx](#)
vid

4.49.1 Detailed Description

Device's configuration.

4.49.2 Member Data Documentation

4.49.2.1 pid

```
unsigned int IruvcDevParam_t::pid
```

pid

4.49.2.2 same_idx

```
unsigned int IruvcDevParam_t::same_idx
```

vid

4.49.2.3 vid

```
unsigned int IruvcDevParam_t::vid
```

vid

The documentation for this struct was generated from the following file:

- [libiruvc.h](#)

4.50 IruvcFrameGetParams_t Struct Reference

```
#include <libiruvc.h>
```

Public Attributes

- int [reserved](#) [32]

4.50.1 Member Data Documentation

4.50.1.1 reserved

```
int IruvcFrameGetParams_t::reserved[32]
```

The documentation for this struct was generated from the following file:

- [libiruvc.h](#)

4.51 IruvcStreamStopParams_t Struct Reference

```
#include <libiruvc.h>
```

Public Attributes

- [cam_side_preview_ctl stop_mode](#)
- `uint8_t reserved [60]`

4.51.1 Member Data Documentation

4.51.1.1 reserved

```
uint8_t IruvcStreamStopParams_t::reserved[60]
```

4.51.1.2 stop_mode

```
cam\_side\_preview\_ctl IruvcStreamStopParams_t::stop_mode
```

The documentation for this struct was generated from the following file:

- [libiruvc.h](#)

4.52 IrVideoHandle_t Struct Reference

```
#include <libircam.h>
```

Public Attributes

- void * [ir_video_handle](#)
- int(* [ir_video_open](#))(void *device_handle, void *params)
- int(* [ir_video_init](#))(void *device_handle, void *params)
- int(* [ir_video_start_stream](#))(void *device_handle, void *params)
- int(* [ir_video_frame_get](#))(void *device_handle, void *params, uint8_t *frame_data, int len)
- int(* [ir_video_stop_stream](#))(void *device_handle, void *params)
- int(* [ir_video_release](#))(void *device_handle, void *params)
- int(* [ir_video_close](#))(void *device_handle)

4.52.1 Member Data Documentation

4.52.1.1 [ir_video_close](#)

```
int (* IrVideoHandle_t::ir_video_close) (void *device_handle)
```

4.52.1.2 [ir_video_frame_get](#)

```
int (* IrVideoHandle_t::ir_video_frame_get) (void *device_handle, void *params, uint8_t *frame_data, int len)
```

4.52.1.3 [ir_video_handle](#)

```
void* IrVideoHandle_t::ir_video_handle
```

4.52.1.4 [ir_video_init](#)

```
int (* IrVideoHandle_t::ir_video_init) (void *device_handle, void *params)
```

4.52.1.5 [ir_video_open](#)

```
int (* IrVideoHandle_t::ir_video_open) (void *device_handle, void *params)
```

4.52.1.6 ir_video_release

```
int (* IrVideoHandle_t::ir_video_release) (void *device_handle, void *params)
```

4.52.1.7 ir_video_start_stream

```
int (* IrVideoHandle_t::ir_video_start_stream) (void *device_handle, void *params)
```

4.52.1.8 ir_video_stop_stream

```
int (* IrVideoHandle_t::ir_video_stop_stream) (void *device_handle, void *params)
```

The documentation for this struct was generated from the following file:

- [libircam.h](#)

4.53 ktbt_correct_param_t Struct Reference

```
#include <libirtemp.h>
```

Public Attributes

- [int16_t K_Temp](#)
ktemp value of calibration algorithm
- [uint16_t B_Temp](#)
btemp value of calibration algorithm
- [uint16_t Address_CA](#)
address_ca of calibration algorithm
- [uint16_t * kt_array](#)
kt array of calibration algorithm
- [int16_t * bt_array](#)
bt array of calibration algorithm
- [uint16_t vtemp](#)
vtemp value of calibration algorithm

4.53.1 Member Data Documentation

4.53.1.1 Address_CA

```
uint16_t ktbt_correct_param_t::Address_CA
```

address_ca of calibration algorithm

4.53.1.2 B_Temp

```
uint16_t ktbt_correct_param_t::B_Temp
```

btemp value of calibration algorithm

4.53.1.3 bt_array

```
int16_t* ktbt_correct_param_t::bt_array
```

bt array of calibration algorithm

4.53.1.4 K_Temp

```
int16_t ktbt_correct_param_t::K_Temp
```

ktemp value of calibration algorithm

4.53.1.5 kt_array

```
uint16_t* ktbt_correct_param_t::kt_array
```

kt array of calibration algorithm

4.53.1.6 vtemp

```
uint16_t ktbt_correct_param_t::vtemp
```

vtemp value of calibration algorithm

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.54 KtBtCoefficient_t Struct Reference

```
#include <libirtemp.h>
```

Public Attributes

- double [P0](#)
- double [P1](#)
- double [P2](#)
- double [P3](#)

4.54.1 Member Data Documentation

4.54.1.1 P0

```
double KtBtCoefficient_t::P0
```

4.54.1.2 P1

```
double KtBtCoefficient_t::P1
```

4.54.1.3 P2

```
double KtBtCoefficient_t::P2
```

4.54.1.4 P3

```
double KtBtCoefficient_t::P3
```

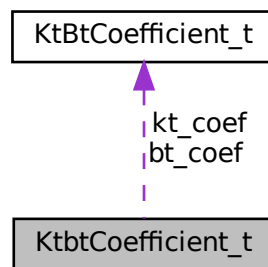
The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.55 KtbtCoefficient_t Struct Reference

```
#include <libirtemp.h>
```

Collaboration diagram for KtbtCoefficient_t:



Public Attributes

- [KtbtCoefficient_t kt_coef](#)
- [KtbtCoefficient_t bt_coef](#)

4.55.1 Member Data Documentation

4.55.1.1 bt_coef

[KtbtCoefficient_t](#) KtbtCoefficient_t::bt_coef

4.55.1.2 kt_coef

[KtbtCoefficient_t](#) KtbtCoefficient_t::kt_coef

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.56 KtInfo_t Struct Reference

The turning address of KT curve.

```
#include <libirtemp.h>
```

Public Attributes

- uint16_t [start_addr](#)
Start turning address of KT curve.
- uint16_t [end_addr](#)
End turning address of KT curve.

4.56.1 Detailed Description

The turning address of KT curve.

4.56.2 Member Data Documentation

4.56.2.1 end_addr

```
uint16_t KtInfo_t::end_addr
```

End turning address of KT curve.

4.56.2.2 start_addr

```
uint16_t KtInfo_t::start_addr
```

Start turning address of KT curve.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.57 Line_t Struct Reference

[Line_t](#) definition.

```
#include <libirtemp.h>
```

Public Attributes

- int [start_x](#)
The start horizontal coordinate of [Line_t](#).
- int [start_y](#)
The start vertical coordinate of [Line_t](#).
- int [end_x](#)
The end horizontal coordinate of [Line_t](#).
- int [end_y](#)
The end vertical coordinate of [Line_t](#).

4.57.1 Detailed Description

[Line_t](#) definition.

4.57.2 Member Data Documentation

4.57.2.1 [end_x](#)

```
int Line_t::end_x
```

The end horizontal coordinate of [Line_t](#).

4.57.2.2 [end_y](#)

```
int Line_t::end_y
```

The end vertical coordinate of [Line_t](#).

4.57.2.3 [start_x](#)

```
int Line_t::start_x
```

The start horizontal coordinate of [Line_t](#).

4.57.2.4 start_y

```
int Line_t::start_y
```

The start vertical coordinate of [Line_t](#).

The documentation for this struct was generated from the following file:

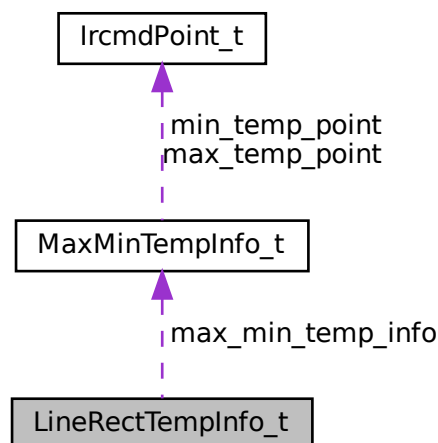
- [libirtemp.h](#)

4.58 LineRectTempInfo_t Struct Reference

Temperature information struct of a line or a rectangle.

```
#include <libircmd.h>
```

Collaboration diagram for LineRectTempInfo_t:



Public Attributes

- float [ave_temp](#)
maximum temperature point's position, Unit: Celsius
- [MaxMinTempInfo_t](#) [max_min_temp_info](#)
Maximum and minimum temperature information struct.

4.58.1 Detailed Description

Temperature information struct of a line or a rectangle.

4.58.2 Member Data Documentation

4.58.2.1 ave_temp

```
float LineRectTempInfo_t::ave_temp
```

maximum temperature point's position,Unit:Celsius

4.58.2.2 max_min_temp_info

```
MaxMinTempInfo_t LineRectTempInfo_t::max_min_temp_info
```

Maximum and minimum temperature information struct.

The documentation for this struct was generated from the following file:

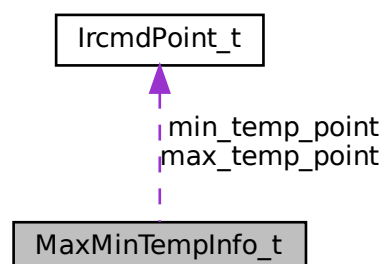
- [libircmd.h](#)

4.59 MaxMinTempInfo_t Struct Reference

Maximum and minimum temperature information struct.

```
#include <libircmd.h>
```

Collaboration diagram for MaxMinTempInfo_t:



Public Attributes

- float [max_temp](#)
maximum temperature value, Unit:Celsius
- float [min_temp](#)
minimum temperature value, Unit:Celsius
- [IrcmdPoint_t](#) [max_temp_point](#)
maximum temperature point's position
- [IrcmdPoint_t](#) [min_temp_point](#)
minimum temperature point's position

4.59.1 Detailed Description

Maximum and minimum temperature information struct.

4.59.2 Member Data Documentation

4.59.2.1 max_temp

```
float MaxMinTempInfo_t::max_temp
```

maximum temperature value, Unit:Celsius

4.59.2.2 max_temp_point

```
IrcmdPoint\_t MaxMinTempInfo_t::max_temp_point
```

maximum temperature point's position

4.59.2.3 min_temp

```
float MaxMinTempInfo_t::min_temp
```

minimum temperature value, Unit:Celsius

4.59.2.4 min_temp_point

`IrcmdPoint_t` `MaxMinTempInfo_t::min_temp_point`

minimum temperature point's position

The documentation for this struct was generated from the following file:

- [libircmd.h](#)

4.60 MultiPointCalibArray_t Struct Reference

```
#include <libirtemp.h>
```

Public Attributes

- `uint16_t` * [default_kt_array](#)
- `int16_t` * [default_bt_array](#)
- `uint16_t` * [default_nuc_table](#)
- `uint16_t` * [correct_table](#)
- `uint32_t` [correct_table_len](#)

4.60.1 Member Data Documentation

4.60.1.1 correct_table

`uint16_t*` `MultiPointCalibArray_t::correct_table`

4.60.1.2 correct_table_len

`uint32_t` `MultiPointCalibArray_t::correct_table_len`

4.60.1.3 default_bt_array

`int16_t*` `MultiPointCalibArray_t::default_bt_array`

4.60.1.4 default_kt_array

```
uint16_t* MultiPointCalibArray_t::default_kt_array
```

4.60.1.5 default_nuc_table

```
uint16_t* MultiPointCalibArray_t::default_nuc_table
```

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.61 MultiPointCalibArray_version3_t Struct Reference

```
#include <libirtemp.h>
```

Public Attributes

- [uint16_t * default_kt_array_low](#)
- [uint16_t * default_kt_array_mid](#)
- [uint16_t * default_kt_array_high](#)
- [int16_t * default_bt_array_low](#)
- [int16_t * default_bt_array_mid](#)
- [int16_t * default_bt_array_high](#)
- [uint16_t * default_nuc_table](#)
- [uint16_t * correct_table](#)
- [uint32_t correct_table_len](#)

4.61.1 Member Data Documentation

4.61.1.1 correct_table

```
uint16_t* MultiPointCalibArray_version3_t::correct_table
```

4.61.1.2 correct_table_len

```
uint32_t MultiPointCalibArray_version3_t::correct_table_len
```

4.61.1.3 default_bt_array_high

```
int16_t* MultiPointCalibArray_version3_t::default_bt_array_high
```

4.61.1.4 default_bt_array_low

```
int16_t* MultiPointCalibArray_version3_t::default_bt_array_low
```

4.61.1.5 default_bt_array_mid

```
int16_t* MultiPointCalibArray_version3_t::default_bt_array_mid
```

4.61.1.6 default_kt_array_high

```
uint16_t* MultiPointCalibArray_version3_t::default_kt_array_high
```

4.61.1.7 default_kt_array_low

```
uint16_t* MultiPointCalibArray_version3_t::default_kt_array_low
```

4.61.1.8 default_kt_array_mid

```
uint16_t* MultiPointCalibArray_version3_t::default_kt_array_mid
```

4.61.1.9 default_nuc_table

```
uint16_t* MultiPointCalibArray_version3_t::default_nuc_table
```

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.62 MultiPointCalibForKtBtParam_t Struct Reference

```
#include <libirtemp.h>
```

Public Attributes

- uint8_t [hotmode](#)
- uint8_t [n](#)
- uint8_t [m](#)
- uint16_t [len](#)
- uint16_t [m_index](#)
- int16_t [Ktemp](#)
- uint16_t [Btemp](#)
- uint16_t [Adress_ca](#)
- uint16_t [H](#) [[MAX_CALI_POINT_NUMBER](#)]
- uint16_t [L](#) [[MAX_CALI_POINT_NUMBER](#)]
- uint16_t [Vtemp](#) [[MAX_CALI_POINT_NUMBER](#)]
- uint16_t [K](#) [[MAX_CALI_POINT_NUMBER](#)]
- int16_t [B](#) [[MAX_CALI_POINT_NUMBER](#)]
- uint16_t [P](#) [[MAX_CALI_POINT_NUMBER](#)]
- float [temp](#) [[MAX_CALI_POINT_NUMBER](#)]

4.62.1 Member Data Documentation

4.62.1.1 [Adress_ca](#)

```
uint16_t MultiPointCalibForKtBtParam_t::Adress_ca
```

4.62.1.2 [B](#)

```
int16_t MultiPointCalibForKtBtParam_t::B[MAX\_CALI\_POINT\_NUMBER]
```

4.62.1.3 [Btemp](#)

```
uint16_t MultiPointCalibForKtBtParam_t::Btemp
```

4.62.1.4 H

```
uint16_t MultiPointCalibForKtBtParam_t::H[MAX_CALI_POINT_NUMBER]
```

4.62.1.5 hotmode

```
uint8_t MultiPointCalibForKtBtParam_t::hotmode
```

4.62.1.6 K

```
uint16_t MultiPointCalibForKtBtParam_t::K[MAX_CALI_POINT_NUMBER]
```

4.62.1.7 Ktemp

```
int16_t MultiPointCalibForKtBtParam_t::Ktemp
```

4.62.1.8 L

```
uint16_t MultiPointCalibForKtBtParam_t::L[MAX_CALI_POINT_NUMBER]
```

4.62.1.9 len

```
uint16_t MultiPointCalibForKtBtParam_t::len
```

4.62.1.10 m

```
uint8_t MultiPointCalibForKtBtParam_t::m
```

4.62.1.11 m_index

```
uint16_t MultiPointCalibForKtBtParam_t::m_index
```

4.62.1.12 n

```
uint8_t MultiPointCalibForKtBtParam_t::n
```

4.62.1.13 P

```
uint16_t MultiPointCalibForKtBtParam_t::P[MAX_CALI_POINT_NUMBER]
```

4.62.1.14 temp

```
float MultiPointCalibForKtBtParam_t::temp[MAX_CALI_POINT_NUMBER]
```

4.62.1.15 Vtemp

```
uint16_t MultiPointCalibForKtBtParam_t::Vtemp[MAX_CALI_POINT_NUMBER]
```

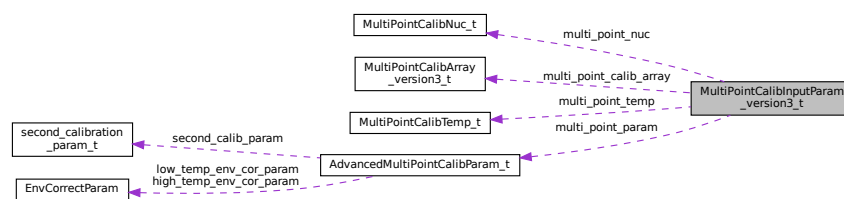
The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.63 MultiPointCalibInputParam_version3_t Struct Reference

```
#include <libirtemp.h>
```

Collaboration diagram for MultiPointCalibInputParam_version3_t:



Public Attributes

- `AdvancedMultiPointCalibParam_t * multi_point_param`
- `MultiPointCalibArray_version3_t * multi_point_calib_array`
- `MultiPointCalibTemp_t * multi_point_temp`
- `MultiPointCalibNuc_t * multi_point_nuc`
- `uint16_t multi_point_num`
- `int product_type`
- `int temp_area`

4.63.1 Member Data Documentation

4.63.1.1 multi_point_calib_array

[MultiPointCalibArray_version3_t*](#) MultiPointCalibInputParam_version3_t::multi_point_calib_array

4.63.1.2 multi_point_nuc

[MultiPointCalibNuc_t*](#) MultiPointCalibInputParam_version3_t::multi_point_nuc

4.63.1.3 multi_point_num

uint16_t MultiPointCalibInputParam_version3_t::multi_point_num

4.63.1.4 multi_point_param

[AdvancedMultiPointCalibParam_t*](#) MultiPointCalibInputParam_version3_t::multi_point_param

4.63.1.5 multi_point_temp

[MultiPointCalibTemp_t*](#) MultiPointCalibInputParam_version3_t::multi_point_temp

4.63.1.6 product_type

int MultiPointCalibInputParam_version3_t::product_type

4.63.1.7 temp_area

int MultiPointCalibInputParam_version3_t::temp_area

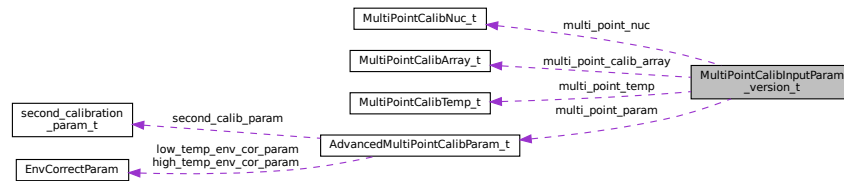
The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.64 MultiPointCalibInputParam_version_t Struct Reference

```
#include <libirtemp.h>
```

Collaboration diagram for MultiPointCalibInputParam_version_t:



Public Attributes

- [AdvancedMultiPointCalibParam_t](#) * multi_point_param
- [MultiPointCalibArray_t](#) * multi_point_calib_array
- [MultiPointCalibTemp_t](#) * multi_point_temp
- [MultiPointCalibNuc_t](#) * multi_point_nuc
- uint16_t multi_point_num
- int product_type

4.64.1 Member Data Documentation

4.64.1.1 multi_point_calib_array

```
MultiPointCalibArray\_t* MultiPointCalibInputParam_version_t::multi_point_calib_array
```

4.64.1.2 multi_point_nuc

```
MultiPointCalibNuc\_t* MultiPointCalibInputParam_version_t::multi_point_nuc
```

4.64.1.3 multi_point_num

```
uint16_t MultiPointCalibInputParam_version_t::multi_point_num
```

4.64.1.4 multi_point_param

```
AdvancedMultiPointCalibParam_t* MultiPointCalibInputParam_version_t::multi_point_param
```

4.64.1.5 multi_point_temp

```
MultiPointCalibTemp_t* MultiPointCalibInputParam_version_t::multi_point_temp
```

4.64.1.6 product_type

```
int MultiPointCalibInputParam_version_t::product_type
```

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.65 MultiPointCalibNuc_t Struct Reference

Temperature information definition.

```
#include <libirtemp.h>
```

Public Attributes

- uint16_t [output_nuc](#)
The camera output nuc value when face object blackbody.
- uint16_t [setting_nuc](#)
The camera setting nuc value corresponding to setting temperature.

4.65.1 Detailed Description

Temperature information definition.

4.65.2 Member Data Documentation

4.65.2.1 output_nuc

```
uint16_t MultiPointCalibNuc_t::output_nuc
```

The camera output nuc value when face object blackbody.

4.65.2.2 setting_nuc

```
uint16_t MultiPointCalibNuc_t::setting_nuc
```

The camera setting nuc value corresponding to setting temperature.

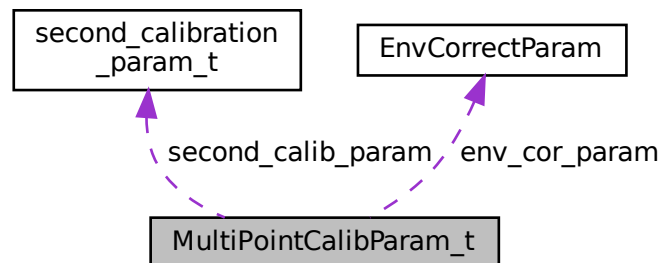
The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.66 MultiPointCalibParam_t Struct Reference

```
#include <libirtemp.h>
```

Collaboration diagram for MultiPointCalibParam_t:



Public Attributes

- [EnvCorrectParam](#) `env_cor_param`
- [second_calibration_param_t](#) `second_calib_param`
- `uint16_t` `cur_vtemp`

4.66.1 Member Data Documentation

4.66.1.1 cur_vtemp

`uint16_t MultiPointCalibParam_t::cur_vtemp`

4.66.1.2 env_cor_param

`EnvCorrectParam MultiPointCalibParam_t::env_cor_param`

4.66.1.3 second_calib_param

`second_calibration_param_t MultiPointCalibParam_t::second_calib_param`

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.67 MultiPointCalibTemp_t Struct Reference

Temperature information definition.

```
#include <libirtemp.h>
```

Public Attributes

- float [output_temp](#)
The camera output temperature when face object blackbody(celcius temperature, unit:celcius)
- float [setting_temp](#)
The object setting temperature(celcius temperature, unit:celcius)

4.67.1 Detailed Description

Temperature information definition.

4.67.2 Member Data Documentation

4.67.2.1 output_temp

```
float MultiPointCalibTemp_t::output_temp
```

The camera output temperature when face object blackbody(celcius temperature, unit:celcius)

4.67.2.2 setting_temp

```
float MultiPointCalibTemp_t::setting_temp
```

The object setting temperature(celcius temperature, unit:celcius)

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.68 NucFactor_t Struct Reference

NUC map factor.

```
#include <libirtemp.h>
```

Public Attributes

- int [P0](#)
The constant term coefficient of T-NUC mapping.
- int [P1](#)
The primary term coefficient of T-NUC mapping.
- int [P2](#)
The quadratic term coefficient of T-NUC mapping.

4.68.1 Detailed Description

NUC map factor.

4.68.2 Member Data Documentation

4.68.2.1 P0

```
int NucFactor_t::P0
```

The constant term coefficient of T-NUC mapping.

4.68.2.2 P1

```
int NucFactor_t::P1
```

The primary term coefficient of T-NUC mapping.

4.68.2.3 P2

```
int NucFactor_t::P2
```

The quadratic term coefficient of T-NUC mapping.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.69 param_attribute_t Struct Reference

attribute of params(the range of value and quantify step size)

```
#include <libircmd.h>
```

Public Attributes

- uint16_t [max_value](#)
- uint16_t [min_value](#)
- uint16_t [step](#)

4.69.1 Detailed Description

attribute of params(the range of value and quantify step size)

4.69.2 Member Data Documentation

4.69.2.1 max_value

```
uint16_t param_attribute_t::max_value
```

4.69.2.2 min_value

```
uint16_t param_attribute_t::min_value
```

4.69.2.3 step

```
uint16_t param_attribute_t::step
```

The documentation for this struct was generated from the following file:

- [libircmd.h](#)

4.70 ReCaliFactor_t Struct Reference

The correction factor after recalibration process.

```
#include <libirtemp.h>
```

Public Attributes

- int [K](#)
The K value of environment estimatimation after double/single point calibration process.
- int [B](#)
The B value of environment estimatimation after double/single point calibration process.

4.70.1 Detailed Description

The correction factor after recalibration process.

4.70.2 Member Data Documentation

4.70.2.1 B

```
int ReCaliFactor_t::B
```

The B value of environment estimatimation after double/single point calibration process.

4.70.2.2 K

```
int ReCaliFactor_t::K
```

The K value of environment estimation after double/single point calibration process.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.71 Rtc_Time_t Struct Reference

rtc time (high accuracy)

```
#include <libircmd.h>
```

Public Attributes

- uint16_t [year](#)
- uint8_t [month](#)
- uint8_t [day](#)
- uint8_t [hour](#)
- uint8_t [minute](#)
- uint8_t [second](#)
- uint16_t [millisecond](#)
- uint16_t [microsecond](#)

4.71.1 Detailed Description

rtc time (high accuracy)

4.71.2 Member Data Documentation

4.71.2.1 day

```
uint8_t Rtc_Time_t::day
```

4.71.2.2 hour

```
uint8_t Rtc_Time_t::hour
```

4.71.2.3 microsecond

```
uint16_t Rtc_Time_t::microsecond
```

4.71.2.4 millisecond

```
uint16_t Rtc_Time_t::millisecond
```

4.71.2.5 minute

```
uint8_t Rtc_Time_t::minute
```

4.71.2.6 month

```
uint8_t Rtc_Time_t::month
```

4.71.2.7 second

```
uint8_t Rtc_Time_t::second
```

4.71.2.8 year

```
uint16_t Rtc_Time_t::year
```

The documentation for this struct was generated from the following file:

- [libircmd.h](#)

4.72 SearchInfo_t Struct Reference

The search information of KT curve.

```
#include <libirtemp.h>
```

Public Attributes

- uint16_t [start_point](#)
Start search point of KT curve.
- uint16_t [end_point](#)
End search point of KT curve.

4.72.1 Detailed Description

The search information of KT curve.

4.72.2 Member Data Documentation

4.72.2.1 end_point

uint16_t SearchInfo_t::end_point

End search point of KT curve.

4.72.2.2 start_point

uint16_t SearchInfo_t::start_point

Start search point of KT curve.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.73 second_calibration_param_t Struct Reference

the parameter of SECOND_CALIBRATION_V1 algorithm

```
#include <libirtemp.h>
```

Public Attributes

- `int16_t K_Temp`
ktemp value of calibration algorithm
- `int16_t B_Temp`
btemp value of calibration algorithm
- `uint16_t Address_CA`
address_ca of calibration algorithm
- `uint16_t kt_len`
kt array length of calibration algorithm
- `uint16_t bt_len`
bt array length of calibration algorithm
- `uint16_t nuc_len`
the length of nuc_t table
- `uint16_t high_vtemp`
the mean vtemp at high temperatures
- `uint16_t low_vtemp`
the mean vtemp at low temperatures
- `float object_temp_high`
the object high temperature (unit: Celsius)
- `float object_temp_low`
the object high temperature (unit: Celsius)
- `float setting_temp_high`
the setting high temperature (unit: Celsius)
- `float setting_temp_low`
the setting low temperature (unit: Celsius)

4.73.1 Detailed Description

the parameter of SECOND_CALIBRATION_V1 algorithm

4.73.2 Member Data Documentation

4.73.2.1 Address_CA

```
uint16_t second_calibration_param_t::Address_CA
```

address_ca of calibration algorithm

4.73.2.2 B_Temp

```
int16_t second_calibration_param_t::B_Temp
```

btemp value of calibration algorithm

4.73.2.3 bt_len

```
uint16_t second_calibration_param_t::bt_len
```

bt array length of calibration algorithm

4.73.2.4 high_vtemp

```
uint16_t second_calibration_param_t::high_vtemp
```

the mean vtemp at high temperatures

4.73.2.5 K_Temp

```
int16_t second_calibration_param_t::K_Temp
```

ktemp value of calibration algorithm

4.73.2.6 kt_len

```
uint16_t second_calibration_param_t::kt_len
```

kt array length of calibration algorithm

4.73.2.7 low_vtemp

```
uint16_t second_calibration_param_t::low_vtemp
```

the mean vtemp at low temperatures

4.73.2.8 nuc_len

```
uint16_t second_calibration_param_t::nuc_len
```

the length of nuc_t table

4.73.2.9 object_temp_high

```
float second_calibration_param_t::object_temp_high
```

the object high temperature (unit: Celsius)

4.73.2.10 object_temp_low

```
float second_calibration_param_t::object_temp_low
```

the object high temperature (unit: Celsius)

4.73.2.11 setting_temp_high

```
float second_calibration_param_t::setting_temp_high
```

the setting high temperature(unit: Celsius)

4.73.2.12 setting_temp_low

```
float second_calibration_param_t::setting_temp_low
```

the setting low temperature (unit: Celsius)

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.74 second_calibration_result_t Struct Reference

the calibration of SECOND_CALIBRATION_V1 algorithm

```
#include <libirtemp.h>
```

Public Attributes

- `uint16_t * kt_array`
kt array of calibration algorithm
- `int16_t * bt_array`
bt array of calibration algorithm

4.74.1 Detailed Description

the calibration of SECOND_CALIBRATION_V1 algorithm

4.74.2 Member Data Documentation

4.74.2.1 bt_array

```
int16_t* second_calibration_result_t::bt_array
```

bt array of calibration algorithm

4.74.2.2 kt_array

```
uint16_t* second_calibration_result_t::kt_array
```

kt array of calibration algorithm

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.75 TempDataRes_t Struct Reference

TempData resolution.

```
#include <libirtemp.h>
```

Public Attributes

- uint16_t [width](#)
resolution width
- uint16_t [height](#)
resolution height

4.75.1 Detailed Description

TempData resolution.

4.75.2 Member Data Documentation

4.75.2.1 height

```
uint16_t TempDataRes_t::height
```

resolution height

4.75.2.2 width

```
uint16_t TempDataRes_t::width
```

resolution width

The documentation for this struct was generated from the following file:

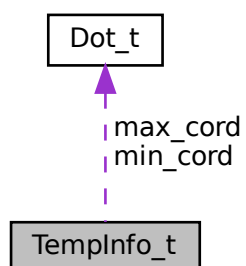
- [libirtemp.h](#)

4.76 TempInfo_t Struct Reference

Temperature information definition.

```
#include <libirtemp.h>
```

Collaboration diagram for TempInfo_t:



Public Attributes

- `uint16_t max_temp`
The maxmium temperature of [Area_t](#) or [Line_t](#).
- `uint16_t min_temp`
The minmium temperature of [Area_t](#) or [Line_t](#).
- `uint16_t avr_temp`
The average temperature of [Area_t](#) or [Line_t](#).
- `Dot_t max_cord`
The maxmium temperature point coordinate of [Area_t](#) or [Line_t](#).
- `Dot_t min_cord`
The minmium temperature point coordinate of [Area_t](#) or [Line_t](#).

4.76.1 Detailed Description

Temperature infomation definition.

4.76.2 Member Data Documentation

4.76.2.1 `avr_temp`

```
uint16_t TempInfo_t::avr_temp
```

The average temperature of [Area_t](#) or [Line_t](#).

4.76.2.2 `max_cord`

```
Dot_t TempInfo_t::max_cord
```

The maxmium temperature point coordinate of [Area_t](#) or [Line_t](#).

4.76.2.3 `max_temp`

```
uint16_t TempInfo_t::max_temp
```

The maxmium temperature of [Area_t](#) or [Line_t](#).

4.76.2.4 min_cord

`Dot_t TempInfo_t::min_cord`

The minmium temperature point coordinate of [Area_t](#) or [Line_t](#).

4.76.2.5 min_temp

`uint16_t TempInfo_t::min_temp`

The minmium temperature of [Area_t](#) or [Line_t](#).

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.77 TempThreshold_t Struct Reference

Temperature Threshold.

```
#include <libirtemp.h>
```

Public Attributes

- int [upper_limit](#)
The upper limit of temperature threshold.
- int [lower_limit](#)
The lower limit of temperature threshold.

4.77.1 Detailed Description

Temperature Threshold.

4.77.2 Member Data Documentation

4.77.2.1 lower_limit

`int TempThreshold_t::lower_limit`

The lower limit of temperature threshold.

4.77.2.2 upper_limit

```
int TempThreshold_t::upper_limit
```

The upper limit of temperature threshold.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.78 TwoPointCalibResult_t Struct Reference

```
#include <libirtemp.h>
```

Public Attributes

- uint16_t * [new_kt_array](#)
- int16_t * [new_bt_array](#)

4.78.1 Member Data Documentation

4.78.1.1 new_bt_array

```
int16_t* TwoPointCalibResult_t::new_bt_array
```

4.78.1.2 new_kt_array

```
uint16_t* TwoPointCalibResult_t::new_kt_array
```

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.79 TwoPointCalibResult_version3_t Struct Reference

```
#include <libirtemp.h>
```

Public Attributes

- uint16_t * [new_kt_array_low](#)
- uint16_t * [new_kt_array_mid](#)
- uint16_t * [new_kt_array_high](#)
- int16_t * [new_bt_array_low](#)
- int16_t * [new_bt_array_mid](#)
- int16_t * [new_bt_array_high](#)

4.79.1 Member Data Documentation

4.79.1.1 new_bt_array_high

```
int16_t* TwoPointCalibResult_version3_t::new_bt_array_high
```

4.79.1.2 new_bt_array_low

```
int16_t* TwoPointCalibResult_version3_t::new_bt_array_low
```

4.79.1.3 new_bt_array_mid

```
int16_t* TwoPointCalibResult_version3_t::new_bt_array_mid
```

4.79.1.4 new_kt_array_high

```
uint16_t* TwoPointCalibResult_version3_t::new_kt_array_high
```

4.79.1.5 new_kt_array_low

```
uint16_t* TwoPointCalibResult_version3_t::new_kt_array_low
```

4.79.1.6 new_kt_array_mid

```
uint16_t* TwoPointCalibResult_version3_t::new_kt_array_mid
```

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.80 UartConDevParams_t Struct Reference

```
#include <libiruart.h>
```

Public Attributes

- int [baudrate](#)
- int [reserved](#) [63]

4.80.1 Member Data Documentation

4.80.1.1 baudrate

```
int UartConDevParams_t::baudrate
```

4.80.1.2 reserved

```
int UartConDevParams_t::reserved[63]
```

The documentation for this struct was generated from the following file:

- [libiruart.h](#)

4.81 UnitRgb_t Struct Reference

```
#include <libirparse.h>
```

Public Attributes

- uint8_t [r](#)
- uint8_t [g](#)
- uint8_t [b](#)

4.81.1 Member Data Documentation

4.81.1.1 b

```
uint8_t UnitRgb_t::b
```

4.81.1.2 g

```
uint8_t UnitRgb_t::g
```

4.81.1.3 r

```
uint8_t UnitRgb_t::r
```

The documentation for this struct was generated from the following file:

- [libirparse.h](#)

4.82 UnitYuv_t Struct Reference

```
#include <libirparse.h>
```

Public Attributes

- [uint8_t y](#)
- [uint8_t u](#)
- [uint8_t v](#)

4.82.1 Member Data Documentation

4.82.1.1 u

```
uint8_t UnitYuv_t::u
```

4.82.1.2 v

```
uint8_t UnitYuv_t::v
```

4.82.1.3 y

```
uint8_t UnitYuv_t::y
```

The documentation for this struct was generated from the following file:

- [libirparse.h](#)

4.83 UpdateFile_t Struct Reference

Information about scattered file in the update package zip.

```
#include <libirupgrade.h>
```

Public Attributes

- char * [update_type](#)
- char * [update_file](#)
- char * [src_file](#)
- char * [init_data_hash](#)
- char * [present_data_hash](#)

4.83.1 Detailed Description

Information about scattered file in the update package zip.

4.83.2 Member Data Documentation

4.83.2.1 init_data_hash

```
char* UpdateFile_t::init_data_hash
```

4.83.2.2 present_data_hash

```
char* UpdateFile_t::present_data_hash
```

4.83.2.3 src_file

```
char* UpdateFile_t::src_file
```

4.83.2.4 update_file

```
char* UpdateFile_t::update_file
```

4.83.2.5 update_type

```
char* UpdateFile_t::update_type
```

The documentation for this struct was generated from the following file:

- [libirupgrade.h](#)

4.84 UpdateFileInfo_t Struct Reference

Information about property file in the update package zip.

```
#include <libirupgrade.h>
```

Public Attributes

- char * [update_file](#)
- char * [src_file](#)
- char * [data_hash](#)

4.84.1 Detailed Description

Information about property file in the update package zip.

4.84.2 Member Data Documentation

4.84.2.1 data_hash

```
char* UpdateFileInfo_t::data_hash
```

4.84.2.2 src_file

```
char* UpdateFileInfo_t::src_file
```

4.84.2.3 update_file

```
char* UpdateFileInfo_t::update_file
```

The documentation for this struct was generated from the following file:

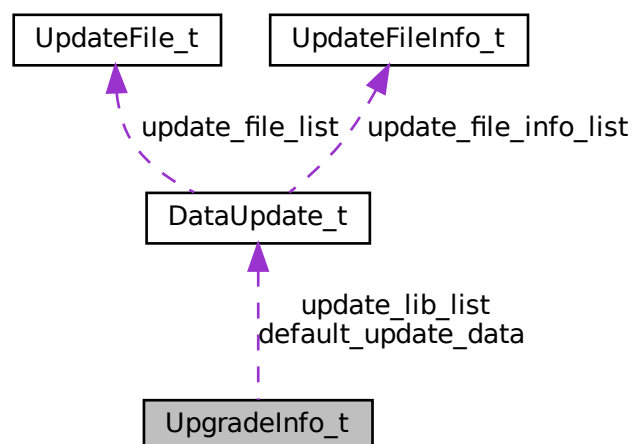
- [libirupgrade.h](#)

4.85 UpgradeInfo_t Struct Reference

Information of version difference in the update package zip.

```
#include <libirupgrade.h>
```

Collaboration diagram for UpgradeInfo_t:



Public Attributes

- char * [generate_time](#)
- char * [device_name](#)
- char * [customer_id](#)
- bool [check_flag](#)
- char * [data_version](#)
- uint32_t [baudrate](#)
- char * [fw_version](#)
- char * [fw_hash](#)
- char * [fw_src_file](#)
- char * [whole_package_version](#)
- char * [whole_version](#)
- bool [has_default_update_data](#)
- int [orgin_data_version_number](#)
- [DataUpdate_t](#) [update_lib_list](#) [UPDATE_LIB_LIST_MAX_NUM]
- [DataUpdate_t](#) [default_update_data](#)

4.85.1 Detailed Description

Information of version difference in the update package zip.

4.85.2 Member Data Documentation

4.85.2.1 baudrate

```
uint32_t UpgradeInfo_t::baudrate
```

4.85.2.2 check_flag

```
bool UpgradeInfo_t::check_flag
```

4.85.2.3 customer_id

```
char* UpgradeInfo_t::customer_id
```

4.85.2.4 data_version

`char* UpgradeInfo_t::data_version`

4.85.2.5 default_update_data

`DataUpdate_t UpgradeInfo_t::default_update_data`

4.85.2.6 device_name

`char* UpgradeInfo_t::device_name`

4.85.2.7 fw_hash

`char* UpgradeInfo_t::fw_hash`

4.85.2.8 fw_src_file

`char* UpgradeInfo_t::fw_src_file`

4.85.2.9 fw_version

`char* UpgradeInfo_t::fw_version`

4.85.2.10 generate_time

`char* UpgradeInfo_t::generate_time`

4.85.2.11 has_default_update_data

`bool UpgradeInfo_t::has_default_update_data`

4.85.2.12 orgin_data_version_number

```
int UpgradeInfo_t::orgin_data_version_number
```

4.85.2.13 update_lib_list

```
DataUpdate_t UpgradeInfo_t::update_lib_list[UPDATE_LIB_LIST_MAX_NUM]
```

4.85.2.14 whole_package_version

```
char* UpgradeInfo_t::whole_package_version
```

4.85.2.15 whole_version

```
char* UpgradeInfo_t::whole_version
```

The documentation for this struct was generated from the following file:

- [libirupgrade.h](#)

4.86 UserCallback_t Struct Reference

User's callback function and parameters.

```
#include <libiruvcl.h>
```

Public Attributes

- [IruvcHandle_t](#) * [iruvc_handle](#)
libiruvcl's handle
- void * [usr_func](#)
user's function
- void * [usr_param](#)
parameters of user's function

4.86.1 Detailed Description

User's callback function and parameters.

4.86.2 Member Data Documentation

4.86.2.1 iruvc_handle

[IruvcHandle_t*](#) UserCallback_t::iruvc_handle

libiruvc's handle

4.86.2.2 usr_func

void* UserCallback_t::usr_func

user's function

4.86.2.3 usr_param

void* UserCallback_t::usr_param

parameters of user's function

The documentation for this struct was generated from the following file:

- [libiruvc.h](#)

4.87 UvcConDevParam_t Struct Reference

```
#include <libiruvc.h>
```

Public Attributes

- int [reserved](#) [64]

4.87.1 Member Data Documentation

4.87.1.1 reserved

```
int UvcConDevParam_t::reserved[64]
```

The documentation for this struct was generated from the following file:

- [libiruv.h](#)

4.88 V4l2ConDevParams_t Struct Reference

```
#include <libirv4l2.h>
```

Public Attributes

- int [reserved](#) [64]

4.88.1 Member Data Documentation

4.88.1.1 reserved

```
int V4l2ConDevParams_t::reserved[64]
```

The documentation for this struct was generated from the following file:

- [libirv4l2.h](#)

4.89 VideoOutputInfo_t Struct Reference

video output info

```
#include <libircmd.h>
```

Public Attributes

- int [video_output_status](#)
- int [video_output_format](#)
- int [video_output_fps](#)
- int [video_output_num](#)

4.89.1 Detailed Description

video output info

4.89.2 Member Data Documentation

4.89.2.1 video_output_format

```
int VideoOutputInfo_t::video_output_format
```

4.89.2.2 video_output_fps

```
int VideoOutputInfo_t::video_output_fps
```

4.89.2.3 video_output_num

```
int VideoOutputInfo_t::video_output_num
```

4.89.2.4 video_output_status

```
int VideoOutputInfo_t::video_output_status
```

The documentation for this struct was generated from the following file:

- [libircmd.h](#)

4.90 VtempInfo_t Struct Reference

Some vtemp constants for vtemp convert to true temperature.

```
#include <libirtemp.h>
```

Public Attributes

- [int16_t Ktemp](#)
Ktemp.
- [int16_t Btemp](#)
Btemp.
- [uint16_t Address_CA](#)
Address_CA.

4.90.1 Detailed Description

Some vtemp constants for vtemp convert to true temperature.

4.90.2 Member Data Documentation

4.90.2.1 Address_CA

```
uint16_t VtempInfo_t::Address_CA
```

Address_CA.

4.90.2.2 Btemp

```
int16_t VtempInfo_t::Btemp
```

Btemp.

4.90.2.3 Ktemp

```
int16_t VtempInfo_t::Ktemp
```

Ktemp.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.91 VTempSet_t Struct Reference

The vtemp of sensor in double point calibration process.

```
#include <libirtemp.h>
```

Public Attributes

- int [VTemp_High](#)
The vtemp of sensor when calibratie high environment temperature in double point calibration process.
- int [VTemp_Low](#)
The vtemp of sensor when calibratie low environment temperature in double point calibration process.

4.91.1 Detailed Description

The vtemp of sensor in double point calibration process.

4.91.2 Member Data Documentation

4.91.2.1 VTemp_High

```
int VTempSet_t::VTemp_High
```

The vtemp of sensor when calibratie high environment temperature in double point calibration process.

4.91.2.2 VTemp_Low

```
int VTempSet_t::VTemp_Low
```

The vtemp of sensor when calibratie low environment temperature in double point calibration process.

The documentation for this struct was generated from the following file:

- [libirtemp.h](#)

4.92 WholePackage_t Struct Reference

Information of target version in the whole package zip.

```
#include <libirupgrade.h>
```

Public Attributes

- char * [generate_time](#)
- char * [device_name](#)
- char * [customer_id](#)
- bool [check_flag](#)
- char * [fw_version](#)
- char * [fw_hash](#)
- char * [fw_src_file](#)
- char * [data_version](#)
- char * [data_hash](#)
- char * [data_src_file](#)
- char * [whole_package_version](#)
- char * [whole_version](#)

4.92.1 Detailed Description

Information of target version in the whole package zip.

4.92.2 Member Data Documentation

4.92.2.1 check_flag

```
bool WholePackage_t::check_flag
```

4.92.2.2 customer_id

```
char* WholePackage_t::customer_id
```

4.92.2.3 data_hash

```
char* WholePackage_t::data_hash
```

4.92.2.4 data_src_file

```
char* WholePackage_t::data_src_file
```

4.92.2.5 data_version

```
char* WholePackage_t::data_version
```

4.92.2.6 device_name

```
char* WholePackage_t::device_name
```

4.92.2.7 fw_hash

```
char* WholePackage_t::fw_hash
```

4.92.2.8 fw_src_file

```
char* WholePackage_t::fw_src_file
```

4.92.2.9 fw_version

```
char* WholePackage_t::fw_version
```

4.92.2.10 generate_time

```
char* WholePackage_t::generate_time
```

4.92.2.11 whole_package_version

```
char* WholePackage_t::whole_package_version
```

4.92.2.12 whole_version

```
char* WholePackage_t::whole_version
```

The documentation for this struct was generated from the following file:

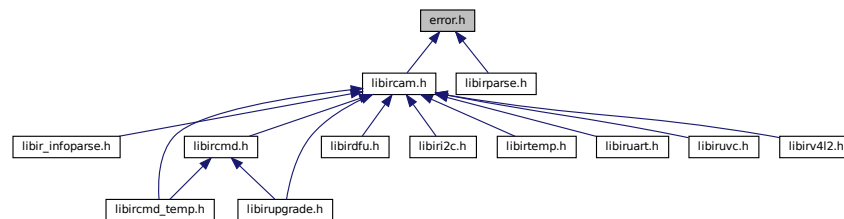
- [libirupgrade.h](#)

Chapter 5

File Documentation

5.1 error.h File Reference

This graph shows which files directly or indirectly include this file:



Enumerations

- enum `IrlibError_e` {
 `IRLIB_SUCCESS` = 0, `IRCAM_PARAM_ERROR` = -100, `IRCAM_MEMORY_ALLOC_FAILED` = -101,
 `IRCMD_PARAM_ERROR` = -200,
 `IRCMD_MEMORY_ALLOC_FAILED` = -201, `IRCMD_CRC_CHECK_FAILED` = -202, `IRCMD_READ_FUNC_EXECUTE_FAILED` = -203, `IRCMD_WRITE_FUNC_EXECUTE_FAILED` = -204,
 `IRCMD_CMD_EXECUTE_FAILED` = -205, `IRCMD_CMD_UNSUPPORTED` = -206, `IRDFU_PARAM_ERROR` = -300, `IRDFU_MEMORY_ALLOC_FAILED` = -301,
 `IRDFU_FIND_DFU_DEV_FAILED` = -302, `IRI2C_PARAM_ERROR` = -400, `IRI2C_MEMORY_ALLOC_FAILED` = -401, `IRI2C_CRC_CHECK_FAILED` = -402,
 `IRI2C_CRC16_CMP_FAILED` = -403, `IRI2C_FUNCTION_UNSUPPORTED` = -404, `IRI2C_CONTROL_TRANSFER_FAILED` = -405, `IRI2C_SET_CMD_HEADER_OR_DATA_FAILED` = -406,
 `IRI2C_GET_CMD_DATA_FAILED` = -407, `IRI2C_FILL_CMD_HEADER_FAILED` = -408, `IRI2C_FETCH_CMD_HEADER_FAILED` = -409, `IRI2C_READ_ONE_TRANSFER_DATA_FAILED` = -410,
 `IRI2C_WRITE_ONE_TRANSFER_DATA_FAILED` = -411, `IRI2C_WRITE_CMD_WITHOUT_RETURN_FAILED` = -412, `IRI2C_GET_STATUS_FAILED` = -413, `IRI2C_GET_CMD_STATUS_TIMEOUT` = -414,
 `IRI2C_READ_STD_CMD_FAILED` = -415, `IRI2C_WRITE_STD_CMD_FAILED` = -416, `IRI2C_WRITE_STD_CMD_WITHOUT_RETURN_FAILED` = -417, `IRI2C_CMD_EXECUTE_STATUS_ERROR` = -418,
 `IRI2C_CMD_EXECUTE_FAILED` = -419, `IRUART_PARAM_ERROR` = -500, `IRUART_MEMORY_ALLOC_FAILED` = -501, `IRUART_CRC_CHECK_FAILED` = -502,
 `IRUART_CRC16_CMP_FAILED` = -503, `IRUART_FUNCTION_UNSUPPORTED` = -504, `IRUART_CONTROL_TRANSFER_FAILED` = -505,
}

= -505, IRUART_CONTROL_TRANSFER_WRITE_FAILED = -506,
IRUART_CONTROL_TRANSFER_READ_TIMEOUT = -507, IRUART_MAGIC_NUM_CHECK_FAILED =
-508, IRUART_OPEN_DEV_FAILED = -509, IRUART_CLOSE_DEV_FAILED = -510,
IRUART_SETCOMMSTATE_FAILED = -511, IRUART_GETCOMMSTATE_FAILED = -512, IRUART_SETUPCOMM_FAILED
= -513, IRUART_CLOSECOMMERROR_FAILED = -514,
IRUART_TCSETATTR_FAILED = -515, IRUART_TCGETATTR_FAILED = -516, IRUART_GET_STATUS_FAILED
= -517, IRUART_CMD_EXECUTE_STATUS_ERROR = -518,
IRUART_CMD_EXECUTE_FAILED = -519, IRUART_SET_CMD_HEADER_OR_DATA_FAILED = -520,
IRUART_GET_CMD_DATA_FAILED = -521, IRUART_FIRMWARE_DOWNLOAD_FAILED = -522,
IRUART_SET_BAUDRATE_FAILED = -523, IRUVC_PARAM_ERROR = -600, IRUVC_MEMORY_ALLOC_FAILED
= -601, IRUVC_CRC_CHECK_FAILED = -602,
IRUVC_CRC16_CMP_FAILED = -603, IRUVC_IO_ERROR = -604, IRUVC_ACCESS_ERROR = -605,
IRUVC_FIND_DEVICE_FAILED = -606,
IRUVC_NOT_FOUND_ERROR = -607, IRUVC_RESOURCE_BUSY = -608, IRUVC_OPERATION_TIMEOUT
= -609, IRUVC_OVERFLOW_ERROR = -610,
IRUVC_PIPE_ERROR = -611, IRUVC_INTERRUPTED_ERROR = -612, IRUVC_NO_MEM_ERROR = -613,
IRUVC_OPERATION_UNSUPPORTED = -614,
IRUVC_UVC_INIT_FAILED = -615, IRUVC_GET_DEV_LIST_FAILED = -616, IRUVC_GET_DEV_INFO_FAILED
= -617, IRUVC_OPEN_DEV_FAILED = -618,
IRUVC_GET_DEV_INDEX_FAILED = -619, IRUVC_GET_DEV_DESCRIPTOR_FAILED = -620, IRUVC_DEV_OPENED
= -621, IRUVC_GET_FORMAT_FAILED = -622,
IRUVC_USER_CALLBACK_EMPTY = -623, IRUVC_START_STREAMING_FAILED = -624, IRUVC_GET_FRAME_TIMEOUT
= -625, IRUVC_CONTROL_TRANSFER_FAILED = -626,
IRUVC_GET_STATUS_FAILED = -627, IRUVC_SET_CMD_HEADER_OR_DATA_FAILED = -628,
IRUVC_GET_CMD_DATA_FAILED = -629, IRUVC_CHECK_DONE_FAILED = -630,
IRUVC_DETECT_DEVICE_FAILED = -631, IRUVC_INVALID_DEV_ERROR = -632, IRUVC_INVALID_MODE_ERROR
= -633, IRUVC_CALLBACK_EXISTS_ERROR = -634,
IRUVC_STATUS_ERROR = -635, IRUVC_OTHER_ERROR = -636, IRV4L2_PARAM_ERROR = -700,
IRV4L2_MEMORY_ALLOC_FAILED = -701,
IRV4L2_CRC_CHECK_FAILED = -702, IRV4L2_CRC16_CMP_FAILED = -703, IRV4L2_FUNCTION_UNSUPPORTED
= -704, IRV4L2_CONTROL_TRANSFER_FAILED = -705,
IRV4L2_OPEN_VIDEO_NODE_FAILED = -706, IRV4L2_CLOSE_VIDEO_NODE_FAILED = -707,
IRV4L2_IOCTL_VIDIOC_QUERYCAP_FAILED = -708, IRV4L2_IOCTL_VIDIOC_REQBUFS_FAILED = -
709,
IRV4L2_IOCTL_VIDIOC_DQBUF_FAILED = -710, IRV4L2_IOCTL_VIDIOC_QBUF_FAILED = -711,
IRV4L2_MMAP_FAILED = -712, IRV4L2_IOCTL_VIDIOC_S_FMT_FAILED = -713,
IRV4L2_IOCTL_VIDIOC_G_FMT_FAILED = -714, IRV4L2_IOCTL_VIDIOC_S_PARAM_FAILED = -715,
IRV4L2_IOCTL_VIDIOC_G_PARAM_FAILED = -716, IRV4L2_IOCTL_VIDIOC_STREAMON_FAILED =
-717,
IRV4L2_SELECT_FRAME_FAILED = -718, IRV4L2_SELECT_FRAME_TIMEOUT = -719, IRV4L2_OPEN_CONTROL_NODE
= -720, IRV4L2_CLOSE_CONTROL_NODE_FAILED = -721,
IRV4L2_SET_CMD_HEADER_OR_DATA_FAILED = -722, IRV4L2_GET_CMD_DATA_FAILED = -723,
IRV4L2_GET_STATUS_FAILED = -724, IRV4L2_CMD_EXECUTE_STATUS_ERROR = -725,
IRV4L2_DEVICE_STATUS_NOT_IDLE = -726, IRV4L2_GET_CMD_STATUS_TIMEOUT = -727,
IRV4L2_CMD_EXECUTE_FAILED = -728, IRUPGRADE_PARAM_ERROR = -900,
IRUPGRADE_MEMORY_ALLOC_FAILED = -901, IRUPGRADE_OPEN_CFG_FILE_FAILED = -902,
IRUPGRADE_CLOSE_CFG_FILE_FAILED = -903, IRUPGRADE_OPEN_DEV_FAILED = -904,
IRUPGRADE_CLOSE_DEV_FAILED = -905, IRUPGRADE_CHARACTER_CHECK_FAILED = -906,
IRUPGRADE_HASH_CHECK_FAILED = -907, IRUPGRADE_VERSION_CHECK_FAILED = -908,
IRUPGRADE_PACKAGE_CHECK_FAILED = -909, IRUPGRADE_CMD_CHANNEL_UNSUPPORTED = -
910, IRUPGRADE_RECORD_DEV_STATUS_FAILED = -911, IRUPGRADE_READ_JSON_DATA_FAILED
= -912,
IRUPGRADE_GET_ZIP_INFO_FAILED = -913, IRUPGRADE_GET_DEV_INFO_FAILED = -914,
IRUPGRADE_SET_DATA_VERSION_FAILED = -915, IRUPGRADE_DOWNLOAD_FIRMWARE_FAILED =
-916,
IRUPGRADE_ENTER_DATA_UPGRADE_MODE_FAILED = -917, IRUPGRADE_LEAVE_DATA_UPGRADE_MODE_FAILED
= -918, IRUPGRADE_PARSE_DEV_INFO_FAILED = -919, IRUPGRADE_PARSE_WHOLE_PACKAGE_JSON_FAILED
= -920,


```

IRUPGRADE_PARSE_UPDATE_PACKAGE_JSON_FAILED = -921, IRUPGRADE_CALC_FILE_HASH_FAILED
= -922, IRUPGRADE_WRITE_SINGLE_FILE_FAILED = -923, IRUPGRADE_UPGRADE_FAILED = -924,
IRUPGRADE_READ_SINGLE_FILE_FAILED = -925, IRUPGRADE_UART_DEVICE_SET_PARAM_FAILED
= -926, IRPARSE_PARAM_ERROR = -1000, IRTEMP_PARAM_ERROR = -1100,
IRTEMP_MEMORY_ALLOC_FAILED = -1101, IRTEMP_OPERATION_FILE_FAILED = -1102, IRTEMP_GET_PIXEL_SQUAR
= -1103, IRTEMP_GET_AVERAGE_VALUE_FAILED = -1104,
IRTEMP_GET_PIXEL_HORZ_NEIGHB_FAILED = -1105, IRTEMP_GET_PIXEL_VERT_NEIGHB_FAILED
= -1106, IRTEMP_GET_PIXEL_SLOPE_NEIGHB_FAILED = -1107, IRTEMP_GET_POINT_OF_SLOPE_LINE_FAILED
= -1108,
IRTEMP_GET_HORZ_LINE_FAILED = -1109, IRTEMP_GET_VERT_LINE_FAILED = -1110, IRTEMP_GET_SLOPE_LINE_FA
= -1111, IRTEMP_REVERSE_TEMP_FAILED = -1112,
IRTEMP_REVERSE_NUC_VALUE_FAILED = -1113, IRTEMP_CALC_ENV_FACTOR_FAILED = -1114,
IRTEMP_GET_KTBT_INDEX_FAILED = -1115, IRTEMP_CORRECT_KTBT_FAILED = -1116,
IRTEMP_UPDATE_KTBT_BY_DOUBLE_POINT_FAILED = -1117, IRTEMP_CALC_SITF_AND_OFFSET_FAILED
= -1118, IRTEMP_CALC_SITF_AND_OFFSET_BEFORE_KTBT_FAILED = -1119, IRTEMP_GET_HUM_INDEX_FAILED
= -1120,
IRTEMP_GET_TEMP_INDEX_FAILED = -1121, IRTEMP_GET_DIST_INDEX_FAILED = -1122, IRTEMP_GET_EMS_FAILED
= -1123, IRTEMP_GET_TAU_FAILED = -1124,
IRTEMP_RECALC_TAU_FAILED = -1125, IRTEMP_READ_TAU_FAILED = -1126, IRTEMP_CORRECT_TEMP_FAILED
= -1127, IRTEMP_REVERSE_TEMP_CORRECT_FAILED = -1128,
IRTEMP_CALC_VAPOR_PRESSURE_FAILED = -1129, IRTEMP_CALC_COEF_FAILED = -1130,
IRTEMP_CALC_MULTI_POINT_FAILED = -1131, IRTEMP_LINEARINTERP_FAILED = -1132,
IRTEMP_REMAP_TEMP_FAILED = -1133, IRPROC_PARAM_ERROR = -1200, IRPROC_MEMORY_ALLOC_FAILED
= -1201, IRPROC_FMT_UNSUPPORTED = -1202,
IRPROC_GET_HISTOGRAM_FAILED = -1203, IRPROC_GET_AGC_LIMIT_FAILED = -1204, IRPROC_GET_LIMITED_ACC
= -1205, IRPROC_GET_NEW_MAP_FAILED = -1206,
IRPROC_GET_GAUSS_KERNEL_FAILED = -1207, IRPROC_GAUSSIAN_FILTER_FAILED = -1208,
IRPROC_CALC_DDE_COEF_FAILED = -1209, IRPROC_EXTRACT_DETAIL_FAILED = -1210,
IRPROC_CONTROL_AUTO_GAIN_FAILED = -1211, IRPROC_ENHANCE_DIGITAL_DETAIL_FAILED =
-1212, IRINFOPARSE_PARAM_ERROR = -1300, IRINFOPARSE_CRC_CHECK_FAILED = -13401,
IRINFOPARSE_CRC8_CMP_FAILED = -1302, IRCMDCHANNEL_PARAM_ERROR = -1400, IRCMDCHANNEL_INIT_FAILED
= -1401, IRCMDCHANNEL_UPGRADE_FAILED = -1402,
IRCMDCHANNEL_SET_VALUE_FAILED = -1403, IRCMDCHANNEL_GET_VALUE_FAILED = -1404,
IRCMDCHANNEL_CAMERA_NOT_INIT = -1405, IRCMDCHANNEL_CAMERA_START_FAILED = -1406,
IRCMDCHANNEL_CAMERA_STOP_FAILED = -1407, IRCMDCHANNEL_CAMERA_RELEASE_FAILED = -
1408, IRCMDCHANNEL_SET_TARGET_DETECT_OPEN_FAILED = -1409, IRCMDCHANNEL_OPEN_SHUTTER_FAILED
= -1410,
IRCMDCHANNEL_CLOSED_SHUTTER_FAILED = -1411, IRCMDCHANNEL_UPDATE_SHUTTER_FAILED
= -1412 }

```

Error type in libraries For the return value of the error report, the first two indicate different libraries, and the last two indicate the type of error. For example -201, 2 means the error of libircmd library, 01 means the error type is dynamic alloc memory failed. For example -1000, 10 means the error of libirupgrade library, 00 means the error type is parameter error.

5.1.1 Enumeration Type Documentation

5.1.1.1 IrlibError_e

```
enum IrlibError_e
```

Error type in libraries For the return value of the error report, the first two indicate different libraries, and the last two indicate the type of error. For example -201, 2 means the error of libircmd library, 01 means the error type is dynamic alloc memory failed. For example -1000, 10 means the error of libirupgrade library, 00 means the error type is parameter error.

Enumerator

IRLIB_SUCCESS	Function execute success
IRCAM_PARAM_ERROR	Invalid parameter
IRCAM_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRCMD_PARAM_ERROR	Invalid parameter
IRCMD_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRCMD_CRC_CHECK_FAILED	Crc check failed
IRCMD_READ_FUNC_EXECUTE_FAILED	Read data function execute failed
IRCMD_WRITE_FUNC_EXECUTE_FAILED	Write data function execute failed
IRCMD_CMD_EXECUTE_FAILED	Command execute failed
IRCMD_CMD_UNSUPPORTED	Command not supported
IRDFU_PARAM_ERROR	Invalid parameter
IRDFU_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRDFU_FIND_DFU_DEV_FAILED	Find DFU capable USB device failed
IRI2C_PARAM_ERROR	Invalid parameter
IRI2C_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRI2C_CRC_CHECK_FAILED	Crc check failed
IRI2C_CRC16_CMP_FAILED	Crc16 compare failed
IRI2C_FUNCTION_UNSUPPORTED	Function not supported
IRI2C_CONTROL_TRANSFER_FAILED	Device control transfer error
IRI2C_SET_CMD_HEADER_OR_DATA_FAILED	Set command header or data failed
IRI2C_GET_CMD_DATA_FAILED	Get command data failed
IRI2C_FILL_CMD_HEADER_FAILED	Fill command header failed
IRI2C_FETCH_CMD_HEADER_FAILED	Fetch command header failed
IRI2C_READ_ONE_TRANSFER_DATA_FAILED	Read one transfer data failed
IRI2C_WRITE_ONE_TRANSFER_DATA_FAILED	Write one transfer data failed
IRI2C_WRITE_CMD_WITHOUT_RETURN_FAILED	Write command with no return status failed
IRI2C_GET_STATUS_FAILED	Get command execute status failed
IRI2C_GET_CMD_STATUS_TIMEOUT	Get command execute status timeout
IRI2C_READ_STD_CMD_FAILED	Standard read command execute failed
IRI2C_WRITE_STD_CMD_FAILED	Standard write command execute failed
IRI2C_WRITE_STD_CMD_WITHOUT_RETURN_FAILED	Standard write command with no status return execute failed
IRI2C_CMD_EXECUTE_STATUS_ERROR	Command execution result is error
IRI2C_CMD_EXECUTE_FAILED	Command execute failed
IRUART_PARAM_ERROR	Invalid parameter
IRUART_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRUART_CRC_CHECK_FAILED	Crc check failed
IRUART_CRC16_CMP_FAILED	Crc16 compare failed
IRUART_FUNCTION_UNSUPPORTED	Function not supported
IRUART_CONTROL_TRANSFER_READ_FAILED	Device read control transfer failed
IRUART_CONTROL_TRANSFER_WRITE_FAILED	Device write control transfer failed
IRUART_CONTROL_TRANSFER_READ_TIMEOUT	Device read control transfer timeout

Enumerator

IRUART_MAGIC_NUM_CHECK_FAILED	Magic number verify failed
IRUART_OPEN_DEV_FAILED	Open device failed
IRUART_CLOSE_DEV_FAILED	Close device failed
IRUART_SETCOMMSTATE_FAILED	Set com configuration failed in windows
IRUART_GETCOMMSTATE_FAILED	Get com configuration failed in windows
IRUART_SETUPCOMM_FAILED	Set the read/write buffer failed
IRUART_CLOSECOMMERROR_FAILED	Clear com failed in libiruart
IRUART_TCSETATTR_FAILED	Set com configuration failed in linux
IRUART_TCGETATTR_FAILED	Get com configuration failed in linux
IRUART_GET_STATUS_FAILED	Get command execute status failed
IRUART_CMD_EXECUTE_STATUS_ERROR	Command execution result is error
IRUART_CMD_EXECUTE_FAILED	Command execute failed
IRUART_SET_CMD_HEADER_OR_DATA_FAILED	Set command header or data failed
IRUART_GET_CMD_DATA_FAILED	Get command data failed
IRUART_FIRMWARE_DOWNLOAD_FAILED	firmware download failed
IRUART_SET_BAUDRATE_FAILED	set baudrate failed
IRUVC_PARAM_ERROR	Parameters error
IRUVC_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRUVC_CRC_CHECK_FAILED	Crc check failed
IRUVC_CRC16_CMP_FAILED	Crc16 compare failed
IRUVC_IO_ERROR	Error io
IRUVC_ACCESS_ERROR	Error access
IRUVC_FIND_DEVICE_FAILED	Find device failed
IRUVC_NOT_FOUND_ERROR	Entity not found
IRUVC_RESOURCE_BUSY	Resource busy
IRUVC_OPERATION_TIMEOUT	Operation timed out
IRUVC_OVERFLOW_ERROR	Overflow
IRUVC_PIPE_ERROR	Pipe error
IRUVC_INTERRUPTED_ERROR	System call interrupted
IRUVC_NO_MEM_ERROR	Insufficient memory
IRUVC_OPERATION_UNSUPPORTED	Operation not supported
IRUVC_UVC_INIT_FAILED	Libirvc's service context initialize failed
IRUVC_GET_DEV_LIST_FAILED	Get device list failed
IRUVC_GET_DEV_INFO_FAILED	Get device info failed
IRUVC_OPEN_DEV_FAILED	Open device fail
IRUVC_GET_DEV_INDEX_FAILED	Get device's index failed
IRUVC_GET_DEV_DESCRIPTOR_FAILED	Get device's descriptor failed
IRUVC_DEV_OPENED	Device is already opened
IRUVC_GET_FORMAT_FAILED	Device doesn't provide a matching stream
IRUVC_USER_CALLBACK_EMPTY	User's callback function is empty
IRUVC_START_STREAMING_FAILED	Device start streaming failed
IRUVC_GET_FRAME_TIMEOUT	Over time when getting frame
IRUVC_CONTROL_TRANSFER_FAILED	Control transfer failed
IRUVC_GET_STATUS_FAILED	Get command execute status failed
IRUVC_SET_CMD_HEADER_OR_DATA_FAILED	Set command header or data failed
IRUVC_GET_CMD_DATA_FAILED	Get command data failed
IRUVC_CHECK_DONE_FAILED	Checking vdcmd send failed
IRUVC_DETECT_DEVICE_FAILED	Detect device failed

Enumerator

IRUVC_INVALID_DEV_ERROR	Device is not UVC-compliant
IRUVC_INVALID_MODE_ERROR	Mode not supported
IRUVC_CALLBACK_EXISTS_ERROR	Resource has a callback (can't use polling and async)
IRUVC_STATUS_ERROR	Status error
IRUVC_OTHER_ERROR	Undefined error
IRV4L2_PARAM_ERROR	Invalid parameter
IRV4L2_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRV4L2_CRC_CHECK_FAILED	Crc check failed
IRV4L2_CRC16_CMP_FAILED	Crc16 compare failed
IRV4L2_FUNCTION_UNSUPPORTED	Function not supported
IRV4L2_CONTROL_TRANSFER_FAILED	Device control transfer error
IRV4L2_OPEN_VIDEO_NODE_FAILED	Open device video node failed
IRV4L2_CLOSE_VIDEO_NODE_FAILED	Close device video node failed
IRV4L2_IOCTL_VIDIOC_QUERYCAP_FAILED	ioctl VIDIOC_QUERYCAP failed
IRV4L2_IOCTL_VIDIOC_REQBUFS_FAILED	ioctl VIDIOC_REQBUFS failed
IRV4L2_IOCTL_VIDIOC_DQBUF_FAILED	ioctl VIDIOC_DQBUF failed
IRV4L2_IOCTL_VIDIOC_QBUF_FAILED	ioctl VIDIOC_QBUF failed
IRV4L2_MMAP_FAILED	Mmap failed
IRV4L2_IOCTL_VIDIOC_S_FMT_FAILED	ioctl VIDIOC_S_FMT failed
IRV4L2_IOCTL_VIDIOC_G_FMT_FAILED	ioctl VIDIOC_G_FMT failed
IRV4L2_IOCTL_VIDIOC_S_PARAM_FAILED	ioctl VIDIOC_S_PARAM failed
IRV4L2_IOCTL_VIDIOC_G_PARAM_FAILED	ioctl VIDIOC_G_PARAM failed
IRV4L2_IOCTL_VIDIOC_STREAMON_FAILED	Start stream failed
IRV4L2_SELECT_FRAME_FAILED	Get frame data failed
IRV4L2_SELECT_FRAME_TIMEOUT	Get frame data timeout
IRV4L2_OPEN_CONTROL_NODE_FAILED	Open device control node failed
IRV4L2_CLOSE_CONTROL_NODE_FAILED	Open device control node failed
IRV4L2_SET_CMD_HEADER_OR_DATA_FAILED	Set command header or data failed
IRV4L2_GET_CMD_DATA_FAILED	Get command data failed
IRV4L2_GET_STATUS_FAILED	Get command execute status failed
IRV4L2_CMD_EXECUTE_STATUS_ERROR	Command execution result is error
IRV4L2_DEVICE_STATUS_NOT_IDLE	Device status is not idle
IRV4L2_GET_CMD_STATUS_TIMEOUT	Get command execute status timeout
IRV4L2_CMD_EXECUTE_FAILED	Command execute failed
IRUPGRADE_PARAM_ERROR	Invalid parameter
IRUPGRADE_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRUPGRADE_OPEN_CFG_FILE_FAILED	Open config file failed
IRUPGRADE_CLOSE_CFG_FILE_FAILED	Close config file failed
IRUPGRADE_OPEN_DEV_FAILED	Open device failed
IRUPGRADE_CLOSE_DEV_FAILED	Close device failed
IRUPGRADE_CHARACTER_CHECK_FAILED	Character check failed
IRUPGRADE_HASH_CHECK_FAILED	Hash check failed
IRUPGRADE_VERSION_CHECK_FAILED	Version check failed
IRUPGRADE_PACKAGE_CHECK_FAILED	Package check failed
IRUPGRADE_CMD_CHANNEL_UNSUPPORTED	Command channel unsupported
IRUPGRADE_RECORD_DEV_STATUS_FAILED	Record device status failed
IRUPGRADE_READ_JSON_DATA_FAILED	Read whole package json data failed

Enumerator

IRUPGRADE_GET_ZIP_INFO_FAILED	Get information from zip package failed
IRUPGRADE_GET_DEV_INFO_FAILED	Get information from device failed
IRUPGRADE_SET_DATA_VERSION_FAILED	Modified data version failed
IRUPGRADE_DOWNLOAD_FIRMWARE_FAILED	Download firmware failed
IRUPGRADE_ENTER_DATA_UPGRADE_MODE↔ _FAILED	Enter data update mode failed
IRUPGRADE_LEAVE_DATA_UPGRADE_MODE↔ FAILED	Leave data update mode failed
IRUPGRADE_PARSE_DEV_INFO_FAILED	Parse device information failed
IRUPGRADE_PARSE_WHOLE_PACKAGE_JSON↔ N_FAILED	Parse whole package json failed
IRUPGRADE_PARSE_UPDATE_PACKAGE_JSON↔ N_FAILED	Parse update package json failed
IRUPGRADE_CALC_FILE_HASH_FAILED	Calculate file hash failed
IRUPGRADE_WRITE_SINGLE_FILE_FAILED	Write single file failed
IRUPGRADE_UPGRADE_FAILED	Upgrade failed
IRUPGRADE_READ_SINGLE_FILE_FAILED	Read single file failed
IRUPGRADE_UART_DEVICE_SET_PARAM_FAI↔ LED	uart device set param failed
IRPARSE_PARAM_ERROR	Invalid parameter
IRTEMP_PARAM_ERROR	Invalid parameter
IRTEMP_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRTEMP_OPERATION_FILE_FAILED	File operation failed
IRTEMP_GET_PIXEL_SQUARE_FAILED	Get pixel square array
IRTEMP_GET_AVERAGE_VALUE_FAILED	Get array average value failed
IRTEMP_GET_PIXEL_HORZ_NEIGHB_FAILED	Get pixel horizontal neighb array failed
IRTEMP_GET_PIXEL_VERT_NEIGHB_FAILED	Get pixel vertical neighb array failed
IRTEMP_GET_PIXEL_SLOPE_NEIGHB_FAILED	Get pixel slope neighb array failed
IRTEMP_GET_POINT_OF_SLOPE_LINE_FAILED	Get point set of slope line failed
IRTEMP_GET_HORZ_LINE_FAILED	Get horizontal line temperture failed
IRTEMP_GET_VERT_LINE_FAILED	Get vertical line temperture failed
IRTEMP_GET_SLOPE_LINE_FAILED	Get slope line temperture failed
IRTEMP_REVERSE_TEMP_FAILED	Reverse enhance distance temp correct failed
IRTEMP_REVERSE_NUC_VALUE_FAILED	Reverse nuc value with nuc table failed
IRTEMP_CALC_ENV_FACTOR_FAILED	Calculate temperature environment factor with distance and environment temperature failed
IRTEMP_GET_KTBT_INDEX_FAILED	Get index of kt array and bt array according vtemp failed
IRTEMP_CORRECT_KTBT_FAILED	Ktbt correct process failed
IRTEMP_UPDATE_KTBT_BY_DOUBLE_POINT↔ FAILED	Update kt and bt array by double point failed
IRTEMP_CALC_SITF_AND_OFFSET_FAILED	Calculate sitf and offset failed in libirtemp
IRTEMP_CALC_SITF_AND_OFFSET_BEFORE↔ KTBT_FAILED	Calculate target sitf and offset before ktbt correct failed
IRTEMP_GET_HUM_INDEX_FAILED	Get humidity read/write index depend on humidity failed
IRTEMP_GET_TEMP_INDEX_FAILED	Get temperature read/write index depend on environment temperature failed

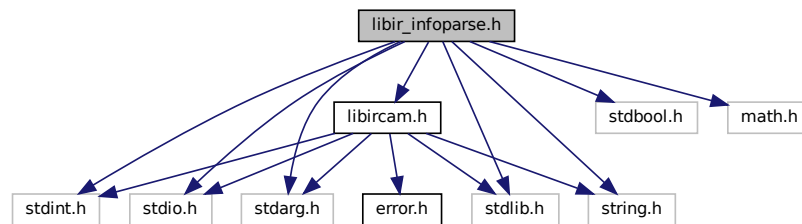
Enumerator

IRTEMP_GET_DIST_INDEX_FAILED	Get distance read/write index depend on distance failed
IRTEMP_GET_EMS_FAILED	Get compatible ems failed
IRTEMP_GET_TAU_FAILED	Get compatible tau with target temperature and distance failed
IRTEMP_RECALC_TAU_FAILED	Recalculate tau failed
IRTEMP_READ_TAU_FAILED	Read tau failed
IRTEMP_CORRECT_TEMP_FAILED	Temperature correct failed
IRTEMP_REVERSE_TEMP_CORRECT_FAILED	Reverse temperature correct failed
IRTEMP_CALC_VAPOR_PRESSURE_FAILED	Calculate vapor pressure failed
IRTEMP_CALC_COEF_FAILED	Calculate coefficient failed
IRTEMP_CALC_MULTI_POINT_FAILED	Calculate new nuc table by multi-point
IRTEMP_LINEARINTERP_FAILED	LinearInterp failed
IRTEMP_REMAP_TEMP_FAILED	Remap temp failed
IRPROC_PARAM_ERROR	Invalid parameter
IRPROC_MEMORY_ALLOC_FAILED	Dynamic alloc memory failed
IRPROC_FMT_UNSUPPORTED	Format unsupported
IRPROC_GET_HISTOGRAM_FAILED	Get histogram result of image_y14
IRPROC_GET_AGC_LIMIT_FAILED	Get agc limit parameters via image_hist failed
IRPROC_GET_LIMITED_ACC_HISTOGRAM_FAILED	Get limited accumulative histogram result failed
IRPROC_GET_NEW_MAP_FAILED	Get new map list of the cumulative histogram failed
IRPROC_GET_GAUSS_KERNEL_FAILED	Get gauss kernel parameters failed
IRPROC_GAUSSIAN_FILTER_FAILED	Gaussian filter failed
IRPROC_CALC_DDE_COEF_FAILED	Dde coefficient calculate failed
IRPROC_EXTRACT_DETAIL_FAILED	Y14 image extract detail failed
IRPROC_CONTROL_AUTO_GAIN_FAILED	y14 image auto gain control failed
IRPROC_ENHANCE_DIGITAL_DETAIL_FAILED	Y14 image digital detail enhance failed
IRINFOPARSE_PARAM_ERROR	Invalid parameter
IRINFOPARSE_CRC_CHECK_FAILED	Crc check failed
IRINFOPARSE_CRC8_CMP_FAILED	Crc16 compare failed
IRCMDCHANNEL_PARAM_ERROR	Invalid parameter
IRCMDCHANNEL_INIT_FAILED	init failed
IRCMDCHANNEL_UPGRADE_FAILED	firmware upgrade failed
IRCMDCHANNEL_SET_VALUE_FAILED	set value failed
IRCMDCHANNEL_GET_VALUE_FAILED	get value failed
IRCMDCHANNEL_CAMERA_NOT_INIT	camera not init
IRCMDCHANNEL_CAMERA_START_FAILED	camera start failed
IRCMDCHANNEL_CAMERA_STOP_FAILED	camera stop failed
IRCMDCHANNEL_CAMERA_RELEASE_FAILED	camera release failed
IRCMDCHANNEL_SET_TARGET_DETECT_OPEN_FAILED	set target detect open failed
IRCMDCHANNEL_OPEN_SHUTTER_FAILED	open shutter failed
IRCMDCHANNEL_CLOSED_SHUTTER_FAILED	closed shutter failed
IRCMDCHANNEL_UPDATE_SHUTTER_FAILED	update shutter failed

5.2 libir_infoparse.h File Reference

```
#include <stdint.h>
#include <stdio.h>
#include <stdarg.h>
#include <stdlib.h>
#include <stdbool.h>
#include <string.h>
#include <math.h>
#include "libircam.h"
```

Include dependency graph for libir_infoparse.h:



Classes

- struct [linfoDeviceInfo_t](#)
- struct [linfoFrameSta_t](#)
- struct [linfoImageSta_t](#)
- struct [linfoTpdSta_t](#)
- struct [linfoShutterSta_t](#)
- struct [linfoRegSta_t](#)
- struct [linfoDeviceSta_t](#)
- struct [linfoPoint_t](#)
- struct [linfoKBCInfo_t](#)
- struct [linfoFrameTpd_t](#)
- struct [linfoPixelTpd_t](#)
- struct [linfoLineRectTpd_t](#)
- struct [linfoNewPixelTpd_t](#)

New temperature measurement points at the software assignable information.

- struct [linfoNewLineRectTpd_t](#)

New temperature measurement rectangle at the software assignable information.

- struct [linfoShapeTpd_t](#)
- struct [linfoKBInfo_t](#)

KB correction module.

- struct [linfoAFInfo_t](#)

Autofocus Module.

- struct [linfoSNRInfo_t](#)

SNR Module.

- struct [linfoAGCInfo_t](#)

AGC Module.

- struct [linfoTpdInfo_t](#)
- struct [linfoTestInfo_t](#)
- struct [linfoStatusInfo_t](#)
- struct [linfoFuncInfo_t](#)

Macros

- `#define _CRT_SECURE_NO_WARNINGS`

Enumerations

- enum `IrinfoparseLogLevels_e` { `IRINFOPARSE_LOG_DEBUG` = 0, `IRINFOPARSE_LOG_ERROR` = 1, `IRINFOPARSE_LOG_NO_PRINT` = 2 }
Log level definition in libir_info parse library.
- enum `IrinfoFrameFormat_e` { `YUV422_FORMAT` = 0, `Y14_FORMAT`, `NV12_FORMAT` }
- enum `IrinfoFrameMode_e` { `ONLY_IMAGE_MODE` = 0, `ONLY_TEMP_MODE`, `IMAGE_AND_TEMP_MODE` }
- enum `IrinfoFrameType_e` { `REAL_TIME_IMAGE` = 0, `FREEZE_IMAGE` }
- enum `IrinfoTempType_e` { `DEGREES_CELSIUS_TEMP` = 0, `FAHRENHEIT_TEMP`, `KELVIN_TEMP` }

Functions

- const DLLEXPORT char * `irinfoparse_version` (void)
Get libir_info parse library's version number.
- DLLEXPORT void `irinfoparse_log_register` (enum `IrinfoparseLogLevels_e` log_level, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)
register log level for libir_info parse library
- DLLEXPORT `IrlibError_e` `irinfoparse_get_irinfo_status_info` (uint8_t *info_data, `IrinfoStatusInfo_t` *status_info)
Get status info from information line.
- DLLEXPORT `IrlibError_e` `irinfoparse_get_irinfo_test_info` (uint8_t *info_data, `IrinfoTestInfo_t` *test_info)
Get test info from information line.
- DLLEXPORT `IrlibError_e` `irinfoparse_get_irinfo_function_info` (uint8_t *info_data, `IrinfoFuncInfo_t` *function_info)
Get function info from information line.

5.2.1 Macro Definition Documentation

5.2.1.1 _CRT_SECURE_NO_WARNINGS

```
#define _CRT_SECURE_NO_WARNINGS
```

5.2.2 Enumeration Type Documentation

5.2.2.1 IrinfoFrameFormat_e

```
enum IrinfoFrameFormat_e
```


Enumerator

YUV422_FORMAT	
Y14_FORMAT	
NV12_FORMAT	

5.2.2.2 IrinfoFrameMode_e

```
enum IrinfoFrameMode_e
```

Enumerator

ONLY_IMAGE_MODE	
ONLY_TEMP_MODE	
IMAGE_AND_TEMP_MODE	

5.2.2.3 IrinfoFrameType_e

```
enum IrinfoFrameType_e
```

Enumerator

REAL_TIME_IMAGE	
FREEZE_IMAGE	

5.2.2.4 IrinfoparseLogLevels_e

```
enum IrinfoparseLogLevels_e
```

Log level definition in libir_infoparse library.

Enumerator

IRINFOPARSE_LOG_DEBUG	print debug and error information
IRINFOPARSE_LOG_ERROR	only print error information
IRINFOPARSE_LOG_NO_PRINT	don't print debug and error information

5.2.2.5 IrinfoTempType_e

enum [IrinfoTempType_e](#)

Enumerator

DEGREES_CELSIUS_TEMP	
FAHRENHEIT_TEMP	
KELVIN_TEMP	

5.2.3 Function Documentation

5.2.3.1 irinfoparse_get_irinfo_function_info()

```
DLLEXPORT IrlibError\_e irinfoparse_get_irinfo_function_info (
    uint8_t * info_data,
    IrinfoFuncInfo\_t * function_info )
```

Get function info from information line.

Parameters

in	<i>info_data</i>	information line
in	<i>function_info</i>	

5.2.3.2 irinfoparse_get_irinfo_status_info()

```
DLLEXPORT IrlibError\_e irinfoparse_get_irinfo_status_info (
    uint8_t * info_data,
    IrinfoStatusInfo\_t * status_info )
```

Get status info from information line.

Parameters

in	<i>info_data</i>	information line
in	<i>status_info</i>	

5.2.3.3 irinfoparse_get_irinfo_test_info()

```
DLLEXPORT IrlibError\_e irinfoparse_get_irinfo_test_info (
```

```
uint8_t * info_data,
IrinfoTestInfo_t * test_info )
```

Get test info from information line.

Parameters

in	<i>info_data</i>	information line
in	<i>test_info</i>	

5.2.3.4 irinfoparse_log_register()

```
DLL_EXPORT void irinfoparse_log_register (
    enum IrinfoparseLogLevels_e log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

register log level for libir_infoparse library

Parameters

in	<i>log_level</i>	log level
in	<i>callback</i>	the callback function for printing log messages
in	<i>priv_data</i>	the private parameter of callback function

5.2.3.5 irinfoparse_version()

```
const DLL_EXPORT char* irinfoparse_version (
    void )
```

Get libir_infoparse library's version number.

Returns

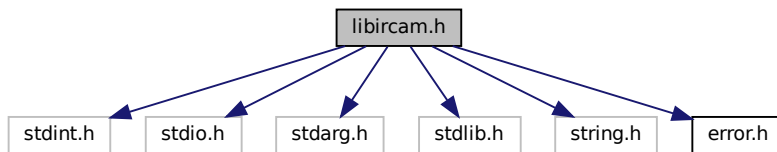
version number's string

5.3 libircam.h File Reference

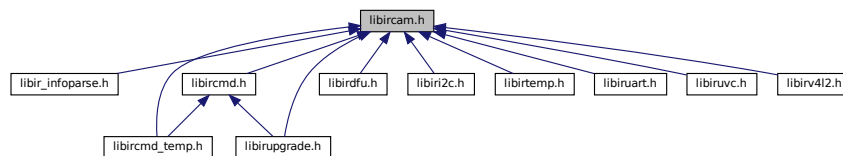
```
#include <stdint.h>
#include <stdio.h>
#include <stdarg.h>
#include <stdlib.h>
#include <string.h>
```

```
#include "error.h"
```

Include dependency graph for libircam.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct [IrVideoHandle_t](#)
- struct [IrControlHandle_t](#)

Macros

- [#define _CRT_SECURE_NO_WARNINGS](#)

Enumerations

- enum [ircam_log_level_t](#) { [IRCAM_LOG_DEBUG](#) = 0, [IRCAM_LOG_ERROR](#) = 1, [IRCAM_LOG_NO_PRINT](#) = 2 }
- Log level definition in libircam library.*
- enum [command_channel_e](#) { [COMMAND_CHANNEL_IDLE](#) = 0, [UVC_USB_COMMAND_CHANNEL](#) = 1, [V4L2_USB_COMMAND_CHANNEL](#) = 2, [UART_COMMAND_CHANNEL](#) = 3, [UVC_USB_I2C_COMMAND_CHANNEL](#) = 4, [V4L2_USB_I2C_COMMAND_CHANNEL](#) = 5, [I2C_COMMAND_CHANNEL](#) = 6 }
- the kind of command channel*

Functions

- DLLEXPORT void [ircam_log_register](#) ([ircam_log_level_t](#) log_level, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)
- const DLLEXPORT char * [ircam_version](#) (void)
- const DLLEXPORT char * [ircam_version_number](#) (void)
- DLLEXPORT [IrlibError_e](#) [ir_video_handle_create](#) ([IrVideoHandle_t](#) **ir_video_handle)
- DLLEXPORT [IrlibError_e](#) [ir_video_handle_delete](#) ([IrVideoHandle_t](#) **ir_video_handle)
- DLLEXPORT [IrlibError_e](#) [ir_control_handle_create](#) ([IrControlHandle_t](#) **ir_control_handle)
- DLLEXPORT [IrlibError_e](#) [ir_control_handle_delete](#) ([IrControlHandle_t](#) **ir_control_handle)

5.3.1 Macro Definition Documentation

5.3.1.1 _CRT_SECURE_NO_WARNINGS

```
#define _CRT_SECURE_NO_WARNINGS
```

5.3.2 Enumeration Type Documentation

5.3.2.1 command_channel_e

```
enum command\_channel\_e
```

the kind of command channel

Enumerator

COMMAND_CHANNEL_IDLE	
UVC_USB_COMMAND_CHANNEL	
V4L2_USB_COMMAND_CHANNEL	
UART_COMMAND_CHANNEL	
UVC_USB_I2C_COMMAND_CHANNEL	
V4L2_USB_I2C_COMMAND_CHANNEL	
I2C_COMMAND_CHANNEL	

5.3.2.2 ircam_log_level_t

```
enum ircam\_log\_level\_t
```

Log level definition in libircam library.

Enumerator

IRCAM_LOG_DEBUG	print debug and error information
IRCAM_LOG_ERROR	only print error information
IRCAM_LOG_NO_PRINT	don't print debug and error information

5.3.3 Function Documentation

5.3.3.1 ir_control_handle_create()

```
DLLEXPORT IrlibError_e ir_control_handle_create (
    IrControlHandle_t ** ir_control_handle )
```

5.3.3.2 ir_control_handle_delete()

```
DLLEXPORT IrlibError_e ir_control_handle_delete (
    IrControlHandle_t ** ir_control_handle )
```

5.3.3.3 ir_video_handle_create()

```
DLLEXPORT IrlibError_e ir_video_handle_create (
    IrVideoHandle_t ** ir_video_handle )
```

5.3.3.4 ir_video_handle_delete()

```
DLLEXPORT IrlibError_e ir_video_handle_delete (
    IrVideoHandle_t ** ir_video_handle )
```

5.3.3.5 ircam_log_register()

```
DLLEXPORT void ircam_log_register (
    ircam_log_level_t log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

5.3.3.6 ircam_version()

```
const DLLEXPORT char* ircam_version (  
    void )
```

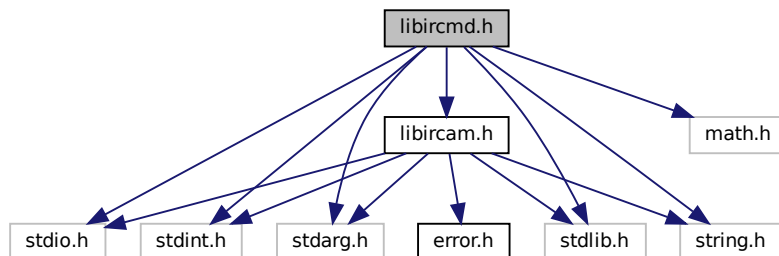
5.3.3.7 ircam_version_number()

```
const DLLEXPORT char* ircam_version_number (  
    void )
```

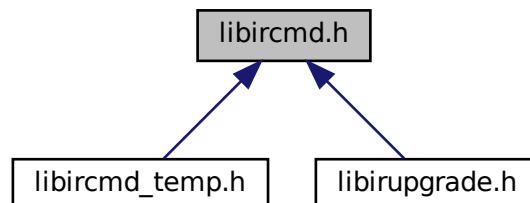
5.4 libircmd.h File Reference

```
#include <stdio.h>  
#include <stdint.h>  
#include <stdarg.h>  
#include <math.h>  
#include <stdlib.h>  
#include <string.h>  
#include "libircam.h"
```

Include dependency graph for libircmd.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct [IrcmdPoint_t](#)
Point struct, start from 0.
- struct [IrcmdRect_t](#)
Rectangle struct, start from 1.
- struct [MaxMinTemplInfo_t](#)
Maximum and minimum temperature information struct.
- struct [LineRectTemplInfo_t](#)
Temperature information struct of a line or a rectangle.
- struct [param_attribute_t](#)
attribute of params(the range of value and quantify step size)
- struct [Rtc_Time_t](#)
rtc time (high accuracy)
- struct [VideoOutputInfo_t](#)
video output info

Macros

- #define [COMMAND_IN_DEVELOPING](#)
- #define [DDE_MAX_LEVEL](#) 100
- #define [SNR_MAX_LEVEL](#) 100
- #define [AGC_MAX_LEVEL](#) 100
- #define [YUV_MAX_FORMAT](#) 3
- #define [TU_TA_MIN_LEVEL](#) 233
- #define [TU_TA_MAX_LEVEL](#) 373
- #define [TAU_EMS_MIN_LEVEL](#) 1
- #define [TAU_EMS_MAX_LEVEL](#) 16384

Typedefs

- typedef struct [_IrcmdHandle_t](#) [IrcmdHandle_t](#)

Enumerations

- enum [IrcmdLogLevel_e](#) { [IRCMD_LOG_DEBUG](#) = 0, [IRCMD_LOG_ERROR](#) = 1, [IRCMD_LOG_NO_PRINT](#) = 2 }
Log level definition in libircmd library.
- enum [callback_mode_e](#) { [UPGRADE_PROCESS_CALLBACK](#) = 0 }
location of callback function calls
- enum [basic_auto_ffc_status_e](#) { [BASIC_AUTO_FFC_DISABLE](#) = 0, [BASIC_AUTO_FFC_ENABLE](#) = 1, [BASIC_AUTO_FFC_SEMI_DISABLE](#) = 2 }
auto shutter status
- enum [basic_auto_ffc_param_e](#) { [BASIC_AUTO_TEMP_THRESHOLD](#) = 0, [BASIC_AUTO_MIN_INTERVAL](#) = 1, [BASIC_AUTO_MAX_INTERVAL](#) = 2 }
auto shutter parameters types
- enum [adv_shutter_status_e](#) { [ADV_SHUTTER_CLOSE_STA](#) = 0, [ADV_SHUTTER_OPEN_STA](#) = 1, [ADV_SHUTTER_UNKNOW_STA](#) = 2 }
current shutter switching status

- enum `basic_all_ffc_status_e` { `BASIC_ALL_FFC_FUNCTION_AVAILABLE` = 0, `BASIC_ALL_FFC_FUNCTION_UNAVAILABLE` = 1, `BASIC_ALL_FFC_FUNCTION_CANCEL` = 2 }
Whether all shutter functions are available (include auto shutter, manual shutter)
- enum `basic_ffc_shutter_behavior_mode_e` { `BASIC_ONLY_B_UPDATE`, `BASIC_ONLY_OOC_UPDATE`, `BASIC_BOTH_B_AND_OOC_UPDATE` }
Property page's shutter behavior mode.
- enum `basic_dev_info_type_e` {
`BASIC_DEV_NAME` = 1, `BASIC_DEV_FW_RELEASE_VERSION` = 2, `BASIC_DEV_CUSTOMER_ID` = 3,
`BASIC_DEV_VENDOR_ID` = 4,
`BASIC_DEV_PRODUCT_ID` = 5, `BASIC_DEV_INFO_GET_PN` = 6, `BASIC_DEV_INFO_GET_SN` = 7,
`BASIC_DEV_ISP_VERSION` = 8,
`BASIC_DEV_ABSTRACT_VERSION` = 9, `BASIC_DEV_SYSTEM_VERSION` = 10, `BASIC_DEV_CALI_VERSION`
= 11, `BASIC_DEV_WHOLE_DATA_VERSION` = 12,
`BASIC_DEV_WHOLE_PACKAGE_VERSION` = 13, `BASIC_DEV_SENSOR_TYPE` = 14, `BASIC_DEV_SENSOR_ID`
= 15 }
the kind of basic device info
- enum `adv_realtime_status_type_e` { `ADV_IR_SENSOR_VTEMP` = 0, `ADV_IR_FPC_TEMP` = 1 }
device realtime status type
- enum `basic_device_status_e` {
`BASIC_STARTUP_MODE` = 0, `BASIC_PREVIEW_MODE` = 1, `BASIC_VIDEO_OUT_MODE` = 2,
`BASIC_UPDATE_MODE` = 3,
`BASIC_DEVICE_ERROR_MODE` = 4 }
current status of device status
- enum `basic_enable_status_e` { `BASIC_DISABLE` = 0, `BASIC_ENABLE` = 1 }
the status of enable
- enum `adv_voltage_type_e` { `ADV_LOW_VOLTAGE` = 0, `ADV_HIGH_VOLTAGE` = 1 }
the type of voltage
- enum `driver_type_e` { `IDLE_DRIVER` = 0, `MIPI_DRIVER` = 1, `USB_DRIVER` = 2, `DVP_DRIVER` = 3 }
driver's type
- enum `adv_mipi_channel_type_e` { `ADV_DOUBLIE_VC_CHANNEL` = 0, `ADV_SINGLE_VC_CHANNEL` = 1 }
Vc channel type when the device gets stream for MIPI.
- enum `adv_stream_source_mode_e` {
`ADV_IR_SOURCE_MODE` = 0, `ADV_KBC_SOURCE_MODE` = 1, `ADV_TNR_SOURCE_MODE` = 2,
`ADV_HBC_DPC_SOURCE_MODE` = 3,
`ADV_VBC_SOURCE_MODE` = 4, `ADV_SNR_SOURCE_MODE` = 5, `ADV_DDE_SOURCE_MODE` = 6,
`ADV_AGC_SOURCE_MODE` = 7,
`ADV_GAMMA_SOURCE_MODE` = 8, `ADV_TPD_SOURCE_MODE` = 9, `ADV_MIRROR_SOURCE_MODE`
= 10, `ADV_PICTURE_SOURCE_MODE` = 11 }
Y16 mode source stream modes.
- enum `adv_output_frame_rate_e` { `ADV_LOW_RATE` = 30, `ADV_HIGH_RATE` = 60 }
the rate of output frame
- enum `adv_mini2256_output_frame_rate_e` { `ADV_RATE_25` = 25, `ADV_RATE_50` = 50 }
the rate of output frame for mini2256
- enum `basic_boot_logo_status_e` { `BASIC_BOOT_LOGO_DISABLE` = 0, `BASIC_BOOT_LOGO_ENABLE` = 1 }
boot logo status
- enum `adv_YUV_format_e` { `ADV_UYVY` = 0, `ADV_VYUY` = 1, `ADV_YUYV` = 2, `ADV_YVYU` = 3 }
the format of YUV
- enum `adv_video_output_format_e` {
`ADV_USB_FORMAT` = 0, `ADV_DVP_FORMAT` = 1, `ADV_BT656_FORMAT` = 2, `ADV_MIPI_FORMAT` = 3,
`ADV_BT656_INTERLACED_FORMAT` = 4, `ADV_NTSC_FORMAT` = 5, `ADV_PAL_FORMAT` = 6,
`ADV_VO_SPI_FORMAT` = 7 }
the format of video output ADV_BT656: Line by line form ADV_BT656_INTERLACED: Interlaced form
- enum `video_output_num_e` { `SINGLE_VIDEO` = 0, `DOUBLE_VIDEO` = 1 }

- the number of video output*
 - enum `adv_preview_info_e` { `ADV_CURRENT_FPS` = 1, `ADV_CURRENT_WIDTH` = 2, `ADV_CURRENT_HEIGHT` = 3 }
 - preview information*
 - enum `basic_mirror_and_flip_status_e` { `BASIC_NO_MIRROR_OR_FLIP` = 0, `BASIC_ONLY_MIRROR`, `BASIC_ONLY_FLIP`, `BASIC_BOTH_MIRROR_AND_FLIP` }
 - image flip and mirror's type*
 - enum `basic_roi_level_e` { `BASIC_ROI_DISABLE` = 0, `BASIC_ROI_THIRD` = 1, `BASIC_ROI_HALF` = 2 }
 - the level of roi*
 - enum `basic_image_scene_mode_e` { `BASIC_DEFAULT_MODE` = 0, `BASIC_PATROL_MODE` = 1, `BASIC_MANUAL_MODE` = 2, `BASIC_LOW_TEMP_MODE` = 3, `BASIC_LINEAR_MODE` = 4, `BASIC_LOW_CONTRAST_MODE` = 5, `BASIC_HIGH_CONTRAST_MODE` = 6, `BASIC_HIGH_LIGHT_MODE` = 7, `BASIC_RESERVE1_MODE` = 8, `BASIC_RESERVE2_MODE` = 9, `BASIC_RESERVE3_MODE` = 10, `BASIC_RESERVE4_MODE` = 11, `BASIC_RESCUE_MODE` = 12, `BASIC_URBAN_MODE` = 13, `BASIC_TEMP_MEASURE_MODE` = 14, `BASIC_OUTDOORS_MODE` = 15 }
 - The mode of the image scene.*
 - enum `adv_profession_mode_e` { `ADV_NORMAL_MODE` = 0, `ADV_PROFESSIONAL_MODE` = 1 }
 - The professional mode.*
 - enum `adv_algorithm_module_e` { `KBC_BYPASS` = 1, `TNR_BYPASS` = 2, `HBC_BYPASS` = 3, `VBC_BYPASS` = 4, `SNR1_BYPASS` = 5, `AGC_BYPASS` = 6, `DDE_BYPASS` = 7, `GAMMA_BYPASS` = 8, `CROSS_BYPASS` = 9, `PIC_MIRROR_BYPASS` = 10, `TMP_MIRRRPR_BYPASS` = 11 }
 - algorithm module*
 - enum `adv_algorithm_module_switch_status_e` { `ADV_ALGORITHM_MODULE_NORMAL` = 0, `ADV_ALGORITHM_MODULE_SWITCH` = 1 }
 - algorithm module switch status*
 - enum `adv_edge_enhance_level_e` { `ADV_ZEROTH_LEVEL` = 0, `ADV_FIRST_LEVEL` = 1, `ADV_SECOND_LEVEL` = 2 }
 - The level of enhance.*
 - enum `adv_isothermal_status_e` { `BASIC_ISOTHERMAL_DISABLE` = 0, `BASIC_ISOTHERMAL_ENABLE` = 1 }
 - isothermal switch status*
 - enum `adv_isothermal_mode_e` { `ADV_IOSTHERMAL_MODE_0`, `ADV_IOSTHERMAL_MODE_1`, `ADV_IOSTHERMAL_MODE_2`, `ADV_IOSTHERMAL_MODE_3`, `ADV_IOSTHERMAL_MODE_4`, `ADV_IOSTHERMAL_MODE_5` }
 - the mode of pseudo color*
 - enum `adv_second_calib_status_e` { `ADV_SECOND_CALI_DISABLE` = 0, `ADV_SECOND_CALI_ENABLE` = 1 }
 - the status of second cali flag*
 - enum `adv_second_calib_flag_status_e` { `ADV_SECOND_CALI_FLAG_TEC_K` = 0, `ADV_SECOND_CALI_FLAG_KT` = 1, `ADV_SECOND_CALI_FLAG_BT` = 2, `ADV_SECOND_CALI_FLAG_RMV` = 3, `ADV_SECOND_CALI_FLAG_NUC_T` = 4, `ADV_SECOND_CALI_FLAG_DPC` = 5 }
 - the type of second cali flag_index*
 - enum `adv_K_calib_status_e` { `BASIC_LOW_TEMP_OBJECT` = 0, `BASIC_HIGH_TEMP_OBJECT` = 1, `BASIC_CALCULATION` = 2 }
 - calib K value*
 - enum `adv_cursor_switch_status_e` { `ADV_CURSOR_DISABLE` = 0, `ADV_CURSOR_ENABLE` = 1 }
 - cursor switch status*
 - enum `basic_gain_status_e` { `BASIC_LOW_GAIN` = 0, `BASIC_HIGH_GAIN`, `BASIC_2LOW_GAIN`, `BASIC_2HIGH_GAIN` }

- the device's gain value*
- enum [adv_info_line_index_e](#) {
[ADV_SW_MODEL_INFO](#) = 0, [ADV_SW_FRAME_INFO](#) = 1, [ADV_SW_MODEL_STATUS](#) = 3,
[ADV_SW_SHUTTER_STATUS](#) = 4,
[ADV_SW_AUTO_SHUTTER_STATUS](#) = 5, [ADV_SW_ISP_INFO_VERSION](#) = 6, [ADV_SW_TPD_FORMAT](#)
= 7, [ADV_SW_MIRROR_ENALBE](#) = 8,
[ADV_SW_FLIP_ENALBE](#) = 9, [ADV_SW_TEMP_AREA](#) = 10 }
- the index of information line*
- enum [basic_sun_detect_status_e](#) { [BASIC_SUN_DETECT_DISABLE](#) = 0, [BASIC_SUN_DETECT_ENABLE](#)
= 1 }
- sun detect switch status*
- enum [basic_restore_type_e](#) {
[BASIC_AI_RECAL_DATA](#) = 0, [BASIC_RESTORE_ALGORITHM_PARAMETERS](#) = 1, [BASIC_RESTORE_SYSTEM_PARAMETERS](#)
= 2, [BASIC_RESTORE_K_VALUE](#) = 3,
[BASIC_RESTORE_DPC_DATA](#) = 4, [BASIC_RESTORE_RMCOVER_DATA](#) = 5, [BASIC_RESTORE_TPD_DATA](#)
= 6, [BASIC_RESTORE_PROFESSION_SCENE_LEVEL](#) = 7,
[BASIC_RESTORE_DOUBLE_TPD_DATA](#) = 8 }
- the type of data to restore*
- enum [basic_save_type_e](#) {
[BASIC_SAVE_ALGORITHM_PARAMETERS](#) = 0, [BASIC_SAVE_SYSTEM_PARAMETERS](#) = 1, [BASIC_SAVE_K_VALUE](#)
= 2, [BASIC_SAVE_DPC_DATA](#) = 3,
[BASIC_SAVE_RMCOVER_DATA](#) = 4, [BASIC_SAVE_TPD_DATA](#) = 5, [BASIC_SAVE_VIDEO_OUTPUT_FORMAT](#)
= 6, [BASIC_SAVE_PROFESSION_SCENE_LEVEL](#) = 7,
[BASIC_SAVE_DOUBLE_TPD_DATA](#) = 8 }
- the type of data to save*

Functions

- const DLLEXPORT char * [ircmd_version](#) (void)
Get libircmd library's version.
- const DLLEXPORT char * [ircmd_version_number](#) (void)
Get libircmd library's version number.
- DLLEXPORT void [ircmd_log_register](#) ([IrcmdLogLevel_e](#) log_level, void *(*callback)(void *callback_data,
void *priv_data), void *priv_data)
Register log print via log_level.
- DLLEXPORT [IrcmdHandle_t](#) * [ircmd_create_handle](#) ([IrcmdHandle_t](#) *ir_control_handle)
Create and initialize [ircmd](#)'s handle.
- DLLEXPORT [IrcmdError_e](#) [ircmd_delete_handle](#) ([IrcmdHandle_t](#) *handle)
Delete and release [ircmd](#)'s handle.
- DLLEXPORT [IrcmdError_e](#) [vdcmd_set_polling_wait_time](#) ([IrcmdHandle_t](#) *handle, uint16_t timeout_ms)
Set single command polling time.
- DLLEXPORT [IrcmdError_e](#) [ircmd_init_callback_function](#) ([IrcmdHandle_t](#) *handle, int mode, void *(*callback)(void
*callback_data, void *priv_data), void *priv_data)
Init callback function.
- DLLEXPORT [IrcmdError_e](#) [ircmd_delete_callback_function](#) ([IrcmdHandle_t](#) *handle, int mode)
Delete callback function.
- DLLEXPORT [IrcmdError_e](#) [ircmd_get_percentage_of_data_transfer](#) ([IrcmdHandle_t](#) *handle, float *percentage)
Get percentage of data transfer.
- DLLEXPORT [IrcmdError_e](#) [basic_ffc_update](#) ([IrcmdHandle_t](#) *handle)
Manual shutter update.
- DLLEXPORT [IrcmdError_e](#) [basic_auto_ffc_status_set](#) ([IrcmdHandle_t](#) *handle, int ffc_status)
Set auto shutter status.

- DLLEXPORT [IrlibError_e basic_auto_ffc_status_get](#) ([IrcmdHandle_t](#) *handle, int *ffc_status)
Get auto shutter status.
- DLLEXPORT [IrlibError_e basic_auto_ffc_current_params_set](#) ([IrcmdHandle_t](#) *handle, int param_type, int value)
Set auto FFC's parameter if param_type is BASIC_AUTO_MAX_INTERVAL, value must more than 5 maximun interval must lower than minimun interval.
- DLLEXPORT [IrlibError_e basic_auto_ffc_current_params_get](#) ([IrcmdHandle_t](#) *handle, int param_type, int *value)
Get current auto FFC's parameter.
- DLLEXPORT [IrlibError_e basic_auto_ffc_params_attribute_get](#) ([IrcmdHandle_t](#) *handle, int param_type, [param_attribute_t](#) *attribute)
Get attribute of auto FFC's parameter.
- DLLEXPORT [IrlibError_e adv_shutter_status_get](#) ([IrcmdHandle_t](#) *handle, int *status)
Get current shutter switching status.
- DLLEXPORT [IrlibError_e adv_shutter_status_set](#) ([IrcmdHandle_t](#) *handle, int status)
Set current shutter switching status.
- DLLEXPORT [IrlibError_e basic_all_ffc_function_status_get](#) ([IrcmdHandle_t](#) *handle, int *status)
Get current all shutter function enable status.
- DLLEXPORT [IrlibError_e basic_all_ffc_function_status_set](#) ([IrcmdHandle_t](#) *handle, int status)
Set current all shutter function enable status, If the status is BASIC_ALL_FFC_FUNCTION_UNAVAILABLE, all shutter functions are disabled.
- DLLEXPORT [IrlibError_e adv_manual_ffc_update](#) ([IrcmdHandle_t](#) *handle, int mode_type)
do FFC update with behavior selected
- DLLEXPORT [IrlibError_e adv_ffc_without_shutter](#) ([IrcmdHandle_t](#) *handle)
do FFC without shutter
- DLLEXPORT [IrlibError_e adv_shutter_tab_close](#) ([IrcmdHandle_t](#) *handle)
Close the shutter tab.
- DLLEXPORT [IrlibError_e adv_shutter_tab_open](#) ([IrcmdHandle_t](#) *handle)
Open the shutter tab.
- DLLEXPORT [IrlibError_e adv_external_trigger_shutterless](#) ([IrcmdHandle_t](#) *handle)
External trigger shutterless mode.
- DLLEXPORT [IrlibError_e basic_device_info_get](#) ([IrcmdHandle_t](#) *handle, int device_info_type, char *data)
Get device's information.
- DLLEXPORT [IrlibError_e adv_powered_time_get](#) ([IrcmdHandle_t](#) *handle, int *time)
Gets the device powered on time.
- DLLEXPORT [IrlibError_e basic_device_temp_get](#) ([IrcmdHandle_t](#) *handle, float *temperature)
Get device's temperature.
- DLLEXPORT [IrlibError_e adv_device_realtime_status_set](#) ([IrcmdHandle_t](#) *handle, int status_type, uint32↵_t data)
Set device realtime status.
- DLLEXPORT [IrlibError_e adv_device_realtime_status_get](#) ([IrcmdHandle_t](#) *handle, int status_type, uint32↵_t *data)
Get device realtime status.
- DLLEXPORT [IrlibError_e basic_firmware_log_data_get](#) ([IrcmdHandle_t](#) *handle, uint8_t *log_data)
Read the firmware log data.
- DLLEXPORT [IrlibError_e adv_command_channel_type_get](#) ([IrcmdHandle_t](#) *handle, int *command_↵channel_type)
Get current command channel type.
- DLLEXPORT [IrlibError_e basic_device_current_status_get](#) ([IrcmdHandle_t](#) *handle, int *status)
Get current device status.
- DLLEXPORT [IrlibError_e adv_temp_coordinate_of_interest_get](#) ([IrcmdHandle_t](#) *handle, [IrcmdRect_t](#) rect↵_pos, [LineRectTempInfo_t](#) *rect_temp_info)

- Get coordinates of the max and min temperature in the region of interest.*
- DLLEXPORT [IrlibError_e adv_device_sleep_set](#) ([IrcmdHandle_t](#) *handle, int status)
- Set device sleep status.*
- DLLEXPORT [IrlibError_e adv_device_sleep_get](#) ([IrcmdHandle_t](#) *handle, int *status)
- Get device sleep status.*
- DLLEXPORT [IrlibError_e basic_rtc_current_time_set](#) ([IrcmdHandle_t](#) *handle, [Rtc_Time_t](#) *rtc_time)
- Set the current time of the device.*
- DLLEXPORT [IrlibError_e basic_rtc_current_time_get](#) ([IrcmdHandle_t](#) *handle, [Rtc_Time_t](#) *time)
- Get the current time of the device.*
- DLLEXPORT [IrlibError_e adv_voltage_change_set](#) ([IrcmdHandle_t](#) *handle, int voltage)
- Set DVP/I2C voltage.*
- DLLEXPORT [IrlibError_e adv_voltage_change_get](#) ([IrcmdHandle_t](#) *handle, int *voltage)
- Get DVP/UART voltage.*
- DLLEXPORT [IrlibError_e adv_algorithm_parameters_write](#) ([IrcmdHandle_t](#) *handle, uint16_t reg_address, uint16_t length, uint8_t *data)
- Write FPGA algorithm parameters data for X3.*
- DLLEXPORT [IrlibError_e adv_algorithm_parameters_read](#) ([IrcmdHandle_t](#) *handle, uint16_t reg_address, uint16_t length, uint8_t *data)
- Read FPGA algorithm parameters data for X3.*
- DLLEXPORT [IrlibError_e adv_fpga_version_get](#) ([IrcmdHandle_t](#) *handle, uint8_t *data)
- Get the version of fpga.*
- DLLEXPORT [IrlibError_e adv_device_open](#) ([IrcmdHandle_t](#) *handle, void *params)
- Open device with its device node.*
- DLLEXPORT [IrlibError_e adv_device_init](#) ([IrcmdHandle_t](#) *handle, void *params)
- init device, use this interface set some useful paramters for device*
- DLLEXPORT [IrlibError_e adv_device_close](#) ([IrcmdHandle_t](#) *handle)
- Close device.*
- DLLEXPORT [IrlibError_e basic_reset_to_rom](#) ([IrcmdHandle_t](#) *handle)
- Reset to rom, prepare for update firmware.*
- DLLEXPORT [IrlibError_e basic_firmware_download](#) ([IrcmdHandle_t](#) *handle, char *file_name, uint8_t *data, uint32_t file_len)
- download firmware*
- DLLEXPORT [IrlibError_e adv_enter_reboot_mode](#) ([IrcmdHandle_t](#) *handle)
- enter reboot mode*
- DLLEXPORT [IrlibError_e adv_cache_clear](#) ([IrcmdHandle_t](#) *handle)
- clear cache to avoid increased boot time*
- DLLEXPORT [IrlibError_e basic_preview_stop](#) ([IrcmdHandle_t](#) *handle)
- Stop stream.*
- DLLEXPORT [IrlibError_e basic_video_stream_continue](#) ([IrcmdHandle_t](#) *handle)
- When the device is in automatic plotting mode, let the stream flow out.*
- DLLEXPORT [IrlibError_e basic_video_stream_pause](#) ([IrcmdHandle_t](#) *handle)
- When the device is in automatic drawing mode, the data stream is temporarily not flowed out.*
- DLLEXPORT [IrlibError_e adv_mipi_channel_type_set](#) ([IrcmdHandle_t](#) *handle, int channel_type)
- Set different vc channel when the device gets stream for MIPI.*
- DLLEXPORT [IrlibError_e adv_stream_source_mode_set](#) ([IrcmdHandle_t](#) *handle, int src_stream_mode)
- Set intermediate stream source mode. For image frame only, the image frame is switched, the default mode is A↔DV_PICTURE_SOURCE_MODE; for double image frame, Non-image frame is switched, it cannot be switched to ADV_PICTURE_SOURCE_MODE.*
- DLLEXPORT [IrlibError_e adv_stream_source_mode_get](#) ([IrcmdHandle_t](#) *handle, int *src_stream_mode)
- Get intermediate stream source mode.*
- DLLEXPORT [IrlibError_e adv_output_frame_rate_set](#) ([IrcmdHandle_t](#) *handle, int rate)

- Set the rate of output frame.*

 - DLLEXPORT [IrlibError_e adv_output_frame_rate_get](#) ([IrcmdHandle_t](#) *handle, int *rate)

Get the rate of output frame.
- DLLEXPORT [IrlibError_e adv_boot_logo_status_set](#) ([IrcmdHandle_t](#) *handle, int status)

Set boot logo status.

 - DLLEXPORT [IrlibError_e adv_boot_logo_status_get](#) ([IrcmdHandle_t](#) *handle, int *status)

Get boot logo status.
- DLLEXPORT [IrlibError_e adv_yuv_format_set](#) ([IrcmdHandle_t](#) *handle, int format)

Set the format of YUV.

 - DLLEXPORT [IrlibError_e adv_yuv_format_get](#) ([IrcmdHandle_t](#) *handle, int *format)

Get the format of YUV.
- DLLEXPORT [IrlibError_e basic_preview_start](#) ([IrcmdHandle_t](#) *handle, [VideoOutputInfo_t](#) output_info)

Choose the appropriate mode to start stream @support Tiny2C USB,DVP,MIPI,VO_SPI.
- DLLEXPORT [IrlibError_e adv_digital_video_output_set](#) ([IrcmdHandle_t](#) *handle, [VideoOutputInfo_t](#) output_info)

Set the digital video output @support MINI2 USB,DVP,BT656,MIPI.

 - DLLEXPORT [IrlibError_e adv_digital_video_output_get](#) ([IrcmdHandle_t](#) *handle, [VideoOutputInfo_t](#) *output_info)

Get the digital video output.
- DLLEXPORT [IrlibError_e adv_analog_video_output_set](#) ([IrcmdHandle_t](#) *handle, int status, int format)

Set the analog video output info @support MINI2 NTSC,PAL.

 - DLLEXPORT [IrlibError_e adv_analog_video_output_get](#) ([IrcmdHandle_t](#) *handle, int *status, int *format)

Get the analog video output info.
- DLLEXPORT [IrlibError_e adv_external_synchronization_set](#) ([IrcmdHandle_t](#) *handle, int status)

Set external synchronization status @support MINI2.

 - DLLEXPORT [IrlibError_e adv_external_synchronization_get](#) ([IrcmdHandle_t](#) *handle, int *status)

Get external synchronization status @support MINI2.
- DLLEXPORT [IrlibError_e adv_7393_analog_set](#) ([IrcmdHandle_t](#) *handle, int status, int format)

Set 7393 analog video output info @support MINI2.

 - DLLEXPORT [IrlibError_e adv_7393_analog_get](#) ([IrcmdHandle_t](#) *handle, int *status, int *format)

Get 7393 analog video output info @support MINI2.
- DLLEXPORT [IrlibError_e adv_picture_freeze_status_set](#) ([IrcmdHandle_t](#) *handle, int status)

Set the status of picture freeze When freeze is enable, the device does not respond to zoom_set, mirror_flip_set and image_scene_set commands.

 - DLLEXPORT [IrlibError_e adv_picture_freeze_status_get](#) ([IrcmdHandle_t](#) *handle, int *status)

Get the status of picture freeze.
- DLLEXPORT [IrlibError_e adv_device_preview_info_get](#) ([IrcmdHandle_t](#) *handle, int preview_info, uint32_t *data)

Get basic preview info.
- DLLEXPORT [IrlibError_e basic_mirror_and_flip_status_set](#) ([IrcmdHandle_t](#) *handle, int value)

Set mirror and flip status.

 - DLLEXPORT [IrlibError_e basic_mirror_and_flip_status_get](#) ([IrcmdHandle_t](#) *handle, int *value)

Get mirror and flip status.
- DLLEXPORT [IrlibError_e basic_center_zoom_set](#) ([IrcmdHandle_t](#) *handle, float scale)

Set the scale.

 - DLLEXPORT [IrlibError_e basic_center_zoom_get](#) ([IrcmdHandle_t](#) *handle, float *scale)

Get the scale.
- DLLEXPORT [IrlibError_e adv_coordinate_zoom_set](#) ([IrcmdHandle_t](#) *handle, [IrcmdPoint_t](#) point_pos, float scale)

Set the scale.

 - DLLEXPORT [IrlibError_e adv_coordinate_zoom_get](#) ([IrcmdHandle_t](#) *handle, [IrcmdPoint_t](#) *point_pos, float *scale)

- Get the scale.*
- DLLEXPORT [IrlibError_e basic_image_brightness_level_set](#) ([IrcmdHandle_t](#) *handle, int level)
- Set image brightness level.*
- DLLEXPORT [IrlibError_e basic_current_brightness_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)
- Get image brightness level.*
- DLLEXPORT [IrlibError_e basic_attribute_of_brightness_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)
- Get attribute of brightness level.*
- DLLEXPORT [IrlibError_e basic_image_contrast_level_set](#) ([IrcmdHandle_t](#) *handle, int level)
- Set image contrast level. The contrast corresponding to brightness.*
- DLLEXPORT [IrlibError_e basic_current_contrast_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)
- Get image contrast level. The contrast corresponding to brightness.*
- DLLEXPORT [IrlibError_e basic_attribute_of_contrast_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)
- Get attribute of contrast level. The contrast corresponding to brightness.*
- DLLEXPORT [IrlibError_e basic_global_contrast_level_set](#) ([IrcmdHandle_t](#) *handle, int level)
- Set global contrast level.*
- DLLEXPORT [IrlibError_e basic_global_contrast_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)
- Get global contrast level.*
- DLLEXPORT [IrlibError_e basic_attribute_of_global_contrast_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)
- Get attribute of global contrast level.*
- DLLEXPORT [IrlibError_e basic_image_detail_enhance_level_set](#) ([IrcmdHandle_t](#) *handle, int level)
- Set image detail enhance level.*
- DLLEXPORT [IrlibError_e basic_current_detail_enhance_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)
- Get image detail enhance level.*
- DLLEXPORT [IrlibError_e basic_attribute_of_detail_enhance_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)
- Get attribute of detail enhance level.*
- DLLEXPORT [IrlibError_e basic_image_noise_reduction_level_set](#) ([IrcmdHandle_t](#) *handle, int level)
- Set level of image noise reduction For tiny2C, this function only works in professional mode.*
- DLLEXPORT [IrlibError_e basic_current_image_noise_reduction_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)
- Get current level of image noise reduction.*
- DLLEXPORT [IrlibError_e basic_attribute_of_noise_reduction_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)
- Get attribute of image noise reduction.*
- DLLEXPORT [IrlibError_e basic_image_roi_level_set](#) ([IrcmdHandle_t](#) *handle, int level)
- Set image parameters roi level.*
- DLLEXPORT [IrlibError_e basic_current_image_roi_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)
- Get image parameters roi level.*
- DLLEXPORT [IrlibError_e basic_attribute_of_roi_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)
- Get attribute of roi level.*
- DLLEXPORT [IrlibError_e basic_image_agc_level_set](#) ([IrcmdHandle_t](#) *handle, int level)
- Set image parameters agc level, support for AC02.*
- DLLEXPORT [IrlibError_e basic_current_agc_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)
- Get image parameters agc level, support for AC02.*
- DLLEXPORT [IrlibError_e basic_attribute_of_agc_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)
- Get attribute of agc level, support for AC02.*
- DLLEXPORT [IrlibError_e basic_image_scene_mode_set](#) ([IrcmdHandle_t](#) *handle, int mode)

- Set image scene mode.*

 - DLLEXPORT [IrlibError_e basic_current_image_scene_mode_get](#) ([IrcmdHandle_t](#) *handle, int *mode)

Get current image scene mode.
- DLLEXPORT [IrlibError_e basic_attribute_of_scene_mode_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)

Get basic attribute of image scene mode.
- DLLEXPORT [IrlibError_e adv_profession_mode_set](#) ([IrcmdHandle_t](#) *handle, int mode)

Set professional mode in scene mode @support Tiny2C.
- DLLEXPORT [IrlibError_e adv_profession_mode_get](#) ([IrcmdHandle_t](#) *handle, int *mode)

Get professional mode in scene mode @support Tiny2C.
- DLLEXPORT [IrlibError_e basic_gamma_rate_set](#) ([IrcmdHandle_t](#) *handle, int rate)

Set the gamma rate.
- DLLEXPORT [IrlibError_e basic_attribute_of_gamma_level_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)

Get the attribute of gamma level.
- DLLEXPORT [IrlibError_e basic_current_gamma_level_get](#) ([IrcmdHandle_t](#) *handle, int *rate)

Get the current gamma level.
- DLLEXPORT [IrlibError_e adv_algorithm_module_bypass_set](#) ([IrcmdHandle_t](#) *handle, [adv_algorithm_module_e](#) module, int status)

Set the algorithm module switch.
- DLLEXPORT [IrlibError_e adv_algorithm_module_bypass_get](#) ([IrcmdHandle_t](#) *handle, [adv_algorithm_module_e](#) module, int *status)

Get the current gamma level.
- DLLEXPORT [IrlibError_e basic_time_noise_reduce_level_set](#) ([IrcmdHandle_t](#) *handle, int level)

Set the time noise reduce level.
- DLLEXPORT [IrlibError_e basic_time_noise_reduce_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)

Get the time noise reduce level.
- DLLEXPORT [IrlibError_e basic_time_noise_reduce_attribute_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)

Get the attribute of time noise reduce level.
- DLLEXPORT [IrlibError_e basic_space_noise_reduce_level_set](#) ([IrcmdHandle_t](#) *handle, int level)

Set the space noise reduce level.
- DLLEXPORT [IrlibError_e basic_space_noise_reduce_level_get](#) ([IrcmdHandle_t](#) *handle, int *level)

Get the space noise reduce level.
- DLLEXPORT [IrlibError_e basic_space_noise_reduce_attribute_get](#) ([IrcmdHandle_t](#) *handle, [param_attribute_t](#) *attribute)

Get the attribute of space noise reduce level.
- DLLEXPORT [IrlibError_e adv_edge_enhance_set](#) ([IrcmdHandle_t](#) *handle, int level)

Set enhance level of objects' edges.
- DLLEXPORT [IrlibError_e adv_edge_enhance_get](#) ([IrcmdHandle_t](#) *handle, int *level)

Get enhance level of objects' edges.
- DLLEXPORT [IrlibError_e adv_cel_level_set](#) ([IrcmdHandle_t](#) *handle, int level)

Set cel level @support Tiny2C.
- DLLEXPORT [IrlibError_e adv_cel_level_restore](#) ([IrcmdHandle_t](#) *handle)

Restore cel level @support Tiny2C.
- DLLEXPORT [IrlibError_e basic_palette_num_get](#) ([IrcmdHandle_t](#) *handle, int *color_number)

Get the total number of palette color tables supported by the device.
- DLLEXPORT [IrlibError_e basic_palette_name_get](#) ([IrcmdHandle_t](#) *handle, int color_index, char *color_name ↵)

Get palette color name.
- DLLEXPORT [IrlibError_e basic_palette_idx_get](#) ([IrcmdHandle_t](#) *handle, int *color_index)

Get current palette color index.

- DLLEXPORT [IrlibError_e basic_palette_idx_set](#) ([IrcmdHandle_t](#) *handle, int color_index)
Set palette color index.
- DLLEXPORT [IrlibError_e adv_palette_switch_set](#) ([IrcmdHandle_t](#) *handle, int palette_switch)
Set palette color switch @support Tiny2C.
- DLLEXPORT [IrlibError_e adv_isothermal_switch_get](#) ([IrcmdHandle_t](#) *handle, int *isothermal_switch_type)
Get isothermal switch.
- DLLEXPORT [IrlibError_e adv_isothermal_switch_set](#) ([IrcmdHandle_t](#) *handle, int isothermal_switch_type)
Set isothermal switch.
- DLLEXPORT [IrlibError_e adv_isothermal_mode_get](#) ([IrcmdHandle_t](#) *handle, int *value)
Get isothermal mode.
- DLLEXPORT [IrlibError_e adv_isothermal_mode_set](#) ([IrcmdHandle_t](#) *handle, int value)
Set isothermal mode.
- DLLEXPORT [IrlibError_e adv_isothermal_limit_get](#) ([IrcmdHandle_t](#) *handle, float *high_temp, float *low_temp)
Get isothermal analysis range.usually low gain:-50~650;high gain:-70~200.
- DLLEXPORT [IrlibError_e adv_isothermal_limit_set](#) ([IrcmdHandle_t](#) *handle, float high_temp, float low_temp)
Set isothermal analysis range.usually low gain:-50~650;high gain:-70~200, suggest to use adv_isothermal_limit_get to get the range Floating point numbers are not currently supported,and floating point numbers will be forced to integers.
- DLLEXPORT [IrlibError_e adv_isothermal_limit_get_high_precision](#) ([IrcmdHandle_t](#) *handle, float *high_temp, float *low_temp)
Get isothermal analysis range.usually low gain:-50~650;high gain:-70~200.
- DLLEXPORT [IrlibError_e adv_isothermal_limit_set_high_precision](#) ([IrcmdHandle_t](#) *handle, float high_temp, float low_temp)
Set isothermal analysis range.usually low gain:-50~650;high gain:-70~200, suggest to use adv_isothermal_limit_get to get the range Floating point numbers are supported.
- DLLEXPORT [IrlibError_e adv_second_cali_flag_set](#) ([IrcmdHandle_t](#) *handle, int flag_index, int status)
Set second cali flag status If status is ADV_SECOND_CALI_DISABLE, the default data is used based on flag_index eg:status is ADV_SECOND_CALI_DISABLE, flag_index is ADV_SECOND_CALI_FLAG_KT, use default kt data If status is ADV_SECOND_CALI_ENABLE, the second cali data is used based on flag_index eg:status is ADV_SECOND_CALI_FLAG_KT, use second cali kt data.
- DLLEXPORT [IrlibError_e adv_second_cali_flag_get](#) ([IrcmdHandle_t](#) *handle, int flag_index, int *status)
Get second cali flag status.
- DLLEXPORT [IrlibError_e adv_k_value_calibration](#) ([IrcmdHandle_t](#) *handle, int status)
K value calibration.
- DLLEXPORT [IrlibError_e adv_k_value_calibration_cancel](#) ([IrcmdHandle_t](#) *handle)
Cancel K value calibration, restore old data.
- DLLEXPORT [IrlibError_e adv_k_value_clear](#) ([IrcmdHandle_t](#) *handle)
Clear K value calibration.
- DLLEXPORT [IrlibError_e adv_auto_rmcover_cali](#) ([IrcmdHandle_t](#) *handle)
Auto remove cover calibration.
- DLLEXPORT [IrlibError_e adv_rmcover_calib_cancel](#) ([IrcmdHandle_t](#) *handle)
Cancel rmcover calib data.
- DLLEXPORT [IrlibError_e adv_rmcover_calib_clear](#) ([IrcmdHandle_t](#) *handle)
Clear rmcover calib data.
- DLLEXPORT [IrlibError_e adv_auto_dpc_cali](#) ([IrcmdHandle_t](#) *handle)
Auto dead pixel calibration.
- DLLEXPORT [IrlibError_e adv_dpc_calib_data_cancel](#) ([IrcmdHandle_t](#) *handle)
Cancel dpc calib data.
- DLLEXPORT [IrlibError_e adv_set_cursor_position_to_dpc](#) ([IrcmdHandle_t](#) *handle, [IrcmdPoint_t](#) point_pos)
Set cursor position to a dead pixel.

- DLLEXPORT [IrlibError_e](#) [adv_set_cursor_position_to_non_dpc](#) ([IrcmdHandle_t](#) *handle, [IrcmdPoint_t](#) point_pos)
Set cursor position to a non-dead pixel.
- DLLEXPORT [IrlibError_e](#) [adv_cursor_switch_status_set](#) ([IrcmdHandle_t](#) *handle, int status)
Set cursor switch status.
- DLLEXPORT [IrlibError_e](#) [adv_cursor_switch_status_get](#) ([IrcmdHandle_t](#) *handle, int *status)
Get cursor switch status.
- DLLEXPORT [IrlibError_e](#) [adv_cursor_position_set](#) ([IrcmdHandle_t](#) *handle, [IrcmdPoint_t](#) point_pos)
Set cursor position.
- DLLEXPORT [IrlibError_e](#) [adv_cursor_position_get](#) ([IrcmdHandle_t](#) *handle, [IrcmdPoint_t](#) *point_pos)
Get cursor position.
- DLLEXPORT [IrlibError_e](#) [adv_dpc_calib_data_clear](#) ([IrcmdHandle_t](#) *handle)
Clear dead pixel cell calib data.
- DLLEXPORT [IrlibError_e](#) [basic_gain_set](#) ([IrcmdHandle_t](#) *handle, int value)
Set gain value of the device.
- DLLEXPORT [IrlibError_e](#) [basic_gain_get](#) ([IrcmdHandle_t](#) *handle, int *gain_value)
Get gain value of the device.
- DLLEXPORT [IrlibError_e](#) [adv_info_line_get](#) ([IrcmdHandle_t](#) *handle, int offset, int index, int length, [uint32_t](#) *data)
Get info line information @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [basic_sun_detect_switch_get](#) ([IrcmdHandle_t](#) *handle, int *status)
Get current sun detect switch status.
- DLLEXPORT [IrlibError_e](#) [basic_sun_detect_switch_set](#) ([IrcmdHandle_t](#) *handle, int status)
Set sun detect switch for sunburn protection.
- DLLEXPORT [IrlibError_e](#) [basic_sun_detect_pixel_ratio_get](#) ([IrcmdHandle_t](#) *handle, int *max_pixel_ratio, int *min_pixel_ratio)
Get the range of the current saturation pixel ratio.
- DLLEXPORT [IrlibError_e](#) [basic_sun_detect_pixel_ratio_set](#) ([IrcmdHandle_t](#) *handle, int max_pixel_ratio, int min_pixel_ratio)
Set the range of saturation pixel ratio for sun detect. The more the ratio, the more saturated pixel. When the saturation pixel ratio is within the setting range, it is determined that the saturation is caused by the sun.
- DLLEXPORT [IrlibError_e](#) [basic_sun_detect_roundness_level_get](#) ([IrcmdHandle_t](#) *handle, int *max_↵roundness_level, int *min_roundness_level)
Get the range of the current roundness level.
- DLLEXPORT [IrlibError_e](#) [basic_sun_detect_roundness_level_set](#) ([IrcmdHandle_t](#) *handle, int max_↵roundness_level, int min_roundness_level)
Set the range of roundness level for sun detect. The lower the level, the more round the object is. When the roundness level of the object is within the setting range, the object is judged to be sun.
- DLLEXPORT [IrlibError_e](#) [basic_restore_default_data](#) ([IrcmdHandle_t](#) *handle, int type)
Restore default data.
- DLLEXPORT [IrlibError_e](#) [basic_save_data](#) ([IrcmdHandle_t](#) *handle, int type)
Save data, include algorithm parameters, system parameters, calib data.

5.4.1 Macro Definition Documentation

5.4.1.1 AGC_MAX_LEVEL

```
#define AGC_MAX_LEVEL 100
```

5.4.1.2 COMMAND_IN_DEVELOPING

```
#define COMMAND_IN_DEVELOPING
```

5.4.1.3 DDE_MAX_LEVEL

```
#define DDE_MAX_LEVEL 100
```

5.4.1.4 SNR_MAX_LEVEL

```
#define SNR_MAX_LEVEL 100
```

5.4.1.5 TAU_EMS_MAX_LEVEL

```
#define TAU_EMS_MAX_LEVEL 16384
```

5.4.1.6 TAU_EMS_MIN_LEVEL

```
#define TAU_EMS_MIN_LEVEL 1
```

5.4.1.7 TU_TA_MAX_LEVEL

```
#define TU_TA_MAX_LEVEL 373
```

5.4.1.8 TU_TA_MIN_LEVEL

```
#define TU_TA_MIN_LEVEL 233
```

5.4.1.9 YUV_MAX_FORMAT

```
#define YUV_MAX_FORMAT 3
```

5.4.2 Typedef Documentation

5.4.2.1 IrcmdHandle_t

```
typedef struct _IrcmdHandle_t IrcmdHandle_t
```

5.4.3 Enumeration Type Documentation

5.4.3.1 adv_algorithm_module_e

```
enum adv_algorithm_module_e
```

algorithm module

Enumerator

KBC_BYPASS	
TNR_BYPASS	
HBC_BYPASS	
VBC_BYPASS	
SNR1_BYPASS	
AGC_BYPASS	
DDE_BYPASS	
GAMMA_BYPASS	
CROSS_BYPASS	
PIC_MIRROR_BYPASS	
TMP_MIRRRPR_BYPASS	

5.4.3.2 adv_algorithm_module_switch_status_e

```
enum adv_algorithm_module_switch_status_e
```

algorithm module switch status

Enumerator

ADV_ALGORITHM_MODULE_NORMAL	
ADV_ALGORITHM_MODULE_BYPASS	

5.4.3.3 adv_cursor_switch_status_e

enum [adv_cursor_switch_status_e](#)

cursor switch status

Enumerator

ADV_CURSOR_DISABLE	
ADV_CURSOR_ENABLE	

5.4.3.4 adv_edge_enhance_level_e

enum [adv_edge_enhance_level_e](#)

The level of enhance.

Enumerator

ADV_ZEROTH_LEVEL	
ADV_FIRST_LEVEL	
ADV_SECOND_LEVEL	

5.4.3.5 adv_info_line_index_e

enum [adv_info_line_index_e](#)

the index of information line

Enumerator

ADV_SW_MODEL_INFO	
ADV_SW_FRAME_INFO	
ADV_SW_MODEL_STATUS	
ADV_SW_SHUTTER_STATUS	
ADV_SW_AUTO_SHUTTER_STATUS	
ADV_SW_ISP_INFO_VERSION	
ADV_SW_TPD_FORMAT	
ADV_SW_MIRROR_ENALBE	
ADV_SW_FLIP_ENALBE	
ADV_SW_TEMP_AREA	

5.4.3.6 adv_isothermal_mode_e

enum `adv_isothermal_mode_e`

the mode of pseudo color

Enumerator

ADV_IOSTHERMAL_MODE↔ _0	Mapping pseudo-color table according to AGC output when isothermal mode.
ADV_IOSTHERMAL_MODE↔ _1	Mapping pseudo-color table according to AGC output.
ADV_IOSTHERMAL_MODE↔ _2	Mapping pseudo-color table according to TPD output.
ADV_IOSTHERMAL_MODE↔ _3	full color compress mode, other points are the same color
ADV_IOSTHERMAL_MODE↔ _4	uncompress color mode, points remain black and white
ADV_IOSTHERMAL_MODE↔ _5	full color compress mode, other points remain black and white

5.4.3.7 adv_isothermal_status_e

enum `adv_isothermal_status_e`

isothermal switch status

Enumerator

BASIC_ISOHERMAL_DISABLE	
BASIC_ISOHERMAL_ENABLE	

5.4.3.8 adv_K_calib_status_e

enum `adv_K_calib_status_e`

calib K value

Enumerator

BASIC_LOW_TEMP_OBJECT	
BASIC_HIGH_TEMP_OBJECT	
BASIC_CALCULATION	

5.4.3.9 adv_mini2256_output_frame_rate_e

enum [adv_mini2256_output_frame_rate_e](#)

the rate of output frame for mini2256

Enumerator

ADV_RATE_25	
ADV_RATE_50	

5.4.3.10 adv_mipi_channel_type_e

enum [adv_mipi_channel_type_e](#)

Vc channel type when the device gets stream for MIPI.

Enumerator

ADV_DOUBLIE_VC_CHANNEL	
ADV_SINGLE_VC_CHANNEL	

5.4.3.11 adv_output_frame_rate_e

enum [adv_output_frame_rate_e](#)

the rate of output frame

Enumerator

ADV_LOW_RATE	
ADV_HIGH_RATE	

5.4.3.12 adv_preview_info_e

enum [adv_preview_info_e](#)

preview infomation

Enumerator

ADV_CURRENT_FPS	current preview fps
ADV_CURRENT_WIDTH	current preview width
ADV_CURRENT_HEIGHT	current preview height

5.4.3.13 adv_profession_mode_e

enum `adv_profession_mode_e`

The professional mode.

Enumerator

ADV_NORMAL_MODE	
ADV_PROFESSIONAL_MODE	

5.4.3.14 adv_realtime_status_type_e

enum `adv_realtime_status_type_e`

device realtime status type

Enumerator

ADV_IR_SENSOR_VTEMP	ir sensor temp
ADV_IR_FPC_TEMP	

5.4.3.15 adv_second_calib_flag_status_e

enum `adv_second_calib_flag_status_e`

the type of second cali flag_index

Enumerator

ADV_SECOND_CALI_FLAG_TEC_K	
ADV_SECOND_CALI_FLAG_KT	
ADV_SECOND_CALI_FLAG_BT	
ADV_SECOND_CALI_FLAG_RMV	
ADV_SECOND_CALI_FLAG_NUC↔ _T	
ADV_SECOND_CALI_FLAG_DPC	

5.4.3.16 adv_second_calib_status_e

enum [adv_second_calib_status_e](#)

the status of second cali flag

Enumerator

ADV_SECOND_CALI_DISABLE	
ADV_SECOND_CALI_ENABLE	

5.4.3.17 adv_shutter_status_e

enum [adv_shutter_status_e](#)

current shutter switching status

Enumerator

ADV_SHUTTER_CLOSE_STA	The shutter is close.
ADV_SHUTTER_OPEN_STA	The shutter is open.
ADV_SHUTTER_UNKNOW_STA	The shutter is in an unknown status. Appears only when first powered up and is more difficult to acquire.

5.4.3.18 adv_stream_source_mode_e

enum [adv_stream_source_mode_e](#)

Y16 mode source stream modes.

Enumerator

ADV_IR_SOURCE_MODE	ir sensor mode source
ADV_KBC_SOURCE_MODE	KB correction mode source.
ADV_TNR_SOURCE_MODE	time noise reduction mode source
ADV_HBC_DPC_SOURCE_MODE	dead pixel correction and horizontal band correction mode source
ADV_VBC_SOURCE_MODE	vertical band correction mode source
ADV_SNR_SOURCE_MODE	space noise reduction mode source
ADV_DDE_SOURCE_MODE	digital detail enhance mode source
ADV_AGC_SOURCE_MODE	automatic gain control mode source
ADV_GAMMA_SOURCE_MODE	GAMMA mode source.
ADV_TPD_SOURCE_MODE	temperature mode source
ADV_MIRROR_SOURCE_MODE	mirror mode source
ADV_PICTURE_SOURCE_MODE	yuv mode source(stream source after all algorithms)

5.4.3.19 adv_video_output_format_e

enum `adv_video_output_format_e`

the format of video output ADV_BT656: Line by line form ADV_BT656_INTERLACED: Interlaced form

Enumerator

ADV_USB_FORMAT	
ADV_DVP_FORMAT	
ADV_BT656_FORMAT	
ADV_MIPI_FORMAT	
ADV_BT656_INTERLACED_FORMAT	
ADV_NTSC_FORMAT	
ADV_PAL_FORMAT	
ADV_VO_SPI_FORMAT	

5.4.3.20 adv_voltage_type_e

enum `adv_voltage_type_e`

the type of voltage

Enumerator

ADV_LOW_VOLTAGE	
ADV_HIGH_VOLTAGE	

5.4.3.21 adv_YUV_format_e

enum `adv_YUV_format_e`

the format of YUV

Enumerator

ADV_UYVY	
ADV_VYUY	
ADV_YUYV	
ADV_YVYU	

5.4.3.22 basic_all_ffc_status_e

enum `basic_all_ffc_status_e`

Whether all shutter functions are available (include auto shutter, manual shutter)

Enumerator

BASIC_ALL_FFC_FUNCTION_AVAILABLE	
BASIC_ALL_FFC_FUNCTION_UNAVAILABLE	
BASIC_ALL_FFC_FUNCTION_CANCEL	

5.4.3.23 basic_auto_ffc_param_e

enum `basic_auto_ffc_param_e`

auto shutter parameters types

Enumerator

BASIC_AUTO_TEMP_THRESHOLD	Vtemp's threshold to trigger shutter update in auto shutter. Do ffc update according to the temperature difference between latest update. unit:cnt(36cnt~=1c)
BASIC_AUTO_MIN_INTERVAL	Trigger auto shutter minimun interval,to prevent updates too frequently. unit:1s
BASIC_AUTO_MAX_INTERVAL	Trigger auto shutter maximun interval,shutter updates automatically at this time. unit:1s

5.4.3.24 basic_auto_ffc_status_e

enum `basic_auto_ffc_status_e`

auto shutter status

Enumerator

BASIC_AUTO_FFC_DISABLE	
BASIC_AUTO_FFC_ENABLE	
BASIC_AUTO_FFC_SEMI_DISABLE	in SEMI_DISALBE status, auto ffc parameters will be set as the semi default parameters by the firmware usually, auto_min_interval and auto_max_interval are set to longer time support for G1280s and CS640

5.4.3.25 basic_boot_logo_status_e

enum `basic_boot_logo_status_e`

boot logo status

Enumerator

BASIC_BOOT_LOGO_DISABLE	
BASIC_BOOT_LOGO_ENABLE	

5.4.3.26 basic_dev_info_type_e

enum `basic_dev_info_type_e`

the kind of basic device info

Enumerator

BASIC_DEV_NAME	device name
BASIC_DEV_FW_RELEASE_VERSION	release version of firmware
BASIC_DEV_CUSTOMER_ID	customer_id
BASIC_DEV_VENDOR_ID	vendor_id
BASIC_DEV_PRODUCT_ID	product_id
BASIC_DEV_INFO_GET_PN	product number
BASIC_DEV_INFO_GET_SN	serial number
BASIC_DEV_ISP_VERSION	the version of ISP algorithm parameter
BASIC_DEV_ABSTRACT_VERSION	the version of abstract data
BASIC_DEV_SYSTEM_VERSION	the version of system config data
BASIC_DEV_CALI_VERSION	the version of calibration data
BASIC_DEV_WHOLE_DATA_VERSION	the version of whole data
BASIC_DEV_WHOLE_PACKAGE_VERSION	the version of whole package
BASIC_DEV_SENSOR_TYPE	the sensor type
BASIC_DEV_SENSOR_ID	the sensor id

5.4.3.27 basic_device_status_e

enum `basic_device_status_e`

current status of device status

Enumerator

BASIC_STARTUP_MODE	Initialization of device data status after start-up.
BASIC_PREVIEW_MODE	The initialization of the device has been completed, the device is out of the map, but the stream is not output.
BASIC_VIDEO_OUT_MODE	Normal output stream of the device.
BASIC_UPDATE_MODE	The device is in upgrade mode.
BASIC_DEVICE_ERROR_MODE	The device is not in the above status or failed to get status.

5.4.3.28 basic_enable_status_e

```
enum basic_enable_status_e
```

the status of enable

Enumerator

BASIC_DISABLE	
BASIC_ENABLE	

5.4.3.29 basic_ffc_shutter_behavior_mode_e

```
enum basic_ffc_shutter_behavior_mode_e
```

Property page's shutter behavior mode.

Enumerator

BASIC_ONLY_B_UPDATE	
BASIC_ONLY_OOC_UPDATE	
BASIC_BOTH_B_AND_OOC_UPDATE	

5.4.3.30 basic_gain_status_e

```
enum basic_gain_status_e
```

the device's gain value

Enumerator

BASIC_LOW_GAIN	low gain
BASIC_HIGH_GAIN	high gain
BASIC_2LOW_GAIN	low-low gain
BASIC_2HIGH_GAIN	high-high gain

5.4.3.31 basic_image_scene_mode_e

enum `basic_image_scene_mode_e`

The mode of the image scene.

Enumerator

BASIC_DEFAULT_MODE	
BASIC_PATROL_MODE	
BASIC_MANUAL_MODE	
BASIC_LOW_TEMP_MODE	
BASIC_LINEAR_MODE	
BASIC_LOW_CONTRAST_MODE	
BASIC_HIGH_CONTRAST_MODE	
BASIC_HIGH_LIGHT_MODE	
BASIC_RESERVE1_MODE	
BASIC_RESERVE2_MODE	
BASIC_RESERVE3_MODE	
BASIC_RESERVE4_MODE	
BASIC_RESCUE_MODE	
BASIC_URBAN_MODE	
BASIC_TEMP_MEASURE_MODE	
BASIC_OUTDOORS_MODE	

5.4.3.32 basic_mirror_and_flip_status_e

enum `basic_mirror_and_flip_status_e`

image flip and mirror's type

Enumerator

BASIC_NO_MIRROR_OR_FLIP	
BASIC_ONLY_MIRROR	
BASIC_ONLY_FLIP	
BASIC_BOTH_MIRROR_AND_FLIP	

5.4.3.33 basic_restore_type_e

enum `basic_restore_type_e`

the type of data to restore

Enumerator

BASIC_All_RECAL_DATA	include dead pxial data, recover data and tpd data, support for 640 and 1280
BASIC_RESTORE_ALGORITHM_PARAMETERS	restore algorithm parameters to default, support for AC02 include denoise level, detail level, roi level, brightness level, contrast level
BASIC_RESTORE_SYSTEM_PARAMETERS	restore the system function parameter, support for mini2384 and P2L for mini2384, system function parameters include: video output format, image data source, uart voltage settings and more for P2L, system function parameters include: scene mode, palette index, shutter param, image data source and more
BASIC_RESTORE_K_VALUE	restore k value to default, support for mini2384 and P2L
BASIC_RESTORE_DPC_DATA	restore dpc calib data to default, support for mini2384 and P2L
BASIC_RESTORE_RMCOVER_DATA	restore rmcover calib data to default, support for mini2384 and P2L
BASIC_RESTORE_TPD_DATA	restore one point tpd calib data to default, support for P2L
BASIC_RESTORE_PROFESSION_SCENE_LEVEL	restore profession scene level, support for Tiny2C
BASIC_RESTORE_DOUBLE_TPD_DATA	restore two points tpd calib data, support for Tiny2C

5.4.3.34 basic_roi_level_e

```
enum basic_roi_level_e
```

the level of roi

Enumerator

BASIC_ROI_DISABLE	
BASIC_ROI_THIRD	
BASIC_ROI_HALF	

5.4.3.35 basic_save_type_e

```
enum basic_save_type_e
```

the type of data to save

Enumerator

BASIC_SAVE_ALGORITHM_PARAMETERS	save changed algorithm parameters, support for AC02 for AC02, algorithm parameters include: denoise level, detail level, roi level, brightness level, contrast level
BASIC_SAVE_SYSTEM_PARAMETERS	save the changed system function parameter, support for mini2384 and P2L for mini2384, system function parameters include: image data source, uart voltage settings and more for P2L, system function parameters include: scene mode, palette index, shutter param, image data source and more
BASIC_SAVE_K_VALUE	save k value after calib, support for mini2384 and P2L
BASIC_SAVE_DPC_DATA	save dpc calib data after calib, support for mini2384 and P2L
BASIC_SAVE_RMCOVER_DATA	save rmcover calib data after calib, support for mini2384 and P2L
BASIC_SAVE_TPD_DATA	save one point tpd calib data after calib, support for P2L
BASIC_SAVE_VIDEO_OUTPUT_FORMAT	save video output format
BASIC_SAVE_PROFESSION_SCENE_LEVLE	save profession scene level, support for Tiny2C
BASIC_SAVE_DOUBLE_TPD_DATA	save two points tpd calib data after calib, support for Tiny2C

5.4.3.36 basic_sun_detect_status_e

enum `basic_sun_detect_status_e`

sun detect switch status

Enumerator

BASIC_SUN_DETECT_DISABLE	
BASIC_SUN_DETECT_ENABLE	

5.4.3.37 callback_mode_e

enum `callback_mode_e`

location of callback function calls

Enumerator

UPGRADE_PROCESS_CALLBACK	
--------------------------	--

5.4.3.38 driver_type_e

enum `driver_type_e`

driver's type

Enumerator

IDLE_DRIVER	
MIPI_DRIVER	
USB_DRIVER	
DVP_DRIVER	

5.4.3.39 IrcmdLogLevel_e

enum `IrcmdLogLevel_e`

Log level definition in libircmd library.

Enumerator

IRCMD_LOG_DEBUG	print debug and error information
IRCMD_LOG_ERROR	only print error information
IRCMD_LOG_NO_PRINT	don't print debug and error information

5.4.3.40 video_output_num_e

enum `video_output_num_e`

the number of video output

Enumerator

SINGLE_VIDEO	
DOUBLE_VIDEO	

5.4.4 Function Documentation

5.4.4.1 adv_7393_analog_get()

```
DLLEXPORT IrlibError\_e adv_7393_analog_get (
    IrcmdHandle\_t * handle,
    int * status,
    int * format )
```

Get 7393 analog video output info @support MINI2.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see basic_enable_status_e
in	<i>format</i>	see adv_video_output_format_e , only support ADV_NTSC_FORMAT and ADV_PAL_FORMAT

Returns

see [IrlibError_e](#)

5.4.4.2 adv_7393_analog_set()

```
DLLEXPORT IrlibError\_e adv_7393_analog_set (
    IrcmdHandle\_t * handle,
    int status,
    int format )
```

Set 7393 analog video output info @support MINI2.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see basic_enable_status_e
in	<i>format</i>	see adv_video_output_format_e , only support ADV_NTSC_FORMAT and ADV_PAL_FORMAT

Returns

see [IrlibError_e](#)

5.4.4.3 adv_algorithm_module_bypass_get()

```
DLLEXPORT IrlibError\_e adv_algorithm_module_bypass_get (
    IrcmdHandle\_t * handle,
    adv\_algorithm\_module\_e module,
    int * status )
```

Get the current gamma level.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>module</i>	select the module you want to get, see adv_algorithm_module_e
out	<i>status</i>	normal or bypass, see adv_algorithm_module_switch_status_e (normal:0,bypass:1)

Returns

see [IrlibError_e](#)

5.4.4.4 [adv_algorithm_module_bypass_set\(\)](#)

```
DLLEXPORT IrlibError\_e adv\_algorithm\_module\_bypass\_set (  
    IrcmdHandle\_t * handle,  
    adv\_algorithm\_module\_e module,  
    int status )
```

Set the algorithm module switch.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>module</i>	select the module to be bypassed, see adv_algorithm_module_e
in	<i>status</i>	normal or bypass, see adv_algorithm_module_switch_status_e (normal:0,bypass:1)

Returns

see [IrlibError_e](#)

5.4.4.5 [adv_algorithm_parameters_read\(\)](#)

```
DLLEXPORT IrlibError\_e adv\_algorithm\_parameters\_read (  
    IrcmdHandle\_t * handle,  
    uint16_t reg_address,  
    uint16_t length,  
    uint8_t * data )
```

Read FPGA algorithm parameters data for X3.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>reg_address</i>	start address of the data in the register
in	<i>length</i>	the length of algorithm parameters data,must be a multiple of 4.range:0-64
out	<i>data</i>	algorithm parameters data

Returns

see `IrllibError_e`

5.4.4.6 adv_algorithm_parameters_write()

```
DLLEXPORT IrllibError_e adv_algorithm_parameters_write (
    IrcmdHandle_t * handle,
    uint16_t reg_address,
    uint16_t length,
    uint8_t * data )
```

Write FPGA algorithm parameters data for X3.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>reg_address</i>	start address of the data in the register
in	<i>length</i>	the length of algorithm parameters data,must be a multiple of 4.range:0-64
in	<i>data</i>	algorithm parameters data

Returns

see `IrllibError_e`

5.4.4.7 adv_analog_video_output_get()

```
DLLEXPORT IrllibError_e adv_analog_video_output_get (
    IrcmdHandle_t * handle,
    int * status,
    int * format )
```

Get the analog video output info.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>status</i>	see <code>basic_enable_status_e</code>
out	<i>format</i>	see <code>adv_video_output_format_e</code> , only support ADV_NTSC_FORMAT and ADV_PAL_FORMAT

Returns

see `IrllibError_e`

5.4.4.8 adv_analog_video_output_set()

```
DLLEXPORT IrlibError\_e adv_analog_video_output_set (
    IrcmdHandle\_t * handle,
    int status,
    int format )
```

Set the analog video output info @support MINI2 NTSC,PAL.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see basic_enable_status_e
in	<i>format</i>	see adv_video_output_format_e , only support ADV_NTSC_FORMAT and ADV_PAL_FORMAT

Returns

see [IrlibError_e](#)

5.4.4.9 adv_auto_dpc_cali()

```
DLLEXPORT IrlibError\_e adv_auto_dpc_cali (
    IrcmdHandle\_t * handle )
```

Auto dead pixel calibration.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.10 adv_auto_rmcover_cali()

```
DLLEXPORT IrlibError\_e adv_auto_rmcover_cali (
    IrcmdHandle\_t * handle )
```

Auto remove cover calibration.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see `IrlibError_e`

5.4.4.11 `adv_boot_logo_status_get()`

```
DLL_EXPORT IrlibError_e adv_boot_logo_status_get (
    IrcmdHandle_t * handle,
    int * status )
```

Get boot logo status.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
out	<i>status</i>	boot logo status, see <code>basic_boot_logo_status_e</code>

Returns

see `IrlibError_e`

5.4.4.12 `adv_boot_logo_status_set()`

```
DLL_EXPORT IrlibError_e adv_boot_logo_status_set (
    IrcmdHandle_t * handle,
    int status )
```

Set boot logo status.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
in	<i>status</i>	boot logo status, see <code>basic_boot_logo_status_e</code>

Returns

see `IrlibError_e`

5.4.4.13 `adv_cache_clear()`

```
DLL_EXPORT IrlibError_e adv_cache_clear (
    IrcmdHandle_t * handle )
```

clear cache to avoid increased boot time

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.14 adv_cel_level_restore()

```
DLLEXPORT IrlibError\_e adv_cel_level_restore (
    IrcmdHandle\_t * handle )
```

Restore cel level @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.15 adv_cel_level_set()

```
DLLEXPORT IrlibError\_e adv_cel_level_set (
    IrcmdHandle\_t * handle,
    int level )
```

Set cel level @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	cel level, range:0-100

Returns

see [IrlibError_e](#)

5.4.4.16 `adv_command_channel_type_get()`

```
DLLEXPORT IrlibError\_e adv_command_channel_type_get (
    IrcmdHandle\_t * handle,
    int * command_channel_type )
```

Get current command channel type.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>command_channel_type</i>	current command channel, see command_channel_e

Returns

see [IrlibError_e](#)

5.4.4.17 `adv_coordinate_zoom_get()`

```
DLLEXPORT IrlibError\_e adv_coordinate_zoom_get (
    IrcmdHandle\_t * handle,
    IrcmdPoint\_t * point_pos,
    float * scale )
```

Get the scale.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>point_pos</i>	the point's position
out	<i>scale</i>	user-set scale, range:1-8

Returns

see [IrlibError_e](#)

5.4.4.18 `adv_coordinate_zoom_set()`

```
DLLEXPORT IrlibError\_e adv_coordinate_zoom_set (
    IrcmdHandle\_t * handle,
    IrcmdPoint\_t point_pos,
    float scale )
```

Set the scale.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>point_pos</i>	the point's position
in	<i>scale</i>	user-set scale, range:1-8

Returns

see [IrlibError_e](#)

5.4.4.19 adv_cursor_position_get()

```
DLL_EXPORT IrlibError_e adv_cursor_position_get (
    IrcmdHandle_t * handle,
    IrcmdPoint_t * point_pos )
```

Get cursor position.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>point_pos</i>	the point's position

Returns

see [IrlibError_e](#)

5.4.4.20 adv_cursor_position_set()

```
DLL_EXPORT IrlibError_e adv_cursor_position_set (
    IrcmdHandle_t * handle,
    IrcmdPoint_t point_pos )
```

Set cursor position.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>point_pos</i>	the point's position

Returns

see [IrlibError_e](#)

5.4.4.21 `adv_cursor_switch_status_get()`

```
DLLEXPORT IrlibError\_e adv_cursor_switch_status_get (
    IrcmdHandle\_t * handle,
    int * status )
```

Get cursor switch status.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>status</i>	see adv_cursor_switch_status_e

Returns

see [IrlibError_e](#)

5.4.4.22 `adv_cursor_switch_status_set()`

```
DLLEXPORT IrlibError\_e adv_cursor_switch_status_set (
    IrcmdHandle\_t * handle,
    int status )
```

Set cursor switch status.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see adv_cursor_switch_status_e

Returns

see [IrlibError_e](#)

5.4.4.23 `adv_device_close()`

```
DLLEXPORT IrlibError\_e adv_device_close (
    IrcmdHandle\_t * handle )
```

Close device.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see `IrllibError_e`

5.4.4.24 `adv_device_init()`

```
DLL_EXPORT IrllibError_e adv_device_init (
    IrcmdHandle_t * handle,
    void * params )
```

init device, use this interface set some useful paramters for device

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
in	<i>params</i>	useful paramters for device, it likes int[64], eg: for uart device the int[0] is baudrate

Returns

see `IrllibError_e`

5.4.4.25 `adv_device_open()`

```
DLL_EXPORT IrllibError_e adv_device_open (
    IrcmdHandle_t * handle,
    void * params )
```

Open device with its device node.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
in	<i>params</i>	device's node

Returns

see `IrllibError_e`

5.4.4.26 `adv_device_preview_info_get()`

```
DLL_EXPORT IrllibError_e adv_device_preview_info_get (
    IrcmdHandle_t * handle,
```

```
int preview_info,  
uint32_t * data )
```

Get basic preview info.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>preview_info</i>	see adv_preview_info_e
out	<i>data</i>	information value

Returns

see [IrlibError_e](#)

5.4.4.27 adv_device_realtime_status_get()

```
DLLEXPORT IrlibError\_e adv_device_realtime_status_get (
    IrcmdHandle\_t * handle,
    int status_type,
    uint32_t * data )
```

Get device realtime status.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status_type</i>	status information type,see adv_realtime_status_type_e
out	<i>data</i>	status information value

Returns

see [IrlibError_e](#)

5.4.4.28 adv_device_realtime_status_set()

```
DLLEXPORT IrlibError\_e adv_device_realtime_status_set (
    IrcmdHandle\_t * handle,
    int status_type,
    uint32_t data )
```

Set device realtime status.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status_type</i>	status information type,see adv_realtime_status_type_e
in	<i>data</i>	status information value

Returns

see `IrlibError_e`

5.4.4.29 `adv_device_sleep_get()`

```
DLL_EXPORT IrlibError_e adv_device_sleep_get (
    IrcmdHandle_t * handle,
    int * status )
```

Get device sleep status.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
out	<i>status</i>	see <code>basic_enable_status_e</code>

Returns

see `IrlibError_e`

5.4.4.30 `adv_device_sleep_set()`

```
DLL_EXPORT IrlibError_e adv_device_sleep_set (
    IrcmdHandle_t * handle,
    int status )
```

Set device sleep status.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
in	<i>status</i>	see <code>basic_enable_status_e</code>

Returns

see `IrlibError_e`

5.4.4.31 `adv_digital_video_output_get()`

```
DLL_EXPORT IrlibError_e adv_digital_video_output_get (
    IrcmdHandle_t * handle,
    VideoOutputInfo_t * output_info )
```

Get the digital video output.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>output_info</i>	see VideoOutputInfo_t

Returns

see [IrlibError_e](#)

5.4.4.32 `adv_digital_video_output_set()`

```
DLLEXPORT IrlibError\_e adv_digital_video_output_set (
    IrcmdHandle\_t * handle,
    VideoOutputInfo\_t output_info )
```

Set the digital video output @support MINI2 USB,DVP,BT656,MIPI.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>output_info</i>	see VideoOutputInfo_t , dont need to fill video_output_num

Returns

see [IrlibError_e](#)

5.4.4.33 `adv_dpc_calib_data_cancel()`

```
DLLEXPORT IrlibError\_e adv_dpc_calib_data_cancel (
    IrcmdHandle\_t * handle )
```

Cancel dpc calib data.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.34 `adv_dpc_calib_data_clear()`

```
DLLEXPORT IrlibError\_e adv_dpc_calib_data_clear (
    IrcmdHandle\_t * handle )
```

Clear dead pixel cell calib data.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.35 `adv_edge_enhance_get()`

```
DLLEXPORT IrlibError\_e adv_edge_enhance_get (
    IrcmdHandle\_t * handle,
    int * level )
```

Get enhance level of objects' edges.

Attention

This function takes effect only when the current scene mode supports it

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	enhance level, the higher the level, the more pronounced the edges of objects, see adv_edge_enhance_level_e

Returns

see [IrlibError_e](#)

5.4.4.36 `adv_edge_enhance_set()`

```
DLLEXPORT IrlibError\_e adv_edge_enhance_set (
    IrcmdHandle\_t * handle,
    int level )
```

Set enhance level of objects' edges.

Attention

This function takes effect only when the current scene mode supports it

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	enhance level, the higher the level, the more pronounced the edges of objects, see adv_edge_enhance_level_e

Returns

see [IrlibError_e](#)

5.4.4.37 adv_enter_reboot_mode()

```
DLL_EXPORT IrlibError_e adv_enter_reboot_mode (
    IrcmdHandle_t * handle )
```

enter reboot mode

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.38 adv_external_synchronization_get()

```
DLL_EXPORT IrlibError_e adv_external_synchronization_get (
    IrcmdHandle_t * handle,
    int * status )
```

Get external synchronization status @support MINI2.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see basic_enable_status_e

Returns

see [IrlibError_e](#)

5.4.4.39 adv_external_synchronization_set()

```
DLLEXPORT IrlibError\_e adv_external_synchronization_set (
    IrcmdHandle\_t * handle,
    int status )
```

Set external synchronization status @support MINI2.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see basic_enable_status_e

Returns

see [IrlibError_e](#)

5.4.4.40 adv_external_trigger_shutterless()

```
DLLEXPORT IrlibError\_e adv_external_trigger_shutterless (
    IrcmdHandle\_t * handle )
```

External trigger shutterless mode.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.41 adv_ffc_without_shutter()

```
DLLEXPORT IrlibError\_e adv_ffc_without_shutter (
    IrcmdHandle\_t * handle )
```

do FFC without shutter

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see `IrlibError_e`

5.4.4.42 adv_fpga_version_get()

```
DLLEXPORT IrlibError_e adv_fpga_version_get (
    IrcmdHandle_t * handle,
    uint8_t * data )
```

Get the version of fpga.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>data[4]</i>	the version of fpga

Returns

see `IrlibError_e`

5.4.4.43 adv_info_line_get()

```
DLLEXPORT IrlibError_e adv_info_line_get (
    IrcmdHandle_t * handle,
    int offset,
    int index,
    int length,
    uint32_t * data )
```

Get info line information @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>offset</i>	the offset against index
in	<i>index</i>	see <code>adv_info_line_index_e</code> , select the index corresponding to the information you want
in	<i>length</i>	the length of data to be read
out	<i>data</i>	store the read data from info line

Returns

see `IrlibError_e`

5.4.4.44 `adv_isothermal_limit_get()`

```
DLLEXPORT IrlibError\_e adv_isothermal_limit_get (
    IrcmdHandle\_t * handle,
    float * high_temp,
    float * low_temp )
```

Get isothermal analysis range.usually low gain:-50~650;high gain:-70~200.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>high_temp</i>	the high temp of isothermal analysis range,unit: Celcius
out	<i>low_temp</i>	the low temp of isothermal analysis range,unit: Celcius

Returns

see [IrlibError_e](#)

5.4.4.45 `adv_isothermal_limit_get_high_precision()`

```
DLLEXPORT IrlibError\_e adv_isothermal_limit_get_high_precision (
    IrcmdHandle\_t * handle,
    float * high_temp,
    float * low_temp )
```

Get isothermal analysis range.usually low gain:-50~650;high gain:-70~200.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>high_temp</i>	the high temp of isothermal analysis range,unit: Celcius
out	<i>low_temp</i>	the low temp of isothermal analysis range,unit: Celcius

Returns

see [IrlibError_e](#)

5.4.4.46 `adv_isothermal_limit_set()`

```
DLLEXPORT IrlibError\_e adv_isothermal_limit_set (
    IrcmdHandle\_t * handle,
    float high_temp,
    float low_temp )
```

Set isothermal analysis range.usually low gain:-50~650;high gain:-70~200, suggest to use `adv_isothermal_limit_get` to get the range Floating point numbers are not currently supported,and floating point numbers will be forced to integers.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>high_temp</i>	the high temp of isothermal analysis range,unit: Celcius
in	<i>low_temp</i>	the low temp of isothermal analysis range,unit: Celcius

Returns

see [IrlibError_e](#)

5.4.4.47 adv_isothermal_limit_set_high_precision()

```
DLLEXPORT IrlibError\_e adv_isothermal_limit_set_high_precision (
    IrcmdHandle\_t * handle,
    float high_temp,
    float low_temp )
```

Set isothermal analysis range.usually low gain:-50~650;high gain:-70~200, suggest to use [adv_isothermal_limit_get](#) to get the range Floating point numbers are supported.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>high_temp</i>	the high temp of isothermal analysis range,unit: Celcius
in	<i>low_temp</i>	the low temp of isothermal analysis range,unit: Celcius

Returns

see [IrlibError_e](#)

5.4.4.48 adv_isothermal_mode_get()

```
DLLEXPORT IrlibError\_e adv_isothermal_mode_get (
    IrcmdHandle\_t * handle,
    int * value )
```

Get isothermal mode.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>value</i>	the mode of isothermal,see adv_isothermal_mode_e

Returns

see `IrllibError_e`

5.4.4.49 adv_isothermal_mode_set()

```
DLLEXPORT IrllibError_e adv_isothermal_mode_set (
    IrcmdHandle_t * handle,
    int value )
```

Set isothermal mode.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
in	<i>value</i>	the mode of isothermal,see <code>adv_isothermal_mode_e</code>

Returns

see `IrllibError_e`

5.4.4.50 adv_isothermal_switch_get()

```
DLLEXPORT IrllibError_e adv_isothermal_switch_get (
    IrcmdHandle_t * handle,
    int * isothermal_switch_type )
```

Get isothermal switch.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
out	<i>isothermal_switch_type</i>	the switch of iosthermal,see <code>adv_isothermal_status_e</code>

Returns

see `IrllibError_e`

5.4.4.51 adv_isothermal_switch_set()

```
DLLEXPORT IrllibError_e adv_isothermal_switch_set (
    IrcmdHandle_t * handle,
    int isothermal_switch_type )
```

Set isothermal switch.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>isothermal_switch_type</i>	the switch of iosthermal,see adv_isothermal_status_e

Returns

see [IrlibError_e](#)

5.4.4.52 adv_k_value_calibration()

```
DLLEXPORT IrlibError\_e adv_k_value_calibration (
    IrcmdHandle\_t * handle,
    int status )
```

K value calibration.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see adv_K_calib_status_e

Returns

see [IrlibError_e](#)

5.4.4.53 adv_k_value_calibration_cancel()

```
DLLEXPORT IrlibError\_e adv_k_value_calibration_cancel (
    IrcmdHandle\_t * handle )
```

Cancel K value calibration, restore old data.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.54 adv_k_value_clear()

```
DLLEXPORT IrlibError\_e adv_k_value_clear (
    IrcmdHandle\_t * handle )
```

Clear K value calibration.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.55 adv_manual_ffc_update()

```
DLLEXPORT IrlibError\_e adv_manual_ffc_update (
    IrcmdHandle\_t * handle,
    int mode_type )
```

do FFC update with behavior selected

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>mode_type</i>	select behavior when ffc updata,see basic_ffc_shutter_behavior_mode_e

Returns

see [IrlibError_e](#)

5.4.4.56 adv_mipi_channel_type_set()

```
DLLEXPORT IrlibError\_e adv_mipi_channel_type_set (
    IrcmdHandle\_t * handle,
    int channel_type )
```

Set different vc channel when the device gets stream for MIPI.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>channel_type</i>	mipi vc channel type,see adv_mipi_channel_type_e

Returns

see `IrllibError_e`

5.4.4.57 adv_output_frame_rate_get()

```
DLLEXPORT IrllibError_e adv_output_frame_rate_get (
    IrcmdHandle_t * handle,
    int * rate )
```

Get the rate of output frame.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
out	<i>rate</i>	see <code>adv_output_frame_rate_e</code>

Returns

see `IrllibError_e`

5.4.4.58 adv_output_frame_rate_set()

```
DLLEXPORT IrllibError_e adv_output_frame_rate_set (
    IrcmdHandle_t * handle,
    int rate )
```

Set the rate of output frame.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
in	<i>rate</i>	mini2256 see <code>adv_mini2256_output_frame_rate_e</code> , others see <code>adv_output_frame_rate_e</code>

Returns

see `IrllibError_e`

5.4.4.59 adv_palette_switch_set()

```
DLLEXPORT IrllibError_e adv_palette_switch_set (
    IrcmdHandle_t * handle,
    int palette_switch )
```

Set palette color switch @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>palette_switch</i>	palette color switch, see basic_enable_status_e

Returns

see [IrlibError_e](#)

5.4.4.60 `adv_picture_freeze_status_get()`

```
DLLEXPORT IrlibError_e adv_picture_freeze_status_get (
    IrcmdHandle_t * handle,
    int * status )
```

Get the status of picture freeze.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>status</i>	see basic_picture_freeze_status_e

Returns

see [IrlibError_e](#)

5.4.4.61 `adv_picture_freeze_status_set()`

```
DLLEXPORT IrlibError_e adv_picture_freeze_status_set (
    IrcmdHandle_t * handle,
    int status )
```

Set the status of picture freeze When freeze is enable, the device does not respond to `zoom_set`, `mirror_flip_set` and `image_scene_set` commands.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	see basic_enable_status_e

Returns

see `IrlibError_e`

5.4.4.62 `adv_powered_time_get()`

```
DLLEXPORT IrlibError_e adv_powered_time_get (
    IrcmdHandle_t * handle,
    int * time )
```

Gets the device powered on time.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
out	<i>time</i>	the powered on time, unit: second

Returns

see `IrlibError_e`

5.4.4.63 `adv_profession_mode_get()`

```
DLLEXPORT IrlibError_e adv_profession_mode_get (
    IrcmdHandle_t * handle,
    int * mode )
```

Get professional mode in scene mode @support Tiny2C.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
out	<i>mode</i>	professional mode in scene mode, see <code>adv_profession_mode_e</code>

Returns

see `IrlibError_e`

5.4.4.64 `adv_profession_mode_set()`

```
DLLEXPORT IrlibError_e adv_profession_mode_set (
    IrcmdHandle_t * handle,
    int mode )
```

Set professional mode in scene mode @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>mode</i>	professional mode in scene mode, see adv_profession_mode_e

Returns

see [IrlibError_e](#)

5.4.4.65 [adv_rmcover_calib_cancel\(\)](#)

```
DLLEXPORT IrlibError\_e adv_rmcover_calib_cancel (
    IrcmdHandle\_t * handle )
```

Cancel rmcover calib data.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.66 [adv_rmcover_calib_clear\(\)](#)

```
DLLEXPORT IrlibError\_e adv_rmcover_calib_clear (
    IrcmdHandle\_t * handle )
```

Clear rmcover calib data.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.67 adv_second_cali_flag_get()

```
DLLEXPORT IrlibError\_e adv_second_cali_flag_get (
    IrcmdHandle\_t * handle,
    int flag_index,
    int * status )
```

Get second cali flag status.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>flag_index</i>	the index of flag that needed to set
out	<i>status</i>	see adv_second_calib_status_e

Returns

see [IrlibError_e](#)

5.4.4.68 adv_second_cali_flag_set()

```
DLLEXPORT IrlibError\_e adv_second_cali_flag_set (
    IrcmdHandle\_t * handle,
    int flag_index,
    int status )
```

Set second cali flag status If status is ADV_SECOND_CALI_DISABLE, the default data is used based on flag_index eg:status is ADV_SECOND_CALI_DISABLE, flag_index is ADV_SECOND_CALI_FLAG_KT, use default kt data If status is ADV_SECOND_CALI_ENABLE, the second cali data is used based on flag_index eg:status is ADV_SECOND_CALI_ENABLE, flag_index is ADV_SECOND_CALI_FLAG_KT, use second cali kt data.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>flag_index</i>	the index of flag that needed to set
in	<i>status</i>	see adv_second_calib_status_e

Returns

see [IrlibError_e](#)

5.4.4.69 adv_set_cursor_position_to_dpc()

```
DLLEXPORT IrlibError\_e adv_set_cursor_position_to_dpc (
    IrcmdHandle\_t * handle,
    IrcmdPoint\_t point_pos )
```

Set cursor position to a dead pixel.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>point_pos</i>	the cursor point's position

Returns

see [IrlibError_e](#)

5.4.4.70 adv_set_cursor_position_to_non_dpc()

```
DLLEXPORT IrlibError\_e adv_set_cursor_position_to_non_dpc (
    IrcmdHandle\_t * handle,
    IrcmdPoint\_t point_pos )
```

Set cursor position to a non-dead pixel.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>point_pos</i>	the cursor point's position

Returns

see [IrlibError_e](#)

5.4.4.71 adv_shutter_status_get()

```
DLLEXPORT IrlibError\_e adv_shutter_status_get (
    IrcmdHandle\_t * handle,
    int * status )
```

Get current shutter switching status.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>status</i>	status of shutter, see adv_shutter_status_e

Returns

see [IrlibError_e](#)

5.4.4.72 adv_shutter_status_set()

```
DLLEXPORT IrlibError\_e adv_shutter_status_set (
    IrcmdHandle\_t * handle,
    int status )
```

Set current shutter switching status.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	status of shutter,only support 0:close or 1:open

Returns

see [IrlibError_e](#)

5.4.4.73 adv_shutter_tab_close()

```
DLLEXPORT IrlibError\_e adv_shutter_tab_close (
    IrcmdHandle\_t * handle )
```

Close the shutter tab.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.74 adv_shutter_tab_open()

```
DLLEXPORT IrlibError\_e adv_shutter_tab_open (
    IrcmdHandle\_t * handle )
```

Open the shutter tab.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see `IrllibError_e`

5.4.4.75 `adv_stream_source_mode_get()`

```
DLLEXPORT IrllibError_e adv_stream_source_mode_get (
    IrcmdHandle_t * handle,
    int * src_stream_mode )
```

Get intermediate stream source mode.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>src_stream_mode</i>	source video streaming mode,see adv_stream_source_mode_e

Returns

see `IrllibError_e`

5.4.4.76 `adv_stream_source_mode_set()`

```
DLLEXPORT IrllibError_e adv_stream_source_mode_set (
    IrcmdHandle_t * handle,
    int src_stream_mode )
```

Set intermediate stream source mode. For image frame only, the image frame is switched, the default mode is `ADV_PICTURE_SOURCE_MODE`; for double image frame, Non-image frame is switched, it cannot be switched to `ADV_PICTURE_SOURCE_MODE`.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>src_stream_mode</i>	source video streaming mode,see adv_stream_source_mode_e

Returns

see `IrllibError_e`

5.4.4.77 `adv_temp_coordinate_of_interest_get()`

```
DLLEXPORT IrlibError\_e adv_temp_coordinate_of_interest_get (
    IrcmdHandle\_t * handle,
    IrcmdRect\_t rect_pos,
    LineRectTempInfo\_t * rect_temp_info )
```

Get coordinates of the max and min temperature in the region of interest.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>rect_pos</i>	start and end coordinates of interest region
out	<i>rect_temp_info</i>	information about the region of interest

Returns

see [IrlibError_e](#)

5.4.4.78 `adv_voltage_change_get()`

```
DLLEXPORT IrlibError\_e adv_voltage_change_get (
    IrcmdHandle\_t * handle,
    int * voltage )
```

Get DVP/UART voltage.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>voltage</i>	see adv_voltage_type_e

Returns

see [IrlibError_e](#)

5.4.4.79 `adv_voltage_change_set()`

```
DLLEXPORT IrlibError\_e adv_voltage_change_set (
    IrcmdHandle\_t * handle,
    int voltage )
```

Set DVP/I2C voltage.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>voltage</i>	see adv_voltage_type_e

Returns

see [IrlibError_e](#)

5.4.4.80 adv_yuv_format_get()

```
DLL_EXPORT IrlibError_e adv_yuv_format_get (
    IrcmdHandle_t * handle,
    int * format )
```

Get the format of YUV.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>format</i>	see adv_YUV_format_e

Returns

see [IrlibError_e](#)

5.4.4.81 adv_yuv_format_set()

```
DLL_EXPORT IrlibError_e adv_yuv_format_set (
    IrcmdHandle_t * handle,
    int format )
```

Set the format of YUV.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>format</i>	see adv_YUV_format_e

Returns

see `IrlibError_e`

5.4.4.82 basic_all_ffc_function_status_get()

```
DLLEXPORT IrlibError_e basic_all_ffc_function_status_get (
    IrcmdHandle_t * handle,
    int * status )
```

Get current all shutter function enable status.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
out	<i>status</i>	whether all shutter functions are in effect,see <code>basic_all_ffc_status_e</code>

Returns

see `IrlibError_e`

5.4.4.83 basic_all_ffc_function_status_set()

```
DLLEXPORT IrlibError_e basic_all_ffc_function_status_set (
    IrcmdHandle_t * handle,
    int status )
```

Set current all shutter function enable status, If the status is `BASIC_ALL_FFC_FUNCTION_UNAVAILABLE`, all shutter functions are disabled.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
in	<i>status</i>	whether all shutter functions are in effect,see <code>basic_all_ffc_status_e</code>

Returns

see `IrlibError_e`

5.4.4.84 basic_attribute_of_agc_level_get()

```
DLLEXPORT IrlibError_e basic_attribute_of_agc_level_get (
    IrcmdHandle_t * handle,
    param_attribute_t * attribute )
```

Get attribute of agc level,support for AC02.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of agc level

Returns

see [IrlibError_e](#)

5.4.4.85 basic_attribute_of_brightness_level_get()

```
DLLEXPORT IrlibError\_e basic_attribute_of_brightness_level_get (
    IrcmdHandle\_t * handle,
    param\_attribute\_t * attribute )
```

Get attribute of brightness level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of brightness level

Returns

see [IrlibError_e](#)

5.4.4.86 basic_attribute_of_contrast_level_get()

```
DLLEXPORT IrlibError\_e basic_attribute_of_contrast_level_get (
    IrcmdHandle\_t * handle,
    param\_attribute\_t * attribute )
```

Get attribute of contrast level. The contrast corresponding to brightness.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of contrast level

Returns

see [IrlibError_e](#)

5.4.4.87 basic_attribute_of_detail_enhance_level_get()

```
DLLEXPORT IrlibError\_e basic_attribute_of_detail_enhance_level_get (
    IrcmdHandle\_t * handle,
    param\_attribute\_t * attribute )
```

Get attribute of detail enhance level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of detail enhance level

Returns

see [IrlibError_e](#)

5.4.4.88 basic_attribute_of_gamma_level_get()

```
DLLEXPORT IrlibError\_e basic_attribute_of_gamma_level_get (
    IrcmdHandle\_t * handle,
    param\_attribute\_t * attribute )
```

Get the attribute of gamma level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of gamma level

Returns

see [IrlibError_e](#)

5.4.4.89 basic_attribute_of_global_contrast_level_get()

```
DLLEXPORT IrlibError\_e basic_attribute_of_global_contrast_level_get (
    IrcmdHandle\_t * handle,
    param\_attribute\_t * attribute )
```

Get attribute of global contrast level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of contrast level

Returns

see `IrllibError_e`

5.4.4.90 basic_attribute_of_noise_reduction_level_get()

```
DLLEXPORT IrllibError_e basic_attribute_of_noise_reduction_level_get (
    IrcmdHandle_t * handle,
    param_attribute_t * attribute )
```

Get attribute of image noise reduction.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
out	<i>attribute</i>	attribute of image noise reduction

Returns

see `IrllibError_e`

5.4.4.91 basic_attribute_of_roi_level_get()

```
DLLEXPORT IrllibError_e basic_attribute_of_roi_level_get (
    IrcmdHandle_t * handle,
    param_attribute_t * attribute )
```

Get attribute of roi level.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
out	<i>attribute</i>	attribute of roi level

Returns

see `IrllibError_e`

5.4.4.92 basic_attribute_of_scene_mode_get()

```
DLLEXPORT IrllibError_e basic_attribute_of_scene_mode_get (
    IrcmdHandle_t * handle,
    param_attribute_t * attribute )
```

Get basic attribute of image scene mode.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of scene mode

Returns

see [IrlibError_e](#)

5.4.4.93 basic_auto_ffc_current_params_get()

```
DLLEXPORT IrlibError\_e basic_auto_ffc_current_params_get (
    IrcmdHandle\_t * handle,
    int param_type,
    int * value )
```

Get current auto FFC's parameter.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>param_type</i>	auto FFC's parameter type,see basic_auto_ffc_param_e
out	<i>value</i>	auto FFC's parameter value

Returns

see [IrlibError_e](#)

5.4.4.94 basic_auto_ffc_current_params_set()

```
DLLEXPORT IrlibError\_e basic_auto_ffc_current_params_set (
    IrcmdHandle\_t * handle,
    int param_type,
    int value )
```

Set auto FFC's parameter if *param_type* is BASIC_AUTO_MAX_INTERVAL, value must more than 5 maximun interval must lower than minimun interval.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>param_type</i>	auto FFC's parameter type,see basic_auto_ffc_param_e
in	<i>value</i>	auto FFC's parameter value

Returns

see `IrlibError_e`

5.4.4.95 basic_auto_ffc_params_attribute_get()

```
DLLEXPORT IrlibError_e basic_auto_ffc_params_attribute_get (
    IrcmdHandle_t * handle,
    int param_type,
    param_attribute_t * attribute )
```

Get attribute of auto FFC's parameter.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>param_type</i>	auto FFC's parameter type,see <code>basic_ffc_auto_shutter_param_e</code>
out	<i>attribute</i>	attribute of auto FFC's parameter

Returns

see `IrlibError_e`

5.4.4.96 basic_auto_ffc_status_get()

```
DLLEXPORT IrlibError_e basic_auto_ffc_status_get (
    IrcmdHandle_t * handle,
    int * ffc_status )
```

Get auto shutter status.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
out	<i>ffc_status</i>	enable status of ffc,see <code>basic_auto_ffc_status_e</code>

Returns

see `IrlibError_e`

5.4.4.97 basic_auto_ffc_status_set()

```
DLLEXPORT IrlibError\_e basic_auto_ffc_status_set (
    IrcmdHandle\_t * handle,
    int ffc_status )
```

Set auto shutter status.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>ffc_status</i>	enable status of shutter,see basic_auto_ffc_status_e

Returns

see [IrlibError_e](#)

5.4.4.98 basic_center_zoom_get()

```
DLLEXPORT IrlibError\_e basic_center_zoom_get (
    IrcmdHandle\_t * handle,
    float * scale )
```

Get the scale.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>scale</i>	user-set scale, range:1-8 (accuracy: 0.1)

Returns

see [IrlibError_e](#)

5.4.4.99 basic_center_zoom_set()

```
DLLEXPORT IrlibError\_e basic_center_zoom_set (
    IrcmdHandle\_t * handle,
    float scale )
```

Set the scale.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>scale</i>	user-set scale, range:1-8 (accuracy: 0.1)

Returns

see `IrlibError_e`

5.4.4.100 basic_current_agc_level_get()

```
DLLEXPORT IrlibError_e basic_current_agc_level_get (
    IrcmdHandle_t * handle,
    int * level )
```

Get image parameters agc level,support for AC02.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
out	<i>level</i>	level of agc,range:0-100

Returns

see `IrlibError_e`

5.4.4.101 basic_current_brightness_level_get()

```
DLLEXPORT IrlibError_e basic_current_brightness_level_get (
    IrcmdHandle_t * handle,
    int * level )
```

Get image brightness level.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
out	<i>level</i>	Brightness level,range:0-100

Returns

see `IrlibError_e`

5.4.4.102 basic_current_contrast_level_get()

```
DLLEXPORT IrlibError_e basic_current_contrast_level_get (
    IrcmdHandle_t * handle,
    int * level )
```

Get image contrast level.The contrast corresponding to brightness.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	Contrast level,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.103 basic_current_detail_enhance_level_get()

```
DLLEXPORT IrlibError\_e basic_current_detail_enhance_level_get (
    IrcmdHandle\_t * handle,
    int * level )
```

Get image detail enhance level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	Detail enhance level,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.104 basic_current_gamma_level_get()

```
DLLEXPORT IrlibError\_e basic_current_gamma_level_get (
    IrcmdHandle\_t * handle,
    int * rate )
```

Get the current gamma level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>rate</i>	gamma rate, range: 0-100

Returns

see [IrlibError_e](#)

5.4.4.105 basic_current_image_noise_reduction_level_get()

```
DLLEXPORT IrlibError\_e basic_current_image_noise_reduction_level_get (
    IrcmdHandle\_t * handle,
    int * level )
```

Get current level of image noise reduction.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	level of noise reduction,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.106 basic_current_image_roi_level_get()

```
DLLEXPORT IrlibError\_e basic_current_image_roi_level_get (
    IrcmdHandle\_t * handle,
    int * level )
```

Get image parameters roi level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	level of roi,range:0-2, see basic_roi_level_e

Returns

see [IrlibError_e](#)

5.4.4.107 basic_current_image_scene_mode_get()

```
DLLEXPORT IrlibError\_e basic_current_image_scene_mode_get (
    IrcmdHandle\_t * handle,
    int * mode )
```

Get current image scene mode.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>mode</i>	current image image mode

Returns

see `IrLibError_e`

5.4.4.108 basic_device_current_status_get()

```
DLLEXPORT IrLibError_e basic_device_current_status_get (
    IrCmdHandle_t * handle,
    int * status )
```

Get current device status.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
out	<i>status</i>	the status of device, see <code>basic_device_status_e</code>

Returns

see `IrLibError_e`

5.4.4.109 basic_device_info_get()

```
DLLEXPORT IrLibError_e basic_device_info_get (
    IrCmdHandle_t * handle,
    int device_info_type,
    char * data )
```

Get device's information.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>device_info_type</i>	the device's information type, see <code>basic_dev_info_type_e</code>
out	<i>data[100]</i>	the device's information data

Returns

see `IrLibError_e`

5.4.4.110 basic_device_temp_get()

```
DLLEXPORT IrlibError\_e basic_device_temp_get (
    IrcmdHandle\_t * handle,
    float * temperature )
```

Get device's temperature.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>temperature</i>	module's temperature, unit: celsius

Returns

see [IrlibError_e](#)

5.4.4.111 basic_ffc_update()

```
DLLEXPORT IrlibError\_e basic_ffc_update (
    IrcmdHandle\_t * handle )
```

Manual shutter update.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.112 basic_firmware_download()

```
DLLEXPORT IrlibError\_e basic_firmware_download (
    IrcmdHandle\_t * handle,
    char * file_name,
    uint8_t * data,
    uint32_t file_len )
```

download firmware

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>file_name</i>	the name of firmware file
in	<i>data</i>	data of firmware file
in	<i>file_len</i>	size of firmware file

Returns

see `IruartError_e`

5.4.4.113 basic_firmware_log_data_get()

```
DLLEXPORT IrlibError_e basic_firmware_log_data_get (
    IrcmdHandle_t * handle,
    uint8_t * log_data )
```

Read the firmware log data.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
out	<i>log_data</i>	firmware log data(length:1024 * 1024Byte)

Returns

see `IrlibError_e`

5.4.4.114 basic_gain_get()

```
DLLEXPORT IrlibError_e basic_gain_get (
    IrcmdHandle_t * handle,
    int * gain_value )
```

Get gain value of the device.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
out	<i>gain_value</i>	the device's gain value,see <code>basic_gain_status_e</code>

Returns

see `IrlibError_e`

5.4.4.115 basic_gain_set()

```
DLLEXPORT IrlibError_e basic_gain_set (
    IrcmdHandle_t * handle,
    int value )
```

Set gain value of the device.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>value</i>	the device's gain value,see basic_gain_status_e

Returns

see [IrlibError_e](#)

5.4.4.116 basic_gamma_rate_set()

```
DLLEXPORT IrlibError\_e basic_gamma_rate_set (
    IrcmdHandle\_t * handle,
    int rate )
```

Set the gamma rate.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>rate</i>	gamma rate, range: 0-100

Returns

see [IrlibError_e](#)

5.4.4.117 basic_global_contrast_level_get()

```
DLLEXPORT IrlibError\_e basic_global_contrast_level_get (
    IrcmdHandle\_t * handle,
    int * level )
```

Get global contrast level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	Contrast level,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.118 basic_global_contrast_level_set()

```
DLLEXPORT IrlibError\_e basic_global_contrast_level_set (
    IrcmdHandle\_t * handle,
    int level )
```

Set global contrast level.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	Contrast level,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.119 basic_image_agc_level_set()

```
DLLEXPORT IrlibError\_e basic_image_agc_level_set (
    IrcmdHandle\_t * handle,
    int level )
```

Set image parameters agc level,support for AC02.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	level of agc,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.120 basic_image_brightness_level_set()

```
DLLEXPORT IrlibError\_e basic_image_brightness_level_set (
    IrcmdHandle\_t * handle,
    int level )
```

Set image brightness level.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	brightness level,range:0-100

Returns

see `IrlibError_e`

5.4.4.121 basic_image_contrast_level_set()

```
DLLEXPORT IrlibError_e basic_image_contrast_level_set (
    IrcmdHandle_t * handle,
    int level )
```

Set image contrast level. The contrast corresponding to brightness.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>level</i>	Contrast level, range: 0-100

Returns

see `IrlibError_e`

5.4.4.122 basic_image_detail_enhance_level_set()

```
DLLEXPORT IrlibError_e basic_image_detail_enhance_level_set (
    IrcmdHandle_t * handle,
    int level )
```

Set image detail enhance level.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>level</i>	detail enhance level, range: 0-100

Returns

see `IrlibError_e`

5.4.4.123 basic_image_noise_reduction_level_set()

```
DLLEXPORT IrlibError_e basic_image_noise_reduction_level_set (
    IrcmdHandle_t * handle,
    int level )
```

Set level of image noise reduction For tiny2C, this function only works in professional mode.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	level of noise reduction ,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.124 basic_image_roi_level_set()

```
DLLEXPORT IrlibError\_e basic_image_roi_level_set (
    IrcmdHandle\_t * handle,
    int level )
```

Set image parameters roi level.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	level of roi,range:0-2, see basic_roi_level_e

Returns

see [IrlibError_e](#)

5.4.4.125 basic_image_scene_mode_set()

```
DLLEXPORT IrlibError\_e basic_image_scene_mode_set (
    IrcmdHandle\_t * handle,
    int mode )
```

Set image scene mode.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>mode</i>	mode of image scene,see basic_image_scene_mode_e

Returns

see [IrlibError_e](#)

5.4.4.126 basic_mirror_and_flip_status_get()

```
DLL_EXPORT IrlibError_e basic_mirror_and_flip_status_get (
    IrcmdHandle_t * handle,
    int * value )
```

Get mirror and flip status.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>value</i>	the mirror and flip status, see basic_mirror_and_flip_status_e

Returns

see [IrlibError_e](#)

5.4.4.127 basic_mirror_and_flip_status_set()

```
DLL_EXPORT IrlibError_e basic_mirror_and_flip_status_set (
    IrcmdHandle_t * handle,
    int value )
```

Set mirror and flip status.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>value</i>	the mirror and flip status, see basic_mirror_and_flip_status_e

Returns

see [IrlibError_e](#)

5.4.4.128 basic_palette_idx_get()

```
DLL_EXPORT IrlibError_e basic_palette_idx_get (
    IrcmdHandle_t * handle,
    int * color_index )
```

Get current palette color index.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
out	<i>color_index</i>	palette color index

Returns

see `IrlibError_e`

5.4.4.129 basic_palette_idx_set()

```
DLLEXPORT IrlibError_e basic_palette_idx_set (
    IrcmdHandle_t * handle,
    int color_index )
```

Set palette color index.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
in	<i>color_index</i>	palette color index

Returns

see `IrlibError_e`

5.4.4.130 basic_palette_name_get()

```
DLLEXPORT IrlibError_e basic_palette_name_get (
    IrcmdHandle_t * handle,
    int color_index,
    char * color_name )
```

Get palette color name.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
in	<i>color_index</i>	palette color index
out	<i>color_name[10]</i>	palette color name.eg:YP0500

Returns

see `IrlibError_e`

5.4.4.131 basic_palette_num_get()

```
DLLEXPORT IrlibError\_e basic_palette_num_get (
    IrcmdHandle\_t * handle,
    int * color_number )
```

Get the total number of palette color tables supported by the device.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>color_number</i>	the total number of palette color tables

Returns

see [IrlibError_e](#)

5.4.4.132 basic_preview_start()

```
DLLEXPORT IrlibError\_e basic_preview_start (
    IrcmdHandle\_t * handle,
    VideoOutputInfo\_t output_info )
```

Choose the appropriate mode to start stream @support Tiny2C USB,DVP,MIPI,VO_SPI.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>output_info</i>	see VideoOutputInfo_t , dont need to fill video_output_fps

Returns

see [IrlibError_e](#)

5.4.4.133 basic_preview_stop()

```
DLLEXPORT IrlibError\_e basic_preview_stop (
    IrcmdHandle\_t * handle )
```

Stop stream.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.134 basic_reset_to_rom()

```
DLLEXPORT IrlibError\_e basic_reset_to_rom (
    IrcmdHandle\_t * handle )
```

Reset to rom, prepare for update firmware.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.135 basic_restore_default_data()

```
DLLEXPORT IrlibError\_e basic_restore_default_data (
    IrcmdHandle\_t * handle,
    int type )
```

Restore default data.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>type</i>	the type of restore data,see basic_restore_type_e

Returns

see [IrlibError_e](#)

5.4.4.136 basic_rtc_current_time_get()

```
DLLEXPORT IrlibError_e basic_rtc_current_time_get (
    IrcmdHandle_t * handle,
    Rtc_Time_t * time )
```

Get the current time of the device.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>time</i>	record the power-on time,see Rtc_Time_t

Returns

see [IrlibError_e](#)

5.4.4.137 basic_rtc_current_time_set()

```
DLLEXPORT IrlibError_e basic_rtc_current_time_set (
    IrcmdHandle_t * handle,
    Rtc_Time_t * rtc_time )
```

Set the current time of the device.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>time</i>	rtc start time,see Rtc_Time_t

Returns

see [IrlibError_e](#)

5.4.4.138 basic_save_data()

```
DLLEXPORT IrlibError_e basic_save_data (
    IrcmdHandle_t * handle,
    int type )
```

Save data, include algorithm parameters, system parameters, calib data.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>type</i>	the type of save data,see basic_save_type_e

Returns

see `IrllibError_e`

5.4.4.139 basic_space_noise_reduce_attribute_get()

```
DLLEXPORT IrllibError_e basic_space_noise_reduce_attribute_get (
    IrcmdHandle_t * handle,
    param_attribute_t * attribute )
```

Get the attribute of space noise reduce level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of space noise reduce level

Returns

see `IrllibError_e`

5.4.4.140 basic_space_noise_reduce_level_get()

```
DLLEXPORT IrllibError_e basic_space_noise_reduce_level_get (
    IrcmdHandle_t * handle,
    int * level )
```

Get the space noise reduce level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	space noise reduce level, range: 0-100

Returns

see `IrllibError_e`

5.4.4.141 basic_space_noise_reduce_level_set()

```
DLLEXPORT IrllibError_e basic_space_noise_reduce_level_set (
    IrcmdHandle_t * handle,
    int level )
```

Set the space noise reduce level.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	space noise reduce level, range: 0-100

Returns

see [IrlibError_e](#)

5.4.4.142 basic_sun_detect_pixel_ratio_get()

```
DLLEXPORT IrlibError_e basic_sun_detect_pixel_ratio_get (
    IrcmdHandle_t * handle,
    int * max_pixel_ratio,
    int * min_pixel_ratio )
```

Get the range of the current saturation pixel ratio.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>max_pixel_ratio</i>	maximum ratio of saturated pixels, must be bigger than min_pixel_ratio ,range:0-100
out	<i>min_pixel_ratio</i>	minimum ratio of saturated pixels, must be smaller than max_pixel_ratio ,range:0-100

Returns

see [IrlibError_e](#)

5.4.4.143 basic_sun_detect_pixel_ratio_set()

```
DLLEXPORT IrlibError_e basic_sun_detect_pixel_ratio_set (
    IrcmdHandle_t * handle,
    int max_pixel_ratio,
    int min_pixel_ratio )
```

Set the range of saturation pixel ratio for sun detect. The more the ratio, the more saturated pixel. When the saturation pixel ratio is within the setting range, it is determined that the saturation is caused by the sun.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>max_pixel_ratio</i>	maximum ratio of saturated pixels, must be bigger than min_pixel_ratio ,range:0-100
in	<i>min_pixel_ratio</i>	minimum ratio of saturated pixels, must be smaller than max_pixel_ratio ,range:0-100

Returns

see `IrlibError_e`

5.4.4.144 basic_sun_detect_roundness_level_get()

```
DLLEXPORT IrlibError_e basic_sun_detect_roundness_level_get (
    IrcmdHandle_t * handle,
    int * max_roundness_level,
    int * min_roundness_level )
```

Get the range of the current roundness level.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
out	<i>max_roundness_level</i>	maximum level of roundness
out	<i>min_roundness_level</i>	minimum level of roundness

Returns

see `IrlibError_e`

5.4.4.145 basic_sun_detect_roundness_level_set()

```
DLLEXPORT IrlibError_e basic_sun_detect_roundness_level_set (
    IrcmdHandle_t * handle,
    int max_roundness_level,
    int min_roundness_level )
```

Set the range of roundness level for sun detect. The lower the level, the more round the object is. When the roundness level of the object is within the setting range, the object is judged to be sun.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>max_roundness_level</i>	maximum level of roundness, must be higher than <code>min_roundness_level</code> , range:0-100
in	<i>min_roundness_level</i>	maximum level of roundness, must be lower than <code>max_roundness_level</code> , range:0-100

Returns

see `IrlibError_e`

5.4.4.146 basic_sun_detect_switch_get()

```
DLLEXPORT IrlibError_e basic_sun_detect_switch_get (
    IrcmdHandle_t * handle,
    int * status )
```

Get current sun detect switch status.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>status</i>	the switch status of sun detect,see basic_sun_detect_status_e

Returns

see [IrlibError_e](#)

5.4.4.147 basic_sun_detect_switch_set()

```
DLLEXPORT IrlibError_e basic_sun_detect_switch_set (
    IrcmdHandle_t * handle,
    int status )
```

Set sun detect switch for sunburn protection.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	the switch status of sun detect,see basic_sun_detect_status_e

Returns

see [IrlibError_e](#)

5.4.4.148 basic_time_noise_reduce_attribute_get()

```
DLLEXPORT IrlibError_e basic_time_noise_reduce_attribute_get (
    IrcmdHandle_t * handle,
    param_attribute_t * attribute )
```

Get the attribute of time noise reduce level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>attribute</i>	attribute of time noise reduce level

Returns

see [IrlibError_e](#)

5.4.4.149 basic_time_noise_reduce_level_get()

```
DLLEXPORT IrlibError\_e basic_time_noise_reduce_level_get (
    IrcmdHandle\_t * handle,
    int * level )
```

Get the time noise reduce level.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>level</i>	time noise reduce level, range: 0-100

Returns

see [IrlibError_e](#)

5.4.4.150 basic_time_noise_reduce_level_set()

```
DLLEXPORT IrlibError\_e basic_time_noise_reduce_level_set (
    IrcmdHandle\_t * handle,
    int level )
```

Set the time noise reduce level.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>level</i>	time noise reduce level, range: 0-100

Returns

see [IrlibError_e](#)

5.4.4.151 basic_video_stream_continue()

```
DLLEXPORT IrlibError_e basic_video_stream_continue (
    IrcmdHandle_t * handle )
```

When the device is in automatic plotting mode, let the stream flow out.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
----	---------------	-----------------------

Returns

see IrlibError_e

5.4.4.152 basic_video_stream_pause()

```
DLLEXPORT IrlibError_e basic_video_stream_pause (
    IrcmdHandle_t * handle )
```

When the device is in automatic drawing mode, the data stream is temporarily not flowed out.

Parameters

in	<i>handle</i>	<i>ircmd's</i> handle
----	---------------	-----------------------

Returns

see IrlibError_e

5.4.4.153 ircmd_create_handle()

```
DLLEXPORT IrcmdHandle_t* ircmd_create_handle (
    IrControlHandle_t * ir_control_handle )
```

Create and initialize *ircmd's* handle.

Parameters

in	<i>ir_control_handle</i>	<i>ir_control's</i> handle
----	--------------------------	----------------------------

Returns

IrcmdHandle_t struct pointer

5.4.4.154 ircmd_delete_callback_function()

```
DLLEXPORT IrlibError\_e ircmd_delete_callback_function (
    IrcmdHandle\_t * handle,
    int mode )
```

Delete callback function.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>mode</i>	location of callback function calls,see callback_mode_e

Returns

see [IrlibError_e](#)

5.4.4.155 ircmd_delete_handle()

```
DLLEXPORT IrlibError\_e ircmd_delete_handle (
    IrcmdHandle\_t * handle )
```

Delete and release [ircmd's](#) handle.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.4.4.156 ircmd_get_percentage_of_data_transfer()

```
DLLEXPORT IrlibError\_e ircmd_get_percentage_of_data_transfer (
    IrcmdHandle\_t * handle,
    float * percentage )
```

Get percentage of data transfer.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>percentage</i>	percentage of data transfer

Returns

see [IrlibError_e](#)

5.4.4.157 ircmd_init_callback_function()

```
DLLEXPORT IrlibError_e ircmd_init_callback_function (
    IrcmdHandle_t * handle,
    int mode,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

Init callback function.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>mode</i>	location of callback function calls,see callback_mode_e
in	<i>callback</i>	the callback function of ircmd
in	<i>callback_data</i>	the parameter of callback function
in	<i>priv_data</i>	the private parameter of callback function

Returns

see [IrlibError_e](#)

5.4.4.158 ircmd_log_register()

```
DLLEXPORT void ircmd_log_register (
    IrcmdLogLevel_e log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

Register log print via *log_level*.

Parameters

in	<i>log_level</i>	debug information level
in	<i>callback</i>	callback function for printing log messages
in	<i>priv_data</i>	the private parameter of callback function

5.4.4.159 ircmd_version()

```
const DLLEXPORT char* ircmd_version (
    void )
```

Get libircmd library's version.

Returns

version's string

5.4.4.160 ircmd_version_number()

```
const DLLEXPORT char* ircmd_version_number (
    void )
```

Get libircmd library's version number.

Returns

version number's string

5.4.4.161 vdcmd_set_polling_wait_time()

```
DLLEXPORT IrlibError_e vdcmd_set_polling_wait_time (
    IrcmdHandle_t * handle,
    uint16_t timeout_ms )
```

Set single command polling time.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>timeout_ms</i>	timeout time,uint:ms

Returns

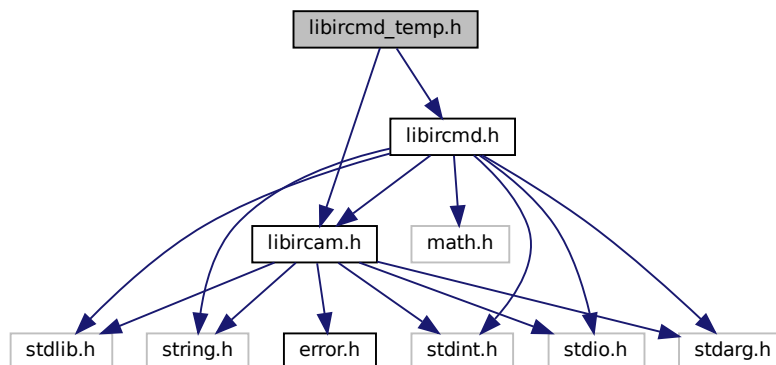
see IrlibError_e

5.5 libircmd_temp.h File Reference

```
#include "libircam.h"
```

```
#include "libircmd.h"
```

Include dependency graph for libircmd_temp.h:



Classes

- struct [IrcmdLine_t](#)
Line struct, start from 1.

Enumerations

- enum [tpd_ktbt_recal_point_idx](#) { [TPD_KTBT_RECAL_P1](#) = 0, [TPD_KTBT_RECAL_P2](#) }
Ktbt recalculate point index.

Functions

- `DLLEXPORT IrlibError\_e basic_point_temp_info_get (IrcmdHandle\_t *handle, IrcmdPoint\_t point_pos, float *point_temp_value)`
Get the temperature value of a point.
- `DLLEXPORT IrlibError\_e basic_line_temp_info_get (IrcmdHandle\_t *handle, IrcmdLine\_t line_pos, LineRectTempInfo\_t *line_temp_info)`
Get the temperature information of a line.
- `DLLEXPORT IrlibError\_e basic_rect_temp_info_get (IrcmdHandle\_t *handle, IrcmdRect\_t rect_pos, LineRectTempInfo\_t *rect_temp_info)`
Get the temperature information of a rectangle.
- `DLLEXPORT IrlibError\_e basic_frame_temp_info_get (IrcmdHandle\_t *handle, MaxMinTempInfo\_t *frame_temp_info)`
Get the temperature information of the whole frame.
- `DLLEXPORT IrlibError\_e adv_show_point_temp (IrcmdHandle\_t *handle, int status, uint8_t point_idx, IrcmdPoint\_t point)`
Display the point's temperature on the screen.
- `DLLEXPORT IrlibError\_e adv_show_frame_temp (IrcmdHandle\_t *handle, int status)`
Display the max and min frame temperature points on the screen.
- `DLLEXPORT IrlibError\_e adv_env_correct_switch_set (IrcmdHandle\_t *handle, int status)`
Set environment correct switch @support Tiny2C.

- DLLEXPORT [IrlibError_e](#) [adv_env_correct_switch_get](#) ([IrcmdHandle_t](#) *handle, int *status)
Get environment correct switch @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_tu_set](#) ([IrcmdHandle_t](#) *handle, int tu)
Set tu in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_tu_get](#) ([IrcmdHandle_t](#) *handle, int *tu)
Get tu in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_ta_set](#) ([IrcmdHandle_t](#) *handle, int ta)
Set ta in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_ta_get](#) ([IrcmdHandle_t](#) *handle, int *ta)
Get ta in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_tau_set](#) ([IrcmdHandle_t](#) *handle, int tau)
Set tau in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_tau_get](#) ([IrcmdHandle_t](#) *handle, int *tau)
Get tau in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_ems_set](#) ([IrcmdHandle_t](#) *handle, int ems)
Set ems in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_env_correct_ems_get](#) ([IrcmdHandle_t](#) *handle, int *ems)
Get ems in environment correct @support Tiny2C.
- DLLEXPORT [IrlibError_e](#) [adv_tpd_point_coord_set](#) ([IrcmdHandle_t](#) *handle, uint8_t index, [IrcmdPoint_t](#) point_pos)
Hide the point's temperature measurement.
- DLLEXPORT [IrlibError_e](#) [adv_tpd_line_coord_set](#) ([IrcmdHandle_t](#) *handle, uint8_t index, [IrcmdLine_t](#) line↔_pos)
Set line coord.
- DLLEXPORT [IrlibError_e](#) [adv_tpd_rect_coord_set](#) ([IrcmdHandle_t](#) *handle, uint8_t index, [IrcmdRect_t](#) rect↔_pos)
Set rect coord.
- DLLEXPORT [IrlibError_e](#) [adv_recal_tpd_by_1point](#) ([IrcmdHandle_t](#) *handle, float temp)
Auto recalculate the kt_bt by 1 point's temperature.
- DLLEXPORT [IrlibError_e](#) [adv_recal_tpd_by_1point_cancel](#) ([IrcmdHandle_t](#) *handle)
Cancel tpd recal data.
- DLLEXPORT [IrlibError_e](#) [adv_recal_tpd_by_1point_clear](#) ([IrcmdHandle_t](#) *handle)
Clear tpd recal data.
- DLLEXPORT [IrlibError_e](#) [adv_recal_tpd_by_2point](#) ([IrcmdHandle_t](#) *handle, int point_index, float temp)
Auto recalculate the kt_bt by 2 points' temperature, set P1(low temp) first, then set P2(high temp)
- DLLEXPORT [IrlibError_e](#) [adv_recal_tpd_by_2point_cancel](#) ([IrcmdHandle_t](#) *handle)
Cancel thre resulet of kt_bt recalibration.
- DLLEXPORT [IrlibError_e](#) [adv_recal_tpd_by_2point_clear](#) ([IrcmdHandle_t](#) *handle)
Clear thre resulet of kt_bt recalibration.

5.5.1 Enumeration Type Documentation

5.5.1.1 tpd_ktbt_recal_point_idx

```
enum tpd\_ktbt\_recal\_point\_idx
```

Ktbt recalculate point index.

Enumerator

TPD_KTBT_RECAL_P1	the first point
TPD_KTBT_RECAL_P2	the second point

5.5.2 Function Documentation

5.5.2.1 `adv_env_correct_ems_get()`

```
DLLEXPORT IrlibError\_e adv_env_correct_ems_get (
    IrcmdHandle\_t * handle,
    int * ems )
```

Get ems in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>ems</i>	target emissivity, range:1~16384

Returns

see [IrlibError_e](#)

5.5.2.2 `adv_env_correct_ems_set()`

```
DLLEXPORT IrlibError\_e adv_env_correct_ems_set (
    IrcmdHandle\_t * handle,
    int ems )
```

Set ems in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>ems</i>	target emissivity, range:1~16384

Returns

see [IrlibError_e](#)

5.5.2.3 adv_env_correct_switch_get()

```
DLLEXPORT IrlibError\_e adv_env_correct_switch_get (
    IrcmdHandle\_t * handle,
    int * status )
```

Get environment correct switch @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	enable or disable, see basic_enable_status_e

Returns

see [IrlibError_e](#)

5.5.2.4 adv_env_correct_switch_set()

```
DLLEXPORT IrlibError\_e adv_env_correct_switch_set (
    IrcmdHandle\_t * handle,
    int status )
```

Set environment correct switch @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	enable or disable, see basic_enable_status_e

Returns

see [IrlibError_e](#)

5.5.2.5 adv_env_correct_ta_get()

```
DLLEXPORT IrlibError\_e adv_env_correct_ta_get (
    IrcmdHandle\_t * handle,
    int * ta )
```

Get ta in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>ta</i>	atmospheric temperature, range:233~373 (unit: kelvin degree celsius)

Returns

see [IrlibError_e](#)

5.5.2.6 adv_env_correct_ta_set()

```
DLLEXPORT IrlibError\_e adv_env_correct_ta_set (
    IrcmdHandle\_t * handle,
    int ta )
```

Set ta in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>ta</i>	atmospheric temperature, range:233~373 (unit: kelvin degree celsius)

Returns

see [IrlibError_e](#)

5.5.2.7 adv_env_correct_tau_get()

```
DLLEXPORT IrlibError\_e adv_env_correct_tau_get (
    IrcmdHandle\_t * handle,
    int * tau )
```

Get tau in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>tau</i>	atmospheric transmittance, range:1~16384

Returns

see [IrlibError_e](#)

5.5.2.8 adv_env_correct_tau_set()

```
DLLEXPORT IrlibError\_e adv_env_correct_tau_set (
    IrcmdHandle\_t * handle,
    int tau )
```

Set tau in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>tau</i>	atmospheric transmittance, range:1~16384

Returns

see [IrlibError_e](#)

5.5.2.9 adv_env_correct_tu_get()

```
DLLEXPORT IrlibError\_e adv_env_correct_tu_get (
    IrcmdHandle\_t * handle,
    int * tu )
```

Get tu in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>tu</i>	reflection temperature, range:233~373 (unit: kelvin degree celsius)

Returns

see [IrlibError_e](#)

5.5.2.10 adv_env_correct_tu_set()

```
DLLEXPORT IrlibError\_e adv_env_correct_tu_set (
    IrcmdHandle\_t * handle,
    int tu )
```

Set tu in environment correct @support Tiny2C.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>tu</i>	reflection temperature, range:233~373 (unit: kelvin degree celsius)

Returns

see `IrLibError_e`

5.5.2.11 `adv_recal_tpd_by_1point()`

```
DLLEXPORT IrLibError_e adv_recal_tpd_by_1point (
    IrcmdHandle_t * handle,
    float temp )
```

Auto recalculate the `kt_bt` by 1 point's temperature.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>temp</i>	the target black body's temperature when calibrating

Returns

see `IrLibError_e`

5.5.2.12 `adv_recal_tpd_by_1point_cancel()`

```
DLLEXPORT IrLibError_e adv_recal_tpd_by_1point_cancel (
    IrcmdHandle_t * handle )
```

Cancel tpd recal data.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
----	---------------	-----------------------------

Returns

see `IrLibError_e`

5.5.2.13 `adv_recal_tpd_by_1point_clear()`

```
DLLEXPORT IrLibError_e adv_recal_tpd_by_1point_clear (
    IrcmdHandle_t * handle )
```

Clear tpd recal data.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.5.2.14 adv_recal_tpd_by_2point()

```
DLLEXPORT IrlibError\_e adv_recal_tpd_by_2point (
    IrcmdHandle\_t * handle,
    int point_index,
    float temp )
```

Auto recalculate the kt_bt by 2 points' temperature, set P1(low temp) first, then set P2(high temp)

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>point_index</i>	choose P1 for low temp, P2 for high temp, see tpd_ktbt_recal_point_idx
in	<i>temp</i>	the target black body's temperature when calibrating

Returns

see [IrlibError_e](#)

5.5.2.15 adv_recal_tpd_by_2point_cancel()

```
DLLEXPORT IrlibError\_e adv_recal_tpd_by_2point_cancel (
    IrcmdHandle\_t * handle )
```

Cancel thre resulet of kt_bt recalibration.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.5.2.16 `adv_recal_tpd_by_2point_clear()`

```
DLLEXPORT IrlibError\_e adv_recal_tpd_by_2point_clear (
    IrcmdHandle\_t * handle )
```

Clear the result of `kt_bt` recalibration.

Parameters

in	<i>handle</i>	ircmd's handle
----	---------------	--------------------------------

Returns

see [IrlibError_e](#)

5.5.2.17 `adv_show_frame_temp()`

```
DLLEXPORT IrlibError\_e adv_show_frame_temp (
    IrcmdHandle\_t * handle,
    int status )
```

Display the max and min frame temperature points on the screen.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	enable or disable, see basic_enable_status_e

Returns

see [IrlibError_e](#)

5.5.2.18 `adv_show_point_temp()`

```
DLLEXPORT IrlibError\_e adv_show_point_temp (
    IrcmdHandle\_t * handle,
    int status,
    uint8_t point_idx,
    IrcmdPoint\_t point )
```

Display the point's temperature on the screen.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>status</i>	enable or disable, see basic_enable_status_e
in	<i>point_idx</i>	point's index(0-4)
in	<i>point</i>	the position information of point

Returns

see `IrlibError_e`

5.5.2.19 adv_tpd_line_coord_set()

```
DLLEXPORT IrlibError_e adv_tpd_line_coord_set (
    IrcmdHandle_t * handle,
    uint8_t index,
    IrcmdLine_t line_pos )
```

Set line coord.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>index</i>	line index,range:0-1
in	<i>line_pos</i>	the line's position

Returns

see `IrlibError_e`

5.5.2.20 adv_tpd_point_coord_set()

```
DLLEXPORT IrlibError_e adv_tpd_point_coord_set (
    IrcmdHandle_t * handle,
    uint8_t index,
    IrcmdPoint_t point_pos )
```

Hide the point's temperature measurement.

Parameters

in	<i>handle</i>	<code>ircmd's</code> handle
in	<i>path</i>	the path of preview
in	<i>point_idx</i>	the position infomation of point

Returns

see `IrlibError_e`

Display the line's temperature measurement.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>path</i>	the path of preview
in	<i>line_idx</i>	line's index.(0-5)
in	<i>line</i>	the position infomation of line

Returns

see [IrlibError_e](#)

Hide the line's temperature measurement.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>path</i>	the path of preview
in	<i>line_idx</i>	the position infomation of line

Returns

see [IrlibError_e](#)

Display the rectangle's temperature measurement.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>path</i>	the path of preview
in	<i>rect_idx</i>	rectangle's index.(0-5)
in	<i>rect</i>	the position infomation of rectangle

Returns

see [IrlibError_e](#)

Hide the rectangle's temperature measurement.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>path</i>	the path of preview
in	<i>rect_idx</i>	the position infomation of rectangle

Returns

see [IrlibError_e](#)

Hide the whole frame temperature measurement.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>path</i>	the path of preview

Returns

see [IrlibError_e](#)

Set point coord

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>index</i>	point index,range:0-8
in	<i>point_pos</i>	the point's position

Returns

see [IrlibError_e](#)

5.5.2.21 adv_tpd_rect_coord_set()

```
DLLEXPORT IrlibError\_e adv_tpd_rect_coord_set (
    IrcmdHandle\_t * handle,
    uint8_t index,
    IrcmdRect\_t rect_pos )
```

Set rect coord.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>index</i>	rect index.range:0-5
in	<i>rect_pos</i>	the rect's position

Returns

see [IrlibError_e](#)

5.5.2.22 basic_frame_temp_info_get()

```
DLLEXPORT IrlibError\_e basic_frame_temp_info_get (
    IrcmdHandle\_t * handle,
    MaxMinTempInfo\_t * frame_temp_info )
```

Get the temperature information of the whole frame.

Parameters

in	<i>handle</i>	ircmd's handle
out	<i>frame_temp_info</i>	the temperature information of the whole frame

Returns

see [IrlibError_e](#)

5.5.2.23 basic_line_temp_info_get()

```
DLL_EXPORT IrlibError_e basic_line_temp_info_get (
    IrcmdHandle_t * handle,
    IrcmdLine_t line_pos,
    LineRectTempInfo_t * line_temp_info )
```

Get the temperature information of a line.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>line_pos</i>	the line's position
out	<i>line_temp_info</i>	the temperature information of the line

Returns

see [IrlibError_e](#)

5.5.2.24 basic_point_temp_info_get()

```
DLL_EXPORT IrlibError_e basic_point_temp_info_get (
    IrcmdHandle_t * handle,
    IrcmdPoint_t point_pos,
    float * point_temp_value )
```

Get the temperature value of a point.

Parameters

in	<i>handle</i>	ircmd's handle
in	<i>point_pos</i>	the point's position
out	<i>point_temp_value</i>	the point temp value,Unit:Celsius

Returns

see `IrLibError_e`

5.5.2.25 `basic_rect_temp_info_get()`

```
DLLEXPORT IrLibError_e basic_rect_temp_info_get (
    IrcmdHandle_t * handle,
    IrcmdRect_t rect_pos,
    LineRectTempInfo_t * rect_temp_info )
```

Get the temperature information of a rectangle.

Parameters

in	<i>handle</i>	<code>ircmd</code> 's handle
in	<i>rect_pos</i>	the rectangle's position
out	<i>rect_temp_info</i>	the temperature information of the rectangle

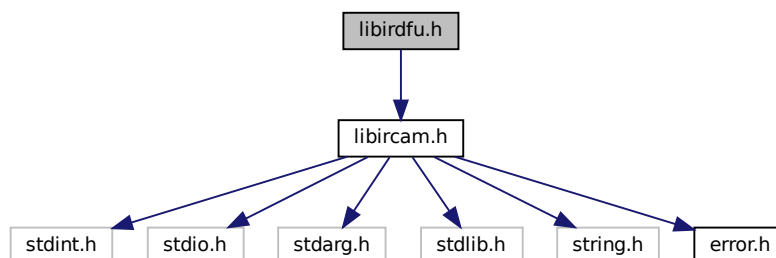
Returns

see `IrLibError_e`

5.6 libirdfu.h File Reference

```
#include "libircam.h"
```

Include dependency graph for libirdfu.h:



Macros

- `#define` `DFU_FUNCTION_IN_DEVELOPING`

Enumerations

- enum [IrdFuLogLevel_e](#) { IRDFU_LOG_DEBUG = 0, IRDFU_LOG_ERROR = 1, IRDFU_LOG_NO_PRINT = 2 }

Log level definition in libirdfu library.

Functions

- const DLLEXPORT char * [irdfu_version](#) (void)
Get libirdfu library's version.
- const DLLEXPORT char * [irdfu_version_number](#) (void)
Get libirdfu library's version number.
- DLLEXPORT void [irdfu_log_register](#) ([IrdFuLogLevel_e](#) log_level, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)
Register log print via log_level.
- DLLEXPORT [IrlibError_e](#) [irdfu_handle_create](#) ([IrControlHandle_t](#) *ir_control_handle)
Assigning values to function pointers.
- DLLEXPORT [IrlibError_e](#) [irdfu_handle_delete](#) ([IrControlHandle_t](#) *ir_control_handle)
Emptying function pointer data.
- DLLEXPORT [IrlibError_e](#) [irdfu_detect_device_status](#) (void *dfu_handle, void *params, int *status)
detect device status with usb interface
- DLLEXPORT [IrlibError_e](#) [irdfu_firmware_download](#) (void *dfu_handle, char *file_name, uint8_t *data, uint32_t file_len, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)
download firmware with usb interface

Variables

- DLLEXPORT typedef void(* [irdfu_callback](#))(void *callback_data, void *priv_data)

5.6.1 Macro Definition Documentation

5.6.1.1 DFU_FUNCTION_IN_DEVELOPING

```
#define DFU_FUNCTION_IN_DEVELOPING
```

5.6.2 Enumeration Type Documentation

5.6.2.1 IrdFuLogLevel_e

```
enum IrdFuLogLevel\_e
```

Log level definition in libirdfu library.

Enumerator

IRDFU_LOG_DEBUG	print debug and error infomation
IRDFU_LOG_ERROR	only print error infomation
IRDFU_LOG_NO_PRINT	don't print debug and error infomation

5.6.3 Function Documentation

5.6.3.1 irdfu_detect_device_status()

```
DLLEXPORT IrlibError_e irdfu_detect_device_status (
    void * dfu_handle,
    void * params,
    int * status )
```

detect device status with usb interface

Parameters

in	<i>dfu_handle</i>	irdfu's handle
in	<i>params</i>	reserved parameter of status detection
out	<i>status</i>	status of devices,0:Unknow status,1:Rom code,2:Cache Code

Returns

see IrlibError_e

5.6.3.2 irdfu_firmware_download()

```
DLLEXPORT IrlibError_e irdfu_firmware_download (
    void * dfu_handle,
    char * file_name,
    uint8_t * data,
    uint32_t file_len,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

download firmware with usb interface

Parameters

in	<i>dfu_handle</i>	irdfu's handle
in	<i>file_name</i>	the name of firmware file
in	<i>data</i>	data of firmware file
in	<i>file_len</i>	size of firmware file
in	<i>callback</i>	callback function of usb firmware download function
in	<i>priv_data</i>	the private parameter of callback function

Returns

see `IrLibError_e`

5.6.3.3 irdfu_handle_create()

```
DLLEXPORT IrLibError_e irdfu_handle_create (
    IrControlHandle_t * ir_control_handle )
```

Assigning values to function pointers.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

see `IrLibError_e`

5.6.3.4 irdfu_handle_delete()

```
DLLEXPORT IrLibError_e irdfu_handle_delete (
    IrControlHandle_t * ir_control_handle )
```

Emptying function pointer data.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

see `IrLibError_e`

5.6.3.5 irdfu_log_register()

```
DLLEXPORT void irdfu_log_register (
    IrdfuLogLevels_e log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

Register log print via `log_level`.

Parameters

in	<i>log_level</i>	debug information level
in	<i>callback</i>	callback function for printing log messages
in	<i>priv_data</i>	the private parameter of callback function

5.6.3.6 irdfu_version()

```
const DLLEXPORT char* irdfu_version (  
    void )
```

Get libirdfu library's version.

Returns

version's string

5.6.3.7 irdfu_version_number()

```
const DLLEXPORT char* irdfu_version_number (  
    void )
```

Get libirdfu library's version number.

Returns

version number's string

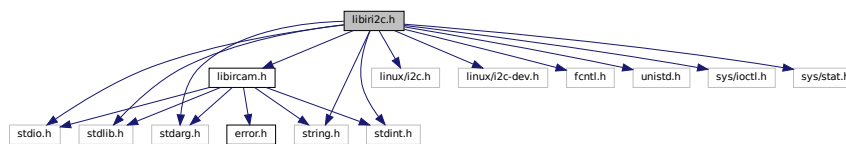
5.6.4 Variable Documentation**5.6.4.1 irdfu_callback**

```
DLLEXPORT typedef void(* irdfu_callback) (void *callback_data, void *priv_data)
```

5.7 libiri2c.h File Reference

```
#include <stdio.h>
#include <linux/i2c.h>
#include <linux/i2c-dev.h>
#include <fcntl.h>
#include <stdlib.h>
#include <stdarg.h>
#include <unistd.h>
#include <sys/ioctl.h>
#include <sys/stat.h>
#include <string.h>
#include <stdint.h>
#include "libircam.h"
```

Include dependency graph for libiri2c.h:



Classes

- struct [I2cParam_t](#)
- struct [I2cConDevParams_t](#)

Typedefs

- typedef struct [_Iri2CHandle_t](#) [Iri2CHandle_t](#)

Enumerations

- enum [Iri2cLogLevels_e](#) { [IRI2C_LOG_DEBUG](#) = 0, [IRI2C_LOG_ERROR](#) = 1, [IRI2C_LOG_NO_PRINT](#) = 2 }
- Log level definition in libiri2c library.*

Functions

- void [iri2c_log_register](#) (enum [Iri2cLogLevels_e](#) log_level, void [*\(*callback\)\(void *callback_data, void *priv_data\)](#), void [*priv_data](#))
register log level for libiri2c library
- const char * [iri2c_version](#) (void)
Get libiri2c library's version.
- const char * [iri2c_version_number](#) (void)
Get libiri2c library's version number.
- [Iri2CHandle_t](#) * [iri2c_handle_create](#) ([Iri2CHandle_t](#) *ir_control_handle)
Create an iri2c's handle.
- void [iri2c_handle_delete](#) ([Iri2CHandle_t](#) *ir_control_handle)

- Delete and release iri2c's handle.*

 - [IrlibError_e iri2c_device_open](#) ([Iri2cHandle_t](#) *i2c_handle, const char *dev_node)

Open an ir i2c device with its device node.
- [IrlibError_e iri2c_device_init](#) ([Iri2cHandle_t](#) *i2c_handle, [I2cConDevParams_t](#) *param)

Initialize the configuration of the control node.
- [IrlibError_e iri2c_data_read](#) ([Iri2cHandle_t](#) *i2c_handle, void *cmd_param, uint8_t *data, int len)

read data from the i2c device
- [IrlibError_e iri2c_data_write](#) ([Iri2cHandle_t](#) *i2c_handle, void *cmd_param, uint8_t *data, int len)

write data to the i2c device
- [IrlibError_e iri2c_device_release](#) ([Iri2cHandle_t](#) *i2c_handle, [I2cConDevParams_t](#) *param)

Release ir2c's configuration of the control node.
- [IrlibError_e iri2c_device_close](#) ([Iri2cHandle_t](#) *i2c_handle)

close an i2c device
- [IrlibError_e iri2c_firmware_download](#) ([Iri2cHandle_t](#) *i2c_handle, char *file_name, uint8_t *data, uint32_t file_len, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)

download firmware with i2c interface
- [IrlibError_e iri2c_detect_device_status](#) ([Iri2cHandle_t](#) *i2c_handle, void *params, int *status)

detect device status with i2c interface
- [IrlibError_e iri2c_command_channel_get](#) ([Iri2cHandle_t](#) *i2c_handle, int *command_channel_type)

Get i2c command channel type.
- [IrlibError_e iri2c_bootloader_download](#) ([Iri2cHandle_t](#) *i2c_handle, char *file_name, uint8_t *data, uint32_t file_len, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)

download bootloader with i2c interface

Variables

- void(* [iri2c_debug_print](#))(const char *fmt,...)
- Error type in libiri2c library.*
- void(* [iri2c_error_print](#))(const char *fmt,...)

5.7.1 Typedef Documentation

5.7.1.1 Iri2cHandle_t

```
typedef struct _Iri2cHandle_t Iri2cHandle_t
```

5.7.2 Enumeration Type Documentation

5.7.2.1 Iri2cLogLevels_e

```
enum Iri2cLogLevels_e
```

Log level definition in libiri2c library.

Enumerator

IRI2C_LOG_DEBUG	print debug and error information
IRI2C_LOG_ERROR	only print error information
IRI2C_LOG_NO_PRINT	don't print debug and error information

5.7.3 Function Documentation

5.7.3.1 iri2c_bootloader_download()

```

IrlibError_e iri2c_bootloader_download (
    Iri2cHandle_t * i2c_handle,
    char * file_name,
    uint8_t * data,
    uint32_t file_len,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )

```

download bootloader with i2c interface

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>file_name</i>	the name of bootloader file
in	<i>data</i>	data of bootloader file
in	<i>file_len</i>	size of bootloader file
in	<i>callback</i>	callback function of i2c bootloader download function
in	<i>priv_data</i>	the private parameter of bootloader download callback function

Returns

see IrlibError_e

5.7.3.2 iri2c_command_channel_get()

```

IrlibError_e iri2c_command_channel_get (
    Iri2cHandle_t * i2c_handle,
    int * command_channel_type )

```

Get i2c command channel type.

Parameters

in	<i>i2c_handle</i>	iri2c's handle
out	<i>command_channel_type</i>	command channel of i2c,see
		command_channel_e

Returns

see `IrlibError_e`

5.7.3.3 iri2c_data_read()

```
IrlibError_e iri2c_data_read (  
    Iri2cHandle_t * i2c_handle,  
    void * cmd_param,  
    uint8_t * data,  
    int len )
```

read data from the i2c device

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>cmd_param</i>	read param
out	<i>data</i>	the pointer for data which prepare to read
in	<i>len</i>	the read length

Returns

see `IrlibError_e`

5.7.3.4 iri2c_data_write()

```
IrlibError_e iri2c_data_write (  
    Iri2cHandle_t * i2c_handle,  
    void * cmd_param,  
    uint8_t * data,  
    int len )
```

write data to the i2c device

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>cmd_param</i>	write param
in	<i>data</i>	the pointer for data which prepare to write
in	<i>len</i>	the write length

Returns

see `IrlibError_e`

5.7.3.5 iri2c_detect_device_status()

```
IrlibError_e iri2c_detect_device_status (
    Iri2cHandle_t * i2c_handle,
    void * params,
    int * status )
```

detect device status with i2c interface

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>params</i>	device parameter
out	<i>status</i>	device status

Returns

see IrlibError_e

5.7.3.6 iri2c_device_close()

```
IrlibError_e iri2c_device_close (
    Iri2cHandle_t * i2c_handle )
```

close an i2c device

Parameters

in	<i>i2c_handle</i>	iri2c's handle
----	-------------------	----------------

Returns

see IrlibError_e

5.7.3.7 iri2c_device_init()

```
IrlibError_e iri2c_device_init (
    Iri2cHandle_t * i2c_handle,
    I2cConDevParams_t * param )
```

Initialize the configuration of the control node.

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>param</i>	the configuration of the control node

Returns

see `IrlibError_e`

5.7.3.8 iri2c_device_open()

```
IrlibError_e iri2c_device_open (
    Iri2cHandle_t * i2c_handle,
    const char * dev_node )
```

Open an ir i2c device with its device node.

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>dev_node</i>	device's node

Returns

see `IrlibError_e`

5.7.3.9 iri2c_device_release()

```
IrlibError_e iri2c_device_release (
    Iri2cHandle_t * i2c_handle,
    I2cConDevParams_t * param )
```

Release ir2c's configuration of the control node.

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>params</i>	the configuration of the control node

Returns

see `IrlibError_e`

5.7.3.10 iri2c_firmware_download()

```
IrlibError_e iri2c_firmware_download (
    Iri2cHandle_t * i2c_handle,
```

```

char * file_name,
uint8_t * data,
uint32_t file_len,
void (*)(void *callback_data, void *priv_data) callback,
void * priv_data )

```

download firmware with i2c interface

Parameters

in	<i>i2c_handle</i>	iri2c's handle
in	<i>file_name</i>	the name of firmware file
in	<i>data</i>	data of firmware file
in	<i>file_len</i>	size of firmware file
in	<i>callback</i>	callback function of i2c firmware download function
in	<i>priv_data</i>	the private parameter of firmware download callback function

Returns

see `IrlibError_e`

5.7.3.11 iri2c_handle_create()

```

Iri2cHandle_t* iri2c_handle_create (
    IrControlHandle_t * ir_control_handle )

```

Create an iri2c's handle.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

Iri2cHandle_t struct pointer

5.7.3.12 iri2c_handle_delete()

```

IrlibError_e iri2c_handle_delete (
    IrControlHandle_t * ir_control_handle )

```

Delete and release iri2c's handle.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

see `IrLibError_e`

5.7.3.13 iri2c_log_register()

```
void iri2c_log_register (
    enum Iri2cLogLevel_e log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

register log level for libiri2c library

Parameters

in	<i>log_level</i>	log level
in	<i>callback</i>	the callback function for printing log messages
in	<i>priv_data</i>	the private parameter of callback function

5.7.3.14 iri2c_version()

```
const char* iri2c_version (
    void )
```

Get libiri2c library's version.

Returns

version's string

5.7.3.15 iri2c_version_number()

```
const char* iri2c_version_number (
    void )
```

Get libiri2c library's version number.

Returns

version number's string

5.7.4 Variable Documentation

5.7.4.1 iri2c_debug_print

```
void(* iri2c_debug_print) (const char *fmt,...)
```

Error type in libiri2c library.

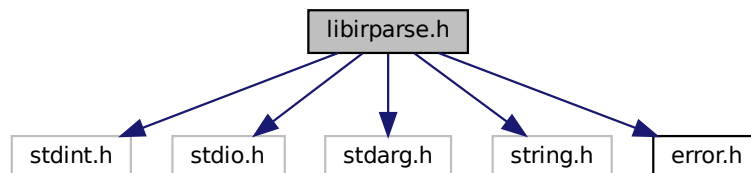
5.7.4.2 iri2c_error_print

```
void(* iri2c_error_print) (const char *fmt,...)
```

5.8 libirparse.h File Reference

```
#include <stdint.h>
#include <stdio.h>
#include <stdarg.h>
#include <string.h>
#include "error.h"
```

Include dependency graph for libirparse.h:



Classes

- struct [UnitYuv_t](#)
- struct [UnitRgb_t](#)
- struct [DataInfo_t](#)

Enumerations

- enum [irparse_log_level_t](#) { [IRPARSE_LOG_DEBUG](#) = 0, [IRPARSE_LOG_ERROR](#) = 1, [IRPARSE_LOG_NO_PRINT](#) = 2 }

Log level definition in libirparse library.

Functions

- DLLEXPORT char * [irparse_version](#) (void)
Get current libirparse library version.
- DLLEXPORT [lrlibError_e y16_to_y14](#) (uint16_t *y16_data, int pixel_num, uint16_t *y14_data)
convert Y16 array to Y14 array
- DLLEXPORT [lrlibError_e y14_to_y8](#) (uint16_t *y14_data, int pixel_num, uint8_t *y8_data)
convert Y14 array to Y8 array
- DLLEXPORT [lrlibError_e y14_to_rgb](#) (uint16_t *packet_data, int pixel_num, uint8_t *rgb_data)
convert Y14 array to RGB array
- DLLEXPORT [lrlibError_e y14_to_yuv444](#) (uint16_t *packet_data, int pixel_num, uint8_t *yuv_data)
convert Y14 array to YUV444 array. Y=Y14toY8 U=128 V=128
- DLLEXPORT [lrlibError_e yuv444_to_yuv422](#) (uint8_t *yuv444_data, int pixel_num, uint8_t *yuv422_data)
convert YUV444 array to YUV422 array
- DLLEXPORT [lrlibError_e yuv422_to_rgb](#) (uint8_t *yuv422, int pixel_num, uint8_t *rgb_data)
convert YUV422 array to RGB array
- DLLEXPORT [lrlibError_e yuv444_to_rgb](#) (uint8_t *packet_data, int pixel_num, uint8_t *rgb_data)
Convert YUV444 array to RGB array.
- DLLEXPORT [lrlibError_e rgb_to_bgr](#) (uint8_t *rgb_data, int pixel_num, uint8_t *bgr_data)
Convert RGB array to BGR array.
- DLLEXPORT [lrlibError_e yuv422_add_y](#) (uint8_t *src, int pixel_num, int y_value, uint8_t *dst)
YUV422 array add a Y value.
- DLLEXPORT [lrlibError_e raw_data_cut](#) (uint8_t *raw_data, int image_byte_size, int temp_byte_size, uint8_t *image_data, uint8_t *temp_data)
Cut the raw frame to image and temperature frames.
- DLLEXPORT void [irparse_log_register](#) ([irparse_log_level_t](#) log_level)
Register log function depend on log level.
- DLLEXPORT [lrlibError_e y14_to_nv12](#) (uint16_t *y14_data, int pixel_num, uint8_t *nv12_data)
Convert Y14 array to NV12 array.Y=Y14toY8 U=128 V=128.
- DLLEXPORT [lrlibError_e y16_to_nv12](#) (uint16_t *y16_data, int pixel_num, uint8_t *nv12_data)
Convert Y16 array to NV12 array.Y=Y16toY8 U=128 V=128.
- DLLEXPORT [lrlibError_e nv12_to_rgb](#) (uint8_t *nv12_data, int im_width, int im_height, uint8_t *rgb_data)
Convert NV12 array to RGB array.
- DLLEXPORT [lrlibError_e uyvy_to_bgra32](#) (uint8_t *uyyv_data, int width, int height, uint8_t *bgra_data)
Convert YUYV array to BGRA array.
- DLLEXPORT [lrlibError_e y16_to_rgb](#) (uint16_t *y16_data, int width, int height, uint8_t *rgb_data)
Convert Y16 array to RGB array.
- DLLEXPORT [lrlibError_e nv16_to_rgb](#) (uint8_t *nv16_data, int width, int height, uint8_t *rgb_data)
Convert NV16 array to RGB array.
- DLLEXPORT [lrlibError_e nv16_to_uyyv](#) (uint8_t *nv16_data, int width, int height, uint8_t *uyyv_data)
Convert NV16 array to YUYV array.
- DLLEXPORT [lrlibError_e uyvy_to_uyyv](#) (uint8_t *uyvy_data, int width, int height, uint8_t *uyyv_data)
Convert UYVY array to YUYV array.
- DLLEXPORT [lrlibError_e vyuy_to_uyyv](#) (uint8_t *vyuy_data, int width, int height, uint8_t *uyyv_data)
Convert VYUY array to YUYV array.
- DLLEXPORT [lrlibError_e yvyu_to_uyyv](#) (uint8_t *yvyu_data, int width, int height, uint8_t *uyyv_data)
Convert YVYU array to YUYV array.
- DLLEXPORT [lrlibError_e uyvy_to_rgb](#) (uint8_t *uyvy_data, int width, int height, uint8_t *rgb_data)
Convert UYVY array to RGB array.
- DLLEXPORT [lrlibError_e rs300_frame_data_cut](#) (uint8_t *raw_data, [DataInfo_t](#) image_data, [DataInfo_t](#) image_info_data, [DataInfo_t](#) temp_data, [DataInfo_t](#) temp_info_data)
Cut the raw frame to different parts.

Variables

- void(* [irparse_debug_print](#))(const char *fmt,...)
- void(* [irparse_error_print](#))(const char *fmt,...)

5.8.1 Enumeration Type Documentation

5.8.1.1 [irparse_log_level_t](#)

enum [irparse_log_level_t](#)

Log level definition in libirparse library.

Enumerator

IRPARSE_LOG_DEBUG	print debug and error information
IRPARSE_LOG_ERROR	only print error information
IRPARSE_LOG_NO_PRINT	don't print debug and error information

5.8.2 Function Documentation

5.8.2.1 [irparse_log_register\(\)](#)

```
DLLEXPORT void irparse_log_register (  
    irparse\_log\_level\_t log_level )
```

Register log function depend on log level.

Parameters

in	<i>log_level</i>	detail see irparse_log_level_t
----	------------------	---

5.8.2.2 [irparse_version\(\)](#)

```
DLLEXPORT char* irparse_version (  
    void )
```

Get current libirparse library version.

Parameters

<i>NULL</i>	
-------------	--

Returns

current libirparse library version

5.8.2.3 nv12_to_rgb()

```
DLLEXPORT IrlibError\_e nv12_to_rgb (
    uint8_t * nv12_data,
    int im_width,
    int im_height,
    uint8_t * rgb_data )
```

Convert NV12 array to RGB array.

Parameters

in	<i>nv12_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>rgb_data</i>	image frame after convert

Returns

see [IrlibError_e](#)

5.8.2.4 nv16_to_rgb()

```
DLLEXPORT IrlibError\_e nv16_to_rgb (
    uint8_t * nv16_data,
    int width,
    int height,
    uint8_t * rgb_data )
```

Convert NV16 array to RGB array.

Parameters

in	<i>nv16_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>rgb_data</i>	image frame after convert

Returns

see `IrllibError_e`

5.8.2.5 nv16_to_yuyv()

```
DLLEXPORT IrllibError_e nv16_to_yuyv (
    uint8_t * nv16_data,
    int width,
    int height,
    uint8_t * yuyv_data )
```

Convert NV16 array to YUYV array.

Parameters

in	<i>nv16_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>yuyv_data</i>	image frame after convert

Returns

see `IrllibError_e`

5.8.2.6 raw_data_cut()

```
DLLEXPORT IrllibError_e raw_data_cut (
    uint8_t * raw_data,
    int image_byte_size,
    int temp_byte_size,
    uint8_t * image_data,
    uint8_t * temp_data )
```

Cut the raw frame to image and temperature frames.

Parameters

in	<i>raw_data</i>	raw data frame from device
in	<i>image_byte_size</i>	image_byte_size of image frame
in	<i>temp_byte_size</i>	temp_byte_size of temperature frame
out	<i>image_data</i>	image frame after cut
out	<i>temp_data</i>	temperature frame after cut

Returns

see `IrllibError_e`

5.8.2.7 `rgb_to_bgr()`

```
DLLEXPORT IrllibError_e rgb_to_bgr (
    uint8_t * rgb_data,
    int pixel_num,
    uint8_t * bgr_data )
```

Convert RGB array to BGR array.

Parameters

in	<i>rgb_data</i>	source RGB image frame
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>bgr_data</i>	BGR image frame after conversion

Returns

see `IrllibError_e`

5.8.2.8 `rs300_frame_data_cut()`

```
DLLEXPORT IrllibError_e rs300_frame_data_cut (
    uint8_t * raw_data,
    DataInfo_t image_data,
    DataInfo_t image_info_data,
    DataInfo_t temp_data,
    DataInfo_t temp_info_data )
```

Cut the raw frame to different parts.

Parameters

in	<i>raw_data</i>	raw data frame from device
	<i>[in/out]</i>	image_data length and data of image frame
	<i>[in/out]</i>	image_info_data length and data of image information line
	<i>[in/out]</i>	temp_data image length and data of temperature frame
	<i>[in/out]</i>	temp_info_data length and data of temperature information line

Returns

see `IrllibError_e`

5.8.2.9 uyvy_to_rgb()

```
DLLEXPORT IrlibError\_e uyvy_to_rgb (
    uint8_t * uyvy_data,
    int width,
    int height,
    uint8_t * rgb_data )
```

Convert UYVY array to RGB array.

Parameters

in	<i>uyvy_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>rgb_data</i>	image frame after convert

Returns

see [IrlibError_e](#)

5.8.2.10 uyvy_to_yuyv()

```
DLLEXPORT IrlibError\_e uyvy_to_yuyv (
    uint8_t * uyvy_data,
    int width,
    int height,
    uint8_t * yuyv_data )
```

Convert UYVY array to YUYV array.

Parameters

in	<i>uyvy_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>yuyv_data</i>	image frame after convert

Returns

see [IrlibError_e](#)

5.8.2.11 vyuy_to_yuyv()

```
DLLEXPORT IrlibError\_e vyuy_to_yuyv (
    uint8_t * vyuy_data,
```

```
int width,  
int height,  
uint8_t * yuyv_data )
```

Convert VYUY array to YUYV array.

Parameters

in	<i>vyuy_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>yuyv_data</i>	image frame after convert

Returns

see `IrlibError_e`

5.8.2.12 y14_to_nv12()

```
DLL_EXPORT IrlibError_e y14_to_nv12 (  
    uint16_t * y14_data,  
    int pixel_num,  
    uint8_t * nv12_data )
```

Convert Y14 array to NV12 array.Y=Y14toY8 U=128 V=128.

Parameters

in	<i>y14_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>nv12_data</i>	image frame after convert

Returns

see `IrlibError_e`

5.8.2.13 y14_to_rgb()

```
DLL_EXPORT IrlibError_e y14_to_rgb (  
    uint16_t * packet_data,  
    int pixel_num,  
    uint8_t * rgb_data )
```

convert Y14 array to RGB array

Parameters

in	<i>packet_data</i>	source Y14 image frame
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>rgb_data</i>	RGB image frame after conversion

Returns

see `IrlibError_e`

5.8.2.14 y14_to_y8()

```
DLLEXPORT IrlibError_e y14_to_y8 (
    uint16_t * y14_data,
    int pixel_num,
    uint8_t * y8_data )
```

convert Y14 array to Y8 array

Parameters

in	<i>y14_data</i>	Y14 data source
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>y8_data</i>	Y8 data after conversion

Returns

see `IrlibError_e`

5.8.2.15 y14_to_yuv444()

```
DLLEXPORT IrlibError_e y14_to_yuv444 (
    uint16_t * packet_data,
    int pixel_num,
    uint8_t * yuv_data )
```

convert Y14 array to YUV444 array. Y=Y14toY8 U=128 V=128

Parameters

in	<i>packet_data</i>	source Y14 image frame
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>yuv_data</i>	YUV444 image frame after conversion

Returns

see `IrllibError_e`

5.8.2.16 y16_to_nv12()

```
DLLEXPORT IrllibError_e y16_to_nv12 (
    uint16_t * y16_data,
    int pixel_num,
    uint8_t * nv12_data )
```

Convert Y16 array to NV12 array. Y=Y16toY8 U=128 V=128.

Parameters

in	<i>y16_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>nv12_data</i>	image frame after convert

Returns

see `IrllibError_e`

5.8.2.17 y16_to_rgb()

```
DLLEXPORT IrllibError_e y16_to_rgb (
    uint16_t * y16_data,
    int width,
    int height,
    uint8_t * rgb_data )
```

Convert Y16 array to RGB array.

Parameters

in	<i>y16_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>rgb_data</i>	image frame after convert

Returns

see `IrllibError_e`

5.8.2.18 y16_to_y14()

```
DLLEXPORT IrlibError\_e y16_to_y14 (
    uint16_t * y16_data,
    int pixel_num,
    uint16_t * y14_data )
```

convert Y16 array to Y14 array

Parameters

in	<i>y16_data</i>	Y16 data source
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>y14_data</i>	Y14 data after conversion

Returns

see [IrlibError_e](#)

5.8.2.19 yuv422_add_y()

```
DLLEXPORT IrlibError\_e yuv422_add_y (
    uint8_t * src,
    int pixel_num,
    int y_value,
    uint8_t * dst )
```

YUV422 array add a Y value.

Parameters

in	<i>src</i>	source image frame
in	<i>pixel_num</i>	pixel number of source image frame
in	<i>y_value</i>	every pixel add <i>y_value</i> to Y component
out	<i>dst</i>	image frame after add <i>y_value</i>

Returns

see [IrlibError_e](#)

5.8.2.20 yuv422_to_rgb()

```
DLLEXPORT IrlibError\_e yuv422_to_rgb (
    uint8_t * yuv422,
    int pixel_num,
    uint8_t * rgb_data )
```

convert YUV422 array to RGB array

Parameters

in	<i>yuv422</i>	source YUV422 image frame
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>rgb_data</i>	RGB image frame after conversion

Returns

see `IrllibError_e`

5.8.2.21 yuv444_to_rgb()

```
DLLEXPORT IrllibError_e yuv444_to_rgb (
    uint8_t * packet_data,
    int pixel_num,
    uint8_t * rgb_data )
```

Convert YUV444 array to RGB array.

Parameters

in	<i>packet_data</i>	source YUV444 image frame
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>rgb_data</i>	RGB image frame after conversion

Returns

see `IrllibError_e`

5.8.2.22 yuv444_to_yuv422()

```
DLLEXPORT IrllibError_e yuv444_to_yuv422 (
    uint8_t * yuv444_data,
    int pixel_num,
    uint8_t * yuv422_data )
```

convert YUV444 array to YUV422 array

Parameters

in	<i>yuv444_data</i>	source YUV444 image frame
in	<i>pixel_num</i>	pixel number of source image frame
out	<i>yuv422_data</i>	YUV422 image frame after conversion

Returns

see `IrllibError_e`

5.8.2.23 yuyv_to_bgra32()

```
DLLEXPORT IrllibError_e yuyv_to_bgra32 (
    uint8_t * yuyv_data,
    int width,
    int height,
    uint8_t * bgra_data )
```

Convert YUYV array to BGRA array.

Parameters

in	<i>yuyv_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>bgra_data</i>	image frame after convert

Returns

see `IrllibError_e`

5.8.2.24 yvyu_to_yuyv()

```
DLLEXPORT IrllibError_e yvyu_to_yuyv (
    uint8_t * yvyu_data,
    int width,
    int height,
    uint8_t * yuyv_data )
```

Convert YVYU array to YUYV array.

Parameters

in	<i>yvyu_data</i>	source image frame
in	<i>width</i>	width of source image frame
in	<i>height</i>	height of source image frame
out	<i>yuyv_data</i>	image frame after convert

Returns

see `IrllibError_e`

5.8.3 Variable Documentation

5.8.3.1 irparse_debug_print

```
void(* irparse_debug_print) (const char *fmt,...)
```

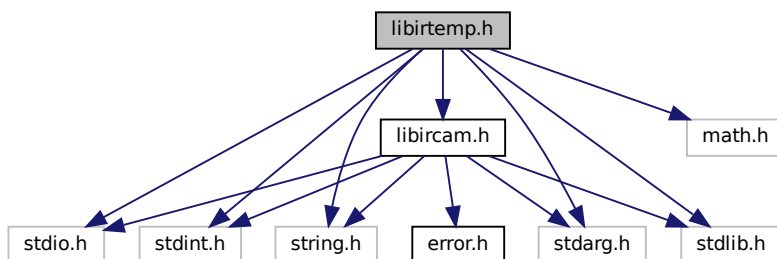
5.8.3.2 irparse_error_print

```
void(* irparse_error_print) (const char *fmt,...)
```

5.9 libirtemp.h File Reference

```
#include <stdio.h>
#include <stdint.h>
#include <string.h>
#include <stdarg.h>
#include <math.h>
#include <stdlib.h>
#include "libircam.h"
```

Include dependency graph for libirtemp.h:



Classes

- struct [TempDataRes_t](#)
TempData resolution.
- struct [SearchInfo_t](#)
The search information of KT curve.
- struct [KtInfo_t](#)
The turning address of KT curve.
- struct [VtemplInfo_t](#)

- Some vtemp constants for vtemp convert to true temperature.

 - struct [CaliPoint_t](#)

The environment temperature point of primary calibration.
 - struct [CaliFactor_t](#)

The normalization coefficient of KT curve.
 - struct [EnvTemp_t](#)

The environment temperature in double point calibration process.
 - struct [VTempSet_t](#)

The vtemp of sensor in double point calibration process.
 - struct [ReCaliFactor_t](#)

The correction factor after recalibration process.
 - struct [EnvParam_t](#)

Environment factor correction parameter.
 - struct [EnvFactor_t](#)

Environment factor correction coefficient.
 - struct [TempThreshold_t](#)

Temperature Threshold.
 - struct [NucFactor_t](#)

NUC map factor.
 - struct [Dot_t](#)

[Dot_t](#) definition.
 - struct [Line_t](#)

[Line_t](#) definition.
 - struct [Area_t](#)

[Area_t](#) definition.
 - struct [TemplInfo_t](#)

Temperature information definition.
 - struct [EnvCorrectParam](#)

Temperature information definition.
 - struct [MultiPointCalibTemp_t](#)

Temperature information definition.
 - struct [MultiPointCalibNuc_t](#)

Temperature information definition.
 - struct [CorrectTable_t](#)

dist correction table information definition
 - struct [second_calibration_param_t](#)

the parameter of SECOND_CALIBRATION_V1 algorithm
 - struct [second_calibration_result_t](#)

the calibration of SECOND_CALIBRATION_V1 algorithm
 - struct [ktbt_correct_param_t](#)
 - struct [MultiPointCalibParam_t](#)
 - struct [AdvancedMultiPointCalibParam_t](#)
 - struct [MultiPointCalibArray_t](#)
 - struct [MultiPointCalibArray_version3_t](#)
 - struct [TwoPointCalibResult_t](#)
 - struct [TwoPointCalibResult_version3_t](#)
 - struct [MultiPointCalibInputParam_version_t](#)
 - struct [MultiPointCalibInputParam_version3_t](#)
 - struct [KtBtCoefficient_t](#)
 - struct [KtbtCoefficient_t](#)
 - struct [MultiPointCalibForKtBtParam_t](#)

Macros

- #define [EPSILON](#) 0.0001
- #define [TEMP_NUM](#) 14
- #define [DIST_NUM](#) 64
- #define [NEW_DIST_NUM](#) 88
- #define [DIST_NUM_V3](#) 88
- #define [TARGET_TEMP_NUM](#) 56
- #define [NEW_TARGET_TEMP_NUM](#) 42
- #define [TARGET_TEMP_NUM_V3](#) 45
- #define [HIGH_GAIN](#) 1
- #define [LOW_GAIN](#) 0
- #define [EMS_MAX](#) 128
- #define [EMS_MIN](#) 0
- #define [TAU_MAX](#) 128
- #define [TAU_MIN](#) 0
- #define [HIGH_GAIN_TA_MAX](#) 430
- #define [HIGH_GAIN_TU_MAX](#) 430
- #define [HIGH_GAIN_TA_MIN](#) 230
- #define [HIGH_GAIN_TU_MIN](#) 230
- #define [LOW_GAIN_TA_MAX](#) 873
- #define [LOW_GAIN_TU_MAX](#) 873
- #define [LOW_GAIN_TA_MIN](#) 230
- #define [LOW_GAIN_TU_MIN](#) 230
- #define [TAU_TEMP_MIN](#) -5
- #define [TAU_TEMP_MAX](#) 55
- #define [TAU_DIST_MAX](#) 50
- #define [TAU_DIST_MIN](#) 0.25
- #define [TARGET_TEMP_MIN](#) 248.15
- #define [TARGET_TEMP_MAX](#) 1623.15
- #define [NUC_T_SIZE](#) 8192
- #define [KT_SIZE](#) 1201
- #define [BT_SIZE](#) 1201
- #define [ADVANCED_NUC_T_SIZE](#) 16384
- #define [ADVANCED_KT_SIZE](#) 3601
- #define [ADVANCED_BT_SIZE](#) 3601
- #define [HEAD_SIZE](#) 128
- #define [MAX_CALI_POINT_NUMBER](#) 20
- #define [ERROR_LOG_TO_FILE](#)(format, ...)

Enumerations

- enum [ProductType_t](#) {
[TINY1B](#) = 0, [TINY1C](#), [TINY1BE](#), [MINI256](#),
[MINI384](#), [MINI640](#), [P2](#), [MINI256_ADVANCED](#) }
- enum [Version_t](#) { [MULTI_POINT_CALIB_V1](#) = 0, [MULTI_POINT_CALIB_V2](#), [MULTI_POINT_CALIB_V3](#) }
- enum [TempArea_t](#) { [LowTempArea](#) = 0, [MidTempArea](#), [HighTempArea](#) }
- enum [ReCaliType_t](#) { [SINGLE_POINT](#) = 0, [DOUBLE_POINT](#) }
- Recalibration type.*
 - enum [AlarmType_t](#) { [TEMP_NORMAL](#) = 0, [OVER_HEAT](#) = 1, [OVER_COLD](#) = 2, [WRONG_TYPE](#) = 3 }
 - Alarm type in libirtemp library.*
- enum [irtemp_log_level_t](#) { [IRTEMP_LOG_DEBUG](#) = 0, [IRTEMP_LOG_ERROR](#) = 1, [IRTEMP_LOG_NO_PRINT](#) = 2 }
- Log level definition in libirtemp library.*
 - enum [irtemp_hotmode_t](#) { [WHITE_HOT](#) = 0, [BLACK_HOT](#) = 1 }
 - hotmode*

Functions

- DLLEXPORT char * [irtemp_version](#) (void)
Get current libirtemp library version.
- DLLEXPORT [IrlibError_e get_point_temp](#) (uint16_t *src, [TempDataRes_t](#) temp_res, [Dot_t](#) point, uint16_t *dst)
Get point's temperature,axis start from 0.
- DLLEXPORT [IrlibError_e get_line_temp](#) (uint16_t *src, [TempDataRes_t](#) temp_res, [Line_t](#) line, [TempInfo_t](#) *temp_info)
Get line's temperature,axis start from 0,and contains the line's boundary.
- DLLEXPORT [IrlibError_e get_rect_temp](#) (uint16_t *src, [TempDataRes_t](#) temp_res, [Area_t](#) rect, [TempInfo_t](#) *temp_info)
Get rectangle's temperature,axis start from 0,and contains the rectangle's boundary.
- DLLEXPORT void [irtemp_log_register](#) ([irtemp_log_level_t](#) log_level)
Register log function depend on log level.
- DLLEXPORT [AlarmType_t point_over_threshold_alarm](#) ([TempThreshold_t](#) temp_threshold, uint16_t cur_temp)
Point temperature over threshold alarm.
- DLLEXPORT [AlarmType_t line_rect_over_threshold_alarm](#) ([TempThreshold_t](#) temp_threshold, [TempInfo_t](#) *temp_info)
[Line_t](#) or rectangle temperature over threshold alarm.
- DLLEXPORT [IrlibError_e find_start_and_end_addr](#) (const uint16_t *kt_array, const [SearchInfo_t](#) *search_info, uint16_t len, [KtInfo_t](#) *kt_info)
Find start and end turning point of KT curve.
- DLLEXPORT [IrlibError_e calculate_Kn_and_Bn](#) (const [VtempInfo_t](#) *vtemp_info, const [KtInfo_t](#) *kt_info, const [CaliPoint_t](#) *cali_point, [CaliFactor_t](#) *cali_factor)
Calculate normalization coefficient Kn and Bn of KT curve.
- DLLEXPORT [IrlibError_e single_point_recalibrate_KB](#) (const [CaliFactor_t](#) *cali_factor, int cur_vtemp, double t_env, [ReCaliFactor_t](#) *recali_factor)
calibrate K and B value with single point calibration process
- DLLEXPORT [IrlibError_e double_point_recalibrate_KB](#) (const [CaliFactor_t](#) *cali_factor, const [VTempSet_t](#) *vtemp_set, const [EnvTemp_t](#) *env_temp, [ReCaliFactor_t](#) *recali_factor)
calibrate K and B value with double point calibration process
- DLLEXPORT [IrlibError_e env_temp_calculate](#) (const [CaliFactor_t](#) *cali_factor, const [ReCaliFactor_t](#) *recali_factor, uint16_t cur_vtemp, double *t_env)
estimate current environment temperature
- DLLEXPORT [IrlibError_e calculate_KE_and_BE](#) (const [EnvParam_t](#) *env_param, const [NucFactor_t](#) *nuc_factor, uint8_t gain_flag, [EnvFactor_t](#) *env_factor)
calculate KE and BE which run in firmware
- DLLEXPORT [IrlibError_e reverse_calc_NUC_with_env_correct](#) (const [NucFactor_t](#) *nuc_factor, double kelvin_temp, uint16_t *nuc_cal)
reverse calculate NUC value with environment correction
- DLLEXPORT [IrlibError_e reverse_calc_NUC_without_env_correct](#) (const [EnvFactor_t](#) *env_factor, uint16_t nuc_cal, uint16_t *nuc_org)
reverse calculate NUC value without environment correction
- DLLEXPORT [IrlibError_e calculate_new_KE_and_BE](#) (const [EnvParam_t](#) *env_param, const [NucFactor_t](#) *nuc_factor, uint8_t gain_flag, [EnvFactor_t](#) *env_factor)
calculate new KE and BE with higher precision
- DLLEXPORT [IrlibError_e recalc_NUC_with_env_correct](#) (const [EnvFactor_t](#) *env_factor, uint16_t nuc_org, uint16_t *nuc_cal)
recalculate new NUC value with environment correction
- DLLEXPORT [IrlibError_e remap_temp](#) (const uint16_t *nuc_table, uint16_t nuc_cal, uint16_t *temp_data)
remap temperature with nuc_table

- DLLEXPORT [IrlibError_e get_NUC_value](#) (uint16_t *src, [TempDataRes_t](#) temp_res, [Dot_t](#) point, uint16_t *nuc_value)
get NUC value of specific point from NUC frame
- DLLEXPORT uint16_t [calculate_tau](#) (uint16_t nuc_dist_high, uint16_t nuc_dist_low, uint16_t nuc_25_high, uint16_t nuc_25_low)
calculate Tau depend on nuc_dist_high,nuc_dist_low,nuc_25_high,nuc_25_low
- DLLEXPORT [IrlibError_e read_tau](#) (const uint16_t *correct_table, float hum, float t_env, float dist, uint16_t *tau)
read tau from correct table
- DLLEXPORT [IrlibError_e write_tau](#) (uint16_t *correct_table, float hum, float t_env, float dist, uint16_t tau)
write specific tau to correct table
- DLLEXPORT [IrlibError_e read_tau_with_target_temp_and_dist](#) (const uint16_t *correct_table, float target_temp, float dist, uint16_t *tau)
read tau from correct table
- DLLEXPORT [IrlibError_e write_tau_with_target_temp_and_dist](#) (uint16_t *correct_table, float target_temp, float dist, uint16_t tau)
write tau to correct table
- DLLEXPORT [IrlibError_e temp_correct](#) (float ems, uint16_t tau, float ta, float org_temp, float *new_temp)
temperature correction with new method
- DLLEXPORT [IrlibError_e reverse_temp_correct](#) (float ems, uint16_t tau, float ta, float new_temp, float *org_temp)
reverse calculate temperature correction with new method
- DLLEXPORT [IrlibError_e calculate_nuc_with_nuc_factor](#) ([NucFactor_t](#) *nuc_factor, float temp, uint16_t *nuc_value)
calculate nuc with nuc factor(P0,P1,P2)
- DLLEXPORT [IrlibError_e generate_high_gain_nuc_t](#) ([NucFactor_t](#) *nuc_factor, uint16_t *nuc_table)
generate NUC_T table in high gain mode with nuc factor(P0,P1,P2)
- DLLEXPORT [IrlibError_e generate_low_gain_nuc_t](#) ([NucFactor_t](#) *nuc_factor, uint16_t *nuc_table)
generate NUC_T table in low gain mode with nuc factor(P0,P1,P2)
- DLLEXPORT [IrlibError_e reverse_calc_NUC_with_nuc_t](#) (uint16_t *nuc_table, float temp, uint16_t *nuc_value)
reverse calculate NUC value with nuc_t table
- DLLEXPORT [IrlibError_e reverse_calc_NUC_with_nuc_t_table_len](#) (uint16_t *nuc_table, float temp, uint16_t *nuc_table_len, uint16_t *nuc_value)
reverse calculate NUC value with nuc_t table
- DLLEXPORT [IrlibError_e calculate_org_KE_and_BE_with_nuc_t](#) (const [EnvParam_t](#) *env_param, const uint16_t *nuc_table, uint8_t gain_flag, [EnvFactor_t](#) *env_factor)
calculate origin KE and BE with NUC-T table.
- DLLEXPORT [IrlibError_e calculate_new_KE_and_BE_with_nuc_t](#) (const [EnvParam_t](#) *env_param, const uint16_t *nuc_table, uint8_t gain_flag, [EnvFactor_t](#) *env_factor)
calculate new KE and BE with higher precision.
- DLLEXPORT [IrlibError_e enhance_distance_temp_correct](#) ([EnvCorrectParam](#) *env_correct_param, uint16_t *correct_table, uint32_t table_len, float org_temp, float *new_temp)
new temp correct method in MINI640/384 compatible old and new correct table
- DLLEXPORT [IrlibError_e reverse_enhance_distance_temp_correct](#) ([EnvCorrectParam](#) *env_correct_param, uint16_t *correct_table, uint32_t table_len, float new_temp, float *org_temp)
reverse new temp correct method in MINI640/384 compatible old and new correct table
- DLLEXPORT [IrlibError_e new_ktbt_recal_double_point_calculate](#) ([MultiPointCalibParam_t](#) *multi_point_param, int product_type, [MultiPointCalibArray_t](#) *multi_point_calib_array, [TwoPointCalibResult_t](#) *two_point_calib_result)
calculate new nuc-t table

- `DLLEXPORT IrlibError_e advanced_ktbt_recal_double_point_calculate` (`AdvancedMultiPointCalibParam_t` *multi_point_param, int product_type, `MultiPointCalibArray_t` *multi_point_calib_array, `TwoPointCalibResult_t` *two_point_calib_result)
calculate new nuc-t table with advanced method
- `DLLEXPORT IrlibError_e new_ktbt_recal_double_point_calculate_with_version` (void *multi_point_calib_↵ input_param, int version, void *two_point_calib_result)
calculate new nuc-t table with version
- `DLLEXPORT IrlibError_e multi_point_calc_user_defined_nuc` (`MultiPointCalibParam_t` *multi_point_param, int product_type, `MultiPointCalibArray_t` *multi_point_calib_array, `TwoPointCalibResult_t` *two_point_calib_↵ _result, `MultiPointCalibTemp_t` *multi_point_temp, `MultiPointCalibNuc_t` *multi_point_nuc)
calculate user defined environment correct coefficient
- `DLLEXPORT IrlibError_e multi_point_calc_user_defined_nuc_in_change_lens` (`MultiPointCalibParam_t` *multi_point_param, `MultiPointCalibArray_t` *multi_point_calib_array, `TwoPointCalibResult_t` *two_point_↵ calib_result, `MultiPointCalibTemp_t` *multi_point_temp, `MultiPointCalibNuc_t` *multi_point_nuc)
calculate nuc before nuc_t for change lens
- `DLLEXPORT IrlibError_e multi_point_calc_user_defined_nuc_with_version` (void *multi_point_calib_input_↵ _param, int version, void *two_point_calib_result)
calculate user defined environment correct coefficient with version
- `DLLEXPORT IrlibError_e multi_point_calc_new_nuc_table` (uint16_t *nuc_table, `MultiPointCalibNuc_t` *multi_point_nuc, uint16_t multi_point_num, uint16_t *new_nuc_table)
calculate new nuc-t table
- `DLLEXPORT IrlibError_e multi_point_calc_new_nuc_table_in_change_lens` (uint16_t *nuc_table, `MultiPointCalibNuc_t` *multi_point_nuc, uint16_t multi_point_num, uint16_t *new_nuc_table)
calculate new nuc-t table for change lens
- `DLLEXPORT IrlibError_e multi_point_calc_new_nuc_table_with_version` (void *multi_point_calib_input_↵ param, int version, uint16_t *new_nuc_table)
calculate new nuc-t table with version
- `DLLEXPORT IrlibError_e multi_point_calc_new_kt_bt` (`MultiPointCalibForKtBtParam_t` *Kt_Bt_Param, uint16_t *org_kt, int16_t *org_bt, uint16_t *new_kt_table, int16_t *new_bt_table)
calculate new kt bt table
- `DLLEXPORT IrlibError_e multi_point_calc_one_point_correct` (uint16_t *nuc_table, `MultiPointCalibNuc_t` *multi_point_nuc, uint16_t multi_point_num, uint16_t *new_nuc_table)
calculate new nuc-t table for one point correct
- `DLLEXPORT IrlibError_e ktbt_correct_process` (`ktbt_correct_param_t` *ktbt_correct_param, uint16_t org_↵ nuc, uint16_t *correct_nuc)
calculate nuc value with ktbt correct algorithm
- `DLLEXPORT IrlibError_e temp_measure_with_NUC_value` (`ktbt_correct_param_t` *ktbt_correct_param, uint16_t *nuc_table, uint16_t org_nuc, uint16_t *temp_data)
calculate temperture with the nuc value which read after the module of snr
- `DLLEXPORT IrlibError_e new_ktbt_recal_double_point_calculate_in_change_lens` (`AdvancedMultiPointCalibParam_t` *multi_point_param, `KtbtCoefficient_t` *ktbt_coef, `MultiPointCalibArray_t` *multi_point_calib_array, `TwoPointCalibResult_t` *two_point_calib_result)
calculate new kt table and bt table for change lens

Variables

- void(* `irtemp_debug_print`)(const char *fmt,...)
- void(* `irtemp_error_print`)(const char *fmt,...)

5.9.1 Macro Definition Documentation

5.9.1.1 ADVANCED_BT_SIZE

```
#define ADVANCED_BT_SIZE 3601
```

5.9.1.2 ADVANCED_KT_SIZE

```
#define ADVANCED_KT_SIZE 3601
```

5.9.1.3 ADVANCED_NUC_T_SIZE

```
#define ADVANCED_NUC_T_SIZE 16384
```

5.9.1.4 BT_SIZE

```
#define BT_SIZE 1201
```

5.9.1.5 DIST_NUM

```
#define DIST_NUM 64
```

5.9.1.6 DIST_NUM_V3

```
#define DIST_NUM_V3 88
```

5.9.1.7 EMS_MAX

```
#define EMS_MAX 128
```

5.9.1.8 EMS_MIN

```
#define EMS_MIN 0
```

5.9.1.9 EPSILON

```
#define EPSILON 0.0001
```

5.9.1.10 ERROR_LOG_TO_FILE

```
#define ERROR_LOG_TO_FILE(  
    format,  
    ... )
```

Value:

```
do{ FILE* fp = NULL;\nfp = fopen("err_log.txt", "a+");\nfprintf(fp, format"\n", __VA_ARGS__);\nfclose(fp); }while(0);
```

5.9.1.11 HEAD_SIZE

```
#define HEAD_SIZE 128
```

5.9.1.12 HIGH_GAIN

```
#define HIGH_GAIN 1
```

5.9.1.13 HIGH_GAIN_TA_MAX

```
#define HIGH_GAIN_TA_MAX 430
```

5.9.1.14 HIGH_GAIN_TA_MIN

```
#define HIGH_GAIN_TA_MIN 230
```

5.9.1.15 HIGH_GAIN_TU_MAX

```
#define HIGH_GAIN_TU_MAX 430
```

5.9.1.16 HIGH_GAIN_TU_MIN

```
#define HIGH_GAIN_TU_MIN 230
```

5.9.1.17 KT_SIZE

```
#define KT_SIZE 1201
```

5.9.1.18 LOW_GAIN

```
#define LOW_GAIN 0
```

5.9.1.19 LOW_GAIN_TA_MAX

```
#define LOW_GAIN_TA_MAX 873
```

5.9.1.20 LOW_GAIN_TA_MIN

```
#define LOW_GAIN_TA_MIN 230
```

5.9.1.21 LOW_GAIN_TU_MAX

```
#define LOW_GAIN_TU_MAX 873
```

5.9.1.22 LOW_GAIN_TU_MIN

```
#define LOW_GAIN_TU_MIN 230
```

5.9.1.23 MAX_CALI_POINT_NUMBER

```
#define MAX_CALI_POINT_NUMBER 20
```

5.9.1.24 NEW_DIST_NUM

```
#define NEW_DIST_NUM 88
```

5.9.1.25 NEW_TARGET_TEMP_NUM

```
#define NEW_TARGET_TEMP_NUM 42
```

5.9.1.26 NUC_T_SIZE

```
#define NUC_T_SIZE 8192
```

5.9.1.27 TARGET_TEMP_MAX

```
#define TARGET_TEMP_MAX 1623.15
```

5.9.1.28 TARGET_TEMP_MIN

```
#define TARGET_TEMP_MIN 248.15
```

5.9.1.29 TARGET_TEMP_NUM

```
#define TARGET_TEMP_NUM 56
```

5.9.1.30 TARGET_TEMP_NUM_V3

```
#define TARGET_TEMP_NUM_V3 45
```

5.9.1.31 TAU_DIST_MAX

```
#define TAU_DIST_MAX 50
```

5.9.1.32 TAU_DIST_MIN

```
#define TAU_DIST_MIN 0.25
```

5.9.1.33 TAU_MAX

```
#define TAU_MAX 128
```

5.9.1.34 TAU_MIN

```
#define TAU_MIN 0
```

5.9.1.35 TAU_TEMP_MAX

```
#define TAU_TEMP_MAX 55
```

5.9.1.36 TAU_TEMP_MIN

```
#define TAU_TEMP_MIN -5
```

5.9.1.37 TEMP_NUM

```
#define TEMP_NUM 14
```

5.9.2 Enumeration Type Documentation

5.9.2.1 AlarmType_t

```
enum AlarmType\_t
```

Alarm type in libirtemp library.

Enumerator

TEMP_NORMAL	normal temperature
OVER_HEAT	over heat
OVER_COLD	over cold
WRONG_TYPE	wrong type

5.9.2.2 irtemp_hotmode_t

```
enum irtemp_hotmode_t
```

hotmode

Enumerator

WHITE_HOT	
BLACK_HOT	

5.9.2.3 irtemp_log_level_t

```
enum irtemp_log_level_t
```

Log level definition in libirtemp library.

Enumerator

IRTEMP_LOG_DEBUG	print debug and error information
IRTEMP_LOG_ERROR	only print error information
IRTEMP_LOG_NO_PRINT	don't print debug and error information

5.9.2.4 ProductType_t

```
enum ProductType_t
```

Enumerator

TINY1B	
TINY1C	
TINY1BE	
MINI256	
MINI384	

Enumerator

MINI640	
P2	
MINI256_ADVANCED	

5.9.2.5 ReCaliType_t

```
enum ReCaliType_t
```

Recalibration type.

Enumerator

SINGLE_POINT	The flag represent single point calibration process.
DOUBLE_POINT	The flag represent double point calibration process.

5.9.2.6 TempArea_t

```
enum TempArea_t
```

Enumerator

LowTempArea	
MidTempArea	
HighTempArea	

5.9.2.7 Version_t

```
enum Version_t
```

Enumerator

MULTI_POINT_CALIB_V1	for RS001 Tiny, {KT}{BT} = 1021 x 2 Byte
MULTI_POINT_CALIB_V2	for RS001 Mini, {KT}{BT} = 1021 x 2 Byte
MULTI_POINT_CALIB_V3	for RS300, 3 temperature area,{KT}{BT} = 3601 x 2 Byte

5.9.3 Function Documentation

5.9.3.1 advanced_ktbt_recal_double_point_calculate()

```
DLLEXPORT IrlibError\_e advanced_ktbt_recal_double_point_calculate (
    AdvancedMultiPointCalibParam\_t * multi_point_param,
    int product_type,
    MultiPointCalibArray\_t * multi_point_calib_array,
    TwoPointCalibResult\_t * two_point_calib_result )
```

calculate new nuc-t table with advanced method

Parameters

in	<i>multi_point_param</i>	advanced multi point calibration parameter
in	<i>product_type</i>	product(refer to ProductType_t)
in	<i>multi_point_calib_array</i>	multi point calibration array (include kt, bt, nuc-t,correct array)
out	<i>two_point_calib_result</i>	second calibration result (include kt array and bt array)

Returns

see [IrlibError_e](#)

5.9.3.2 calculate_KE_and_BE()

```
DLLEXPORT IrlibError\_e calculate_KE_and_BE (
    const EnvParam\_t * env_param,
    const NucFactor\_t * nuc_factor,
    uint8_t gain_flag,
    EnvFactor\_t * env_factor )
```

calculate KE and BE which run in firmware

Parameters

in	<i>env_param</i>	The environment factor correction parameter read from device
in	<i>nuc_factor</i>	The NUC map factor read from device
in	<i>gain_flag</i>	Gain flag for environment factor validity check
out	<i>env_factor</i>	Environment factor correction coefficient contains K_E and B_E

Returns

see [IrlibError_e](#)

5.9.3.3 calculate_Kn_and_Bn()

```
DLLEXPORT IrlibError_e calculate_Kn_and_Bn (
    const VtempInfo_t * vtemp_info,
    const KtInfo_t * kt_info,
    const CaliPoint_t * cali_point,
    CaliFactor_t * cali_factor )
```

Calculate normalization coefficient Kn and Bn of KT curve.

Parameters

in	<i>vtemp_info</i>	Vtemp Information contains Ktemp, Btemp, Address_CA
in	<i>kt_info</i>	KT Information contains start and end turning address
in	<i>cali_point</i>	CaliPoint_t contains high and low environment temperature point of primary calibration
out	<i>cali_factor</i>	The normalization coefficient of KT curve contains Kn and Bn

Returns

see [IrlibError_e](#)

5.9.3.4 calculate_new_KE_and_BE()

```
DLLEXPORT IrlibError_e calculate_new_KE_and_BE (
    const EnvParam_t * env_param,
    const NucFactor_t * nuc_factor,
    uint8_t gain_flag,
    EnvFactor_t * env_factor )
```

calculate new KE and BE with higher precision

Parameters

in	<i>env_param</i>	The new environment factor correction parameter(EMS,TAU,Ta,Tu)
in	<i>nuc_factor</i>	The NUC map factor read from device
in	<i>gain_flag</i>	Gain flag for for environment factor validity check
out	<i>env_factor</i>	Environment factor correction coefficient contains K_E and B_E

Returns

see [IrlibError_e](#)

5.9.3.5 calculate_new_KE_and_BE_with_nuc_t()

```
DLLEXPORT IrlibError_e calculate_new_KE_and_BE_with_nuc_t (
    const EnvParam_t * env_param,
```

```

    const uint16_t * nuc_table,
    uint8_t gain_flag,
    EnvFactor_t * env_factor )

```

calculate new KE and BE with higher precision.

Parameters

in	<i>env_param</i>	The new environment factor correction parameter(EMS,TAU,Ta,Tu)
in	<i>nuc_table</i>	NUC_T table (uin16_t*8192)
in	<i>gain_flag</i>	Gain flag for for environment factor validity check
out	<i>env_factor</i>	Environment factor correction coefficient contains K_E and B_E

Returns

see `IrLibError_e`

5.9.3.6 calculate_nuc_with_nuc_factor()

```

DLEXPORt IrLibError_e calculate_nuc_with_nuc_factor (
    NucFactor_t * nuc_factor,
    float temp,
    uint16_t * nuc_value )

```

calculate nuc with nuc factor(P0,P1,P2)

Parameters

in	<i>nuc_factor</i>	nuc factor contain(P0,P1,P2)
in	<i>temp</i>	target temperature(unit: Celcius)
out	<i>nuc_value</i>	nuc_value

Returns

see `IrLibError_e`

5.9.3.7 calculate_org_KE_and_BE_with_nuc_t()

```

DLEXPORt IrLibError_e calculate_org_KE_and_BE_with_nuc_t (
    const EnvParam_t * env_param,
    const uint16_t * nuc_table,
    uint8_t gain_flag,
    EnvFactor_t * env_factor )

```

calculate origin KE and BE with NUC-T table.

Parameters

in	<i>env_param</i>	The origin environment factor correction parameter(EMS,TAU,Ta,Tu)
in	<i>nuc_table</i>	NUC_T table (uin16_t*8192)
in	<i>gain_flag</i>	Gain flag for for environment factor validity check
out	<i>env_factor</i>	Environment factor correction coefficient contains K_E and B_E

Returns

see IrlibError_e

5.9.3.8 calculate_tau()

```
DLLEXPORT uint16_t calculate_tau (
    uint16_t nuc_dist_high,
    uint16_t nuc_dist_low,
    uint16_t nuc_25_high,
    uint16_t nuc_25_low )
```

calculate Tau depend on nuc_dist_high,nuc_dist_low,nuc_25_high,nuc_25_low

Parameters

in	<i>nuc_dist_low</i>	NUC value at dist(unit:m) facing the high temperature blackbody
in	<i>nuc_dist_high</i>	NUC value at dist(unit:m) facing the low temperature blackbody
in	<i>nuc_25_high</i>	NUC value at 0.25m facing the high temperature blackbody
in	<i>nuc_25_low</i>	NUC value at 0.25m facing the low temperature blackbody

Returns

NUC value

5.9.3.9 double_point_recalibrate_KB()

```
DLLEXPORT IrlibError_e double_point_recalibrate_KB (
    const CaliFactor_t * cali_factor,
    const VTempSet_t * vtemp_set,
    const EnvTemp_t * env_temp,
    ReCaliFactor_t * recal_i_factor )
```

calibrate K and B value with double point calibration process

Parameters

in	<i>cali_factor</i>	The normalization coefficient of KT curve contains Kn and Bn
in	<i>vtemp_set</i>	current vtemp from sensor
Generated by <i>env_temp</i>		current environment temperature (unit:Celcius)
out	<i>recali_factor</i>	The correction factor after recalibration process contains K and B

Returns

see `IrLibError_e`

5.9.3.10 enhance_distance_temp_correct()

```
DLLEXPORT IrLibError_e enhance_distance_temp_correct (
    EnvCorrectParam * env_correct_param,
    uint16_t * correct_table,
    uint32_t table_len,
    float org_temp,
    float * new_temp )
```

new temp correct method in MINI640/384 compatible old and new correct table

Parameters

in	<i>env_correct_param</i>	The new environment factor correction parameter
in	<i>correct_table</i>	The correct table of module
in	<i>table_len</i>	The length of table
in	<i>org_temp</i>	The origin temperature of target(celcius temperature, unit:celcius)
out	<i>new_temp</i>	The new temperature after correct(celcius temperature, unit:celcius)

Returns

see `IrLibError_e`

5.9.3.11 env_temp_calculate()

```
DLLEXPORT IrLibError_e env_temp_calculate (
    const CaliFactor_t * cali_factor,
    const ReCaliFactor_t * recal_i_factor,
    uint16_t cur_vtemp,
    double * t_env )
```

estimate current environment temperature

Parameters

in	<i>cali_factor</i>	The normalization coefficient of KT curve contains Kn and Bn
in	<i>recali_factor</i>	The correction factor after recalibration process contains K and B
in	<i>cur_vtemp</i>	current vtemp from sensor
out	<i>t_env</i>	current environment temperature (unit:Celcius)

Returns

see `IrllibError_e`

5.9.3.12 find_start_and_end_addr()

```
DLLEXPORT IrllibError_e find_start_and_end_addr (
    const uint16_t * kt_array,
    const SearchInfo_t * search_info,
    uint16_t len,
    KtInfo_t * kt_info )
```

Find start and end turning point of KT curve.

Parameters

in	<i>KT_array</i>	KT array
in	<i>search_info</i>	The search information of KT curve contains start_point and end_point
in	<i>len</i>	the length of KT array
out	<i>kt_info</i>	The turning address of KT curve contains start and end turning address

Returns

see `IrllibError_e`

5.9.3.13 generate_high_gain_nuc_t()

```
DLLEXPORT IrllibError_e generate_high_gain_nuc_t (
    NucFactor_t * nuc_factor,
    uint16_t * nuc_table )
```

generate NUC_T table in high gain mode with nuc factor(P0,P1,P2)

Parameters

in	<i>nuc_factor</i>	nuc factor contain(P0,P1,P2)
out	<i>nuc_table</i>	NUC_T table (uin16_t*8192)

Returns

see `IrllibError_e`

5.9.3.14 generate_low_gain_nuc_t()

```
DLLEXPORT IrlibError\_e generate_low_gain_nuc_t (
    NucFactor\_t * nuc_factor,
    uint16\_t * nuc_table )
```

generate NUC_T table in low gain mode with nuc factor(P0,P1,P2)

Parameters

in	<i>nuc_factor</i>	nuc factor contain(P0,P1,P2)
out	<i>nuc_table</i>	NUC_T table (uin16_t *8192)

Returns

see [IrlibError_e](#)

5.9.3.15 get_line_temp()

```
DLLEXPORT IrlibError\_e get_line_temp (
    uint16\_t * src,
    TempDataRes\_t temp_res,
    Line\_t line,
    TempInfo\_t * temp_info )
```

Get line's temperautre,axis start from 0,and contains the line's boundary.

Parameters

in	<i>src</i>	source temperature frame
in	<i>temp_res</i>	input temperature frame resolution, detail see TempDataRes_t
in	<i>line</i>	input line required to get temperature, detail see Line_t
out	<i>temp_info</i>	Temperature infomation of the input line

Returns

see [IrlibError_e](#)

5.9.3.16 get_NUC_value()

```
DLLEXPORT IrlibError\_e get_NUC_value (
    uint16\_t * src,
    TempDataRes\_t temp_res,
    Dot\_t point,
    uint16\_t * nuc_value )
```

get NUC value of specific point from NUC frame

Parameters

in	<i>src</i>	Source NUC frame
in	<i>temp_res</i>	Input temperature frame resolution, detail see TempDataRes_t
in	<i>point</i>	The point required to get NUC value, detail see Dot_t
out	<i>nuc_value</i>	The nuc value at specific point

Returns

see [IrlibError_e](#)

5.9.3.17 `get_point_temp()`

```
DLLEXPORT IrlibError_e get_point_temp (
    uint16_t * src,
    TempDataRes_t temp_res,
    Dot_t point,
    uint16_t * dst )
```

Get point's temperature,axis start from 0.

Parameters

in	<i>src</i>	source temperature frame
in	<i>temp_res</i>	input temperature frame resolution, detail see TempDataRes_t
in	<i>point</i>	input point required to get temperature, detail see Dot_t
out	<i>dst</i>	object temperature value of the input point(kelvin_temp*16)

Returns

see [IrlibError_e](#)

5.9.3.18 `get_rect_temp()`

```
DLLEXPORT IrlibError_e get_rect_temp (
    uint16_t * src,
    TempDataRes_t temp_res,
    Area_t rect,
    TempInfo_t * temp_info )
```

Get rectangle's temperature,axis start from 0,and contains the rectangle's boundary.

Parameters

in	<i>src</i>	source temperature frame
in	<i>temp_res</i>	input temperature frame resolution, detail see TempDataRes_t
Generated by DocGen	<i>rect</i>	input area required to get temperature, detail see Area_t
out	<i>temp_info</i>	Temperature information of the input area

Returns

see `IrLibError_e`

5.9.3.19 irtemp_log_register()

```
DLLEXPORT void irtemp_log_register (
    irtemp_log_level_t log_level )
```

Register log function depend on log level.

Parameters

in	<i>log_level,detail</i>	see <code>irtemp_log_level_t</code>
----	-------------------------	--

5.9.3.20 irtemp_version()

```
DLLEXPORT char* irtemp_version (
    void )
```

Get current libirtemp library version.

Parameters

<i>NULL</i>	
-------------	--

Returns

current libirtemp library version

5.9.3.21 ktbt_correct_process()

```
DLLEXPORT IrLibError_e ktbt_correct_process (
    ktbt_correct_param_t * ktbt_correct_param,
    uint16_t org_nuc,
    uint16_t * correct_nuc )
```

calculate nuc value with ktbt correct algorithm

Parameters

in	<i>ktbt_correct_param</i>	ktbt correct param which contain all the parameter and data
in	<i>org_nuc</i>	orginal nuc value read after the process of space noise reduction
out	<i>correct_nuc</i>	corrected nuc value after the process of ktbt algorithm

Returns

see [IrlibError_e](#)

5.9.3.22 line_rect_over_threshold_alarm()

```
DLLEXPORT AlarmType_t line_rect_over_threshold_alarm (
    TempThreshold_t temp_threshold,
    TempInfo_t * temp_info )
```

[Line_t](#) or rectangle temperature over threshold alarm.

Parameters

in	<i>temp_threshold</i>	Temperature threshold,include upper limit and lower limit
in	<i>temp_info</i>	current temperature. Units:K(integer)

Returns

see [AlarmType_t](#) (max or average value greater than upper limit return OVER_HEAT/min or average value less than lower limit return OVER_COLD)

5.9.3.23 multi_point_calc_new_kt_bt()

```
DLLEXPORT IrlibError_e multi_point_calc_new_kt_bt (
    MultiPointCalibForKtBtParam_t * Kt_Bt_Param,
    uint16_t * org_kt,
    int16_t * org_bt,
    uint16_t * new_kt_table,
    int16_t * new_bt_table )
```

calculate new kt bt table

Parameters

in	<i>Kt_Bt_Param</i>	see MultiPointCalibForKtBtParam_t
in	<i>org_kt</i>	original kt table
in	<i>org_bt</i>	original bt table
out	<i>new_kt_table</i>	new kt table
out	<i>new_bt_table</i>	new bt table

Returns

see [IrlibError_e](#)

5.9.3.24 multi_point_calc_new_nuc_table()

```
DLLEXPORT IrlibError\_e multi_point_calc_new_nuc_table (
    uint16_t * nuc_table,
    MultiPointCalibNuc\_t * multi_point_nuc,
    uint16_t multi_point_num,
    uint16_t * new_nuc_table )
```

calculate new nuc-t table

Parameters

in	<i>nuc_table</i>	the nuc-t table which read from device
in	<i>multi_point_nuc</i>	the nuc value array of setting temperature and output tmperature
in	<i>multi_point_num</i>	the number of multi_point calibration
out	<i>new_nuc_table</i>	the correct table of

Returns

see [IrlibError_e](#)

5.9.3.25 multi_point_calc_new_nuc_table_in_change_lens()

```
DLLEXPORT IrlibError\_e multi_point_calc_new_nuc_table_in_change_lens (
    uint16_t * nuc_table,
    MultiPointCalibNuc\_t * multi_point_nuc,
    uint16_t multi_point_num,
    uint16_t * new_nuc_table )
```

calculate new nuc-t table for change lens

Parameters

in	<i>nuc_table</i>	the nuc-t table which read from device
in	<i>multi_point_nuc</i>	the nuc value array of setting temperature and output tmperature
in	<i>multi_point_num</i>	the number of multi_point calibration
out	<i>new_nuc_table</i>	the correct table of

Returns

see [IrlibError_e](#)

5.9.3.26 multi_point_calc_new_nuc_table_with_version()

```
DLLEXPORT IrlibError\_e multi_point_calc_new_nuc_table_with_version (
    void * multi_point_calib_input_param,
```

```
int version,
uint16_t * new_nuc_table )
```

calculate new nuc-t table with version

Parameters

in	<i>multi_point_calib_input_param</i>	multi point calibration input parameter(refer to MultiPointCalibInputParam_version_t), \ which is determined by version
in	<i>version</i>	version(refer to Version_t)
out	<i>new_nuc_table</i>	the correct table

Returns

see IrLibError_e

5.9.3.27 multi_point_calc_one_point_correct()

```
DLLEXPORT IrLibError_e multi_point_calc_one_point_correct (
    uint16_t * nuc_table,
    MultiPointCalibNuc_t * multi_point_nuc,
    uint16_t multi_point_num,
    uint16_t * new_nuc_table )
```

calculate new nuc-t table for one point correct

Parameters

in	<i>nuc_table</i>	the nuc-t table which read from device
in	<i>multi_point_nuc</i>	the nuc value array of setting temperature and output tmperature.

Note

the first and sencond temperature point is accurate temperature point and the third temperature point is the temperature point to be revised.

the sencond temperature point must greater than the first temperature point.

Parameters

in	<i>multi_point_num</i>	the number of multi_point calibration , here number is 3
out	<i>new_nuc_table</i>	the correct table of

Returns

see IrLibError_e

5.9.3.28 multi_point_calc_user_defined_nuc()

```
DLLEXPORT IrlibError_e multi_point_calc_user_defined_nuc (
    MultiPointCalibParam_t * multi_point_param,
    int product_type,
    MultiPointCalibArray_t * multi_point_calib_array,
    TwoPointCalibResult_t * two_point_calib_result,
    MultiPointCalibTemp_t * multi_point_temp,
    MultiPointCalibNuc_t * multi_point_nuc )
```

calculate user defined environment correct coefficient

Parameters

in	<i>multi_point_param</i>	The parameter for multi_point_calibration
in	<i>product_type</i>	product(unused now)
in	<i>multi_point_calib_array</i>	multi point calibration array (include kt, bt, nuc-t,correct array)
in	<i>two_point_calib_result</i>	second calibration result (include kt array and bt array)
in	<i>multi_point_temp</i>	The setting target temperature and output target temperature
out	<i>multi_point_nuc</i>	The setting target nuc value and output target nuc value

Returns

see IrlibError_e

5.9.3.29 multi_point_calc_user_defined_nuc_in_change_lens()

```
DLLEXPORT IrlibError_e multi_point_calc_user_defined_nuc_in_change_lens (
    MultiPointCalibParam_t * multi_point_param,
    MultiPointCalibArray_t * multi_point_calib_array,
    TwoPointCalibResult_t * two_point_calib_result,
    MultiPointCalibTemp_t * multi_point_temp,
    MultiPointCalibNuc_t * multi_point_nuc )
```

calculate nuc before nuc_t for change lens

Parameters

in	<i>multi_point_param</i>	The parameter for multi_point_calibration
in	<i>multi_point_calib_array</i>	multi point calibration array (include kt, bt, nuc-t,correct array)
in	<i>two_point_calib_result</i>	second calibration result (include kt array and bt array)
in	<i>multi_point_temp</i>	The setting target temperature and output target temperature
out	<i>multi_point_nuc</i>	The setting target nuc value and output target nuc value

Returns

see IrlibError_e

5.9.3.30 multi_point_calc_user_defined_nuc_with_version()

```
DLLEXPORT IrlibError\_e multi_point_calc_user_defined_nuc_with_version (
    void * multi_point_calib_input_param,
    int version,
    void * two_point_calib_result )
```

calculate user defined environment correct coefficient with version

Parameters

in	<i>multi_point_calib_input_param</i>	multi point calibration input parameter(refer to MultiPointCalibInputParam_version_t), \ which is determined by version
in	<i>version</i>	version(refer to Version_t)
in	<i>two_point_calib_result</i>	second calibration result (refer to TwoPointCalibResult_t),which is determined by version

Returns

see [IrlibError_e](#)

5.9.3.31 new_ktbt_recal_double_point_calculate()

```
DLLEXPORT IrlibError\_e new_ktbt_recal_double_point_calculate (
    MultiPointCalibParam\_t * multi_point_param,
    int product_type,
    MultiPointCalibArray\_t * multi_point_calib_array,
    TwoPointCalibResult\_t * two_point_calib_result )
```

calculate new nuc-t table

Parameters

in	<i>multi_point_param</i>	multi point calibration parameter
in	<i>product_type</i>	product(refer to ProductType_t)
in	<i>multi_point_calib_array</i>	multi point calibration array (include kt, bt, nuc-t,correct array)
out	<i>two_point_calib_result</i>	second calibration result (include kt array and bt array)

Returns

see [IrlibError_e](#)

5.9.3.32 new_ktbt_recal_double_point_calculate_in_change_lens()

```
DLLEXPORT IrlibError\_e new_ktbt_recal_double_point_calculate_in_change_lens (
    AdvancedMultiPointCalibParam\_t * multi_point_param,
```

```

KtbtCoefficient_t * ktbt_coef,
MultiPointCalibArray_t * multi_point_calib_array,
TwoPointCalibResult_t * two_point_calib_result )

```

calculate new kt table and bt table for change lens

Parameters

in	<i>multi_point_param</i>	multi point calibration parameter
in	<i>product_type</i>	product(refer to ProductType_t)
in	<i>ktbt_coef</i>	KT, BT third-order fitting coefficients when changing lenses
in	<i>multi_point_calib_array</i>	multi point calibration array (include kt, bt, nuc-t,correct array)
out	<i>two_point_calib_result</i>	second calibration result (include kt array and bt array)

Returns

see [IrlibError_e](#)

5.9.3.33 new_ktbt_recal_double_point_calculate_with_version()

```

DLLEXPORT IrlibError_e new_ktbt_recal_double_point_calculate_with_version (
    void * multi_point_calib_input_param,
    int version,
    void * two_point_calib_result )

```

calculate new nuc-t table with version

Parameters

in	<i>multi_point_calib_input_param</i>	multi point calibration input parameter(refer to MultiPointCalibInputParam_version_t), \ which is determined by version
in	<i>version</i>	version(refer to Version_t)
out	<i>two_point_calib_result</i>	second calibration result (refer to TwoPointCalibResult_t), which is determined by version

Returns

see [IrlibError_e](#)

5.9.3.34 point_over_threshold_alarm()

```

DLLEXPORT AlarmType_t point_over_threshold_alarm (
    TempThreshold_t temp_threshold,
    uint16_t cur_temp )

```

Point temperature over threshold alarm.

Parameters

in	<i>temp_threshold</i>	Temperature threshold,include upper limit and lower limit
in	<i>cur_temp</i>	current temperature. Units:K(integer)

Returns

see AlarmType_t (greater than upper limit return OVER_HEAT/less than lower limit return OVER_COLD)

5.9.3.35 read_tau()

```
DLLEXPORT IrlibError_e read_tau (
    const uint16_t * correct_table,
    float hum,
    float t_env,
    float dist,
    uint16_t * tau )
```

read tau from correct table

Parameters

in	<i>correct</i>	table that store tau at different humidity,environment temperature and object distance
in	<i>hum</i>	humidity(0-1)
in	<i>t_env</i>	environment temperature (unit: Celcius)
in	<i>dist</i>	object distance (0.25-49.99) (unit: m)
out	<i>tau</i>	tau after double linear interpolation

Returns

see IrlibError_e

5.9.3.36 read_tau_with_target_temp_and_dist()

```
DLLEXPORT IrlibError_e read_tau_with_target_temp_and_dist (
    const uint16_t * correct_table,
    float target_temp,
    float dist,
    uint16_t * tau )
```

read tau from correct table

Parameters

in	<i>correct</i>	table that store tau at different target distance and target temperature
in	<i>target_temp</i>	target temperature (unit: Celcius)
in	<i>dist</i>	target distance (0.25-49.99) (unit: m)
out	<i>tau</i>	tau after double linear interpolation

Returns

see `IrllibError_e`

5.9.3.37 recalc_NUC_with_env_correct()

```
DLLEXPORT IrllibError_e recalc_NUC_with_env_correct (
    const EnvFactor_t * env_factor,
    uint16_t nuc_org,
    uint16_t * nuc_cal )
```

recalculate new NUC value with environment correction

Parameters

in	<i>env_factor</i>	Environment factor correction coefficient contains K_E and B_E
in	<i>nuc_org</i>	The NUC value without environment correction
out	<i>nuc_cal</i>	The NUC value with environment correction

Returns

see `IrllibError_e`

5.9.3.38 remap_temp()

```
DLLEXPORT IrllibError_e remap_temp (
    const uint16_t * nuc_table,
    uint16_t nuc_cal,
    uint16_t * temp_data )
```

remap temperature with `nuc_table`

Parameters

in	<i>nuc_table</i>	NUC_T table
in	<i>nuc_cal</i>	The NUC value with environment correction
out	<i>temp_data</i>	Temperature data (kelvin_temp*16)

Returns

see `IrllibError_e`

5.9.3.39 reverse_calc_NUC_with_env_correct()

```
DLLEXPORT IrlibError\_e reverse_calc_NUC_with_env_correct (
    const NucFactor\_t * nuc_factor,
    double kelvin_temp,
    uint16_t * nuc_cal )
```

reverse calculate NUC value with environment correction

Parameters

in	<i>nuc_factor</i>	The NUC map factor read from device
in	<i>kelvin_temp</i>	object temperature(unit:K)
out	<i>nuc_cal</i>	NUC value with environment correction

Returns

see [IrlibError_e](#)

5.9.3.40 reverse_calc_NUC_with_nuc_t()

```
DLLEXPORT IrlibError\_e reverse_calc_NUC_with_nuc_t (
    uint16_t * nuc_table,
    float temp,
    uint16_t * nuc_value )
```

reverse calculate NUC value with nuc_t table

Parameters

in	<i>nuc_table</i>	NUC_T table (uin16_t*8192)
in	<i>temp</i>	celcius temperature
out	<i>nuc_value</i>	nuc value after reverse calculate.

Returns

see [IrlibError_e](#)

5.9.3.41 reverse_calc_NUC_with_nuc_t_table_len()

```
DLLEXPORT IrlibError\_e reverse_calc_NUC_with_nuc_t_table_len (
    uint16_t * nuc_table,
    float temp,
    uint16_t nuc_table_len,
    uint16_t * nuc_value )
```

reverse calculate NUC value with nuc_t table

Parameters

in	<i>nuc_table</i>	NUC_T table
in	<i>temp</i>	celcius temperature
out	<i>nuc_value</i>	nuc value after reverse calculate.

Returns

see `IrllibError_e`

5.9.3.42 reverse_calc_NUC_without_env_correct()

```
DLLEXPORT IrllibError_e reverse_calc_NUC_without_env_correct (
    const EnvFactor_t * env_factor,
    uint16_t nuc_cal,
    uint16_t * nuc_org )
```

reverse calculate NUC value without environment correction

Parameters

in	<i>env_factor</i>	factor correction coefficient contains K_E and B_E
in	<i>nuc_cal</i>	NUC value with environment correction
out	<i>nuc_org</i>	NUC value without environment correction

Returns

see `IrllibError_e`

5.9.3.43 reverse_enhance_distance_temp_correct()

```
DLLEXPORT IrllibError_e reverse_enhance_distance_temp_correct (
    EnvCorrectParam * env_correct_param,
    uint16_t * correct_table,
    uint32_t table_len,
    float new_temp,
    float * org_temp )
```

reverse new temp correct method in MINI640/384 compatible old and new correct table

Parameters

in	<i>env_correct_param</i>	The new environment factor correction parameter
in	<i>correct_table</i>	The correct table of module
in	<i>table_len</i>	The length of table
in	<i>new_temp</i>	The new temperature after correct(celcius temperature, unit:celcius)
out	<i>org_temp</i>	The origin temperature of target(celcius temperature, unit:celcius)

Returns

see `IrLibError_e`

5.9.3.44 reverse_temp_correct()

```
DLLEXPORT IrLibError_e reverse_temp_correct (
    float ems,
    uint16_t tau,
    float ta,
    float new_temp,
    float * org_temp )
```

reverse calculate temperature correction with new method

Parameters

in	<i>ems</i>	Target emissivity(0-1)
in	<i>tau</i>	Atmospheric transmittance(0-16384)
in	<i>ta</i>	Atmospheric temperature(unit: Celcius)
in	<i>new_temp</i>	new temperature after correction(unit: Celcius)
out	<i>org_temp</i>	origin temperature before correction(unit: Celcius)

Returns

see `IrLibError_e`

5.9.3.45 single_point_recalibrate_KB()

```
DLLEXPORT IrLibError_e single_point_recalibrate_KB (
    const CaliFactor_t * cali_factor,
    int cur_vtemp,
    double t_env,
    ReCaliFactor_t * recali_factor )
```

calibrate K and B value with single point calibration process

Parameters

in	<i>cali_factor</i>	The normalization coefficient of KT curve contains Kn and Bn
in	<i>cur_vtemp</i>	current vtemp from sensor
in	<i>t_env</i>	current environment temperature (unit:Celcius)
out	<i>recali_factor</i>	The correction factor after recalibration process contains K and B

Returns

see `IrllibError_e`

5.9.3.46 temp_correct()

```
DLLEXPORT IrllibError_e temp_correct (
    float ems,
    uint16_t tau,
    float ta,
    float org_temp,
    float * new_temp )
```

temperature correction with new method

Parameters

in	<i>ems</i>	Target emissivity(0-1)
in	<i>tau</i>	Atmospheric transmittance(0-16384)
in	<i>ta</i>	Atmospheric temperature(unit: Celcius)
in	<i>org_temp</i>	origin temperature before correction(unit: Celcius)
out	<i>new_temp</i>	new temperature after correction(unit: Celcius)

Returns

see `IrllibError_e`

5.9.3.47 temp_measure_with_NUC_value()

```
DLLEXPORT IrllibError_e temp_measure_with_NUC_value (
    ktbt_correct_param_t * ktbt_correct_param,
    uint16_t * nuc_table,
    uint16_t org_nuc,
    uint16_t * temp_data )
```

calculate tempertature with the nuc value which read after the module of snr

Parameters

in	<i>ktbt_correct_param</i>	Ktbt correct param which contain all the parameter and data
in	<i>nuc_table</i>	The nuc-t table(uint16_t*8192)
in	<i>org_nuc</i>	Orginal nuc value read after the process of space noise reduction(0-16383)
out	<i>temp_data</i>	Tempertature data (kelvin_temp*16)

Returns

see `IrLibError_e`

5.9.3.48 write_tau()

```
DLLEXPORT IrLibError_e write_tau (
    uint16_t * correct_table,
    float hum,
    float t_env,
    float dist,
    uint16_t tau )
```

write specific tau to correct table

Parameters

in	<i>correct</i>	table that store tau at different humidity,environment temperature and distance
in	<i>hum</i>	humidity(0-1)
in	<i>t_env</i>	environment temperature (unit: Celcius)
in	<i>dist</i>	object distance (0.25-49.99) (unit: m)
out	<i>tau</i>	tau that will be written

Returns

see `IrLibError_e`

5.9.3.49 write_tau_with_target_temp_and_dist()

```
DLLEXPORT IrLibError_e write_tau_with_target_temp_and_dist (
    uint16_t * correct_table,
    float target_temp,
    float dist,
    uint16_t tau )
```

write tau to correct table

Parameters

in	<i>correct</i>	table that store tau at different target distance and target temperature
in	<i>target_temp</i>	target temperature (unit: Celcius)
in	<i>dist</i>	target distance (0.25-49.99) (unit: m)
out	<i>tau</i>	tau after double linear interpolation

Returns

see `lrlibError_e`

5.9.4 Variable Documentation

5.9.4.1 `irtemp_debug_print`

```
void(* irtemp_debug_print) (const char *fmt,...)
```

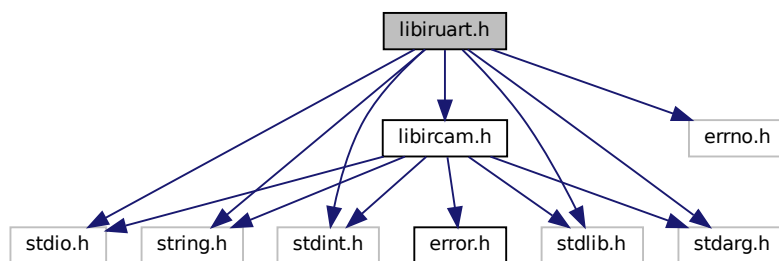
5.9.4.2 `irtemp_error_print`

```
void(* irtemp_error_print) (const char *fmt,...)
```

5.10 libiruart.h File Reference

```
#include <stdio.h>
#include <string.h>
#include <stdint.h>
#include <stdlib.h>
#include <errno.h>
#include <stdarg.h>
#include "libircam.h"
```

Include dependency graph for `libiruart.h`:



Classes

- struct [UartConDevParams_t](#)

Macros

- `#define _CRT_SECURE_NO_WARNINGS`
- `#define _LIBUART_H_`

Typedefs

- `typedef struct _IruartHandle_t IruartHandle_t`

Enumerations

- enum `IruartLogLevels_e` { `IRUART_LOG_DEBUG` = 0, `IRUART_LOG_ERROR` = 1, `IRUART_LOG_NO_PRINT` = 2 }
- Log level definition in libiruart library.*

Functions

- `DLLEXPORT void iruart_log_register` (enum `IruartLogLevels_e` log_level, void `*(callback)(void *callback_data, void *priv_data)`, void `*priv_data`)
register log level for libiruart library
- `const DLLEXPORT char * iruart_version` (void)
Get libiruart library's version.
- `const DLLEXPORT char * iruart_version_number` (void)
Get libiruart library's version number.
- `DLLEXPORT IruartHandle_t * iruart_handle_create` (`IrControlHandle_t *ir_control_handle`)
Create an iruart's handle.
- `DLLEXPORT IrlibError_e iruart_handle_delete` (`IrControlHandle_t *ir_control_handle`)
Delete and release iruart's handle.
- `DLLEXPORT IrlibError_e iruart_open` (`IruartHandle_t *uart_handle`, const char `*dev_node`)
Open an ir uart device with its device node.
- `DLLEXPORT IrlibError_e iruart_init` (`IruartHandle_t *uart_handle`, `UartConDevParams_t *param`)
Initialize the configuration of the control node.
- `DLLEXPORT IrlibError_e iruart_data_read` (`IruartHandle_t *uart_handle`, void `*cmd_param`, uint8_t `*data`, uint32_t len)
read data from the infiray uart device
- `DLLEXPORT IrlibError_e iruart_data_write` (`IruartHandle_t *uart_handle`, void `*cmd_param`, uint8_t `*data`, uint32_t len)
write data to the infiray uart device
- `DLLEXPORT IrlibError_e iruart_data_write_without_read_return_status` (`IruartHandle_t *uart_handle`, void `*cmd_param`, uint8_t `*data`, uint32_t len)
write data to the infiray uart device without reading return status
- `DLLEXPORT IrlibError_e iruart_release` (`IruartHandle_t *uart_handle`, `UartConDevParams_t *param`)
Release libiruart's configuration of the control node.
- `DLLEXPORT IrlibError_e iruart_close` (`IruartHandle_t *uart_handle`)
close an infiray uart device
- `DLLEXPORT IrlibError_e iruart_firmware_download` (`IruartHandle_t *uart_handle`, char `*file_name`, uint8_t `*data`, uint32_t file_len, void `*(callback)(void *callback_data, void *priv_data)`, void `*priv_data`)
download firmware with uart interface
- `DLLEXPORT IrlibError_e iruart_detect_device_status` (`IruartHandle_t *uart_handle`, void `*params`, int `*status`)
detect device status with uart interface
- `DLLEXPORT IrlibError_e iruart_command_channel_get` (`IruartHandle_t *uart_handle`, int `*command_channel_type`)
Get uart command channel type.

5.10.1 Macro Definition Documentation

5.10.1.1 `_CRT_SECURE_NO_WARNINGS`

```
#define _CRT_SECURE_NO_WARNINGS
```

5.10.1.2 `_LIBUART_H_`

```
#define _LIBUART_H_
```

5.10.2 Typedef Documentation

5.10.2.1 `IruartHandle_t`

```
typedef struct _IruartHandle_t IruartHandle_t
```

5.10.3 Enumeration Type Documentation

5.10.3.1 `IruartLogLevels_e`

```
enum IruartLogLevels_e
```

Log level definition in libiruart library.

Enumerator

<code>IRUART_LOG_DEBUG</code>	print debug and error information
<code>IRUART_LOG_ERROR</code>	only print error information
<code>IRUART_LOG_NO_PRINT</code>	don't print debug and error information

5.10.4 Function Documentation

5.10.4.1 iruart_close()

```
DLLEXPORT IrlibError_e iruart_close (
    IruartHandle_t * uart_handle )
```

close an infiray uart device

Parameters

in	<i>uart_handle</i>	iruart's handle
----	--------------------	-----------------

Returns

see IrlibError_e

5.10.4.2 iruart_command_channel_get()

```
DLLEXPORT IrlibError_e iruart_command_channel_get (
    IruartHandle_t * uart_handle,
    int * command_channel_type )
```

Get uart command channel type.

Parameters

in	<i>uart_handle</i>	iruart's handle
out	<i>command_channel_type</i>	command channel of uart,see command_channel_e

Returns

see IrlibError_e

5.10.4.3 iruart_data_read()

```
DLLEXPORT IrlibError_e iruart_data_read (
    IruartHandle_t * uart_handle,
    void * cmd_param,
    uint8_t * data,
    uint32_t len )
```

read data from the infiray uart device

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>cmd_param</i>	read param
out	<i>data</i>	the pointer for data which prepare to read
in	<i>len</i>	the read length

Returns

see `IrLibError_e`

5.10.4.4 iruart_data_write()

```
DLLEXPORT IrLibError_e iruart_data_write (
    IruartHandle_t * uart_handle,
    void * cmd_param,
    uint8_t * data,
    uint32_t len )
```

write data to the infiray uart device

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>cmd_param</i>	write param
in	<i>data</i>	the pointer for data which prepare to write
in	<i>len</i>	the write length

Returns

see `IrLibError_e`

5.10.4.5 iruart_data_write_without_read_return_status()

```
DLLEXPORT IrLibError_e iruart_data_write_without_read_return_status (
    IruartHandle_t * uart_handle,
    void * cmd_param,
    uint8_t * data,
    uint32_t len )
```

write data to the infiray uart device without reading return status

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>cmd_param</i>	write param
in	<i>data</i>	the pointer for data which prepare to write
in	<i>len</i>	the write length

Returns

see `IrLibError_e`

5.10.4.6 iruart_detect_device_status()

```
DLLEXPORT IrlibError_e iruart_detect_device_status (
    IruartHandle_t * uart_handle,
    void * params,
    int * status )
```

detect device status with uart interface

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>params</i>	reserved parameter of status detection
out	<i>status</i>	status of devices,0:Unknow status,1:Rom code,2:Cache Code

Returns

see IrlibError_e

5.10.4.7 iruart_firmware_download()

```
DLLEXPORT IrlibError_e iruart_firmware_download (
    IruartHandle_t * uart_handle,
    char * file_name,
    uint8_t * data,
    uint32_t file_len,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

download firmware with uart interface

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>file_name</i>	the name of firmware file
in	<i>data</i>	data of firmware file
in	<i>file_len</i>	size of firmware file
in	<i>callback</i>	callback function of uart firmware download function
in	<i>priv_data</i>	the private parameter of firmware download callback function

Returns

see IrlibError_e

5.10.4.8 iruart_handle_create()

```
DLLEXPORT IruartHandle_t* iruart_handle_create (
    IrControlHandle_t * ir_control_handle )
```

Create an iruart's handle.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

IruartHandle_t struct pointer

5.10.4.9 iruart_handle_delete()

```
DLLEXPORT IrlibError_e iruart_handle_delete (
    IrControlHandle_t * ir_control_handle )
```

Delete and release iruart's handle.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

see IrlibError_e

5.10.4.10 iruart_init()

```
DLLEXPORT IrlibError_e iruart_init (
    IruartHandle_t * uart_handle,
    UartConDevParams_t * param )
```

Initialize the configuration of the control node.

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>param</i>	the configuration of the control node

Returns

see IrlibError_e

5.10.4.11 iruart_log_register()

```
DLLEXPORT void iruart_log_register (
    enum IruartLogLevels_e log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

register log level for libiruart library

Parameters

in	<i>log_level</i>	log level
in	<i>callback</i>	the callback function for printing log messages
in	<i>priv_data</i>	the private parameter of callback function

5.10.4.12 iruart_open()

```
DLLEXPORT IrlibError_e iruart_open (
    IruartHandle_t * uart_handle,
    const char * dev_node )
```

Open an ir uart device with its device node.

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>dev_node</i>	device's node

Returns

see IrlibError_e

5.10.4.13 iruart_release()

```
DLLEXPORT IrlibError_e iruart_release (
    IruartHandle_t * uart_handle,
    UartConDevParams_t * param )
```

Release libiruart's configuration of the control node.

Parameters

in	<i>uart_handle</i>	iruart's handle
in	<i>params</i>	the configuration of the control node

Returns

see `lrlibError_e`

5.10.4.14 `iruart_version()`

```
const DLLEXPORT char* iruart_version (  
    void )
```

Get libiruart library's version.

Returns

version number's string

5.10.4.15 `iruart_version_number()`

```
const DLLEXPORT char* iruart_version_number (  
    void )
```

Get libiruart library's version number.

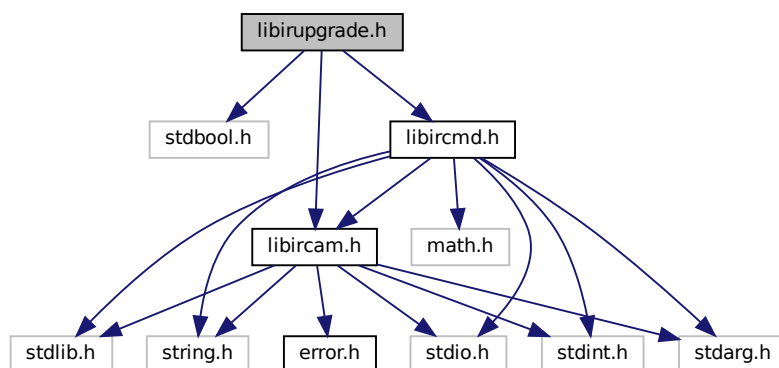
Returns

version number's string

5.11 libirupgrade.h File Reference

```
#include "stdlib.h"  
#include "libircmd.h"  
#include "libircam.h"
```

Include dependency graph for `libirupgrade.h`:



Classes

- struct [UpdateFile_t](#)
Information about scattered file in the update package zip.
- struct [UpdateFileInfo_t](#)
Information about property file in the update package zip.
- struct [DataUpdate_t](#)
Data difference information from the original version to the target version in the update package zip.
- struct [UpgradeInfo_t](#)
Information of version difference in the update package zip.
- struct [WholePackage_t](#)
Information of target version in the whole package zip.

Macros

- `#define _CRT_SECURE_NO_WARNINGS`
- `#define _LIBIRUPGRADE_H_`
- `#define STRING_MAX_LENGTH 300`
- `#define DATA_UPDATE_MAX_NUM 1000`
- `#define SINGLE_FILE_MAX_LENGTH 12000000`
- `#define FILE_INFO_LIST_MAX_NUM 10`
- `#define DATA_OFFSET_LIST_MAX_NUM 50`
- `#define UPDATE_FILE_LIST_MAX_NUM 100`
- `#define UPDATE_FILE_INFO_LIST_MAX_NUM 5`
- `#define UPDATE_LIB_LIST_MAX_NUM 100`

Enumerations

- enum [IrupgradeLogLevels_e](#) { [IRUPGRADE_LOG_DEBUG](#) = 0, [IRUPGRADE_LOG_ERROR](#) = 1, [IRUPGRADE_LOG_NO_PRINT](#) = 2 }
- Log level definition in libirupgrade library.*

Functions

- const DLLEXPORT char * [irupgrade_version](#) (void)
Get libirupgrade library's version.
- const DLLEXPORT char * [irupgrade_version_number](#) (void)
Get libirupgrade library's version number.
- DLLEXPORT void [irupgrade_log_register](#) (enum [IrupgradeLogLevels_e](#) log_level, void *(*callback)(void *callback_data, void *priv_data), void *priv_data)
register log level for libirupgrade library
- DLLEXPORT [IrlibError_e](#) [upgrade_package_function](#) ([IrcmdHandle_t](#) *ircmd_handle, char *dev_info, char *upgrade_pack_path, void *param, void *(*upgrade_process_callback)(void *callback_data, void *priv_data), void *priv_data)
Data upgrade using upgrade package zip Before calling this API, you need to make sure that [ircmd_handle](#) has been initialized.
- DLLEXPORT [IrlibError_e](#) [single_file_write](#) ([IrcmdHandle_t](#) *ircmd_handle, char *target_file_path, void *local_file_data, int local_file_length, void *(*upgrade_process_callback)(void *callback_data, void *priv_data), void *priv_data)
Single file download function Before calling this API, you need to make sure that the device is connected and [ircmd_handle](#) has been initialized.

- DLLEXPORT [IrlibError_e single_file_read](#) ([IrcmdHandle_t](#) *ircmd_handle, char *target_file_path, uint8_t *file_data, int *file_length, void *(*upgrade_process_callback)(void *callback_data, void *priv_data), void *priv_data)

Single file read function Before calling this API, you need to make sure that the device is connected and [ircmd_handle](#) has been initialized.

- DLLEXPORT [IrlibError_e parse_device_information](#) ([command_channel_e](#) command_type, char *dev_info, void **dev_param)

Parse device information to obtain valid device parameters.

5.11.1 Macro Definition Documentation

5.11.1.1 `_CRT_SECURE_NO_WARNINGS`

```
#define _CRT_SECURE_NO_WARNINGS
```

5.11.1.2 `_LIBIRUPGRADE_H_`

```
#define _LIBIRUPGRADE_H_
```

5.11.1.3 `DATA_OFFSET_LIST_MAX_NUM`

```
#define DATA_OFFSET_LIST_MAX_NUM 50
```

5.11.1.4 `DATA_UPDATE_MAX_NUM`

```
#define DATA_UPDATE_MAX_NUM 1000
```

5.11.1.5 `FILE_INFO_LIST_MAX_NUM`

```
#define FILE_INFO_LIST_MAX_NUM 10
```

5.11.1.6 SINGLE_FILE_MAX_LENGTH

```
#define SINGLE_FILE_MAX_LENGTH 12000000
```

5.11.1.7 STRING_MAX_LENGTH

```
#define STRING_MAX_LENGTH 300
```

5.11.1.8 UPDATE_FILE_INFO_LIST_MAX_NUM

```
#define UPDATE_FILE_INFO_LIST_MAX_NUM 5
```

5.11.1.9 UPDATE_FILE_LIST_MAX_NUM

```
#define UPDATE_FILE_LIST_MAX_NUM 100
```

5.11.1.10 UPDATE_LIB_LIST_MAX_NUM

```
#define UPDATE_LIB_LIST_MAX_NUM 100
```

5.11.2 Enumeration Type Documentation

5.11.2.1 IrupgradeLogLevels_e

```
enum IrupgradeLogLevels_e
```

Log level definition in libirupgrade library.

Enumerator

IRUPGRADE_LOG_DEBUG	print debug and error infomation
IRUPGRADE_LOG_ERROR	only print error infomation
IRUPGRADE_LOG_NO_PRINT	don't print debug and error infomation

5.11.3 Function Documentation

5.11.3.1 irupgrade_log_register()

```
DLLEXPORT void irupgrade_log_register (
    enum IrupgradeLogLevels_e log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

register log level for libirupgrade library

Parameters

in	<i>log_level</i>	log level
in	<i>callback</i>	the callback function for printing log messages
in	<i>priv_data</i>	the private parameter of callback function

5.11.3.2 irupgrade_version()

```
const DLLEXPORT char* irupgrade_version (
    void )
```

Get libirupgrade library's version.

Returns

version number's string

5.11.3.3 irupgrade_version_number()

```
const DLLEXPORT char* irupgrade_version_number (
    void )
```

Get libirupgrade library's version number.

Returns

version number's string

5.11.3.4 parse_device_information()

```
DLLEXPORT IrlibError_e parse_device_information (
    command_channel_e command_type,
    char * dev_info,
    void ** dev_param )
```

Parse device information to obtain valid device parameters.

Parameters

in	<i>command_type</i>	type of command channel, see <code>command_channel_e</code>
in	<i>dev_info</i>	information of device node, <code>dev_info</code> and <code>command_type</code> together determine the open device node
out	<i>dev_param</i>	Used to store valid device parameters

Returns

see `IrlibError_e`

5.11.3.5 `single_file_read()`

```
DLLEXPORT IrlibError_e single_file_read (
    IrcmdHandle_t * ircmd_handle,
    char * target_file_path,
    uint8_t * file_data,
    int * file_length,
    void (*)(void *callback_data, void *priv_data) upgrade_process_callback,
    void * priv_data )
```

Single file read function Before calling this API, you need to make sure that the device is connected and `ircmd↵_handle` has been initialized.

Parameters

in	<i>ircmd_handle</i>	<code>ircmd's</code> handle
in	<i>target_file_path</i>	The target file name of the operation recognized by the firmware
out	<i>file_data</i>	Used to store the read file data
out	<i>file_length</i>	Length of read file data(unit:Bytes)
in	<i>upgrade_process_callback</i>	Callback function for displaying progress information
in	<i>priv_data</i>	the private parameter of upgrade process callback function

Returns

see `IrlibError_e`

5.11.3.6 `single_file_write()`

```
DLLEXPORT IrlibError_e single_file_write (
    IrcmdHandle_t * ircmd_handle,
    char * target_file_path,
    void * local_file_data,
    int local_file_length,
```

```
void (*)(void *callback_data, void *priv_data) upgrade_process_callback,  
void * priv_data )
```

Single file download function Before calling this API, you need to make sure that the device is connected and [ircmd_handle](#) has been initialized.

Parameters

in	<i>ircmd_handle</i>	<i>ircmd</i> 's handle
in	<i>target_file_path</i>	The target file name of the operation recognized by the firmware
in	<i>local_file_data</i>	File data intended to be downloaded to device
in	<i>local_file_length</i>	Length of file data(unit:Bytes)
in	<i>upgrade_process_callback</i>	Callback function for displaying progress information
in	<i>priv_data</i>	the private parameter of upgrade process callback function

Returns

see `IrlibError_e`

5.11.3.7 upgrade_package_function()

```

DLLEXPORT IrlibError_e upgrade_package_function (
    IrcmdHandle_t * ircmd_handle,
    char * dev_info,
    char * upgrade_pack_path,
    void * param,
    void (*)(void *callback_data, void *priv_data) upgrade_process_callback,
    void * priv_data )

```

Data upgrade using upgrade package zip Before calling this API, you need to make sure that *ircmd_handle* has been initialized.

Parameters

in	<i>ircmd_handle</i>	<i>ircmd</i> 's handle
in	<i>dev_info</i>	information of device node, <i>dev_info</i> and <i>command_type</i> together determine the open device node, The formats to be filled in are as follows UVC_USB_COMMAND_CHANNEL: "UVC_USB:pid=0x0020,vid=0x3474,sameidx=0;" V4L2_USB_COMMAND_CHANNEL: "V4L2_USB:/dev/video30;" UART_COMMAND_CHANNEL: "UART:\\\\.\\COM10;"(windows) or "UART:/dev/ttyUSB1;" UVC_USB_I2C_COMMAND_CHANNEL: "UVC_USB_I2C:pid=0x0020,vid=0x3474,sameidx=0;" V4L2_USB_I2C_COMMAND_CHANNEL: "V4L2_USB_I2C:/dev/video40;" I2C_COMMAND_CHANNEL: "ONLY_I2C:/dev/i2c-1;"
in	<i>upgrade_pack_path</i>	local path to find upgrade package zip
in	<i>param</i>	reserved parameters
in	<i>upgrade_process_callback</i>	Callback function for displaying progress information
in	<i>priv_data</i>	the private parameter of upgrade process callback function

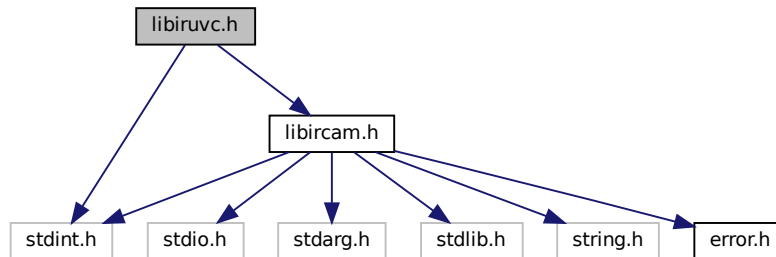
Returns

see `IrlibError_e`

5.12 libiruvc.h File Reference

Libiruvc library header file.

```
#include <stdint.h>
#include "libircam.h"
Include dependency graph for libiruvc.h:
```



Classes

- struct [DevCfg_t](#)
Device's configuration.
- struct [UserCallback_t](#)
User's callback function and parameters.
- struct [IruvcDevParam_t](#)
Device's configuration.
- struct [CameraParam_t](#)
Camera's one stream parameter.
- struct [IruvcCamStreamParams_t](#)
- struct [IruvcStreamStopParams_t](#)
- struct [IruvcFrameGetParams_t](#)
- struct [CameraStreamInfo_t](#)
Camera's supported stream information.
- struct [UvcConDevParam_t](#)

Macros

- #define [FORMAT_YUY2](#) "YUY2"
- #define [FORMAT_MJPEG](#) "MJPEG"
- #define [FPS_CNT_MAX](#) 32

Typedefs

- typedef struct [_IruvcHandle_t](#) [IruvcHandle_t](#)

Enumerations

- enum `cam_side_preview_ctl` { `CLOSE_CAM_SIDE_PREVIEW` = 0, `KEEP_CAM_SIDE_PREVIEW` = 1 }
Control camera side's preview.
- enum `IrUvcLogLevel_t` { `IRUVC_LOG_DEBUG` = 0, `IRUVC_LOG_ERROR` = 1, `IRUVC_LOG_NO_PRINT` = 2 }
Log message levels.

Functions

- const DLLEXPORT char * `iruvc_version` (void)
Get libiruvc library's version.
- const DLLEXPORT char * `iruvc_version_number` (void)
Get libiruvc library's version number.
- DLLEXPORT void `iruvc_log_register` (`IrUvcLogLevel_t` log_level, void (*)(void *callback_data, void *priv_data), void *priv_data)
Register log print via log_level.
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_handle_create` (`IrVideoHandle_t` *ir_video_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_handle_create_with_exist_instance` (`IrVideoHandle_t` *ir_video_handle, `IrUvcHandle_t` *iruvc_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_handle_delete` (`IrUvcHandle_t` *iruvc_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_open` (`IrUvcHandle_t` *iruvc_handle, `IrUvcDevParam_t` *dev_param)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_init` (`IrUvcHandle_t` *iruvc_handle, `IrUvcDevParam_t` *params)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_start_stream` (`IrUvcHandle_t` *iruvc_handle, `IrUvcCamStreamParams_t` *params)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_frame_get` (`IrUvcHandle_t` *iruvc_handle, `IrUvcFrameGetParams_t` *params, uint8_t *frame_data, int len)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_stop_stream` (`IrUvcHandle_t` *iruvc_handle, `IrUvcStreamStopParams_t` *params)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_release` (`IrUvcHandle_t` *iruvc_handle, `IrUvcDevParam_t` *params)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_camera_close` (`IrUvcHandle_t` *iruvc_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_handle_create` (`IrControlHandle_t` *ir_control_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_handle_create_with_exist_instance` (`IrControlHandle_t` *ir_control_handle, `IrUvcHandle_t` *iruvc_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_device_open` (`IrUvcHandle_t` *iruvc_handle, `IrUvcDevParam_t` *dev_param)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_device_init` (`IrUvcHandle_t` *iruvc_handle, `UvcConDevParam_t` *param)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_data_read` (`IrUvcHandle_t` *iruvc_handle, void *i2c_usb_cmd_param, uint8_t *data, uint16_t len)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_data_write` (`IrUvcHandle_t` *iruvc_handle, void *i2c_usb_cmd_param, uint8_t *data, uint16_t len)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_device_release` (`IrUvcHandle_t` *iruvc_handle, `UvcConDevParam_t` *param)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_device_close` (`IrUvcHandle_t` *iruvc_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_handle_delete` (`IrUvcHandle_t` *iruvc_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_firmware_download` (`IrUvcHandle_t` *iruvc_handle, char *file_name, uint8_t *data, uint32_t file_len, void (*)(void *callback_data))
- DLLEXPORT `IrUvcHandle_t` * `iruvc_i2c_usb_command_channel_get` (`IrUvcHandle_t` *iruvc_handle, int *command_channel_type)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_usb_handle_create` (`IrControlHandle_t` *ir_control_handle)
- DLLEXPORT `IrUvcHandle_t` * `iruvc_usb_handle_create_with_exist_instance` (`IrControlHandle_t` *ir_control_handle, `IrUvcHandle_t` *iruvc_handle)

- DLLEXPORT [IrlibError_e](#) [iruvc_usb_device_open](#) ([IruvcHandle_t](#) *iruvc_handle, [IruvcDevParam_t](#) *dev_param)
- DLLEXPORT [IrlibError_e](#) [iruvc_usb_device_init](#) ([IruvcHandle_t](#) *iruvc_handle, [UvcConDevParam_t](#) *param)
- DLLEXPORT [IrlibError_e](#) [iruvc_usb_data_read](#) ([IruvcHandle_t](#) *iruvc_handle, void *usb_cmd_param, uint8_t *data, uint16_t len)
- DLLEXPORT [IrlibError_e](#) [iruvc_usb_data_write](#) ([IruvcHandle_t](#) *iruvc_handle, void *usb_cmd_param, uint8_t *data, uint16_t len)
- DLLEXPORT [IrlibError_e](#) [iruvc_data_write_without_read_return_status](#) ([IruvcHandle_t](#) *iruvc_handle, void *usb_cmd_param, uint8_t *data, uint16_t len)
- DLLEXPORT [IrlibError_e](#) [iruvc_usb_device_release](#) ([IruvcHandle_t](#) *iruvc_handle, [UvcConDevParam_t](#) *param)
- DLLEXPORT [IrlibError_e](#) [iruvc_usb_device_close](#) ([IruvcHandle_t](#) *iruvc_handle)
- DLLEXPORT [IrlibError_e](#) [iruvc_usb_handle_delete](#) ([IruvcHandle_t](#) *iruvc_handle)
- DLLEXPORT [IrlibError_e](#) [iruvc_usb_command_channel_get](#) ([IruvcHandle_t](#) *uart_handle, int *command_channel_type)

5.12.1 Detailed Description

Libiruvc library header file.

5.12.2 Macro Definition Documentation

5.12.2.1 FORMAT_MJPEG

```
#define FORMAT_MJPEG "MJPEG"
```

5.12.2.2 FORMAT_YUY2

```
#define FORMAT_YUY2 "YUY2"
```

5.12.2.3 FPS_CNT_MAX

```
#define FPS_CNT_MAX 32
```

5.12.3 Typedef Documentation

5.12.3.1 IruvcHandle_t

```
typedef struct _IruvcHandle_t IruvcHandle_t
```

5.12.4 Enumeration Type Documentation

5.12.4.1 cam_side_preview_ctl

```
enum cam_side_preview_ctl
```

Control camera side's preview.

Enumerator

CLOSE_CAM_SIDE_PREVIEW	close camera side's preview
KEEP_CAM_SIDE_PREVIEW	keep camera side's preview

5.12.4.2 IruvcLogLevel_t

```
enum IruvcLogLevel_t
```

Log message levels.

Enumerator

IRUVC_LOG_DEBUG	all debug&error infomation
IRUVC_LOG_ERROR	all error information
IRUVC_LOG_NO_PRINT	no print information(default)

5.12.5 Function Documentation

5.12.5.1 iri2c_usb_firmware_download()

```
DLL_EXPORT IrlibError_e iri2c_usb_firmware_download (
    IruvcHandle_t * iruvc_handle,
    char * file_name,
    uint8_t * data,
    uint32_t file_len,
    void (*)(void *callback_data) callback )
```

5.12.5.2 iruvc_camera_close()

```
DLLEXPORT IrlibError\_e iruvc_camera_close (
    IruvcHandle\_t * iruvc_handle )
```

5.12.5.3 iruvc_camera_frame_get()

```
DLLEXPORT IrlibError\_e iruvc_camera_frame_get (
    IruvcHandle\_t * iruvc_handle,
    IruvcFrameGetParams\_t * params,
    uint8\_t * frame_data,
    int len )
```

5.12.5.4 iruvc_camera_handle_create()

```
DLLEXPORT IruvcHandle\_t* iruvc_camera_handle_create (
    IrVideoHandle\_t * ir_video_handle )
```

5.12.5.5 iruvc_camera_handle_create_with_exist_instance()

```
DLLEXPORT IrlibError\_e iruvc_camera_handle_create_with_exist_instance (
    IrVideoHandle\_t * ir_video_handle,
    IruvcHandle\_t * iruvc_handle )
```

5.12.5.6 iruvc_camera_handle_delete()

```
DLLEXPORT IrlibError\_e iruvc_camera_handle_delete (
    IruvcHandle\_t * iruvc_handle )
```

5.12.5.7 iruvc_camera_init()

```
DLLEXPORT IrlibError\_e iruvc_camera_init (
    IruvcHandle\_t * iruvc_handle,
    IruvcDevParam\_t * params )
```

5.12.5.8 iruvc_camera_open()

```
DLLEXPORT IrlibError_e iruvc_camera_open (
    IruvcHandle_t * iruvc_handle,
    IruvcDevParam_t * dev_param )
```

5.12.5.9 iruvc_camera_release()

```
DLLEXPORT IrlibError_e iruvc_camera_release (
    IruvcHandle_t * iruvc_handle,
    IruvcDevParam_t * params )
```

5.12.5.10 iruvc_camera_start_stream()

```
DLLEXPORT IrlibError_e iruvc_camera_start_stream (
    IruvcHandle_t * iruvc_handle,
    IruvcCamStreamParams_t * params )
```

5.12.5.11 iruvc_camera_stop_stream()

```
DLLEXPORT IrlibError_e iruvc_camera_stop_stream (
    IruvcHandle_t * iruvc_handle,
    IruvcStreamStopParams_t * params )
```

5.12.5.12 iruvc_data_write_without_read_return_status()

```
DLLEXPORT IrlibError_e iruvc_data_write_without_read_return_status (
    IruvcHandle_t * iruvc_handle,
    void * usb_cmd_param,
    uint8_t * data,
    uint16_t len )
```

5.12.5.13 iruvc_i2c_usb_command_channel_get()

```
DLLEXPORT IrlibError_e iruvc_i2c_usb_command_channel_get (
    IruvcHandle_t * uart_handle,
    int * command_channel_type )
```

5.12.5.14 iruvc_i2c_usb_data_read()

```
DLLEXPORT IrlibError\_e iruvc_i2c_usb_data_read (
    IruvcHandle\_t * iruvc_handle,
    void * i2c_usb_cmd_param,
    uint8_t * data,
    uint16_t len )
```

5.12.5.15 iruvc_i2c_usb_data_write()

```
DLLEXPORT IrlibError\_e iruvc_i2c_usb_data_write (
    IruvcHandle\_t * iruvc_handle,
    void * i2c_usb_cmd_param,
    uint8_t * data,
    uint16_t len )
```

5.12.5.16 iruvc_i2c_usb_device_close()

```
DLLEXPORT IrlibError\_e iruvc_i2c_usb_device_close (
    IruvcHandle\_t * iruvc_handle )
```

5.12.5.17 iruvc_i2c_usb_device_init()

```
DLLEXPORT IrlibError\_e iruvc_i2c_usb_device_init (
    IruvcHandle\_t * iruvc_handle,
    UvcConDevParam\_t * param )
```

5.12.5.18 iruvc_i2c_usb_device_open()

```
DLLEXPORT IrlibError\_e iruvc_i2c_usb_device_open (
    IruvcHandle\_t * iruvc_handle,
    IruvcDevParam\_t * dev_param )
```

5.12.5.19 iruvc_i2c_usb_device_release()

```
DLLEXPORT IrlibError\_e iruvc_i2c_usb_device_release (
    IruvcHandle\_t * iruvc_handle,
    UvcConDevParam\_t * param )
```

5.12.5.20 iruvc_i2c_usb_handle_create()

```
DLLEXPORT IruvcHandle_t* iruvc_i2c_usb_handle_create (
    IrControlHandle_t * ir_control_handle )
```

5.12.5.21 iruvc_i2c_usb_handle_create_with_exist_instance()

```
DLLEXPORT IrlibError_e iruvc_i2c_usb_handle_create_with_exist_instance (
    IrControlHandle_t * ir_control_handle,
    IruvcHandle_t * iruvc_handle )
```

5.12.5.22 iruvc_i2c_usb_handle_delete()

```
DLLEXPORT IrlibError_e iruvc_i2c_usb_handle_delete (
    IruvcHandle_t * iruvc_handle )
```

5.12.5.23 iruvc_log_register()

```
DLLEXPORT void iruvc_log_register (
    IruvcLogLevel_t log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

Register log print via log_level.

Parameters

in	<i>log_level</i>	debug information level
in	<i>callback</i>	the callback function of iruvc

5.12.5.24 iruvc_usb_command_channel_get()

```
DLLEXPORT IrlibError_e iruvc_usb_command_channel_get (
    IruvcHandle_t * uart_handle,
    int * command_channel_type )
```

5.12.5.25 iruvc_usb_data_read()

```
DLLEXPORT IrlibError\_e iruvc_usb_data_read (
    IruvcHandle\_t * iruvc_handle,
    void * usb_cmd_param,
    uint8\_t * data,
    uint16\_t len )
```

5.12.5.26 iruvc_usb_data_write()

```
DLLEXPORT IrlibError\_e iruvc_usb_data_write (
    IruvcHandle\_t * iruvc_handle,
    void * usb_cmd_param,
    uint8\_t * data,
    uint16\_t len )
```

5.12.5.27 iruvc_usb_device_close()

```
DLLEXPORT IrlibError\_e iruvc_usb_device_close (
    IruvcHandle\_t * iruvc_handle )
```

5.12.5.28 iruvc_usb_device_init()

```
DLLEXPORT IrlibError\_e iruvc_usb_device_init (
    IruvcHandle\_t * iruvc_handle,
    UvcConDevParam\_t * param )
```

5.12.5.29 iruvc_usb_device_open()

```
DLLEXPORT IrlibError\_e iruvc_usb_device_open (
    IruvcHandle\_t * iruvc_handle,
    IruvcDevParam\_t * dev_param )
```

5.12.5.30 iruvc_usb_device_release()

```
DLLEXPORT IrlibError\_e iruvc_usb_device_release (
    IruvcHandle\_t * iruvc_handle,
    UvcConDevParam\_t * param )
```


5.12.5.31 iruvc_usb_handle_create()

```
DLLEXPORT IruvcHandle_t* iruvc_usb_handle_create (
    IrControlHandle_t * ir_control_handle )
```

5.12.5.32 iruvc_usb_handle_create_with_exist_instance()

```
DLLEXPORT IrlibError_e iruvc_usb_handle_create_with_exist_instance (
    IrControlHandle_t * ir_control_handle,
    IruvcHandle_t * iruvc_handle )
```

5.12.5.33 iruvc_usb_handle_delete()

```
DLLEXPORT IrlibError_e iruvc_usb_handle_delete (
    IruvcHandle_t * iruvc_handle )
```

5.12.5.34 iruvc_version()

```
const DLLEXPORT char* iruvc_version (
    void )
```

Get libiruvc library's version.

Returns

version's string

5.12.5.35 iruvc_version_number()

```
const DLLEXPORT char* iruvc_version_number (
    void )
```

Get libiruvc library's version number.

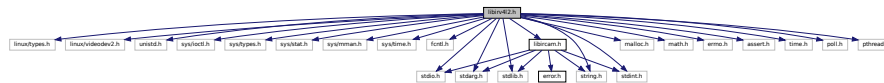
Returns

version number's string

5.13 libirv4l2.h File Reference

```
#include <linux/types.h>
#include <linux/videodev2.h>
#include <unistd.h>
#include <sys/ioctl.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <sys/mman.h>
#include <sys/time.h>
#include <fcntl.h>
#include <stdio.h>
#include <stdarg.h>
#include <stdlib.h>
#include <malloc.h>
#include <math.h>
#include <string.h>
#include <stdint.h>
#include <errno.h>
#include <assert.h>
#include <time.h>
#include <poll.h>
#include <pthread.h>
#include "libircam.h"
```

Include dependency graph for libirv4l2.h:



Classes

- struct [CamDevParams_t](#)
- struct [CamStreamParams_t](#)
- struct [FrameGetParams_t](#)
- struct [V4l2ConDevParams_t](#)

Macros

- #define [CMD_MAGIC](#) 0xEF
- #define [CMD_MAX_NR](#) 3
- #define [CMD_GET_IOWR](#)([CMD_MAGIC](#), 1, UvcCmdParam_t)
- #define [CMD_SET_IOW](#)([CMD_MAGIC](#), 2, UvcCmdParam_t)
- #define [CMD_KBUF_IO](#)([CMD_MAGIC](#), 3)

Typedefs

- typedef struct [_lrv4l2VideoHandle_t](#) [lrv4l2VideoHandle_t](#)
Video Streaming Handle.
- typedef struct [_lrv4l2ControlHandle_t](#) [lrv4l2ControlHandle_t](#)
Control command handle.

Enumerations

- enum [irv4l2_log_level_e](#) { [IRV4L2_LOG_DEBUG](#) = 0, [IRV4L2_LOG_ERROR](#) = 1, [IRV4L2_LOG_NO_PRINT](#) = 2 }

Log level definition in libircmd library.

Functions

- void [irv4l2_log_register](#) ([irv4l2_log_level_e](#) log_level, void [*\(*callback\)](#)(void [*callback_data](#), void [*priv_data](#)), void [*priv_data](#))
register log level for libirv4l2 library
- const char [* libv4l2_version](#) (void)
Get libirv4l2 library's version number.
- [Irv4l2VideoHandle_t](#) [* irv4l2_handle_create](#) ([IrVideoHandle_t](#) [*ir_video_handle](#))
Create an irv4l2's video handle.
- [IrlibError_e](#) [irv4l2_handle_delete](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#))
Delete and release irv4l2's video handle.
- [IrlibError_e](#) [irv4l2_camera_open](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#), char [*dev_node](#))
Open an video device with its device node.
- [IrlibError_e](#) [irv4l2_camera_init](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#), [CamDevParams_t](#) [*params](#))
Initialize the device by the format of the video stream.
- [IrlibError_e](#) [irv4l2_camera_start_stream](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#), [CamStreamParams_t](#) [*params](#))
Start stream with specific one stream parameter of the camera.
- [IrlibError_e](#) [irv4l2_camera_frame_get](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#), [FrameGetParams_t](#) [*params](#), [uint8_t](#) [*frame_data](#), int len)
Polling frame transfered from camera.
- [IrlibError_e](#) [irv4l2_camera_stop_stream](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#), [CamStreamParams_t](#) [*params](#))
Stop the video stream.
- [IrlibError_e](#) [irv4l2_camera_release](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#), [CamDevParams_t](#) [*params](#))
Release libirv4l2's service context.
- [IrlibError_e](#) [irv4l2_camera_close](#) ([Irv4l2VideoHandle_t](#) [*v4l2_handle](#))
Close the device.
- [Irv4l2ControlHandle_t](#) [* irv4l2_i2c_usb_handle_create](#) ([IrControlHandle_t](#) [*ir_control_handle](#))
Create an irv4l2's control handle with the i2c_usb path.
- [IrlibError_e](#) [irv4l2_i2c_usb_handle_delete](#) ([Irv4l2ControlHandle_t](#) [*irv4l2_handle](#))
Delete and release irv4l2's control handle with the i2c_usb path.
- [IrlibError_e](#) [irv4l2_i2c_usb_device_open](#) ([Irv4l2ControlHandle_t](#) [*irv4l2_handle](#), char [*dev_node](#))
Open an ir i2c_usb device with its device node.
- [IrlibError_e](#) [irv4l2_i2c_usb_device_init](#) ([Irv4l2ControlHandle_t](#) [*irv4l2_handle](#), [V4l2ConDevParams_t](#) [*param](#))
Initialize i2c_usb's configuration of the control node.
- [IrlibError_e](#) [irv4l2_i2c_usb_data_read](#) ([Irv4l2ControlHandle_t](#) [*irv4l2_handle](#), void [*cmd_param](#), [uint8_t](#) [*data](#), [uint32_t](#) len)
read data from the ir i2c_usb device
- [IrlibError_e](#) [irv4l2_i2c_usb_data_write](#) ([Irv4l2ControlHandle_t](#) [*irv4l2_handle](#), void [*cmd_param](#), [uint8_t](#) [*data](#), [uint32_t](#) len)
write data to the ir i2c_usb device
- [IrlibError_e](#) [irv4l2_i2c_usb_device_release](#) ([Irv4l2ControlHandle_t](#) [*irv4l2_handle](#), [V4l2ConDevParams_t](#) [*param](#))
Release i2c_usb's configuration of the control node.

- [IrlibError_e irv4l2_i2c_usb_device_close](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle)
close an ir i2c_usb device
- [Irv4l2ControlHandle_t * irv4l2_usb_handle_create](#) ([IrControlHandle_t](#) *ir_control_handle)
Create an irv4l2's control handle with the usb path.
- [IrlibError_e irv4l2_usb_handle_delete](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle)
Delete and release irv4l2's control handle with the usb path.
- [IrlibError_e irv4l2_usb_device_open](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle, char *dev_node)
Open an ir usb device with its device node.
- [IrlibError_e irv4l2_usb_device_init](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle, [V4l2ConDevParams_t](#) *param)
Initialize usb's configuration of the control node.
- [IrlibError_e irv4l2_usb_device_release](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle, [V4l2ConDevParams_t](#) *param)
Release usb's configuration of the control node.
- [IrlibError_e irv4l2_usb_device_close](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle)
close an ir usb device
- [IrlibError_e irv4l2_usb_data_read](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle, void *cmd_param, uint8_t *data, uint32_t len)
read data from the ir usb device
- [IrlibError_e irv4l2_usb_data_write](#) ([Irv4l2ControlHandle_t](#) *irv4l2_handle, void *cmd_param, uint8_t *data, uint32_t len)
write data to the ir usb device

5.13.1 Macro Definition Documentation

5.13.1.1 CMD_GET

```
#define CMD_GET __IOWR(CMD_MAGIC, 1, UvcCmdParam_t)
```

5.13.1.2 CMD_KBUF

```
#define CMD_KBUF __IO(CMD_MAGIC, 3)
```

5.13.1.3 CMD_MAGIC

```
#define CMD_MAGIC 0xEF
```

5.13.1.4 CMD_MAX_NR

```
#define CMD_MAX_NR 3
```

5.13.1.5 CMD_SET

```
#define CMD_SET _IOW(CMD_MAGIC, 2, UvcCmdParam_t)
```

5.13.2 Typedef Documentation

5.13.2.1 Irv4l2ControlHandle_t

```
typedef struct _Irv4l2ControlHandle_t Irv4l2ControlHandle_t
```

Control command handle.

5.13.2.2 Irv4l2VideoHandle_t

```
typedef struct _Irv4l2VideoHandle_t Irv4l2VideoHandle_t
```

Video Streaming Handle.

5.13.3 Enumeration Type Documentation

5.13.3.1 irv4l2_log_level_e

```
enum irv4l2_log_level_e
```

Log level definition in libircmd library.

Enumerator

IRV4L2_LOG_DEBUG	print debug and error infomation
IRV4L2_LOG_ERROR	only print error infomation
IRV4L2_LOG_NO_PRINT	don't print debug and error infomation

5.13.4 Function Documentation

5.13.4.1 irv4l2_camera_close()

```
IrlibError_e irv4l2_camera_close (
    Irv4l2VideoHandle_t * v4l2_handle )
```

Close the device.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
----	--------------------	-----------------------

Returns

see IrlibError_e

5.13.4.2 irv4l2_camera_frame_get()

```
IrlibError_e irv4l2_camera_frame_get (
    Irv4l2VideoHandle_t * v4l2_handle,
    FrameGetParams_t * params,
    uint8_t * frame_data,
    int len )
```

Polling frame transfered from camera.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
in	<i>params</i>	reserved params
out	<i>frame_data</i>	the data of one frame
in	<i>len</i>	the length of one frame

Returns

see IrlibError_e

5.13.4.3 irv4l2_camera_init()

```
IrlibError_e irv4l2_camera_init (
    Irv4l2VideoHandle_t * v4l2_handle,
    CamDevParams_t * params )
```

Initialize the device by the format of the video stream.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
in	<i>params</i>	format of video streams

Returns

see `IrlibError_e`

5.13.4.4 irv4l2_camera_open()

```
IrlibError_e irv4l2_camera_open (
    Irv4l2VideoHandle_t * v4l2_handle,
    char * dev_node )
```

Open an video device with its device node.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
in	<i>dev_node</i>	device's node

Returns

see `IrlibError_e`

5.13.4.5 irv4l2_camera_release()

```
IrlibError_e irv4l2_camera_release (
    Irv4l2VideoHandle_t * v4l2_handle,
    CamDevParams_t * params )
```

Release libirv4l2's service context.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
in	<i>params</i>	format of video streams

Returns

see `IrlibError_e`

5.13.4.6 irv4l2_camera_start_stream()

```
IrlibError_e irv4l2_camera_start_stream (
    Irv4l2VideoHandle_t * v4l2_handle,
    CamStreamParams_t * params )
```

Start stream with specific one stream parameter of the camera.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
in	<i>params</i>	camera's one selected stream parameter

Returns

see IrlibError_e

5.13.4.7 irv4l2_camera_stop_stream()

```
IrlibError_e irv4l2_camera_stop_stream (
    Irv4l2VideoHandle_t * v4l2_handle,
    CamStreamParams_t * params )
```

Stop the video stream.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
in	<i>params</i>	camera's one selected stream parameter

Returns

see IrlibError_e

5.13.4.8 irv4l2_handle_create()

```
Irv4l2VideoHandle_t* irv4l2_handle_create (
    IrVideoHandle_t * ir_video_handle )
```

Create an irv4l2's video handle.

Parameters

in	<i>ir_video_handle</i>	ir camera video handle
----	------------------------	------------------------

Returns

Irv4l2VideoHandle_t struct pointer

5.13.4.9 irv4l2_handle_delete()

```
IrlibError_e irv4l2_handle_delete (
    Irv4l2VideoHandle_t * v4l2_handle )
```

Delete and release irv4l2's video handle.

Parameters

in	<i>v4l2_handle</i>	irv4l2's video handle
----	--------------------	-----------------------

Returns

see IrlibError_e

5.13.4.10 irv4l2_i2c_usb_data_read()

```
IrlibError_e irv4l2_i2c_usb_data_read (
    Irv4l2ControlHandle_t * irv4l2_handle,
    void * cmd_param,
    uint8_t * data,
    uint32_t len )
```

read data from the ir i2c_usb device

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>cmd_param</i>	read param
out	<i>data</i>	the pointer for data which prepare to read
in	<i>len</i>	the read length

Returns

see IrlibError_e

5.13.4.11 irv4l2_i2c_usb_data_write()

```
IrlibError_e irv4l2_i2c_usb_data_write (
    Irv4l2ControlHandle_t * irv4l2_handle,
```

```
void * cmd_param,  
uint8_t * data,  
uint32_t len )
```

write data to the ir i2c_usb device

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>cmd_param</i>	write param
in	<i>data</i>	the pointer for data which prepare to write
in	<i>len</i>	the write length

Returns

see `IrlibError_e`

5.13.4.12 irv4l2_i2c_usb_device_close()

```
IrlibError_e irv4l2_i2c_usb_device_close (  
    Irv4l2ControlHandle_t * irv4l2_handle )
```

close an ir i2c_usb device

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
----	----------------------	-------------------------

Returns

see `IrlibError_e`

5.13.4.13 irv4l2_i2c_usb_device_init()

```
IrlibError_e irv4l2_i2c_usb_device_init (  
    Irv4l2ControlHandle_t * irv4l2_handle,  
    V4l2ConDevParams_t * param )
```

Initialize i2c_usb's configuration of the control node.

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>param</i>	the configuration of the control node

Returns

see `IrlibError_e`

5.13.4.14 irv4l2_i2c_usb_device_open()

```
IrlibError_e irv4l2_i2c_usb_device_open (
    Irv4l2ControlHandle_t * irv4l2_handle,
    char * dev_node )
```

Open an `ir_i2c_usb` device with its device node.

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>dev_node</i>	device's node

Returns

see `IrlibError_e`

5.13.4.15 irv4l2_i2c_usb_device_release()

```
IrlibError_e irv4l2_i2c_usb_device_release (
    Irv4l2ControlHandle_t * irv4l2_handle,
    V4l2ConDevParams_t * param )
```

Release `i2c_usb`'s configuration of the control node.

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>param</i>	the configuration of the control node

Returns

see `IrlibError_e`

5.13.4.16 irv4l2_i2c_usb_handle_create()

```
Irv4l2ControlHandle_t* irv4l2_i2c_usb_handle_create (
    IrControlHandle_t * ir_control_handle )
```

Create an `irv4l2`'s control handle with the `i2c_usb` path.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

Irv4l2ControlHandle_t struct pointer

5.13.4.17 irv4l2_i2c_usb_handle_delete()

```
IrlibError_e irv4l2_i2c_usb_handle_delete (
    Irv4l2ControlHandle_t * irv4l2_handle )
```

Delete and release irv4l2's control handle with the i2c_usb path.

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
----	----------------------	-------------------------

Returns

see IrlibError_e

5.13.4.18 irv4l2_log_register()

```
void irv4l2_log_register (
    irv4l2_log_level_e log_level,
    void (*)(void *callback_data, void *priv_data) callback,
    void * priv_data )
```

register log level for libirv4l2 library

Parameters

in	<i>log_level</i>	log level
in	<i>callback</i>	the callback function for printing log messages
in	<i>priv_data</i>	the private parameter of callback function

5.13.4.19 irv4l2_usb_data_read()

```
IrlibError_e irv4l2_usb_data_read (
    Irv4l2ControlHandle_t * irv4l2_handle,
```

```
void * cmd_param,  
uint8_t * data,  
uint32_t len )
```

read data from the ir usb device

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>cmd_param</i>	read param
out	<i>data</i>	the pointer for data which prepare to read
in	<i>len</i>	the read length

Returns

see `IrlibError_e`

5.13.4.20 irv4l2_usb_data_write()

```
IrlibError_e irv4l2_usb_data_write (   
    Irv4l2ControlHandle_t * irv4l2_handle,   
    void * cmd_param,   
    uint8_t * data,   
    uint32_t len )
```

write data to the ir usb device

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>cmd_param</i>	write param
in	<i>data</i>	the pointer for data which prepare to write
in	<i>len</i>	the write length

Returns

see `IrlibError_e`

5.13.4.21 irv4l2_usb_device_close()

```
IrlibError_e irv4l2_usb_device_close (   
    Irv4l2ControlHandle_t * irv4l2_handle )
```

close an ir usb device

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
----	----------------------	-------------------------

Returns

see `IrlibError_e`

5.13.4.22 irv4l2_usb_device_init()

```
IrlibError_e irv4l2_usb_device_init (
    Irv4l2ControlHandle_t * irv4l2_handle,
    V4l2ConDevParams_t * param )
```

Initialize usb's configuration of the control node.

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>param</i>	the configuration of the control node

Returns

see `IrlibError_e`

5.13.4.23 irv4l2_usb_device_open()

```
IrlibError_e irv4l2_usb_device_open (
    Irv4l2ControlHandle_t * irv4l2_handle,
    char * dev_node )
```

Open an ir usb device with its device node.

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>dev_node</i>	device's node

Returns

see `IrlibError_e`

5.13.4.24 irv4l2_usb_device_release()

```
IrLibError_e irv4l2_usb_device_release (
    Irv4l2ControlHandle_t * irv4l2_handle,
    V4l2ConDevParams_t * param )
```

Release usb's configuration of the control node.

Parameters

in	<i>irv4l2_handle</i>	irv4l2's control handle
in	<i>param</i>	the configuration of the control node

Returns

see IrLibError_e

5.13.4.25 irv4l2_usb_handle_create()

```
Irv4l2ControlHandle_t* irv4l2_usb_handle_create (
    IrControlHandle_t * ir_control_handle )
```

Create an irv4l2's control handle with the usb path.

Parameters

in	<i>ir_control_handle</i>	ir camera control handle
----	--------------------------	--------------------------

Returns

Irv4l2ControlHandle_t struct pointer

5.13.4.26 irv4l2_usb_handle_delete()

```
IrLibError_e irv4l2_usb_handle_delete (
    Irv4l2ControlHandle_t * irv4l2_handle )
```

Delete and release irv4l2's control handle with the usb path.

Parameters

in	<i>handle</i>	irv4l2's control handle
----	---------------	-------------------------

Returns

see `lrlibError_e`

5.13.4.27 libv4l2_version()

```
const char* libv4l2_version (  
    void )
```

Get libirv4l2 library's version number.

Returns

version number's string

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